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EGYPTIAN CODE

FOR

DESIGN AND CONSTRUCTION OF CONCRETE STRUCTURES (ECP 203- 2007)

EGYPTIAN CODE STANDING COMMITTEE
FOR
DESIGN AND CONSTRUCTION OF CONCRETE STRUCTURES
(ECP 203- 2007)

Preface

This document is an unofficial translation of the formalized "Egyptian Code for the Design and Construction of Concrete Structures, ECP 203-2007".

The original document is written in Arabic language which is considered to be the official version of the code. Accordingly, for any differences in the contents or interpretations of any provisions of the code between the original and the translated versions, the contents of the Arabic version shall prevail and govern.

It is noted that the translation of the code was carried out by members of the Egyptian code committees.

Currently, the English translation of the code was technically reviewed by representatives of the Egyptian standing committee of the code.

Subsequently, the translated version of the code shall be presented to the standing committee of the code for an overall review and approval as the official English translation of the code

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APPENDICES:

APPENDIX I (SI) SYSTEM – METRIC SYSTEM (KG.CM) CONVERSIONS

APPENDIX II VALUES OF MECHANICAL PROPERTIES OF PRESTRESSING STEEL IN ACCORDANCE WITH INTERNATIONAL CODES

APPENDIX III NOTATION

APPENDIX IV STANDING COMMITTEE AND TECHNICAL COMMITTEES OF THE CODE

CHAPTER 1
SCOPE AND DESIGN FUNDAMENTALS

1-1 Scope

1 - This code is the formal building code for the design and construction of concrete structures in Egypt. It provides the minimum acceptable requirements for the design, construction, review and quality control for all concrete buildings. For special types of concrete structures such as bridges, tanks, bins, silos, chimneys, blast resistant structures, shell structures, as well as, structures that require special or unconventional construction techniques, the provisions of the code shall govern where applicable and after taking into consideration the more stringent requirements for the design and construction of these types of structures.

2 - The design, supervision and inspection of the construction of concrete structures shall be performed and approved by an experienced syndicated engineer.

3 - The code provides the provisions for design, construction, quality control and inspection of concrete structures, as well as the properties of concrete constituent materials.

4 - The code does not address the following types of structures:
- Light –weight concrete structures
- Ultra- high strength concrete structures

5 - Compliance with the requirements of the design and construction provisions of this code does not relieve the engineer of record of a project from any liabilities and legal responsibilities.

1-2 Objectives of the code

The objectives of this code are to present the requirements necessary to guarantee the integrity and robustness of the structures and parts thereof that can ensure safety against distress, collapse, and instability, as well as, shall provide adequate control of deformations and cracking.

1-3 Design fundamentals

Design of concrete members shall be carried out using one of the following two design methods:

1 - Limit states design method

2 - Elastic design method (Working stress design method)

The design fundamentals of the two design methods are governed by the following:

1 - The properties and strengths of constituent materials used for plain, reinforced, and prestressed concrete works and their characteristic strengths values. The properties, characteristic strengths, and quality control for these materials are given in Chapters 2 and 8 of the code, respectively.

2 - Service loads; including dead, live, moving loads, as well as, the effects of temperature, creep, shrinkage and movements of supports of the structure. Service loads shall be in accordance with the Egyptian code for loads on Structures, ECP 201. The structure shall be designed for adequate performance under the service loads and shall be proportioned for adequate strength using ultimate loads and material strength reduction factors specified in Chapter 3 of this code.

3 - The resultant internal forces and moments in the structural elements (i.e. bending moments, shearing forces, twisting moments and axial forces), that shall be determined using the theory of elastic analysis.

4 - The structure shall be designed such that robustness and integrity of the structure are guaranteed while possessing the capability of preventing the possibility of the occurrence of progressive and total collapses.

1-4 Limit states design method

Limit states design Method comprises the following limit states:

1 - Ultimate strength limit state:

The satisfaction of this limit state will provide the structure and structural members thereof with adequate strength in compliance with the safety requirements stipulated in the code.

2 - Stability limit state:

This limit state is intended to safeguard the structure against the possibility of structural instabilities resulting from sliding, overturning or floating of the structure, as well as, against bucking of elements thereof.

3 - Serviceability limit states

These limit states are intended to ensure adequate performance of the structure under service loads, as follows:

A - CRACKING LIMIT STATE : This limit state is intended to control the adverse effects of cracking of concrete.

B - DEFLECTION LIMIT STATE : This limit state is intended to control the deformation of the structural members.

CHAPTER 2
MATERIALS AND MIXTURES FOR REINFORCED AND PRESTRESSED CONCRETE

2-1 General

This chapter deals with the materials and concrete mixtures for reinforced and pre-stressed concrete with respect to properties, ingredients proportions according to exposure conditions, and required quality for both fresh and hardened concrete stages. Laboratory testing shall be performed in accordance with Appendix (3) and its modification, as well as the Egyptian Standards. In cases that require testing and specifications not specified in this Code, relevant standards shall be used with the approval of all contractual parties.

The following is a list of relevant Egyptian Standards, (ES):

	Standard No	Standard Title
Cement	ES 4756–1/ 2007	Cement– Part 1: Composition, Specifications and Conformity Criteria for Common Cements
	ES 2421–1/ 2005 ISO 9597/ 1989	Cement– Physical and Mechanical Testing– Part 1: Determination of Setting Time and Soundness
	ES 2421–2/ 2005	Cement– Physical and Mechanical Testing– Part 2: Determination of Fineness
	ES 2421–3/ 2007	Cement– Physical and Mechanical Testing– Part 3: Determination of Compressive Strength
	ES 2421–4/ 2005	Cement– Physical and Mechanical Testing– Part 4: Autoclave Expansion of Portland Cement
	ES 2421–6/ 2005	Cement– Physical and Mechanical Testing– Part 6: Heat of Hydration Solution Method
	ES 2421–7/ 2006 ISO 679/ 1989	Cement– Physical and Mechanical Testing– Part 7: Determination of Strength– Prism Method
	ES 2421–8/ 2006	Cement– Physical and Mechanical Testing– Part 8: Method of Testing Fly Ash– Determination of Free Calcium Oxide Content
	ES 2421–9/ 2005	Cement– Physical and Mechanical Testing– Part 9: Heat of Hydration– Semi-Adiabatic Method…EN 196-9/2005

	Standard No	Standard Title
Cement (cont.)	ES 5325/ 2006	Standard Methods for Chemical Analysis of Cement
	ES 583/ 2005	Sulfate Resistant Portland Cement
	ES 2149/ 2005	Moderate Heat Portland Cement
Aggregate	ES 1109/ 2002	Concrete Aggregates from Natural Sources
Admixtures	ES 1899–1/ 2006	Admixtures for Concrete, Mortar and Grout– Part 1: Concrete Admixtures – Definitions, Requirements, Conformity, Marking and Labeling
	ES 1899–2/ 2006	Admixtures for Concrete, Mortar and Grout– Part 2: Reference Concrete and Reference Mortar for Testing EN480-1/1997
	ES 1899–3/ 2006	Admixtures for Concrete, Mortar and Grout– Part 3: Reference Masonry Mortar for Testing Mortar Admixtures
Steel	ES 262/ 2000	Steel for the Reinforcement of Concrete
	ES 76/ 2001	Metallic Materials– Tensile Testing
	ISO 6935–3/ 1992	Steel for the Reinforcement of Concrete– Part 3: Welded Fabric
Concrete	ES 1658–1/ 2006 ISO 1920–1/ 2004	Testing of Concrete– Part 1: Sampling of Fresh Concrete
	ES 1658–2/ 2006 ISO 1920–2/ 2005	Testing of Concrete– Part 2: Properties of Fresh Concrete
	ES 1658–4/ 2006 ISO 1920–3/ 2004	Testing of Concrete– Part 4: Making and Curing Test Specimens
	ES 1658–9/ 2006 ISO 1920–5/ 2004	Testing of Concrete– Part 9: Properties of Hardened Concrete other than Strength

2-2 Properties of materials

2-2-1 Cement

1 - Cement used shall be Portland Cement CEM I (ES 4756-1/2007) or sulfate resisting Portland cement (ES 583/2005) or moderate heat Portland cement (ES2149/2005).

2 - Portland cement containing limestone powder (CEM II/A-LL, CEM II/A-L, CEM II/B-LL, CEM II/B/L) or Portland cement containing by-pass dust shall not be used in concrete.

3 - In case of using cement types other than those mentioned in item (1), previous successful experience shall be required, and it shall comply with the relevant ES and the requirements stated in this Code.

4 - Chloride content in cement shall not exceed 0.06% by weight of cement.

5 - On using different types of Pozzolanic cement – as a precaution to limit alkali aggregate silica reaction or in high sulfate environments – the chemical composition of the pozzolanic portion of these cements shall comply with ES requirements (ES 4765-1/2007), as well as it shall be in a glassy form to assure its reactivity with cement.

6 - In case of using active silica aggregate, the cement alkali content, expressed as equivalent Sodium Oxide, shall not exceed 0.6% by weight of cement.

2-2-2 Aggregates

2-2-2-1 General

River beds, desert and sea beaches are the most common sources for natural aggregates. It should be noted that aggregates from sea beaches shall only be used after passing the salt contamination test or after controlling its salt contamination. Crushed stones and rocks are other major sources for natural aggregates with variable properties depending on their geological origin and properties of parent stone or rock.

2-2-2-2 Aggregate requirements

1 - Aggregate shall comply with the Egyptian Standard ES1109/2002 and the additional requirements mentioned herein in tables (2-1) and (2-2) of this code.

2 - Aggregate particles shall be hard and free from any deleterious materials. Also, aggregate particles shall not contain any materials harmful to concrete and steel reinforcement such as iron pyrite and

coal, and shall not contain any organic impurities that can interfere with the setting and hardening processes of concrete, or adversely affects concrete strength, concrete durability, and steel reinforcement. Previous data and test results for aggregate may be used, and relevant complementary tests for the type of aggregate used shall be conducted in accordance with the Egyptian Standards, ES.

3 - Carbonate aggregates shall be free from siliceous or active carbonate components that have the ability for alkali aggregate reaction causing expansion and cracking. Quarries shall conduct X-ray diffraction and petrographic analysis together with testing given in Section (2-3-4-8).

4 - Artificial or recycled aggregates may be used in concrete as long as it complies with Egyptian Standards and project specifications. The approval of the consultant shall be required prior to usage.

5 - The fineness modulus of fine aggregate shall not be less than 2.6 when used in pre-stressed concrete.

6 - In case of unavailability of aggregate grading which complies with the Egyptian Standards, suitable grading curves, based on previous laboratory and site data may be used after carrying out trial mixture designs and strength assurance mixtures and after receiving the approval of the engineer of record of the project.

6 - The nominal maximum size shall not be more than one fifth the minimum shuttering dimension, one third slab thickness and three quarters the clear distance between reinforcing bars.

7 - The nominal maximum size shall not be more than 40mm for reinforced concrete, and 25mm for pre-stressed concrete applications.

Table (2-1) Allowable limits for some physical and mechanical properties of aggregates

Property*	Maximum Allowable Limit	
	Coarse Aggregate	Fine Aggregate
1- Weight % of fine materials, passing 75μm sieve (sieve #200)	Gravel and crushed gravel 1% Crushed stone 3%**	Natural sand 3% Fine sand from crushed stone 5%**
2- Weight % for clay and friable materials	Gravel and crushed gravel 1% Crushed stone 3%	3%
3- Los Angeles hardness value (passing % from 1.17mm sieve after 500 revolutions)	Gravel and crushed gravel 20% Crushed stone 30%	———
4- Flakiness Index	25%***	———
5- Elongation Index	25%***	———
6- Natural absorption % (24 hours)****	Gravel and crushed gravel 1% Crushed stone 2.5%	2%
7- Crushing value	Concrete surface exposed to abrasion 25% Concrete surface un-exposed to abrasion 30%	———
8- Impact value	Concrete surface exposed to abrasion 30% Concrete surface un-exposed to abrasion 45%	———

* Properties according to Egyptian Standard Specification, testing procedure appendix, and this code.

** Shall be free from clay, silt and friable materials

*** In case flakiness index and elongation index are high this shall be considered in mix design

**** In case absorption % is more than 2.5% this shall be taken into consideration in the mix design

Table (2-2) Allowable limits for chloride and sulfate contents and soundness of aggregates

Property*	Maximum Allowable Limit by Weight % of Aggregate	
	Coarse Aggregate	Fine Aggregate
1- Water soluble chloride ion content (Cl^-)**	0.04%	0.06%
2- Total sulfate content as SO_3	0.4%	0.4%
3- Soundness (expressed as % loss in weight)		
a- Exposure to 5 cycles in Na_2SO_4	12	10
b- Exposure to 5 cycles in $MgSO_4$	18	15

* Properties according to Egyptian Standard Specification and/or testing procedure appendix.

** For pre-stressed concrete, water soluble chlorides shall not be more than 0.01% by weight of all-in aggregate (i.e. combined aggregate)

2-2-3 Mixing and curing water

1 - Water used in mixing shall be clean and free from deleterious materials such as oil, acids, salts, organic materials, silt and clay and any materials which have detrimental effects on both the concrete and reinforcing steel. The salt content in mixing water shall not exceed the values given in item (2).

2 - The maximum allowable salt and harmful materials contents are as follows:

Total dissolved salts	= 2.00 gm/lit
Chloride salts as (Cl^-)	= 0.50 gm/lit
Sulfate salts as (SO_3)	= 0.30 gm/lit
Carbonate and bicarbonate salts	= 1.00 gm/lit
Sodium sulfide salts	= 0.10 gm/lit
Organic materials	= 0.20 gm/lit
Inorganic materials; clay and suspended materials	= 2.00 gm/lit

3 - The pH value of mixing water shall not be less than 7.0. In case of using water other than drinking water, tests shall be carried out to know the actual value before using the water.

4 - Drinking water – excluding bacteriological requirements- is accepted for mixing and curing concrete. Water from other sources may be used for mixing and curing concrete as long as it conforms to the previous requirements in addition to the following requirements:

   a - Initial setting time for cement using the water shall not be more than initial setting time of cement using drinking water by more than 30 minutes, and shall not be less than 45 minutes.

   b - Standard compressive strength, at 7 and 28 days of age, of standard cement mortar specimens using the used water shall not be less than 90% of the compressive strength of cement mortar using drinking water at the same age.

5 - Sea water shall not be used in mixing any type of reinforced concrete.

6 - In case of necessity, sea water may be used in plain concrete which does not contain any reinforcement. The concrete mixture shall be designed using the same water content, and the cement content shall be determined to achieve the required strength. This concrete shall not be in direct contact with reinforced concrete unless suitable insulating material is applied in between. Also, previous experience in using sea water successfully shall be required.

7 - Water suitable for mixing concrete is also suitable for curing concrete.

8 - Used water shall not cause any efflorescence or salt sedimentation or any unacceptable appearance of concrete surface.

2-2-4 Admixtures

Admixtures are used in concrete mixtures in predetermined dosages to improve certain concrete properties or to develop new properties. This is achieved either by their physical or chemical effect. The used admixture shall not affect the concrete volume except air-entraining and mineral admixtures. Also, admixtures shall not have an adverse effect on concrete durability.

Most common admixtures used in concrete mixtures could be classified as follows (table 2-3):

- Chemical admixtures which include, setting time accelerators, and retarding admixture, and normal range and high range water reducers. These admixtures could also be manufactured to have more than one effect such as retarding and normal range water reducer, retarding and high range water reducers, and accelerating and water reducers.

- Air-entraining admixtures.
- Pozzolanic admixtures such as high blast furnace slag, fly ash, silica fume, natural pozzolanic ash. All of these admixtures have pozzolanic action where they react with cement hydration products.
- Other admixtures such as corrosion inhibitor admixtures and coloring admixtures.

The following requirements shall be considered on using admixtures:

1 - Admixtures shall comply with Egyptian Standards, (ES) for each admixture type by testing in accredited laboratory.

2 - Admixtures which do not follow an Egyptian or International Standards may be used based on previous data, experience and test results in accredited laboratories, and shall fulfill project specifications.

3 - Manufacturer shall provide recommendations on the procedure of admixture usage and admixture addition to the mixture, as well as the possibility of splitting the admixture dosage either during mixing or before casting according to temperature, haul distance and working conditions.

4 - Admixtures shall have no adverse effects on concrete and reinforcing steel, especially durability.

5 - Admixtures used in reinforced concrete, pre-stressed concrete and concrete containing any embedded metals shall have no chloride content.

6 - Admixtures shall be used in site trial mixtures to check the performance of the fresh and hardened concrete using the mixture constituents, and to avoid any undesirable effects such as prolonged retardation.

7 - Periodical compatibility and performance checks shall be carried out using the admixture and the available concrete constituents and shall be compared with control mixtures with no admixtures.

8 - The compressive, tensile and bond to reinforcement strengths for concrete mixtures utilizing admixtures shall not be lower than control mixtures without admixtures. In special circumstances; where certain properties are required, a reduction not more than 10% in the concrete strengths will be allowed and with the approval of the designer.

9 - Any admixture consignment shall be accepted by conducting uniformity tests stated in the Egyptian Standards and shall meet those for the accepted sample.

10 - Concrete mixtures with admixtures shall have air content not more than 3%, but not more than 2% above that of the control mixture without admixtures. Concrete mixtures utilizing air-entraining admixtures are excluded.

11 - It is preferable to use one type of admixture in the mix. If situation requires the use of more than one admixture in the same mixture, it is important to have full data about their compatibility which shall be checked by accredited laboratory testing, as well as the approval of the engineer of record of the project.

12 - On using more than one admixture in the concrete mixture, they shall not be mixed together and shall be preferably added to the mixture separately during mixing.

13 - The temperature of fresh concrete containing the admixture shall not be more than $5^{o}C$ above that of the control mixture without the admixture.

14 - The chemical stability of natural or artificial pozzolanic admixtures shall be ascertained before using in concrete mixtures.

15 - Cement manufacturers producing cement types containing any form of admixtures shall announce this information clearly on the cement bag. These cements shall be tested similar to testing concrete mixtures with admixtures.

16 - Climate variability, especially temperature, shall be taken into consideration with all the previous requirements.

Table (2-3) ES 1899-1, 2, 3/2006 requirements for concrete admixtures

1- Performance criteria for concrete mixtures with admixtures

Property	Admixture type						
	Type (A) NRWR	**Type (B) Accelerators**	**Type (C) Retarding**	**Type (D) NRWR + Retarding**	**Type (E) NRWR + Accelerating**	**Type (F) HRWR**	**Type (G) HRWR + Retarding**
a- Fresh concrete							
- Max. water content as % of control mix	95%	---	---	95%	95%	88%	88%
- Increase in air content	≤ 2%	≤ 2%	≤ 2%	≤ 2%	≤ 2%	≤ 2%	≤ 2%
- Total air content	≤ 3%	≤ 3%	≤ 3%	≤ 3%	≤ 3%	≤ 3%	≤ 3%
- Initial set (penetration at 0.5N/mm^2)	Within 1 hour from control mix	More than 1 hour from control mix	At least 1 hour less than control mix	At least 1 hour more than control mix	At least 1 hour less than control mix	Within 1 hour from control mix	At least 1 hour more than control mix
- Final set (penetration at 3.5 N/mm^2)	Within 1 hour from control mix	More than 1 hour from control mix	At least 1 hour less than control mix	-------	At least 1 hour less than control mix	Within 1 hour from control mix	At least 1 hour more than control mix
b- Hardened concrete							
- Min. compressive strength as % of control mix:							
1 day	---	---	125	---	125	140	125
3 days	110	90	125	110	125	125	125
7 days	110	90	100	110	110	115	115
28 days	110	90	100	110	110	110	110
6 months	100	90	90	100	100	100	100
- Min. flexural strength as % of control mix at 28 days	100	90	90	100	100	100	100

Table (2-3) ES 1899-1, 2, 3/2006 requirements for concrete admixtures (cont'd)

2 - Uniformity criteria for performance between tested sample and the sample taken from the consignment and the values stated by the manufacturer

Property	Requirements
- Solid content	- Difference shall not be more than 5% by weight for liquid and solid admixtures
- Ash content	- Difference shall not be more than 1% by weight
- Relative density	- Difference shall not be more than 0.02 for liquid admixtures
- pH value	- Comparison between the two numbers shall be made
- Chloride ion content	- Difference shall not be more than 5% or 0.2% by weight of the admixture whichever is lerger
- Infra-red spectrometer	- Shall be identical to manufacturer data

2-2-5 Steel reinforcement

2-2-5-1 Reinforcing steel types

1 - Concrete is reinforced using steel reinforcement which complies with the Egyptian Standards (ES 262-2000). In case of using welded steel mesh it shall comply with ISO 6935-3/1992

2 - Common types of steel reinforcement are:

a - Mild steel grade 240/350 or 280/450 and it is denoted (ϕ)

b - High tensile steel and it has two grades:
Grade 360/520 and is denoted (ϕ)
Grade 400/600 and is denoted (Φ)
High tensile steel is cold formed or hot drawn steel. High tensile steel produced from mild steel by cold forming shall not be plain bars and shall have ribs which comply with the Egyptian Standards requirements (ES 262/2000), to produce the necessary bond with concrete.

c - Welded steel mesh from plain or deformed or indented bars with mild steel grades (240/350) or (280/450) cold formed to produce steel grade (450/520) denoted as (#). The steel mesh shall be arc welded.

2 - Egyptian Standards shall be used for bar marking and identification.

2-2-5-2 Nominal bar diameters

Nominal bar diameter shall be determined from weight per unit length for reinforcing bars with continuous ribs. The smaller diameter shall be considered in case of reinforcing bars where crossed ribs are used. A maximum of 5% is allowed as tolerance between the nominal unit weight and the actual unit weight.

2-2-5-3 Mechanical properties for reinforcing steel to be used in design

1 - Yield Stress: is the stress at yield plateau for mild steel and high tensile steel which shows a yield phenomenon.

2 - Proof Stress: is the stress that causes a permanent strain value of 0.2% on removing the stress and it is used for high tensile steel which does not show a yield phenomenon.

3 - Ultimate Tensile Strength: is the stress produced in the steel bar by dividing the maximum tensile load by the bar cross sectional area.

4 - Modulus of Elasticity: is the slope of the linear portion of the stress-strain relationship in the elastic region.

5 - Elongation Percent at Failure: is the percentage of elongation at failure load with respect to the gauge length.

The mechanical properties shall be determined according to ES 262/2000. The minimum mechanical properties for reinforcing steel, confirmed by manufacturer's certificate and verified by accredited laboratory testing, shall not be lower than the values given in table (2-4).

2-2-5-4 Steel stress-strain curve

Stress-strain curve obtained from test shall be used. Idealized stress-strain curve given in figure (4-1) can be used by designers as a guide.

2-2-5-5 Steel characteristic strength

The minimum values of the mechanical properties shall not be lower than the values given in table (2-4)

2-2-5-6 Welding of steel bars

Welding of reinforcing steel bars shall comply with specifications set by project consultant and taking into consideration the requirements mentioned in Section (4-2-5-4-3).

2-2-6 Steel reinforcement for pre-stressed concrete

Section (10-2-2) gives all the types and properties for steel reinforcement used in pre-stressed concrete.

Table (2-4) Minimum mechanical properties for different types of steel reinforcement

Steel Type	Grade	Bar Type	Yield Stress or 0.2% Proof Stress (N/mm^2)	Tensile Strength (N/mm^2)	Elongation % (L=10D)*	Cold Bend Test	
						Bar Diameter (mm)	Bending Radius
Mild steel	240/350	Plain bars	240	350	20	D≤25 D>25	2D 3D
	280/450		280	450	18	D≤25 D>25	2D 3D
High tensile steel	360/520	Deformed bars	360	452	12	D≤20 20<D≤36	4D 5D
	400/600		400	600	10	D≤25 20< D ≤25 25<D ≤36	4D 5D 6D
Cold formed welded steel mesh**	450/520	Plain or deformed or indented bars	450	520	8	—	—

* L = gauge length (mm), D = test specimen diameter (mm)
** Not allowed structurally to use steel mesh with bar diameter less than 5mm

2-3 Concrete Properties

2-3-1 Fresh concrete properties

2-3-1-1 Bulk density of concrete

In case no accurate data is available, guide values for the fresh concrete bulk density are as follows:

- 22kN/m^3 for plain concrete using calcareous aggregate.
- 23 kN/m^3 for plain concrete using siliceous aggregate.
- 25 kN/m^3 for normal reinforced concrete, it may be increased for heavily reinforced concrete taking into consideration aggregate type.

2-3-1-2 Concrete consistency

Fresh concrete consistency and workability greatly affect its compactability which in turn influences its homogeneity, and reduces air content and tendency for honeycombing. Slump test is the most common test used on site for determining concrete consistency.

Table (2-5) gives slump values to be used as a guide in determining the suitable slump value for different structural elements. Project specifications shall be referred to in case other test is used to measure the concrete consistency.

Table (2-5) Guide values for concrete slump

Element	Slump (mm)*	Compacting Method
Mass concrete	25-50	Mechanical
Foundation, lightly-reinforced concrete sections (steel reinforcement < 80kg/m^3)***	50-75	Mechanical
Medium to highly reinforced concrete sections (steel reinforcement 80-150 kg/m^3)***	75-125	Mechanical or Manual
Highly reinforced concrete sections (steel reinforcement > 150 kg/m^3)***	125-150**	Light Compaction
Deep foundation and pump concrete	125-200**	Light Compaction

* Slump decreases gradually with time after mixing, time of test after mixing and temperature are among the main factors affecting slump loss; the values indicated in the table are required immediately before casting.
** Slump value is achieved using chemical admixtures
*** Guide values

2-3-1-3 Temperature of fresh concrete

Temperature of fresh concrete shall not exceed 35°C for concrete mixture with or without admixtures. Necessary precautions shall be considered to avoid the increase of the concrete temperature over the required value.

2-3-2 Mechanical properties of hardened concrete

2-3-2-1 Compressive strength

Characteristic Strength (fcu): is the compressive strength at 28 days of age, below which not more than 5% of site test results shall fall below it. It is also known as concrete grade.

Standard cube (150x150x150 mm) is used to determine the compressive strength. Concrete grade for plain concrete shall not be lower than 15 N/mm^2, and for pre-stressed concrete not less than 30 N/mm^2. Table (2-6) gives different concrete grades for reinforced and pre-stressed concrete.

Table (2-6) Concrete grade for reinforced and pre-stressed concrete

Reinforced Concrete (N/mm^2)	20	25	30	35	40	45	50	55	60
Pre-stressed Concrete (N/mm^2)			30	35	40	45	50	55	60

In case of using standard concrete cylinders (150x300 mm) or any specimens with different dimensions, guide correction factors given in table (2-7) may be used to obtain the equivalent standard cube compressive strength.

To have a relation between compressive strength at ages less than 28 days and the characteristic strength, the contractor shall provide sufficient number of cube specimens before the commencement of project to obtain the relation between the characteristic strength and early compressive strength at 3 days or 7 days.

Table (2-7) Guide correction factor to obtain equivalent cube compressive strength for substandard specimens

Mold Shape	Mold Dimensions (mm)	Correction Factor
Cube	100x100x100	0.97
	150x150x150	1.00
	200x200x200	1.05
	300x300x300	1.12
Cylinder	100x200	1.20
	150x300	1.25
	250x500	1.30

* The guide values are for concrete grade less than 40 N/mm^2

* Concrete grade greatly influences the correction factor on changing mould shape and dimensions, thus laboratory tests are required to determine the exact value for the correction factor

2-3-2-2 Axial direct tensile strength

Axial direct tensile strength value may be considered as one of the following two values determined experimentally:

- 0.85 from indirect splitting tensile strength.
- 0.60 from tensile strength by pure bending.

2-3-2-3 Bond strength with reinforcing steel

Bond strength between reinforcing steel and concrete increases with the presence of bar ribs and indentations. Also, it increases by increasing concrete density, cement content, reducing water content, surface texture of reinforcing bars, and the cleanliness of their surface from any paints, oil deposits, bitumen or any other materials that can adversely affect the bond strength with concrete. Section (4-2-5) gives guide values for bond strength. In case of using corrosion protective coating for steel reinforcement, the bond strength shall not be lower than 90% of the bond strength between the concrete and the same reinforcing steel without the protective coating, and shall conform to the design requirements as well as Egyptian standards for the use and application of protective coatings.

2-3-3 Dimensional changes of concrete

2-3-3-1 Modulus of elasticity

Modulus of elasticity shall be determined from equation (2-1):

$$Ec = 4400 \times \sqrt{fcu} \ \text{N/mm}^2 \qquad (2\text{-}1)$$

Where;

Ec = modulus of elasticity (N/mm^2)

fcu =characteristic concrete strength as given in Section 2-3-2-1 (N/mm^2)

2-3-3-2 Transverse deformation (poisson's ratio)

It is the ratio between transverse strain and longitudinal strain for standard specimen. In elastic deformation the ratio (υ) shall be taken as follows:

$\nu = 0.20$ for un-cracked concrete (2-2-a)

$\nu = 0.00$ for cracked concrete (2-2-b)

2-3-3-3 Coefficient of thermal expansion

Coefficient of thermal expansion of plain concrete depends on mixture composition and aggregate type as follows:

- Concrete using siliceous aggregate varies from 1.20 to 1.30 x 10^{-5}

- Concrete using limestone aggregate varies from 0.60 to 0.90 x 10^{-5}
- Concrete using sandstone aggregate varies from 0.90 to 1.20 x 10^{-5}
- Concrete using granite aggregate varies from 0.70 to 0.95 x 10^{-5}
- Concrete using basalt aggregate varies from 0.80 to 0.95 x 10^{-5}

2-3-3-4 drying shrinkage

It is the shrinkage caused by drying of concrete after hardening. It depends on many factors such as ambient relative humidity, time, volume and surface area of concrete element (i.e. nominal dimension B). The nominal dimension is calculated as follows:

$$B = \frac{2\,Ac}{Pc} \qquad (2\text{-}3)$$

Where;

B = Nominal dimension of section (mm)
Ac = Cross sectional area of the concrete element (mm2)
Pc = Perimeter of the concrete section subjected to drying (mm)

Also, drying shrinkage depends on air temperature, w/c ratio, aggregate properties, cement content, and ratio between aggregate and cement mortar contents. Table (2-8-a) gives guide values for drying shrinkage strain.

Table (2-8-a) Guide values for final drying shrinkage strain (x10^{-3})

Air Condition	Dry Air (relative humidity ≈ 55%)*			Humid Air (relative humidity ≈ 75%)*		
Age after which shrinkage starts (days)	Nominal Dimension B (mm)			Nominal Dimension B (mm)		
	B ≥600	200<B<600	B≤200	B ≥600	200<B<600	B≤200
3-7	0.31	0.38	0.43	0.21	0.23	0.26
7-60	0.30	0.31	0.32	0.21	0.22	0.23
> 60	0.28	0.25	0.19	0.20	0.19	0.16

* It is preferred to use the table for relative humidity ranging from 40 to 85%. In case the relative humidity differs from that in table, shrinkage strain values could be deduced proportionally.

2-3-3-5 Creep

It is the inelastic strain that occurs under the effect of all or part of the working loads and depends on time. It depends on many factors such as: the ratio between applied stress to concrete strength, w/c ratio, and concrete age at the start of loading, cross section properties, surrounding relative humidity value, and ratio between aggregate and cement mortar contents. The total strain value caused by creep and elastic instantaneous strain are as follows:

$$\varepsilon t = \varepsilon o\,(1 + \phi) \qquad (2\text{-}4\text{-a})$$

$$\varepsilon t = \frac{fo}{Ect}\,(1 + \phi) \qquad (2\text{-}4\text{-b})$$

Where;

εt = Total strain at time t = ∞

εo = Instantaneous strain caused by load = $\frac{fo}{Ect}$

ϕ εo = Creep strain

ϕ = Creep coefficient

fo = Concrete stress at the start of loading

Ect = Concrete modulus of elasticity at the start of loading

Table (2-8-b) gives guide values for the creep coefficient (ϕ) with respect to the relative humidity, age at the start of loading, and the nominal dimension of the concrete section (B) (previously described in Article 2-3-3-4).

Table (2-8-b) Guide values for final creep coefficient (ϕ)

Air Condition	Dry Air (relative humidity ≈ 55%)*			Humid Air (relative humidity ≈ 75%)*		
Age after which loading starts (days)	Nominal Dimension B (mm)			Nominal Dimension B (mm)		
	B ≥600	200<B<600	B≤200	B ≥600	200<B<600	B≤200
3-7	2.90	3.20	3.80	2.10	2.40	2.70
7-60	2.50	2.80	3.00	1.90	2.00	2.20
> 60	2.00	1.90	1.70	1.70	1.60	1.40

* It is preferred to use the table for relative humidity ranging from 40 to 85%. In case the relative humidity differs from that in table, creep coefficient values could be deduced proportionally.

2-3-4 Durability of concrete

2-3-4-1 General

Concrete structures are affected by chemicals such as oil, fats, sugar solutions, also some organic materials, acids, sulfate and chloride solutions, sea water, and underground water, as well as solutions and vapor in coastal and industrial areas. Concrete properties change due to the exposure to such materials. Also, concrete structures are adversely affected by alkali aggregate reaction, in addition to some mechanical processes such as abrasion and erosion.

Concrete durability for some structures or parts of structures has the priority before concrete mechanical properties. In these structures several factors shall be taken into consideration, which are:

- Mixture ingredients
- Cement type and content
- Aggregate type
- Exposure conditions (i.e. type of aggressive material)
- Shape and size of concrete element
- Concrete permeability to water and liquids
- Concrete permeability to gases
- Harmful material in concrete ingredients
- Concrete construction starting from mixing up to the use of the structure (i.e. a major factor that improves concrete durability is quality control during construction especially during casting, compaction and curing to achieve dense, homogeneous concrete with low permeability)

Concrete durability could be improved by considering the following:

2-3-4-2 Maximum water/cement (w/c) ratio

Table (2-9) may be used to determine the maximum w/c ratio for concrete mixtures using Portland cement and according to exposure conditions.

Table (2-9) Maximum w/c ratio, minimum cement content, and minimum characteristic compressive strength for concrete mixtures exposed to aggressive environments

Exposure Condition	Minimum Cement Content (kg/m^3)* Aggregate Nominal Maximum Size (mm)***			Maximum w/c Ratio**	Minimum Characteristic Compressive Strength (N/mm^2)
	32	20	10		
Concrete is totally isolated from the aggressive surrounding environment	350	350	350	0.50	25
Concrete is exposed to aggressive environment but continuously submerged in water	350	350	400	0.45	30
Concrete is exposed to aggressive environment or sea water, or wetting and drying cycles, or gases, etc.****	350	400	450	0.40	40

* Values in table are for reinforced concrete and pre-stressed concrete, cement contents may be reduced by $50kg/m^3$ for plain unreinforced concrete.

** Normal range and high range water reducers may be used to reduce the w/c ratio and to obtain the desired consistency.

*** If the aggregate nominal maximum size lies between two values in the table the cement content for the smaller nominal size shall be considered.

**** Special precautions shall be considered to avoid shrinkage and thermal stresses cracking.

2-3-4-3 Minimum and maximum cement content

Table (2-9) may be used to determine the minimum cement content for concrete mixtures using Portland cement according to exposure conditions. Generally, cement content in concrete mixtures shall not exceed 450 kg/m^3. In case of using cement content more than 450 kg/m^3, special considerations shall be taken into the design to avoid shrinkage or thermal stresses cracking.

2-3-4-4 Maximum salt and deleterious materials contents in mixing water

In mixing water the salt content and deleterious materials content shall not exceed the values given in Section (2-2-3).

2-3-4-5 Maximum chloride ion content in concrete

To protect the reinforcing steel from corrosion, the chloride ion content at age 28 days for the concrete mixture shall not exceed the values given in table (2-10). The sources of chloride ions are mixing water, aggregate, cement and admixtures and not the surrounding environment.

Table (2-10) Maximum chloride ion content in concrete mixtures to protect reinforcing steel from corrosion

Concrete Type	Exposure Condition	Maximum Chloride Ion Content in Concrete as Percentage of Cement Weight	
		Water Soluble	Acid Soluble
Reinforced Concrete	Exposed to Chlorides	0.15	0.20
	Not Exposed to Chlorides	0.30	0.40
Pre-stressed Concrete	All Conditions	0.06	0.10

2-3-4-6 Maximum sulfate content in concrete

Total sulfate content at age 28 days for the concrete mixture, determined as SO_3, shall not exceed 4% of cement weight in the mixture. The sources of sulfates are mixing water, aggregate, cement and admixtures and not the surrounding environment.

2-3-4-7 Determination of chloride and sulfate contents in concrete

The chloride and sulfate contents for the concrete mixture shall be determined using the procedure outlined in the testing manual (Appendix 3). Three standard cubes shall be prepared during concrete casting, and shall be kept after de-molding away from water or any salt contamination. The chloride and sulfate contents shall be determined at 28 days of age as percentage of cement weight in the mixture.

2-3-4-8 Alkali aggregate reaction

2-3-4-8-1 Alkali-silica reaction

Some aggregate contains various forms of active silica such as; Opal and Cristobalite that can chemically react with the alkalis (such as Sodium Oxide Na_2O, and Potassium Oxide K_2O) found in the cement and other mixture ingredients. The reaction results in a gel-like production (i.e. alkali-silica gel) around aggregate particles that swell upon water

absorption causing internal stresses which result in cracking and deterioration of concrete. Table (2-11) gives the sequential testing procedure that shall be conducted on the aggregate by the quarry before using.

The following recommendations shall be adopted to mitigate the effect of alkali-silica reaction in concrete:

- Equivalent Sodium Oxide content of the Portland cement shall not exceed 0.6% by weight.
- Total alkalis in the concrete mixture expressed as equivalent sodium oxide shall not exceed 3kg/m^3.
- Using supplementary cementing materials (SCM) such as; granulated ground blast furnace slag, silica fume, fly ash, to replace portion of the cement content after the approval of the consultant on the type and replacement level (the effectiveness of the used materials in reducing ASR effect shall be proven by laboratory tests).
- Take necessary precautions to reduce the water permeability of the concrete by using suitable isolation or water proofing membranes.
- The nominal maximum size of the used aggregate shall not exceed 25mm

In case of absence of conformity certificate from quarry, suitability of aggregate for use in reinforced and pre-stressed concrete shall be confirmed.

Table (2-11) Test procedure to detect aggregate alkali-silica reaction

Test	Procedure	Analysis & Results	Precaution
1- Accelerated Mortar Bar test (Test 2-27)* (ASTM C1260-01)	Expansion of mortar bar is measured after 14 days	Aggregate could be used if expansion is less than 0.1%	- If expansion is between 0.1% and 0.2%, test (2) shall be carried out - Aggregate shall be rejected if expansion is greater than 0.2%
2- Concrete Prism Test for Alkali Reactivity (ASTM C1293-01)	Expansion is measured after 1 year	Aggregate could be used if expansion is less than 0.04%	Aggregate shall be rejected if expansion is greater than 0.04%

* The test number refers to the test number in the testing manual (Appendix 3)

2-3-4-8-2 Alkali-carbonate reaction

Some carbonate aggregates react with cement alkalis producing compounds that could cause with time expansion and cracking of concrete resulting in reducing its durability. When Petrographic and X-ray diffraction test results show the likelihood of this phenomenon, then the testing outlined in table (2-12) shall be followed. The testing shall be conducted by the quarry. The cement shall have alkali content, expressed as equivalent sodium oxide, not more than 0.4% by cement weight. Since several factors affect this phenomenon such as aggregate mineral composition, aggregate texture, calcite to dolomite ratio, clay minerals, etc., specialists shall be consulted on using carbonate aggregates in concrete to determine the limits of such factors in affecting the phenomenon.

In case of absence of conformity certificate from quarry, suitability of aggregate for use in reinforced and pre-stressed concrete shall be confirmed.

Table (2-12) Test procedure to determine aggregate alkali-carbonate reaction

Test	Procedure	Analysis & Results	Precaution
1- Potential Alkali Reactivity of Carbonate Rocks – Rock Cylinder Method (Test 2-26)* (ASTM C586-99)	Expansion is measured after 1 year	Aggregate could be used if expansion is less than 0.1%	If expansion is greater than 0.1% Test (2) shall be carried out
2- Alkali Reactivity Using Concrete Prism (ASTM C1105-95)	Expansion is measured after 1 year	Aggregate could be used if expansion is less than: 0.015% after 3 months 0.025% after 6 months 0.03% after 1 year	Aggregate is rejected if expansion is greater than 0.04%

* The test number refers to the test number in the testing manual (Appendix 3)

2-3-4-9 Concrete exposed to acidic medium

Care in selecting concrete ingredients and its production shall be practiced if the concrete is being exposed to acidic medium (pH value < 7.0). This shall include increasing cement content, reducing w/c ratio, reducing sand content, full compaction, increasing concrete cover, and using protective coatings or membranes against acid attack. These

precautions may be used for ordinary Portland cement and sulfate resisting Portland cement. When the pH value of the acidic medium is equal to or less than 5.50, the use of high slag cement could improve the resistance to the acidic medium, and also the use of protective coatings and membranes is essential.

2-3-4-10 Concrete exposed to sulfates

When concrete is exposed to sulfate salts found in the soil or underground water (as magnesium, or sodium, or potassium, or calcium sulfates), the cement type and content shall be considered carefully, as well as aggregate type, nominal maximum size, w/c ratio, and minimum characteristic strength. Values given in table (2-13) could be used to determine these factors.

On using the values in table (2-13), consider the following:

- Values are applicable for concrete using natural aggregate.
- Values are applicable for concrete subjected to underground water with pH value ranging from 6 to 9.
- For small concrete sections or when concrete is subjected to water pressure from one side or partially submerged it is necessary to reduce w/c and/or increase cement content values given in table (2-13) to ensure the minimum concrete permeability.

Table (2-13) Requirements for concrete subjected to sulfates*

Sulfate Concentration as SO_3			Cement Type	Minimum Cement Content			Maximum w/c Ratio**	Minimum Characteristic Strength (N/mm^2)
In Soil		Ground Water		Aggregate Nominal Maximum Size (mm)				
Total SO_3 %	SO_3 (gm/lit) in a mixture of water and soil (2:1)	ppm		32	20	10		
< 0.2	< 1.0	< 300	CEM I	350	400	400	0.52	—
0.2 to 0.35	1.0 to 1.50	300 to 700	CEM I or Moderate Heat	350	400	400	0.50	25
0.35 to 0.50	1.5 to 1.9	700 to 1200	Sulfate Resisting or Moderate Heat	350	400	400	0.45	30
0.50 to 1.0	1.9 to 3.1	1200 to 2500	Sulfate Resisting	400	450	450	0.43	35
1.0 to 2.0	3.1 to 5.6	2500 to 5000	Sulfate Resisting +Protective coating	400	450	450	0.40	40

* Refer to article (2-3-4-11) for dual action of chloride and sulfate salts.
** Dry aggregate.
*** If the aggregate nominal maximum size lies between two values in the table the cement content for the smaller nominal size shall be considered.

2-3-4-11 Concrete exposed to dual action of chlorides and sulfates

Occasionally, concrete is subjected to high concentration of sulfate and chloride salts such as sea water or underground water. Concrete durability is greatly affected by these exposure conditions besides reinforcing steel corrosion. The exposure condition could be by total submersion or exposure to cycles of wetting and drying.

The following precautions shall be considered:

- The used aggregate shall be inert and non-reactive with cement alkalis.
- Using cement with C3A content ranging from 5% to 8%. Portland cement CEM I, or moderate heat Portland cement, or high slag cements, or pozzolanic cement could be used.
- The water soluble chloride in concrete shall not exceed 0.1% of cement weight.
- For sea-water structures exposed to freezing and thawing, air-entraining admixture shall be used.
- The concrete cover shall not be smaller than 50mm for structures submerged in water or exposed to air.
- The concrete cover shall not be smaller than 70mm for structures exposed to wetting and drying cycles.
- Using dense concrete. Table (2-9) shall be used to determine minimum cement content, maximum w/c ratio and minimum characteristic strength while achieving optimum compaction.

2-3-4-12 Freezing and thawing

The use of air-entraining admixture shall improve the concrete durability when exposed to freezing and thawing. The air content shall be determined and accordingly the mixture composition shall be adjusted by the consultant, and by using guide technical data and laboratory results. The following are guide values for average air content for fresh concrete at the time of casting:

- 7% on using aggregate with nominal maximum size of 10mm.
- 6% on using aggregate with nominal maximum size of 15mm.
- 5% on using aggregate with nominal maximum size of 20mm.
- 4% on using aggregate with nominal maximum size of 40mm.

2-3-4-13 protecting reinforcing steel

Protecting reinforcing steel is achieved by the alkaline environment around bars and by sufficient concrete cover. In addition, all the precautions and requirements previously mentioned in Section (2-3-47), especially the minimum cement content and maximum w/c ratio shall be strictly followed. The concrete cover depends on the exposure conditions of

the concrete tension side. Tables (4-13) (10-6) and (10-7) give the recommended concrete cover values.

2-4 Fire resistance of concrete

Concrete resistance to fire is the time of exposure to direct fire, according to standard tests, before disintegration and/or failure of the concrete element. The "Egyptian Code for Design and Construction Requirements for Structures Exposed to Fire" shall be the basis for designing concrete to resist fire.

Herein in this code, it is important to set the principal limitations that shall be taken into consideration to increase the concrete resistance to fire. The most important limitations are:

- Type and dimensions of structural element.
- Concrete cover and reinforcing steel protection.
- Concrete and aggregate type.
- Reinforcement type, steel type and reinforcement arrangement.
- Construction method and structure type.

These limitations shall all be considered in the design to achieve concrete fire resistance for each concrete element to fulfill its goal. To achieve concrete resistance to fire for a time period ranging from half an hour to four hours, the guide values given in tables (2-14-a) and (2-14-b) could be used.

Table (2-14-a) Minimum concrete dimensions in (mm) for fire resistance of reinforced concrete

Fire Period (hour)			0.5	1.0	1.5	2.0	3.0	4.0
Minimum Concrete Dimension (mm)	Column Smallest Dimension		200	200	250	300	400	450
	Simple Beam Breadth		120	120	150	200	240	280
	Continuous Beam Breadth		120	120	120	150	200	240
	Slab Thickness (simple or continuous)		80	100	110	130	150	170
	Concrete Wall	μ*<0.4%	150	150	180	---	---	---
		0.4%<μ<1%	120	120	140	160	200	240
		μ>1%	120	120	120	120	150	180
Concrete Cover Beyond Stirrups (mm)	Column Cover		20**	20	20	25	25	25
	Simple Beam Cover		20**	20	30	45	60	70
	Continuous Beam Cove		20**	20**	25	40	50	60
	Simple Slab Cover		15	20	25	35	45	55
	Continuous Slab Cover		15	20	20	25	35	45
	Concrete Wall Cover	0.4%<μ<1%	25	25	25	25	25	25
		μ>1%	15	15	25	25	25	25

* μ is the longitudinal steel percentage in the concrete wall

** The concrete cover could be reduced to 15mm if the aggregate nominal maximum size is smaller than 15mm

The following requirements shall be considered:

a - The minimum concrete cover shall not smaller than neither the values stated in article (4-3-2-3-b) nor the larger bar diameter used.

b - The concrete cover could spall, if it is more than 40mm. In such case, extra precautions shall be taken to protect the concrete cover by using plastering layer above the concrete surface or using extra reinforcement mesh at a depth of 20mm from concrete surface.

c - On using external plastering layer, it can be used as part of the concrete cover as follows:

1 - When using gypsum or cement mortar as plastering layer, the equivalent cover is equal to 0.6 the actual plastering thickness.

2 - When using lightweight plastering such as vermiculite, the equivalent cover is equal to the complete plastering thickness.

Table (2-14-b) Minimum concrete dimensions in (mm) for fire resistance of pre-stressed concrete

Fire Period (hour)		0.5	1.0	1.5	2.0	3.0	4.0
Minimum Concrete Dimension (mm)	Simple Beam Breadth	120	120	150	200	240	280
	Continuous Beam Breadth	120	120	120	150	200	240
	Slab Thickness (Simple or Continuous)	100	100	110	120	140	150
Concrete Cover Beyond Stirrups (mm)	Simple Beam Cover	25	40	55	70	80	90
	Continuous Beam Cover	20*	30	40	55	70	80
	Simple Slab Cover	25	35	45	55	65	75
	Continuous Slab Cover	20*	25	35	45	55	65

* The concrete cover could be reduced to 15mm if the aggregate nominal maximum size is smaller than 15mm

2-5 Concrete exposed to abrasion and wear

2-5-1 General

Abrasion resistance of concrete is the ability of its surface to resist frictional wear. Concrete compressive strength could be considered as an indication to its abrasion resistance.

Abrasion resistance of concrete could be evaluated by determining the percentage loss in weight or volume or thickness; also visual inspection may be used to evaluate quality of concrete surface.

2-5-2 Requirements for abrasion and wear resistant concrete

1 - Concrete grade shall not be lower than 30 N/mm^2. This is achieved by:

a - Low w/c ratio.

b - Good aggregate grading and nominal maximum size not more than 25mm.

c - For roads, runways, and coastal structures, the aggregate hardness shall comply with the limits stated in table (2-1) when gravel or crushed gravel is used. When using crushed stone the maximum percent passing from sieve 1.7mm shall not exceed 25% when exposed to 500 revolutions in Loss Angeles machine.

d - Concrete slump shall not be more than 75mm.

e - On using more than one concrete layer, the concrete slump for the surface layer shall not exceed 25mm.

f - Air content shall not exceed 3%.

2 - For concrete surfaces exposed to highly abrasive load such as heavy truck loads, erosion in sea water structures, bridge piers, tunnels, the concrete surface layer shall have a characteristic strength not less than 35 N/mm^2, the slump shall not be more 25mm, and the aggregate nominal maximum shall not be more than 12mm. In addition the aggregate shall comply with all the requirements stated in table (2-1).

3 - Care in concrete surface finish by delaying compaction and surface finish until concrete gets rid of all bleed water.

4 - Bleed water on concrete surface could be dewatered by vacuum.

5 - Surface topping could be used in severe exposure conditions; manufacturer data specifications shall be referred to.

6 - Good curing of concrete surface for a period not less than 7 days, curing shall start immediately after surface finish.

2-6 Basiscs of concrete mixture design

2-6-1 General

The aim of mixture design is to determine the constituents' proportions in order to fulfill both the fresh concrete and hardened concrete requirements. That is why mixture design shall principally include the determination of w/c ratio to achieve the required compressive strength and durability at the same time. For those mixtures using low w/c ratios for strength or durability requirements, the use of normal range and high range water reducers shall be considered to obtain the desired fresh concrete workability. All precautions during construction shall be implemented to assure that w/c ratio shall not be increased during concrete manufacture as it will be a violation of the mixture design principals and will have adverse effect on both strength and durability.

Concrete mixture design requires initial information such as properties of ingredient materials, strength requirement, exposure conditions (which will determine the maximum w/c ratio, minimum cement content and cement type). Structure dimensions and construction facilities will dictate required workability level.

Mixture design is highly related to strength evaluation processes, as mixture adjustment will be implemented during construction based on such evaluation.

2-6-2 Mixture design requirements

The engineer may use any suitable mixture design method taking into consideration the following:

2-6-2-1 Compressive strength requirements

Concrete mixtures are designed in order that the mean target compressive strength (fm) is equal to the characteristic compressive strength (fcu) (i.e. Section 2-3-2-1) plus the margin of safety (M), according to the following relationship:

$$fm = fcu + M \quad (2\text{-}5)$$

Margin of safety is calculated as follows:

$$M = K \times s \quad (2\text{-}6)$$

Where;

M = Margin of safety

K = Factor determined based on percentage defects allowed in compressive strength test results, equal to 1.64 to achieve the characteristic compressive strength fcu stated in article (2-3-2-1)

S = Standard deviation for compressive strength test results previously carried out by the contractor

In case of availability of more than 40 compressive strength test results for similar concrete mixtures (i.e. using similar materials and under similar production circumstances), the standard deviation for these test results shall be used as long as the following requirements are met:

- s shall not be less than 4N/mm^2 for fcu ≥ 20N/mm^2
- s shall no be less than 20% of the characteristic strength for fcu <20N/mm^2

In case no data is available the standard deviation is calculated as follows:

- 8N/mm^2 for concrete mixtures with fcu ≥ 20N/mm^2
- 40% of the characteristic strength for concrete mixtures with fcu < 20N/mm^2

The mean target compressive strength (fm) shall be adjusted with work progress and the availability of 40 compressive strength test results or more from which the standard deviation shall be calculated. The calculated standard deviation shall not be lower than 3.5N/mm^2.

Designer has to determine w/c ratio that will achieve the mean target compressive strength for preliminary mixture design and accordingly with every adjustment in the standard deviation afterwards. Consequently this will result in adjusting all other mixture ingredients proportions.

2-6-2-2 Durability requirements

During concrete mixture design steps, all durability requirements shall be fulfilled. This includes water tightness, maximum chloride and sulfate contents in the mixture.

a - Water tightness

Designers shall abide by the maximum w/c ratio and minimum cement content stated in table (2-10) to achieve the required water tightness and those which will achieve mean target compressive strength. Concrete mixtures may contain pozzolanic materials as supplementary cementing material to help accomplish the needed water tightness level in case of high performance concrete mixtures.

All curing provisions shall be followed to assure the continuation of hydration and improve water impermeability and concrete durability.

b - Chloride and sulfate contents

- Chloride ion content in the concrete mixtures shall not exceed the values stated in table (2-10).
- Sulfate content shall not exceed the values stated in article (2-3-4-6).
- Table (8-4-b) gives the frequency of testing.

2-6-2-3 Workability requirements

Concrete workability shall be suitable to the structure dimensions and the available compaction method as given in Section (2-3-1-2). Also, other factors shall be taken into consideration such as mixing, transportation and casting methods and the time between water addition to the mixture and final surface finish. Due to high variability in these factors, designers may

have to use admixtures to improve workability or to develop new properties; article (2-2-4) shall be addressed with respect to this issue.

2-6-3 Assurance trial mixtures

Before the approval of the concrete mixture, it shall be carried out in the laboratory to confirm the requirements of the concrete mixture and to make any necessary adjustments to comply with project specifications. Field trial mixtures shall be carried out to check the performance of site mixing and the achievement of the required compressive strength with the mixture proportions undertaken in the laboratory.

2-6-3-1 Laboratory trial mixtures

Laboratory trial mixtures shall be conducted using the same materials included in the mixture design in order to check out workability, density and compressive strength which shall not be lower than the mean target compressive strength (fm) under laboratory conditions, and to make necessary adjustments to achieve desired properties. Laboratory trial mixtures are mandatory to concrete mixture design procedure and its acceptance.

For the approval of the concrete mixture design, the following information shall be submitted:

1 - Characteristic compressive strength, mean target compressive strength and margin of safety used in mixture design.

2 - Properties of used materials: cement, admixtures and aggregate.

3 - Ingredients proportions by weight to produce one cubic meter with the condition that the aggregate is in a saturated surface dry condition, thus all the mixing water is available for cement hydration.

4 - Slump of trial mixtures.

5 - Average compressive strength results at 28 days of age and its comparison with the mean target compressive strength (fm).

6 - Chloride and sulfate content in hardened concrete.

2-6-3-2 Compulsory assurance field mixtures

Concrete manufacturer, either in site or in ready mixture plant, has to perform full scale three separate trial concrete mixtures using the same materials intended to be used in actual production and which are used in the mixture design. Each mixture is preferred to be carried out separately.

Concrete workability is measured for each mixture and at least nine standard cubes shall be prepared for each mixture to be tested in compression. Three of these cubes could be tested at earlier age such as 3 days or 7 days, while the rest shall be tested at age 28 days. Each set of cubes represents one test. A test result is the average of the tested standard cubes taken from one mixture according to the preparation and testing procedures outlined in the testing manual of this code (Appendix 3) or according to the requirements of the project specifications.

The test results of all the three mixtures shall comply with the following requirements:

1 - The average compressive strength of the three mixtures at age 28 days shall not be lower than 95% of the compressive strength of the laboratory trial mixture using the same material consignments and proportions.

2 - The average compressive strength shall not be lower than the characteristic compressive strength plus 6.5N/mm^2.

3 - Not a single cube shall be lower than the characteristic compressive strength.

4 - The difference between the lowest cube result and the highest cube result in each test shall not be more than 15% of the average strength of that test.

2-6-3-3 Additional assurance mixtures

The contractor or the concrete supplier shall make additional trial mixtures upon the request of the consultant or on making substantial changes to the used materials or mixture proportions. The request of these additional trial mixtures shall not be considered in case of adjusting mixture proportions according to the quality control program with the intention of changing lower strength limits to achieve the mean target strength. Also these additional trial mixtures shall not be requested to check minimum cement content or maximum w/c ratio. As well, the routine testing according to the quality control program, as outlined in Section (8-9-3), shall not be a part of these additional trial mixtures.

2-6-4 Ready mixture concrete

Occasionally, long hauls of ready mixture concrete in truck mixers results in loss of the concrete workability, which dictates the use of set retarding admixtures. Under all circumstances, ready mixture concrete shall

be handled on its delivery at the job site according to the requested specifications for fresh and hardened concrete (i.e., consistency - air content – preparation for strength specimens).

2-6-5 Principles of concrete mixture evaluation

Consultants shall perform continuous evaluation for concrete quality on site using the daily acquired data for concrete workability and strength at different ages. These evaluations could be used to predict in the future any divergence in concrete properties and mixture composition than the approved mixture. In case any divergence in concrete properties or mixture composition is detected, the causes shall be found out, and re-evaluation of the produced concrete, both in the fresh and hardened states, shall be as follows:

2-6-5-1 Fresh concrete evaluation

Upon using slump test values as acceptance criteria for fresh concrete, the difference between test values and project specifications shall not exceed the following values:

- ± 30mm for concrete with slump values > 80mm
- ± 20mm for concrete with slump values < 80mm

2-6-5-2 Hardened concrete evaluation during construction

a - Preliminary evaluation

This procedure is used to evaluate preliminary compressive strength values. The test results shall be accepted and considered to fulfill the characteristic compressive strength if the following two conditions are met:

1 - The test result shall not be lower than the characteristic strength by more than 10%

2 - The average of the test and the previous three test values shall exceed the characteristic strength by a minimum of 10%.

b - Final evaluation

When 40 or more test results are available, the final strength evaluation for the compressive strength shall be conducted. The concrete shall be accepted and considered to fulfill the characteristic strength if the percentage of test results falling below the characteristic strength is less than or equal to 5% of the total number of tested specimens.

In case the conditions in articles (a) and (b) are not fulfilled, the consultant shall be referred to decide the necessary action.

2-7 Ready mix concrete requirements

The ready mix specifications prepared by the Housing and Building National Research Center shall be considered when dealing with fresh concrete delivered to the site. The specifications give details on materials selection and mixture design according to requested specifications to accomplish required concrete quality. These specifications do not cover casting, compaction and curing of concrete after its delivery.

The specifications deal with quality assurance in ready mix batching plants and its technical requirements, and the evaluation procedure for its environmental impact and professional safety.

2-8 Self-compacting concrete requirements

The self-compacting concrete (SCC) specifications prepared by the Housing and Building National Research Center shall be considered when dealing with this type of concrete. The specifications cover the following technical points:

- Requirements to produce SCC using local materials.
- Properties of mixture constituent requirements.
- Mixture design fundamentals.
- Concrete production and casting and curing requirements.
- Performance evaluation procedures.
- Quality control and quality assurance

The specifications appendices include guides for designers, manufacturers, contractors, consultants and testing laboratories.

2-9 Hot-weather concreting requirements

The specifications for hot-weather concreting prepared by the Housing and Building National Research Center shall be considered when dealing with concrete in hot weather. The specifications cover the following technical points:

- Climate in Egypt and definition of hot weather.

- Main problems during hot-weather concreting for fresh and hardened concrete.
- Ambient factors that affect fresh and hardened concrete properties in hot-weather.
- Methods of selecting concrete materials and mixture design.
- Special requirements for hot-weather concrete production such as; materials' consignment, mixing, transportation, form-works, casting, compaction, surface finishing and curing.
- Quality control and quality assurance procedures in hot-weather concreting to produce concrete which fulfills both required strength and durability.
- Different cooling systems.

CHAPTER 3
GENERAL DESIGN CONSIDERATIONS

3-1 Design methods

This chapter is intended to present the main principles for the design of concrete structures and elements thereof that based on statistical concepts shall ensure adequate performance of structures and shall fulfil all the serviceability and safety requirements against all probable types of failures when subjected to all applicable loads. The design can be performed using one of the following two design methods:

1 - Limit States Design Method

2 - Elastic(Working Stress) Design Method

3-1-1 Limit states design method

In compliance with the requirements of section 3-1, members shall be proportioned for adequate strength under ultimate loads using material strength reduction factors that account for all factors that could adversely affect the strength of concrete members. The ultimate load and material strength reduction factors are specified in sections 3-2-1-1 and 3-2-1-2 of this code, respectively. The structure shall also be designed for adequate performance under service loads.

3-1-1-1 Ultimate strength limit state

It is the limit that based on statistical concepts, shall ensure that the structure and elements thereof shall be safe against failure. This limit state shall also control the probable mode of failure (section 4-2)

3-1-1-2 Stability limit state

It is the limit that based on statistical concepts, shall ensure that the structure and elements thereof shall be safe against buckling (section 6-4), overturning, uplift and sliding.

3-1-1-3 Serviceability limit states

They are the limits that based on statistical concepts, shall ensure adequate performance and durability of structures and elements thereof under service loads by controlling the adverse effects that could hinder the intended use of the structure. They are classified as follows:

a - Deformation and deflection limit state

It is the limit that based on statistical concepts, shall control the deflection and deformations that could hider the efficient use of the structure and elements thereof (section 4-3-1).

b - Cracking limit states

It is the limit that based on statistical concepts, shall control the cracking that could hider efficient use and durability of the structure and elements thereof (section 4-3-2)

3-1-2 Elastic (working stress) design method

In compliance with the requirements of section 3-1, members shall be proportioned to satisfy allowable stresses that shall ensure the satisfaction of the safety requirements of structures against failure, including the satisfaction of the stability limit state of section 3-1-1-2, as well as, the serviceability limit states specified in section (3-1-1-3)

3-2 Safety provisions

3-2-1 Safety provisions for limit states design method

3-2-1-1 Loads and load combinations

a - Service loads

Service loads can be defined as those values of the loads which based on statistical concepts have an accepted probability of not being exceeded during the life span of the structure by more than 5%. The probable values of the various types of loads on structures shall be in accordance with the Egyptian Code for Loads on structures (ECP 201). The loads include dead, live, wind, seismic, and dynamic loads. The ECP 201 code also specifies provisions for soil and liquid pressures, as well as, thermal, creep and shrinkage, settlements effects.

b - Ultimate loads

Ultimate loads are evaluated by multiplying the service loads by appropriate load factors for the following load combinations:

1 - For members subject to live loads:

$$U = 1.4\ D + 1.6\ L \qquad (3\text{-}1)$$

Where,

D = Dead Loads

L= Live Loads

2 - For members subject to live loads where the magnitudes of the live loads do not exceed 75% of those of dead loads:

$$U = 1.5\ (D + L) \qquad (3\text{-}2)$$

3 - For members subject to live loads and lateral pressures resulting from soil and fluids:

$$U = 1.4\ D + 1.6\ (E + L) \qquad (3\text{-}3)$$

Where,

E = Lateral pressures

The resulting ultimate load shall be less than that obtained from Equation 3-1. For water tanks, the term 1.6 E in Equations 3-3 and 3-7 shall be replaced by 1.4 E.

4 - For members subject to either wind load, W or seismic load, S the ultimate load shall be taken equal to the greater value obtained from the following two equations:

$$U = 0.8\ (1.4\ D + 1.6\ L + 1.6\ W) \qquad (3\text{-}4)$$

$$U = 1.12\ D + \alpha\ L + S \qquad (3\text{-}5)$$

The resulting ultimate loads shall be less than that obtained from Equation 3-1.

Where,

S = Ultimate seismic loads. The load shall be evaluated in accordance with the provisions of Egyptian Code for Loads on structures (ECP 201).

α = A factor that accounts for the effects of the sustained live loads on the structure during seismic activities.

$\alpha = {}^{1}/_{4}$; for residential buildings

$\alpha = {}^{1}/_{2}$; for public buildings including, schools, hospitals, garages, theatre halls, commercial and office buildings

$\alpha = 1$; for structures subject to loads acting for a long duration of time which include but are not limited to, silos, bins, water tanks, libraries, storage buildings

5 - For the cases of loadings where a higher dead load values would increase the stability of the structure, and where a reduction of dead load values would increase the effects of other loads, the ultimate loads given in items 1, 3 and 4 shall be replaced, respectively, by the following:

$$U = 0.9\ D \qquad (3-6)$$

$$U = 0.9\ D + 1.6\ E \qquad (3-7)$$

$$U = 0.9\ D + 1.3\ W \qquad (3-8)$$

$$U = 0.9\ D + S \qquad (3-9)$$

6 - For the cases where the temperature effects are taken into consideration:

$$U = 0.8\ (1.4\ D + 1.6L + 1.4\ T) \qquad (3-10)$$

But shall not be taken less than

$$U = 1.4\ (D + L) \qquad (3-11)$$

Temperature effects shall be evaluated in accordance with section 3-3.

7 - Dynamic loads, K can be treated as equivalent static load as follows:

$$U = 1.4D + 1.6\ L + 1.6\ K \qquad (3-12)$$

With due consideration of Equation 3-6

c - Loads factors for serviceability limit states and working stress design method

1 - The loads for the cases of serviceability limit states (sections 3-1-1-3) and when the working stress design method is used (section 3-2-1) shall be taken equal to the service loads (section 3-2-1-1-A). The load combinations shall be taken equal to the following:

$$D+L \qquad (3-13-a)$$

$$D + L + W \qquad (3-13-b)$$

$$D+\frac{\alpha L}{1.2}+\frac{S}{1.4} \qquad (3-13-c)$$

2 - For the cases of loadings where a higher dead load values would increase the stability of the structure, and where a reduction of dead load values would increase the effects of other loads the following load combinations shall be applied:

$$0.9\ D \qquad (3-14-a)$$

$$[0.9\ D + W] \quad \text{or} \quad [0.9\ D + (S/1.40)] \qquad (3-14-b)$$

3-2-1-2 Material strength reduction factors

The purpose of the material strength reduction factors, γ_c and γ_s for concrete and steel, respectively, are to allow for the probability of understrength members due to variations in material strengths and dimensions that are considered to be within the allowable tolerances, and to allow for inaccuracies in design equations and assumptions. Their values are dependent on the degree of ductility and required reliability of the member under load effects being considered (such as moment, shear, eccentric compression, etc). Their values also reflect the importance of the member in the structure. They are taken as follows:

1 - Ultimate strength limit state

A - Material Strength reduction factors for concrete, γ_c and for steel, γ_s for applications involving;

- Axial tension and eccentric tension
- Bending moments
- Shearing forces and twisting moments
- Bearing
- Bond

shall be taken as follows:

$$1.5 = \gamma_c \qquad (3\text{-}15\text{-a})$$

$$1.15 = \gamma_s \qquad (3\text{-}15\text{-b})$$

B - Material Strength reduction factors for concrete, γ_c and for steel, γ_s for axial and eccentric compression force applications shall be taken as follows :

$$\gamma_c = 1.5\left\{\left(\frac{7}{6}\right) - \frac{(e/t)}{3}\right\} \geq 1.5 \qquad (3\text{-}16\text{-a})$$

$$\gamma_s = 1.15\left\{\left(\frac{7}{6}\right) - \frac{(e/t)}{3}\right\} \geq 1.15 \qquad (3\text{-}16\text{-b})$$

Where;

$$\left(\frac{e}{t}\right) \geq 0.05$$

2 - Serviceability limit states

Material Strength reduction factors for concrete, γ_c and for steel, γ_s for applications involving:

- Deflection
- Deformations
- Cracking

shall be taken as follows:

$$\gamma_c = \gamma_s = 1.0 \qquad (3\text{-}17)$$

3-2-2 Safety provisions for working stress design method

The safety provisions for working stress design method shall be taken in accordance with the provisions of section 3-2-1-1-C and the provisions of Chapter 5.

3-3 Internal effects

A - Shrinkage of concrete: shall comply with section 2-3-3-4 of the Egyptian Code for Loads on Structures, ECP 201.

B - Temperature effects: shall comply with section 2-3-3-3 of the Egyptian Code for Loads on Structures, ECP 201.

C - Creep of concrete: shall comply with section 2-3-3-5 of this code.

CHAPTER 4
LIMIT STATES DESIGN METHOD

4-1 General considerations

This chapter presents the main principles for the design of reinforced concrete structures in accordance with the limit states design method introduced in Chapter 3. Satisfying such limit states shall guarantee safety of the structure against failure (Section 4-2), as well as shall guarantee that all the serviceability requirements specified in Section 4-3 shall be satisfied.

4-2 Ultimate strength limit state

This section presents the ultimate strength limit state design of sections subject to flexure or eccentric forces (Section 4-2-1) , sections subject to shear forces (Section 4-2-2), and sections subject to torsional moments (Section 4-2-3), as well as the designs for bearing strength (Section 4-2-4) and the bond strength (Section 4-2-5).

4-2-1 Ultimate strength limit state: flexure or eccentric forces

The design of the sections subject to flexure or eccentric forces in compliance with the ultimate strength limit state shall be carried out according to the provisions of this section.

4-2-1-1 Basic assumptions and general considerations

The strength limit state of the sections subject to flexure or flexure combined with axial forces shall satisfy the equilibrium conditions and the compatibility of strains, in addition to the following assumptions and general considerations:

1 - Strains are linearly distributed over the cross-section and consequently, the strains in the reinforcement and concrete are directly proportional to the distance from the neutral axis. This assumption is valid for all members except deep beams in which strains are distributed nonlinearly.

2 - The stress-strain relationship of steel is taken according to the idealized curve shown in Fig. (4-1). The design values of yield strength shall not exceed the upper limits specified on the yield strength for crack control, as given in the fourth assumption.

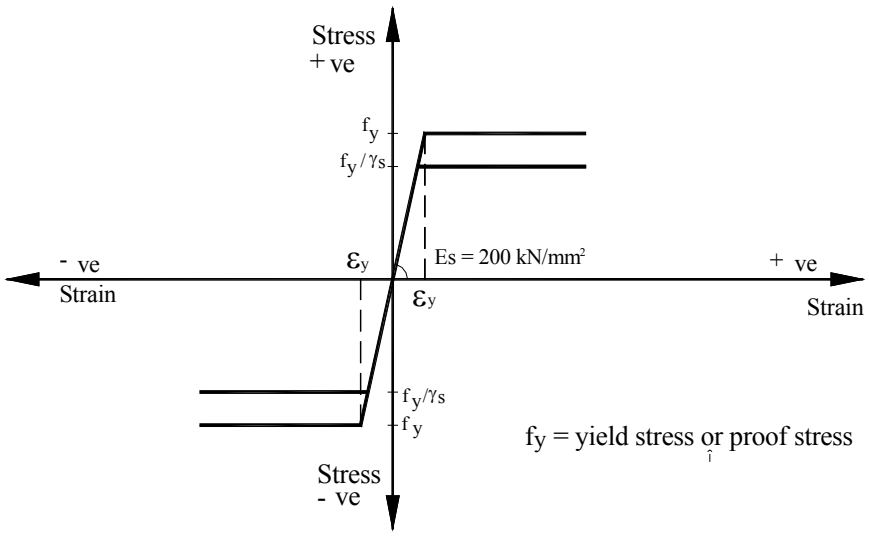


Figure (4-1) : Idealized stress – strain curve for reinforcing steel

3-a The values of f_y for high-tensile steel shall conform with the Egyptian Standards, (ES). For welded wire fabrics values of yield strength shall not exceed 400N/mm^2.

b - When test result values of smooth bars exceed 240 N/mm^2, the design value of f_y shall not be taken higher than 280 N/mm^2.

c - The value of the design yield stress f_y for smooth cold formed welded wire fabrics shall be taken not more than 300 N/mm^2.

4 - The design stresses for steel shall be governed by the cracking limit state requirements of section (4-3-2)

5 - The tensile strength of concrete shall be neglected. Accordingly; all tensile stresses shall be resisted by the reinforcing steel only.

6 - The distribution of stresses in the compression zone of the section shall be taken according to the stress-strain curve obtained from standard laboratory tests. It is also permitted to use the idealized curve shown in Fig. (4-2)

7 - The maximum usable compressive strain in the concrete (ε_{cu}) shall be taken equal to 0.003 for members subject to flexure or flexure combined with axial forces resulting in a part of the cross-section in tension. For

sections subject to axial compression forces at the plastic centriod of the section, where the plastic centroid is the point of application of the ultimate compression force that results in uniform compression on the section, the maximum usable strain shall be taken $\varepsilon_{cu} = 0.002$

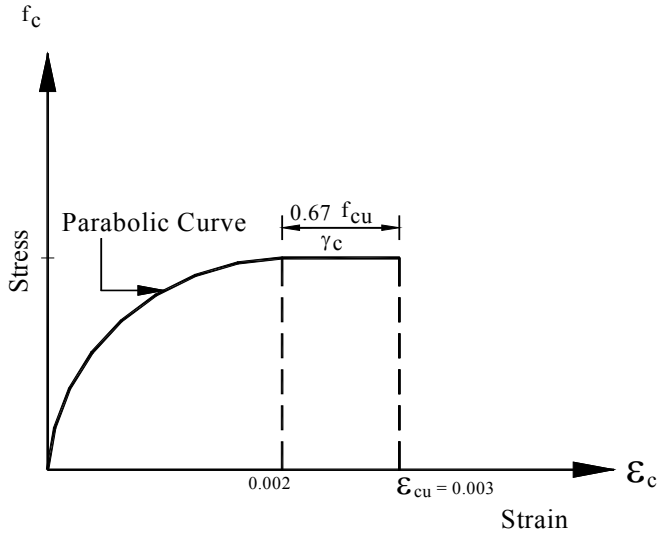


Figure (4-2) : Idealized stress – strain curve for concrete in compression

8 - Based on the sixth and seventh assumptions, the distribution of the ultimate compression stresses on the section shall be as shown in Fig.(4-3).

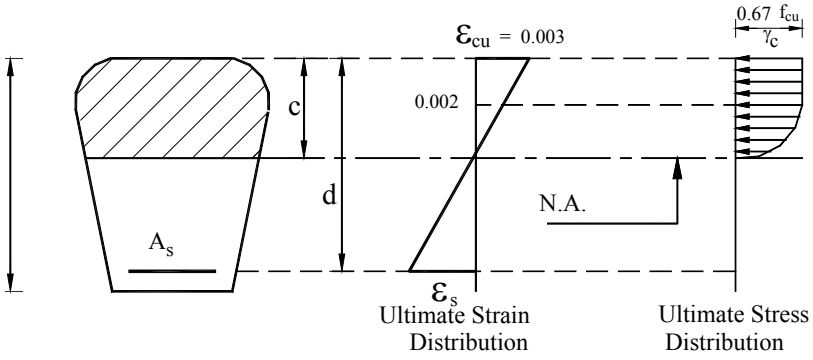


Figure (4-3) : Distribution of strains and ultimate stresses

9 - The requirements of assumptions (6, and 7) shall be considered satisfied for rectangular sections, T-section and trapezoidal sections, as shown in Fig. (4-4), by assuming uniformly distributed compressive stress of the concrete having an equivalent zone limited by the edge of the fibers subjected to the ultimate strain in the compression zone and by a line parallel to the neutral axis and located at a distance a = 0.8c from that edge; where c is the distance between the neutral axis and the edge subjected to the maximum compression. The value of uniform compression stresses is equal to $0.67\, f_{cu} / \gamma_c$. Such a stress distribution is called an Equivalent rectangular stress block.

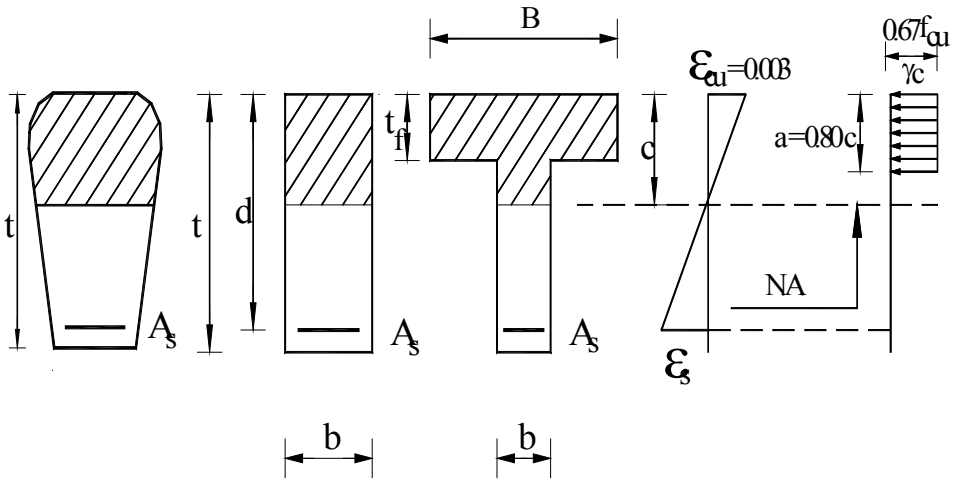


Figure (4-4) : Equivalent rectangular stress block

10 - For circular sections and other sections that are not previously mentioned, the ultimate stresses in the section shall be distributed according to the distribution of ultimate stresses shown in Fig. (4-3). As an alternate distribution, the depth of the equivalent stress block can be determined for such cases by satisfying the conditions of equality of the area of the equivalent stress block and the area of the ultimate stresses with the condition of their centroids coincide .

11 - For sections subject to biaxial bending moments as well as sections subject to biaxial bending moments accompanied by axial forces, the distribution of the ultimate compression stresses shall be according to Fig. (4-3). As an alternate distribution, the depth of the compression stress block can be calculated according to sub-clause 10.

4-2-1-2 Sections subject to flexure

4-2-1-2-a Sections with tension reinforcement only

For sections with tension reinforcement only in beams and in solid slabs and for T-sections in which the neutral axis is located inside the thickness of the slab, the ultimate limit moment shall be determined from the following equation:

$$M_u = \left(\frac{A_s . f_y}{\gamma_s}\right)\left(d - \frac{a}{2}\right) \qquad (4\text{-}1)$$

The depth of the compression stress block shall be calculated from the relation:

$$a = \frac{\left(\frac{A_s . f_y}{\gamma_s}\right)}{\left(\frac{0.67 f_{cu}}{\gamma_c}\right) b} \qquad (4\text{-}2)$$

The ratio a/d shall not be less than 0.1, the lever arm y_{ct} shall not be less than 0.95d, and the minimum reinforcement ratio requirements of Section (4-2-1-2-h) shall also be satisfied. The ratio a/d shall not exceed the values given in Section (4-2-1-2-c).

4-2-1-2-b Balanced sections

For sections with tension reinforcement only, there exist a limiting condition (balanced condition) between brittle failure mode and ductile failure mode. Such a limit occurs when the ultimate tensile strain in steel reaches a value that equals to $f_y y/E_s$ and the ultimate strain in concrete ε_{cu} reaches a value that equals to 0.003 simultaneously. To ensure providing sufficient ductility, the amount of the reinforcing steel shall be less than the value of the balanced condition. Section (4-2-1-2-c) gives the equations and the limits related to avoiding reaching the balanced condition, namely; the maximum percentage of tension reinforcement in a reinforced concrete section $\mu_{\max}$ and the corresponding maximum admissible value of ultimate bending moment in singly reinforced section $M_{u\max}$ as well as the maximum value of the ratio of the distance from the extreme compression fibers to the effective depth of the section. For compatibility with the nominal stress-

strain curve of reinforcing steel shown in Fig. (4-1), a nominal value for the yield strain of steel is taken equals to:

$$\varepsilon_y = \frac{f_y}{\gamma_s . E_s} \qquad (4\text{-}3)$$

4-2-1-2-c Upper limits to values of M_{umax} and μ_{max} for concrete sections with tension reinforcement only and subject to bending moment

$$M_{umax} = \frac{R_{max} . f_{cu} . b.d^2}{\gamma_c} \qquad (4\text{-}4)$$

$$\mu_{max} = \frac{A_{smax}}{b.d} = \frac{\left(\frac{0.67 f_{cu}}{\gamma_c}\right)\left(\frac{a_{max}}{d}\right)}{\left(\frac{f_y}{\gamma_s}\right)} \qquad (4\text{-}5)$$

Tables (4-1) and (4-2) give the values of R_{max} and μ_{max} for different values of steel grades for the case of redistribution of moments. Table (4-1) gives the maximum limits for the case in which moment redistribution is not permitted. In other words, the values of the bending moments shall be taken according to the theory of elasticity in statically indeterminate beams, slabs and frames that are loaded with various ultimate load cases including differential settlement with due confirmation of achieving the design value of f_{cu} after construction, and according to the conditions mentioned in Chapter 6 of this code. In such a case, it shall be recommended to determine the values of the bending moments in statically indeterminate beams and slabs through utilizing an accurate evaluation of the values of the relative values of the stiffness of the structural elements as well as through utilizing supporting systems that are compatible with the design assumptions. It shall be also recommended to satisfy the conditions of deformation and cracking.

Whenever redistribution of moments shall be permitted by an amount of $\pm$ 10%, the values of R_{max} and μ_{max} shall not exceed those given in table (4-2).

Table (4-1) : Values of R_{max}, μ_{max} and c_{max}/d for sections with Tension Steel Only

Steel Grade*	c_{max} /d	μ_{max}	R_{max}
240/350	0.50	$8.56x10^{-4} f_{cu}$	0.214
280/450	0.48	$7.00x10^{-4} f_{cu}$	0.208
360/520	0.44	$5.00x10^{-4} f_{cu}$	0.194
400/600	0.42	$4.31x10^{-4} f_{cu}$	0.187
450/520**	0.40	$3.65x10^{-4} f_{cu}$	0.180

*According to table (2-3) and f_{cu} in N/mm^2.
** For steel net and complying with item (4-2-1-1-3)

Table (4-2) : Values of R_{max}, μ_{max} and c_{max}/d for sections with Tension Steel Only for the Case of Moment Redistribution by an amount of ± 10%

رتبة الصلب*	c_{max}/d	μ_{max}	R_{max}
240/350	0.40	$6.85x10^{-4} f_{cu}$	0.180
280/450	0.38	$5.58x10^{-4} f_{cu}$	0.173
360/520	0.34	$3.88x10^{-4} f_{cu}$	0.157
400/600	0.32	$3.29x10^{-4} f_{cu}$	0.150
450/520**	0.30	$2.74x10^{-4} f_{cu}$	0.142

*According to table (2-3) and f_{cu} in N/mm^2.
** For steel net and complying with item (4-2-1-1-3)

For carrying out moment redistribution by an amount of ±10% according to Table (4-2), the following additional conditions shall be satisfied:

1 - Ensuring the satisfaction of the equilibrium conditions after carrying out the moment redistribution.

2 - Ensuring the satisfaction of the conditions of deformation and cracking.

3 - Ensuring that the summation of the values of the hogging and sagging moments in a single span is not less than 1.2 of the value of M_o as shown in Fig. (4-5) , where, M_o is the maximum moment in a simply supported beam.

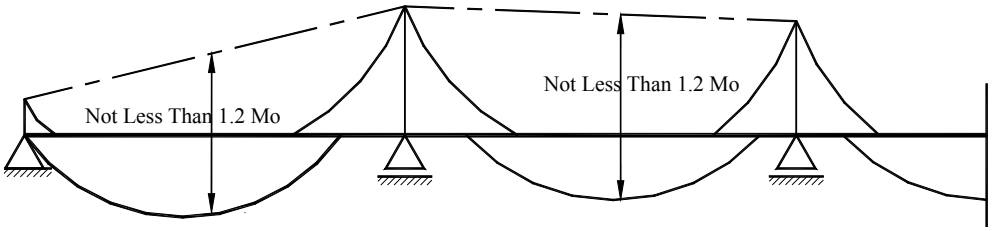


Figure (4-5): Moment distribution in beams

In case of using grades of steel different from those given in Tables (4-1) and (4-2), the values of μ_{max} and c_{max}/d shall be determined based on satisfying the conditions of equilibrium and strain compatibility such that value of c_{max} shall not exceed 0.67 $c_{balanced}$ and with due consideration of Section (4-2-1-1-3-a).

4-2-1-2-d Rectangular sections subject to bending moments with tension and compression reinforcement

The strength of sections can be increased more than the values given in the previous section (4-2-1-2-c) by using compression reinforcement (Fig. 4-6). In such cases, the ultimate strength of the section is calculated from the following equations:

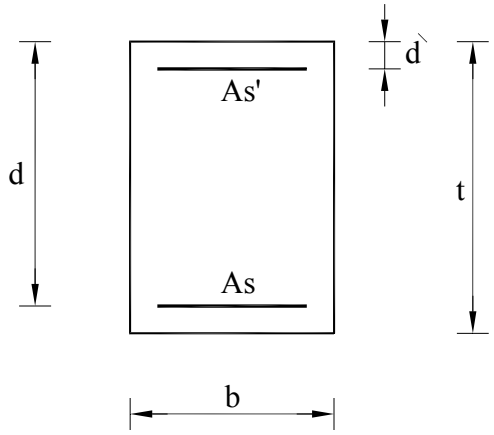


Figure (4-6) : Section with tension and compression reinforcement

$$M_u = R_{max}\left(\frac{f_{cu}}{\gamma_c}\right) b.d^2 + \left(\frac{f_y}{\gamma_s}\right) A'_s\,(d - d') \qquad (4\text{-}6)$$

where

$$A_s\left(\frac{f_y}{\gamma_s}\right)=\frac{0.67\,a_{max}.b.f_{cu}}{\gamma_c}+\frac{A'_s.f_y}{\gamma_s} \qquad (4\text{-}7)$$

The use of compression reinforcement shall be subject to the satisfaction of the following conditions;

1 - The value of strain in the concrete at the level of the compression reinforcement exceeds the yield strain of steel reinforcement. This condition shall be satisfied for the following cases:
d'/d ⟨ 0.20 for the case of mild steel
d'/d ⟨ 0.15 for the case of 360/520 steel
d'/d ⟨ 0.10 for the case of 400/600 steel
If the strain in the concrete at the level of the compression reinforcement is less than the yield strain of steel reinforcement the ultimate strength of the section shall be determined by using the applicable conditions of equilibrium and compatibility of strains.

2 - The use of stirrups at distances not exceeding 15 times the diameter of compression reinforcement in order to prevent buckling of the reinforcement.

3 - Satisfying the conditions of deformation and cracking.

4 - It shall be preferred to limit the area of the compression reinforcement, $A_s^{'}$ in a section subject to flexure to a value less than 40% of the area of the tension reinforcement, A_s.

5 - Generally, the area of compression reinforcement shall not be less than 10% of that in the tension side, in order to enhance the long term deflection of flexure members.

4-2-1-2-e T- and L-shaped sections with compression flange having a depth of the equivalent rectangular stress block exceeding the flange thickness

The design of such sections shall be based on the satisfaction of applicable conditions of equilibrium and compatibility of strains according to Section (4-2-1-1). Alternatively, the design could be carried out assuming that the ultimate moment shall be resisted only by the compression flange of the while neglecting contribution of the web in the ultimate moment resistance. For such cases, the design shall be carried out using the smaller value obtained from the use of the following two equations:

$$M_u = 0.67\left(\frac{f_{cu}}{\gamma_c}\right) B.t_f \left[d - \left(\frac{t_f}{2}\right)\right] \qquad (4\text{-}8\text{-}a)$$

$$M_u = A_s\left(\frac{f_y}{\gamma_s}\right)\left[d - \left(\frac{t_f}{2}\right)\right] \qquad (4\text{-}8\text{-}b)$$

Where;

t_f = thickness of the flange ;

B= effective width of the compression flange

The effective width of the compression flange of the section, B, shall be taken according to Section (6-3-1-9).

4-2-1-2-f Sections having shapes other than those listed in sections (4-2-1-2d & e) and subject to single bending

The design of such sections shall be based on the satisfaction of applicable conditions of equilibrium and compatibility of strains according to Section (4-2-1-1)

4-2-1-2-g Sections subject to biaxial bending

The design of such sections shall be based on the satisfaction of applicable conditions of equilibrium and compatibility of strains according to Section (4-2-1-1). Alternatively, for rectangular sections, the simplified method presented in Section (6-4-6) can be used.

4-2-1-2-h Minimum longitudinal reinforcement for sections subject to flexure

1 - For control of cracking of singly reinforced beams subject to bending moments and in order to guarantee sufficient ductility, the minimum percentage of tension reinforcement in the section shall not be less than the smaller of the following values:

$$\mu_{min} = 0.225\frac{\sqrt{f_{cu}}}{f_y} \geq \frac{1.1}{f_y} \qquad (4\text{-}9)$$

Where f_{uc} and f_y are in N/mm^2. The preceding requirement need not be applied if at every section the area of tensile reinforcement provided is at least one-third greater than that required by analysis according to Section (4-2-1-2-a).

In all cases, minimum percentage of tension reinforcement in the section shall not be less than 0.25% for normal mild steel or 0.15% for high grade steel. For T- and L-shape sections the value of μ_{min} is calculated using the web width.

2 - In case the flange of a T-section is located on the tension side, it is permitted to distribute part of the reinforcing bars not exceeding one-third of the area of the main reinforcement in the effective width of the flange according to Section (6-3-1-9) of this code or a width equals 0.1 of the clear span of the beam, whichever is smaller.

3 - The minimum percentage of tension reinforcement μ_{min} in slabs shall be taken equal to the values given in Sections (6-2-1-2-3) and (6-2-1-3-4) of this code.

4-2-1-3 Sections subject to combined flexure and axial compression

This section deals with the design of concrete sections subject to uniaxial bending or biaxial bending combined with axial compression forces acting at the plastic centroid of the section.

a - The design of concrete section subject to uniaxial bending or biaxial bending combined with axial compression forces shall be carried out using the method of strain compatibility that depends on satisfying the conditions of the equilibrium and the compatibility of the strains of the section resulting from the axial forces and bending moments (Section 4-2-1-1).

b - For sections subject to single moments and an ultimate axial compression load having a value not exceeding either the ultimate load given in equation 4-10 or the balanced compression load, P_b,

$$P_u \leq 0.04\, f_{cu}\, A_c; \qquad (4\text{-}10)$$

the effect of the axial force shall be neglected and the section shall be designed to resist the moment only according to section (4-1-2-4).

The balanced compression load of the section, P_b is defined as the eccentric compression load at which the strain in tension steel reaches a value equals $\varepsilon_y = f_y / E_s \gamma_s$ concurrently with strain in compressed concrete equals $\varepsilon_{cu} = 0.003$.

c - Concrete section subject to an axial compression force and simple bending having a value of the moment less than (P_u. e_{min}) shall be designed for a minimum eccentricity, e_{min} where:

$$e_{min} = \frac{M_u}{P_u} = 0.05\,t \qquad (4\text{-}11)$$

or 20 mm whichever is bigger. For such a case, the design of the section shall conform to Equation 4-12 which is considered to be the maximum value of the compression strength of the section:

1 - For columns with tie reinforcement:

$$P_u = 0.35\,f_{cu}\,A_c + 0.67\,f_y\,A_{sc} \qquad (4\text{-}12\text{-}a)$$

2 - For columns with spiral reinforcement satisfying the requirements of Section (6-4-7-i-j-k), the maximum strength is the smaller value of:

$$P_u = 0.35\,f_{cu}\,A_k + 0.67\,f_y\,A_{sc} + 1.38\,f_{yp}\,V_{sp} \qquad (4\text{-}12\text{-}b)$$

$$P_u = 1.14\,(0.35\,f_{cu}\,A_c + 0.67\,f_y\,A_{sc})$$

$$= 0.40\,f_{cu}\,A_c + 0.76\,f_y\,A_{sc} \qquad (4\text{-}12\text{-}c)$$

Where:

- A_c cross sectional area of the concrete section
- A_k area of concrete core enclosed by the spiral stirrups
- A_{sc} area of longitudinal reinforcement
- f_y yield stress of longitudinal reinforcement
- f_{yp} yield stress of spiral stirrups
- V_{sp} volume of spiral steel reinforcement for unit length of column and equals to:

$$V_{sp} = \frac{\pi\,A_{sp}\,D_k}{p} \qquad (4\text{-}12\text{-}d)$$

Where:

- A_{sp} cross sectional area of spiral reinforcing steel
- D_k diameter of the concrete core enclosed by the centerline of spiral stirrups

P pitch of spiral stirrups and it ranges from 30mm to 80mm according to Section (6-4-7-k)

The percentage of the volume of the spiral steel reinforcement μ_{sp} related to the volume of the concrete core limited by the diameter of the spiral stirrup shall not be less than the value determined by the following equation:

$$\mu_{sp} \geq 0.36\left(\frac{f_{cu}}{f_{yp}}\right)\left[\left(\frac{A_c}{A_k}\right)-1\right] \qquad (4\text{-}12\text{-}c)$$

Where;

$$\mu_{sp} = \frac{V_{sp}}{A_k} \qquad (4\text{-}12\text{-}f)$$

4-2-1-4 Sections subject to axial tension or combined flexure and axial tension

a - Sections subject to axial tension or to tension forces acting inside a distance equals to (d-d') of the section shall be designed on the bases that the tensile force shall be resisted by the steel reinforcement only.

b - The design Concrete sections other than those mentioned in the item, a and subjected to axial tension forces and bending moments shall be based on the satisfaction of applicable conditions of equilibrium and compatibility of strains according to Section (4-2-1-1).

c - In all cases the conditions of the cracking limit state requirement shall be satisfied according to Section (4-3-2).

4-2-2 Ultimate shear strength limit state

4-2-2-1 Beams

4-2-2-1-1 Nominal ultimate shear force in beams

a - For calculating the shear stresses, it shall be generally assumed that the largest shear force shall be the one calculated at the faces of the supports, as shown Fig. (4-7-c) and Fig. (6-21) . For the cases of direct supports underneath the beams, where a normal compression is resulting at the bottom edge of the beam, (Fig. (4-7-a,b) and Fig. (6-22)) , it shall be permitted to calculate the shear stress and to design the web reinforcement based on the shear force acting at a distance equals (d/2) from the face of the support, where (d) is the effective depth of the beam.

b - If a concentrated load P_u is located at a distance (a) from the face of the support, which is less than or equal to twice the effective depth of the beam ($a \leq 2d$), it shall be permitted to calculate the shear stress resulting from this load using the shear force equals to the original shear force multiplied by a/2d (Fig. 4-7) provided that the shear stress calculated according to the original shear force without reduction shall not be more than the value given by Equation (4-16), but not more than 4.0 N/mm^2.

c - Values of the shear force acting at a distance between the location of the largest shear force and the face of the support for the cases in which the critical section is located at a distance, d/2 from the face of the support shall be considered to have a constant value equals to the largest force calculated according to items (a & b) as shown in Fig. (4-7).

4-2-2-1-2 Nominal ultimate shear strength

a - For beams and slabs with constant depth, the nominal ultimate shear stress at any section is calculated from the following relationship:

$$q_u = \frac{Q_u}{b.d} \qquad (4\text{-}13)$$

Where;

Qu = ultimate shear force

b - For beams and corbels with variable depth in which the thickness of the section increases with the increase of the bending moment, the shear force Q_u shall be replaced by the value Q_{ur} conforming to the following equation:

$$Q_{ur} = Q_u - \frac{(M_u . \tan\beta)}{d} \qquad (4\text{-}14)$$

Where β in the angle of inclination of the variation of the depth measured from the beam axis, provided that the value $\tan\beta$ is not more than 0.33.

For beams with variable depth in which the thickness of the cross-section decreases with the increase of the bending moment, the shear force Q_u is replaced by the value Q_{ur} conforming to the following equation:

$$Q_{ur} = Q_u + \frac{(M_u . \tan\beta)}{d} \qquad (4\text{-}15)$$

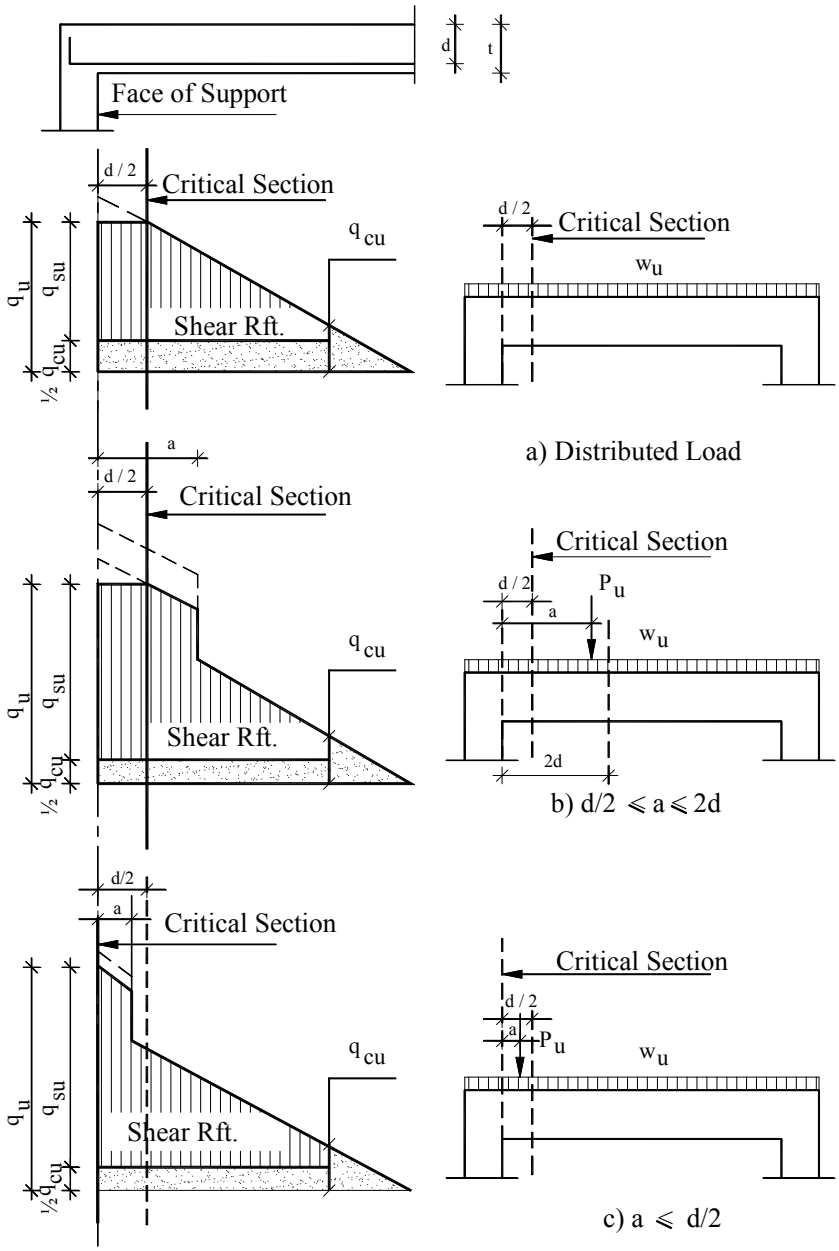


Figure (4-7) : Shear stress distribution and critical sections in beams

c - The nominal shear stress for members subject to shear forces not accompanied by torsional moments shall not be more than the value given by the following equation:

$$q_{umax} = 0.70\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad N/mm^2 \quad < 4\ N/mm^2 \qquad (4\text{-}16)$$

The values of q_{umax} can be taken from Table (4-3).

table (4-3) : Values of ultimate shear stress permitted for cross-sections subject to shear force not accompanied by torsional moments according to equation (4-16)

f_{cu} N/mm^2	20	25	30	35	40	50	60
q_{umax} N /mm^2	2.56	2.86	3.13	3.38	3.60	4.00	4.00

d - For members subject to shear forces accompanied by ultimate torsional moments M_{tu}, the effect of the torsional moments can be neglected if the value of the shear stresses resulting from such moments q_{tu} calculated from equation (4-47) is less than q_{tu} calculated according to equation (4-17) or Table (4-4). Otherwise, the concrete dimensions for cross-sections subjected to combined shear forces and torsional moments shall be determined based on the satisfaction of equation (4-48) for solid sections and equation (4-49) for box-sections.

$$q_{tu} = 0.06\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}17)$$

Table (4-4) : Values of ultimate shear stresses resulting from torsional moments below which torsion can be neglected according to equation (4-17)

f_{cu} N/mm^2	20	25	30	35	40	50	60
q_{tu} N/mm^2	0.22	0.25	0.27	0.29	0.31	0.34	0.38

4-2-2-1-3 Ultimate shear strength provided by concrete

a - For sections subject to shear forces or to shear force accompanied by torsional moments that result in shear stresses having values less than

those given by equation (4-17), the value of the concrete ultimate shear strength shall be calculated using the following equation:

$$q_{cu} = 0.24\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}18)$$

b - For sections subject to compression force, P_u, the value given by equation (4-18) shall be increased by using the multiplier δ_c given by:

$$\delta_c = 1 + 0.07\left(\frac{P_u}{A_c}\right) \qquad (4\text{-}19)$$

c - For sections subject to axial tension force P_u, the ultimate shear strength provided by concrete shall be neglected, unless otherwise calculated by multiplying the concrete contribution of equation (4-18) by the reduction factor δ_t given by:

$$\delta_t = 1 - 0.30\left(\frac{P_u}{A_c}\right) \qquad (4\text{-}20)$$

4-2-2-1-4 Nominal shear strength provided by web reinforcement in beams

a - If the value q_u calculated according to Section (4-2-2-1-2) is greater than the value of the concrete ultimate shear strength q_{cu}, shear reinforcement of one or more of the following types shall be used:

1 - Stirrups normal to the axis of the member.

2 - Inclined stirrups or bent-up bars with an angle not less than 30° with axis of the member accompanied by stirrups normal to the axis of the member.

b - The nominal shear strength provided by shear reinforcement shall be given by:

$$q_{su} = q_u - 0.5\,q_{cu} \qquad (4\text{-}21)$$

Figure (4-7) shown the regions that need web reinforcement with due consideration of the minimum percentage of web reinforcement in the other regions specified in section (4-2-2-1-6).

4-2-2-1-5 Web reinforcement in beams

a - In case of using stirrups normal to the axis of the member without web reinforcement, the web reinforcement shall be calculated from the following equation:

$$\mu_{st} = \frac{A_{st}}{b.s} = \frac{q_{su}}{\left(\frac{f_y}{\gamma_s}\right)} \qquad \textbf{(4-22)}$$

where:

A_{st} = total cross-sectional area of stirrups resisting the shear force

μ_{st} = percentage of stirrups normal to the axis of the member

b = width of the web

s = spacing between stirrups in axis direction

b - In case of using inclined stirrups or bent-up bars inclined to the axis of the member with an angle, α the web reinforcement shall be calculated according to the following equation:

$$\frac{A_{sb}}{b.s} = \frac{q_{sub}}{\left[\left(\frac{f_y}{\gamma_s}\right)(\sin\alpha + \cos\alpha)\right]} \qquad (4\text{-}23)$$

Where

$$q_{sub} = q_{su} - q_{sus} \qquad (4\text{-}24)$$

A_{sb} = cross section area of inclined stirrups or bent-up bars that are inclined with an angle α to axis of the member

q_{sub} = nominal ultimate shear strength of bent-up bars

q_{sus} = nominal ultimate shear strength of stirrups normal to the axis of the member

If the angle $\alpha = 45^{o}$, equation (4-23) can be written in the following form:

$$\frac{A_{sb}}{b.s} = \frac{q_{sub}}{\left[\sqrt{2}\left(\frac{f_y}{\gamma_s}\right)\right]} \qquad (4\text{-}25)$$

b - In case of using one row of bent-up bars that are inclined with an angle α to the axis of the member, the nominal shear strength of the bent-up bars shall be calculated using the following equation:

$$\frac{A_{sb}}{b.d} = \frac{q_{sub}}{\left[\left(\frac{f_y}{\gamma_s}\right)(\sin\alpha)\right]} \qquad (4\text{-}26)$$

In such a case, the value of q_{sub} shall not exceed the following value:

$$q_{sub} \le 0.24\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}27)$$

4-2-2-1-6 General requirements for web reinforcement

a - The minimum percentage of web reinforcement in beams shall not be less than:

$$\mu_{min} = \frac{0.4}{f_y} \qquad (4\text{-}28)$$

In which f_y is an N/mm^2

The percentage μ_{min} shall not be less than the following values:

0.15 for normal mild steel 240/350
0.10 for high grade steel

In all cases, the stirrups shall not be less $5\phi 6$ mm/m

b - For beams with web width greater than or equal to 400mm, as well as for beams having width greater than the height, stirrups having four branches shall be provided such that the distance between the branches shall not be more than 250mm.

c - For beams in which the width is more than the effective depth, the minimum percentage of web reinforcement given by equation (4-28) shall be reduced to the value given by:

$$\mu_{min} = \frac{A_{stmin}}{b.s} = \left(\frac{0.40}{f_y}\right)\left(\frac{q_u}{q_{cu}}\right) \qquad (4\text{-}29)$$

Where

$$\frac{q_u}{q_{cu}} < 1$$

d - The following structural members shall be designed and the thickness their cross-section shall be determined based on the assumption that shear strength shall be provided by concrete only according to equation (4-30):

1 - Footings and slabs.

2 - Beams in which the thickness is not more than 250mm or 2.5 times the flange thickness or one-half the web thickness, whichever is greater. Such a case represents the cases of hidden beams and hollow-block slabs.

$$q_{cu} = 0.16\sqrt{\frac{f_{cu}}{\gamma_c}} \quad \geq q_u \qquad (4\text{-}30)$$

e - The yield stress of steel used as web reinforcement shall not be taken more than 400 N/mm^2.

f - The horizontal distance between stirrups, measured in the direction of member axis, shall not be more than 200mm. For bent-up bars, this distance shall not be more than effective depth, d.

g - The horizontal distance between bent-up bars can be increased to 1.5d provided that the nominal shear stress is not more than one-and-half the shear strength of concrete and it can be increased to 2d if the nominal shear stress is not more than twice the shear strength of concrete.

h - The web reinforcement shall be considered to be effective in the case any line inclined at an angle 45o extended from the mid-depth of the beam to the face of the support crosses one of the bent-up bars through its effective length as shown in Fig. (4-8).

i - Construction joints shall not be made at regions of high shear stresses.

j - In cases where the load is applied near the bottom of the concrete section, sufficient amount of stirrups shall be provided to transmit the load to the top surface of the section. Such hanger reinforcement shall be in addition to the shear reinforcement.

k - In case of providing a construction joints at a section subjected to shear force, the joint shall be designed according to Section (4-2-2-4).

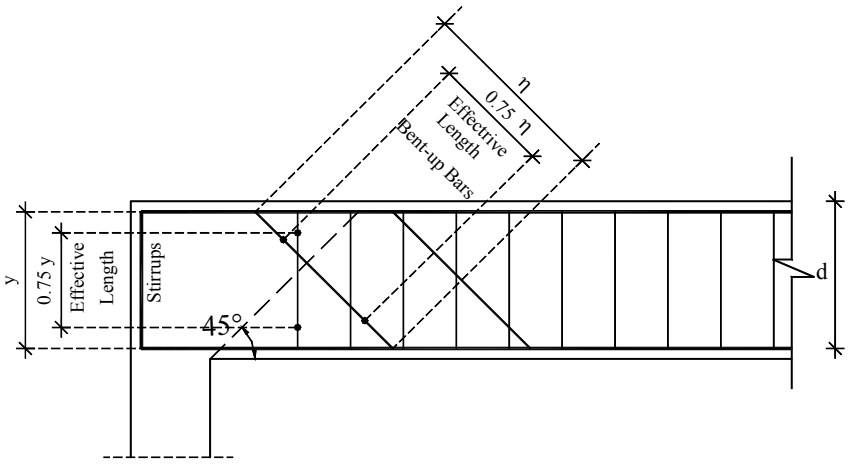


Figure (4-8) : Effective web reinforcement

4-2-2-1-7 D-regions

d - Regions in beams such as locations of openings, cross-section change and locations of concentrated loads shall be designed using the strut-and-tie method according to Section (6-11).

4-2-2-2 Slabs and footings

1 - Slabs and footings shall be designed and their thicknesses shall be determined based on having the shear strength provided by concrete only according to equation (4-30).

2 - The punching shear strength is calculated according to Section (4-2-2-3).

4-2-2-3 Punching shear

a - The critical section for calculating the punching shear stresses around concentrated loads in slabs and footings is located at a distance d/2 from the perimeter of the concentrated force.

b - The punching shear stress shall be calculated from the following equation:

$$q_{up} = \frac{Q_{up}}{(b_o . d)} \qquad (4\text{-}31)$$

Where, b_o is the length of the perimeter of the critical section as shown in Fig. (4-9).

c - When calculating the punching shear stress, the effects of the moments transferred from the flat slabs to the columns shall be considered according to section (6-2-5-8).

d - The nominal concrete punching shear strength shall be calculated as the smaller value of the following:

$$q_{cup} = 0.8\left(\frac{\alpha . d}{b_o} + 0.2\right)\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}32\text{-a})$$

$$q_{cup} = 0.316\left[0.5 + \left(\frac{a}{b}\right)\right]\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}32\text{-b})$$

Where a, b are the smaller and bigger dimensions of the rectangular loaded surface, respectively. For other loading surfaces that are not rectangular in shape, the values of the dimensions, (a and b) shall be determined considering an effective loading surface. Such an effective loading surface shall be taken as the smallest one and the bigger dimension, b is the longest dimension of the effective loading surface. The smaller dimension, a shall be the longest dimension normal to the dimension b from the loading surface, b_o is the length of the perimeter of the critical section and d is the effective depth of the slab as shown in Fig. (4-9-d) for an L-shaped loading surface. The factor α equals 4 for an interior column, 3 for and exterior column and 2 for a corner column. The value q_{cup} shall not be more than the following value:

$$q_{cup} \le 0.316\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}33)$$

With a limiting value of 1.6 N/mm^2.

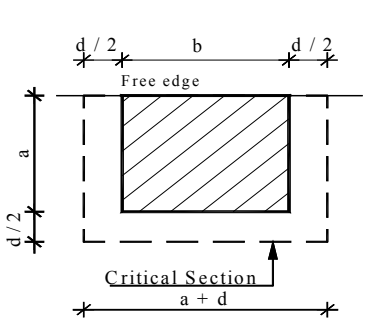


Fig. (4.9.1.B) Edge Column

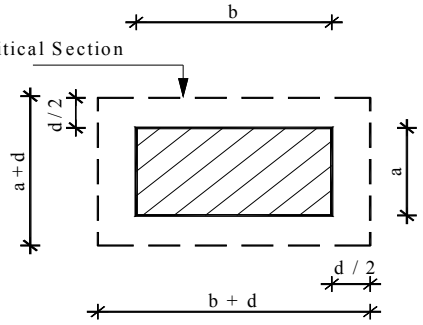


Fig. (4.9.1.A) Interior Column

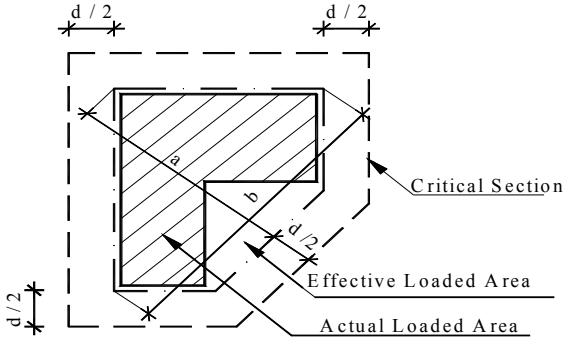


Fig. (4.9.1.D) Non-Rectangular Column

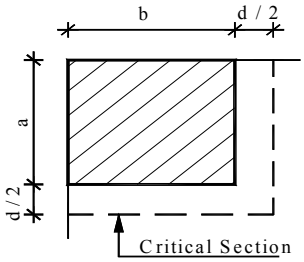


Fig. (4.9.1.C) Corner Column

Figure (4-9) : Critical sections for punching shear

e - The thickness of a slab or a footing required for resisting the punching shear shall be determined based on having the punching shear resisted by concrete only without any contribution provided by reinforcing steel, as follows:

$$q_{cup} \geq q_{up} \qquad (4\text{-}34)$$

4-2-2-4 Shear friction

a - The provisions of this section shall be applied in the cases where the shear forces are transmitted by friction such as in the cases of construction joints.

b - Concrete shear strength shall be neglected and the full amount of the shear force shall be transmitted through reinforcing steel conforming to the following equation:

1 - In case of placing the reinforcing steel normal to the shear plane:

$$A_{sf} = \frac{Q_u}{\mu_f \left(\frac{f_y}{\gamma_s}\right)} + \frac{N_u}{\left(\frac{f_y}{\gamma_s}\right)} \qquad (4\text{-}35)$$

Where μ_f is the coefficient of friction given in the following subsection (c) and N_u is the force normal to the shear plane taken positive if tensile and negative if compressive.

2 - In case of placing the reinforcing steel resisting the shear friction at an angle α_f with the shear plane:

$$A_{sf} = \frac{Q_u}{\left[\left(\frac{f_y}{\gamma_s}\right)\left(\mu_f \sin\alpha_f + \cos\alpha_f\right)\right]} + \frac{N_u}{\left[\left(\frac{f_y}{\gamma_s}\right)\sin\alpha_f\right]} \qquad (4\text{-}36)$$

Where;

$$0 < \alpha_f < 90^o$$

c - The coefficient of shear friction μ_f shall be taken as follows:

- For concrete cast monolithically $\mu_f = 1.2$
- For concrete cast at construction or casting joints provided that roughening the surface such that the roughening depth is within 5mm $\mu_f = 0.8$
- Similar to the previous condition but the roughening depth is less than 5mm and also for the case of fixing structural steel elements on concrete elements $\mu_f = 0.5$

d - In addition to the preceding conditions, the stress due to shear friction Q_u/A_c shall not be more than the value, $0.225\, f_{cu}/\gamma_c$,where Ac in the area of the concrete section resisting shear with a limiting value of 5.0 N/mm^2.

e - The value of f_y shall not be taken more than 400 N/mm^2.

f - In case of sections subject to tension force in addition to the shear force, the area of the reinforcing steel resisting shear shall be increased by the value needed to resist the tension force according to equations (4-35) and (4-36).

4-2-2-5 Brackets and corbels (short cantilevers)

a - Brackets and corbels are cantilevers having shear span-to-depth ratios (a/d) not greater than unity. The provisions of this section apply to short cantilevers in which the depth at the end (outside edge of the bearing area) is not less than one-half its value at the face of the support.

b - Main reinforcement in short cantilevers shall be taken as the greater value of:

$$A_s = A_n + A_f \qquad (4\text{-}37\text{-a})$$

$$A_s = A_n + (2/3)\, A_{sf} \qquad (4\text{-}37\text{-b})$$

The ratio of the main reinforcement $\left(\mu = \frac{As}{bd}\right)$ shall not be less than $0.03\frac{f_{cu}}{f_y}$ In which A_f = Area of main flexural steel reinforcement of the cross-section of the corbel at the face of the support that resists a bending moment having a value of:

$$M_u = Q_u . a + N_u \left(t + \Delta - d\right) \qquad (4\text{-}38)$$

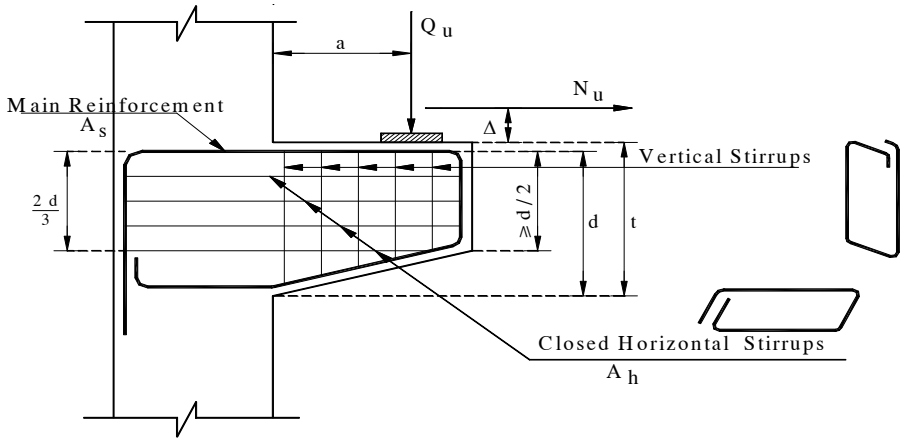


Figure (4-10) Short cantilevers

Such an area shall be determined in accordance with Section (4-2-1-2) for cross-sections subject to bending moments. Q_u is the value of the maximum shear force that shall not exceed the value given in Section (4-2-2-4-d).

A_n = Area of reinforcing steel required to resist a tension force N_u and is calculated from the following relation:

$$A_n = \frac{N_u}{\left(\frac{f_y}{\gamma_s}\right)} \qquad (4\text{-}39)$$

The force N_u shall be regarded as a live load and its design value shall not be less than $0.2Q_u$. The effect of the breaking force shall be taken into account in calculating the torsional moment and the bending moment.

A_{sf} = Area of reinforcing steel required resisting shear force Q_u through friction and its value shall be determined in accordance to Section (4-2-2-4-b).

c - The horizontal reinforcement (A_h) parallel to the main reinforcement: Horizontal closed stirrups shall be uniformly distributed within the upper two-thirds of the effective depth (Fig. 4-10). The required area of such reinforcement shall be given by:

$$A_h = 0.5\ (A_s - A_n) \qquad (4\text{-}40)$$

d - For cases where short cantilevers subject to torsional moments resulting from eccentric vertical or horizontal loading, vertical stirrups satisfying the requirements of torsional resistance of the cross-section shall be provided. In all cases, web reinforcement shall not be less than the minimum code requirements given in Section (4-2-3).

e - Bearing strength underneath the loading plate shall be checked in accordance to Section (4-2-4). Bearing area of load on bracket or corbel shall not project beyond straight portion of main tension bars as shown in Fig. (4-10).

f - Short cantilevers can be designed using the Strut-and-Tie method according to Section (6-11) with due consideration of the

requirements of items (b) and (c) while satisfying equations (4-39) and (4-40).

4-2-2-6 Deep beams in shear

The provisions of this section shall apply to beams with an effective span-to-depth ratio equal to or less than four ($L/d \leq 4.0$).

4-2-2-6-1 Web reinforcement in deep beams using the empirical design method

a - The requirements of this section shall apply to deep beams defined in Section (6-3-2-2) with L/d ≤ 1.25 for simply supported beams and L/d $\leq$ 2.5 for continuous beams and for cases of loading in which loads are applied on the top of the beam as well as for cases of loading in which loads are applied on the compression sides.

b - Critical sections for shear are taken at the following distances, measured from the face of the support,:

1- 0.15Ln for uniform loads where Ln is the clear span of the beam.

2- 0.5a for a concentrated load at a distance (a) from the face of the support.

In both cases, the distance shall not exceed d/2 where d is the effective depth.

c - Nominal ultimate shear stress shall be calculated from:

$$q_u = \frac{Q_u}{(b.g)} \qquad (4\text{-}41)$$

Where g is the effective depth or the clear beam span, whichever is smaller

d - The value of qu shall not be more than the value given by equation (4-16), multiplied by the coefficient δ_d, which shall be given by:

$$\delta_d = \left(\frac{1}{3}\right)\left[2 + \left(\frac{0.4\,L_n}{d}\right)\right] \qquad (4\text{-}42)$$

e - Shear strength of concrete shall be calculated by multiplying the value given by equation (4-18) for ordinary beams by the coefficient δ_{dc} given by:

$$\delta_{dc} = 3.5 - 2.5\left[\frac{M_u}{Q_u . d}\right] \qquad (4\text{-}43)$$

where;

M_u = Value of the maximum moment at the critical section for shear and $1.0 \langle \delta_{dc} \langle 1.9$

The value of q_{cu} in deep beams shall not be more than:

$$q_{cu} \le 0.46\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad N/mm^2 \qquad (4\text{-}44)$$

f - If the value of the nominal ultimate shear stress is more than the value of the concrete ultimate shear strength, the shear strength of the web reinforcement shall be calculated from:

$$q_{su} = q_u - 0.5\ q_{cu} \qquad (4\text{-}45)$$

The web reinforcement shall be calculated according to equation (4-46).

g - Web reinforcement shall be designed according to the following relation:

$$q_{su} = \delta_v . q_{suv} + \delta_h . q_{suh} \qquad (4\text{-}46)$$

where q_{suv} and q_{suh} shall be calculated as follows:

$$q_{suh} = \left(\frac{A_h}{s_h}\right)\left(\frac{f_y}{b.\gamma_s}\right) \qquad (4\text{-}46\text{-b})$$

$$q_{suv} = \left(\frac{A_v}{s_v}\right)\left(\frac{f_y}{b.\gamma_s}\right) \qquad (4\text{-}46\text{-c})$$

The coefficients δ_v and δ_h shall be calculated as follows:

$$\delta_h = \frac{11 - \left(\frac{L_n}{d}\right)}{12} \qquad (4\text{-}46\text{-}d)$$

$$\delta_v = \frac{1 + \left(\frac{L_n}{d}\right)}{12} \qquad (4\text{-}46\text{-}e)$$

Where;

A_h = Area of horizontal web reinforcement parallel to the primary reinforcement
A_v = Area of vertical web reinforcement perpendicular to the primary reinforcement
S_h = Spacing between horizontal web reinforcement
S_v = Spacing between vertical web reinforcement
L_n = Clear span of a deep beam

h - Web reinforcement required for resisting shear stresses shall be continued uninterrupted along the span of the deep beam.

i - The minimum percentage of web reinforcement in deep beams satisfying the conditions of this section (4-2-2-6-1) shall not be less than the following:

1 - Vertical reinforcing steel perpendicular to the beam axis

- Normal-mild steel 240/350 $\frac{A_v}{b.s_v} \geq 0.0020$

  The value of s_v shall not be more than 200mm

- High-tensile steel $\frac{A_v}{b.s_v} \geq 0.0015$

  The value of s_v shall not be more than 200mm

2 - Horizontal reinforcing steel parallel to the beam axis

- Normal-mild steel 240/350 $\frac{A_h}{b.s_h} \geq 0.0030$

  The value of s_h shall not be more than 200mm

- High-tensile steel $\frac{A_h}{b.s_h} \geq 0.0015$

  The value of s_h shall not be more than 200mm

4-2-2-6-2 Web reinforcement in deep beams analyzed according to the strut-and-tie model

a - Deep beams having effective span-to-depth ratio less than or equal to 4.0 can be designed using strut-and-tie model according to sections (6-11) and (3-2-3-6) with due consideration to Sections (4-2-2-6-2) and (4-2-2-6-3).

b - The minimum percentage of web reinforcement in deep beams satisfying the conditions of this section (4-2-2-6-1) shall not be less than the values given in Section (4-2-2-6-1-i).

c - The minimum percentage of web reinforcement in deep beams in which the effective span-to-depth ratio satisfies the relation $1.25 < L/d \leq 4.0$ for simple beams and the relation $2.5 < L/d \leq 4.0$ for deep beams shall be as follows:

1 - Vertical reinforcing steel perpendicular to the beam axis

- Normal-mild steel 240/350 $\frac{A_v}{b.s_v} \geq 0.0030$

  The value of s_v shall not be more than 200mm
- High-tensile steel $\frac{A_v}{b.s_v} \geq 0.0025$

  The value of s_v shall not be more than 200mm

2 - Horizontal reinforcing steel parallel to the beam axis

- Normal-mild steel 240/350 $\frac{A_h}{b.s_h} \geq 0.0020$

  The value of s_h shall not be more than 200mm
- High-tensile steel $\frac{A_h}{b.s_h} \geq 0.0015$

  The value of s_h shall not be more than 200mm

d - For deep beams subjected to concentrated loads, the minimum percentage of web reinforcement in regions having shear span-to-depth ratio less than two (a/d $<$ 2) shall not be less than the values given in Section c provided that such values are not less than those given in Section b for regions in which a/d $<$ 1.0.

4-2-2-6-3 Deep beams supporting loads resulting in tension at the loaded edges

a - deep beams supporting loads resulting in tension at the loaded edges shall be provided with vertical web reinforcement sufficient to transfer the load to a height that is equal at least to one-half of the span. Such vertical reinforcement shall be added to web reinforcement needed for top loaded deep beam. The strut-and-tie model can also be used for the design of such cases.

b - Bottom loaded deep beams subjected to tension at their loaded edges can be designed for shear as shallow beams according to Section (4-2-2-1-2).

4-2-3 Ultimate torsion strength limit state

4-2-3-1 Sections subject to torsion

For calculating the ultimate shear stresses resulting from ultimate torsional moments, critical sections shall be taken at the location of maximum torsion. For cases in which the maximum torsion is at the support, the critical section can be considered located at d/2 from the face of the support.

4-2-3-2 Nominal ultimate shear stresses resulting from torsion

a - The nominal shear stresses developed in a solid section subject to torsional moment shall be evaluated using the following relation:

$$q_{tu} = \frac{M_{tu}}{(2\,A_o \,.\, t_e)} \qquad (4\text{-}47)$$

Where A_o is the area enclosed by the shear flow path and t_e is the wall thickness of a box-section that is equivalent to the original solid section. In lieu of the availability of more exact procedures, the value of A_o can be taken equal to 0.85 A_{oh} where A_{oh} is the area enclosed by the centerlines of the outermost closed stirrups utilized for resisting the torsional moment and $t_e = A_{oh}/P_h$, where P_h is the perimeter of the centerlines of the outermost closed stirrups utilized for resisting the torsional moment (Fig. 4-11).

b - The value of the nominal ultimate shear stresses for T- or L-sections can be calculated by neglecting the contribution of the effective part of the slab and treating the section as a rectangular section.

c - In case of considering the effect of the effective part of the slab when calculating the nominal ultimate shear stresses for T- or an L-sections, the following measures shall be taken:

- The effective width of the slab, measured from the face of the web, shall not be more than three times the slab thickness as shown in Fig. (4-12).
- The effective part of the slab shall be provided with web reinforcement to ensure its efficiency in resisting torsion.

b - Box-Section

The nominal ultimate shear stresses shall be calculated for box-sections using equation (4-47) in which the smaller thickness of $t_e = A_{oh}/P_h$ or the actual smallest wall thickness shall be used.

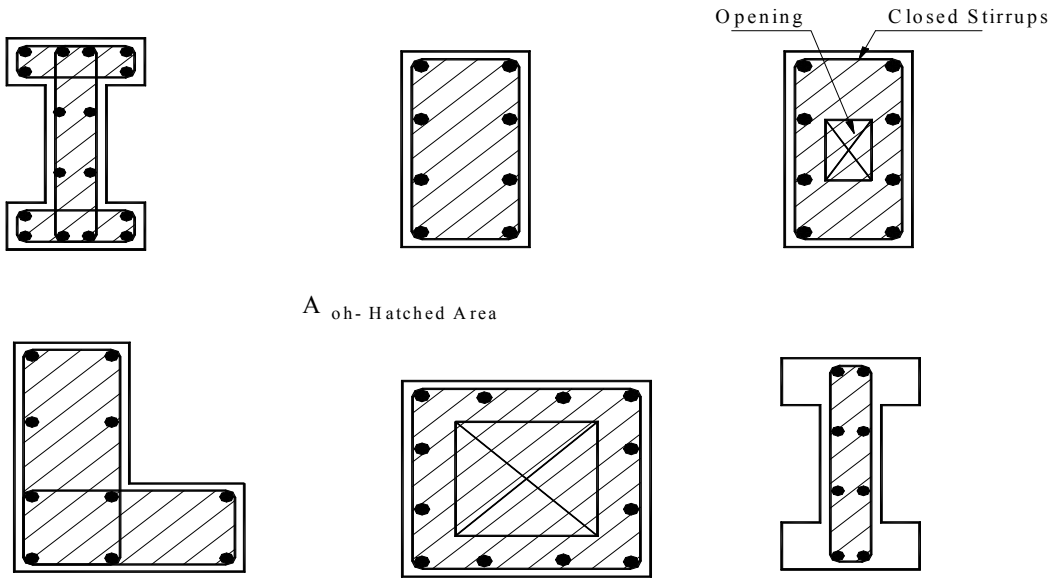


Fig. (4-11) Definition of A_{oh}

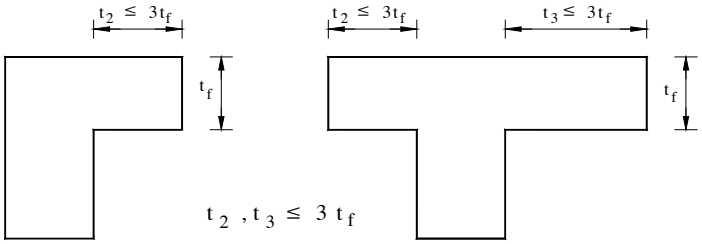


Fig. (4-12) The effective width of the slab

4-2-3-3 The effect of torsion shall be neglected for cases in which the nominal ultimate shear stresses resulting from the ultimate torsion are less than the values given by equation (4-17) or table (4-4).

4-2-3-4 The concrete dimensions of sections subject to combined shear and torsion provided with web reinforcement as well as longitudinal reinforcement shall satisfy the following relationship:

For solid sections:

$$\sqrt{(q_u)^2 + (q_{tu})^2} \le q_{umax} \qquad (4\text{-}48)$$

For box sections:

$$q_u + q_{tu} \le q_{umax} \qquad (4\text{-}49)$$

q_u and q_{tu} shall be calculated using equations (4-13) and (4-47), respectively, and q_{umax} shall be evaluated from equation (4-16) or from Table (4-3).

4-2-3-5 Reinforcing steel for resisting the shear stresses resulting from combined shear and torsion

a - If the value of stresses q_{tu} calculated from equation (4-47) of Section (4-2-3-2) is more than the value evaluated from equation (4-17) of Section (4-2-2-1-2-d) but not more than the value q_{umax} evaluated from equation (4-48) or equation (4-49) of Section (4-2-3-4), reinforcing steel consisting of closed stirrups normal to the axis of the member and longitudinal steel shall be used for resisting

the torsional moment. Such reinforcing steel shall be added to any other reinforcing steel needed for resisting bending moments, axial forces and shear forces according to Table (4-5).

table (4-5) Transverse reinforcement for resisting combined shear and torsion

	$q_{tu} \le 0.06\sqrt{\frac{f_{cu}}{\gamma_c}}$ N/mm^2	$q_{tu} > 0.06\sqrt{\frac{f_{cu}}{\gamma_c}}$ N/mm^2
$q_u < q_{cu}$	Min. shear reinforcement according to item (4-2-2-1-6)	Reinforcement for Resisting q_{tu}
$q_u > q_{cu}$	Reinforcement for Resisting $(q_u - q_{cu}/2)$	Reinforcement for Resisting q_{tu} and $(q_u - q_{cu}/2)$

b - Transverse reinforcing steel required to resist torsion shall be in the form of closed stirrups or welded-wire fabrics. The area of one branch of the stirrup shall be calculated from:

$$A_{str} = \frac{M_{tu} \cdot s}{2\, A_o \left(\frac{f_{yst}}{\gamma_s}\right)} \qquad (4\text{-}50)$$

Where;

A_o = 0.85 A_{oh} as previously defined in Section (4-2-3-2) and A_{oh} is the area enclosed by the centerline of the outermost transverse reinforcement used to resist torsion.

For rectangular sections, equation (4-50) shall take the following form:

$$A_{str} = \frac{M_{tu} \cdot s}{1.7\,(x_1 \cdot y_1)\left(\frac{f_{yst}}{\gamma_s}\right)} \qquad (4\text{-}51)$$

Where:

A_{str} = cross-sectional area of one branch of a stirrup needed to resist torsion (mm^2)

f_{yst} = Yield stress of reinforcing steel used for stirrups resisting torsion with a limiting value of 400 N/mm^2.

X1 = Width of a rectangular stirrup measured between its

axes (Fig. 4-13)

Y1 = Length of a rectangular stirrup measured between its axes (Fig. 4-13)

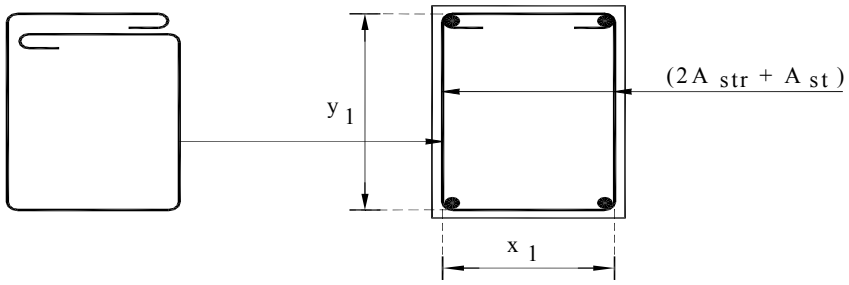


Fig. (4-13) : Detailing of Reinforcement Resisting Torsional Moments and Shear forces (Tow-branch stirrups)

With due consideration of the following:

- The required cross-sectional area of the stirrups, calculated in accordance with section (4-2-3-5), needed to resist shear and torsion shall not be less than the area evaluated from the following equation:

$$\left(2\,A_{str} + A_{st}\right) \geq 0.4\,\frac{(s.b)}{f_{yst}} \qquad (4\text{-}52)$$

  Where f_{yst} is in N/mm^2 and b is the width of the solid section or the summation of the widths of the webs of box sections.

- The distance, s between the stirrups shall not be more than $P_h/8$ or 200 mm, whichever is smaller, where P_h is the length of the perimeter of the transverse reinforcement used for resisting torsion.

- For cross sections having stirrups with more than two branches, only the outer two-branch stirrup shall be considered effective in resisting torsion as shown in Fig. (4-14).

  For box sections, it is permitted to use transverse and longitudinal reinforcement on the inner as well as the outer perimeters as long as the wall thickness t_w is less than or equal to one-sixth the overall width of the cross section. If the wall thickness is more than one-sixth the overall width of the cross section, torsion shall be resisted by reinforcement arranged on the outer perimeter only.

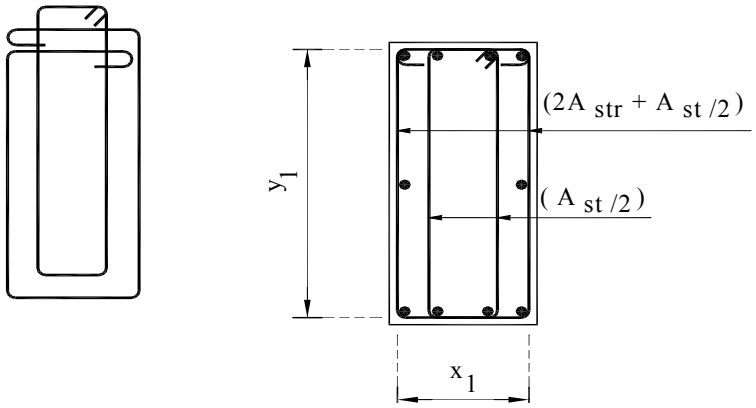


Figure (4-14) Detailing of reinforcement resisting torsional moments and shear forces (stirrups having more than two branches)

c - Additional longitudinal reinforcement for resisting torsion: The area of the additional longitudinal reinforcement shall be determined from equation (4-53-a,b)

$$A_{sl} = \left(\frac{A_{str} \cdot p_h}{s}\right)\left(\frac{f_{yst}}{f_y}\right) \qquad (4\text{-}53\text{-a})$$

The area of the longitudinal reinforcement shall not be less than:

$$A_{slmin} = \frac{0.4\ \sqrt{\frac{f_{cu}}{\gamma_c}}\ A_{cp}}{f_y/\gamma_s} - \left(\frac{A_{str}}{s}\right) p_h \left(\frac{f_{yst}}{f_y}\right) \qquad (4\text{-}53\text{-b})$$

Where ;

A_{cp} is the gross area of the cross section including any openings and f_{cu}, f_y, f_{yst} are in N/mm^2.

The value of A_{str}/s shall not be less than $\frac{1}{6}\frac{b}{f_{yst}}$

The longitudinal reinforcement shall be distributed on the internal perimeter of the exterior stirrup with due consideration of the following:

- The diameter of the bars used as longitudinal reinforcement shall not be less than the distance between the bars divided by 15 or 12mm, whichever is smaller.
- The additional longitudinal reinforcement shall be uniformly distributed inside the perimeter of the exterior stirrups such that the distance between the bars is not more 300mm.
- A longitudinal bar shall be placed at each corner of the cross-section.
- The longitudinal reinforcement required for torsion shall be added to that required for bending moments.
- Both the transverse and the longitudinal reinforcement required to resist torsion shall be extended beyond the last critical section b a distance equal to one-half the length of the perimeter of the stirrups.

4-2-3-6 Redistribution of torsion in statically indeterminate structures

The design of the cross sections and the calculation of the reinforcing steel shall follow the provisions of this chapter with due consideration of the following:

a - Redistribution of torsion shall not permitted in statically indeterminate structures in which torsion is necessary for achieving equilibrium.

b - Redistribution of torsion shall be permitted in statically indeterminate structures in which torsion is not necessary for achieving equilibrium. In such case, torsion results from compatibility of strains and the ultimate torsion can be reduced to a value equals to the cracking torsion according to the following equation:

$$M_{tu} = 0.316 \left(\frac{A^2_{cp}}{p_{cp}} \right) \sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (4\text{-}54)$$

Where;

A_{cp} is the gross area of the cross section including any openings and P_{cp} is the outer perimeter of the cross section. In such a case, bending moments and shear forces shall be redistributed in the adjacent spans.

4-2-3-7 Torsional rigidity of a concrete section

a - The torsional rigidity of a rectangular section (G.C) can be calculated by considering the value of the shear modulus (G) equals 42% of the value of the modulus of elasticity of concrete calculated according to Section (2-3-3-1) and by evaluating the torsion constant (c) according to equation (4-55-a). The torsion constant (c) can be evaluated for L- , T- and box-sections by dividing them into a number of rectangles according to equation (4-55-b).

$$C = \beta\, b^3\, t\, \eta \qquad (4\text{-}55\text{-a})$$

$$C = \Sigma \beta\, b^3\, t\, \eta \qquad (4\text{-}55\text{-b})$$

Where;

η = 0.70 for rectangular sections before cracking resulting from torsional moment not exceeding cracking torsion calculated according to equation (4-54).

η = 0.20 for rectangular sections after cracking

β = Coefficient depending on t/b ratio given in Table (4-6)

Table (4-6) : Values of the coefficient β for calculating the torsional rigidity

t / b	1	1.5	2	3	5	>5
β	0.14	0.20	0.23	0.26	0.29	0.33

b - For cases required more accuracy, torsional rigidity can be calculated using theories of structural mechanics.

4-2-4 Ultimate bearing strength limit state

4-2-4-1 Design ultimate bearing strength

The design ultimate bearing strength limit state shall not exceed $0.67\, A_1 \frac{f_{cu}}{f_y}$

Where;

A1 is the loaded bearing area

Cases mentioned in Sections (4-2-4-2) and (4-2-4-3) shall be excluded.

4-2-4-2 When the supporting surface is wider on all sides than the loaded area, the design ultimate bearing strength on the loaded area shall be equal to the value given in Section (4-2-4-1) multiplied by the factor $\sqrt{A_2/A_1}$ but not more than 2.

Where A_2 = the maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area.

4-2-4-3 When the supporting surface is sloped or stepped, A_2 may be taken as the area of the lower base of the largest frustum of a right pyramid or cone having for its upper base the loaded area, and having side slopes of 1 vertical to 2 horizontal contained wholly within the support (Fig. 4-15).

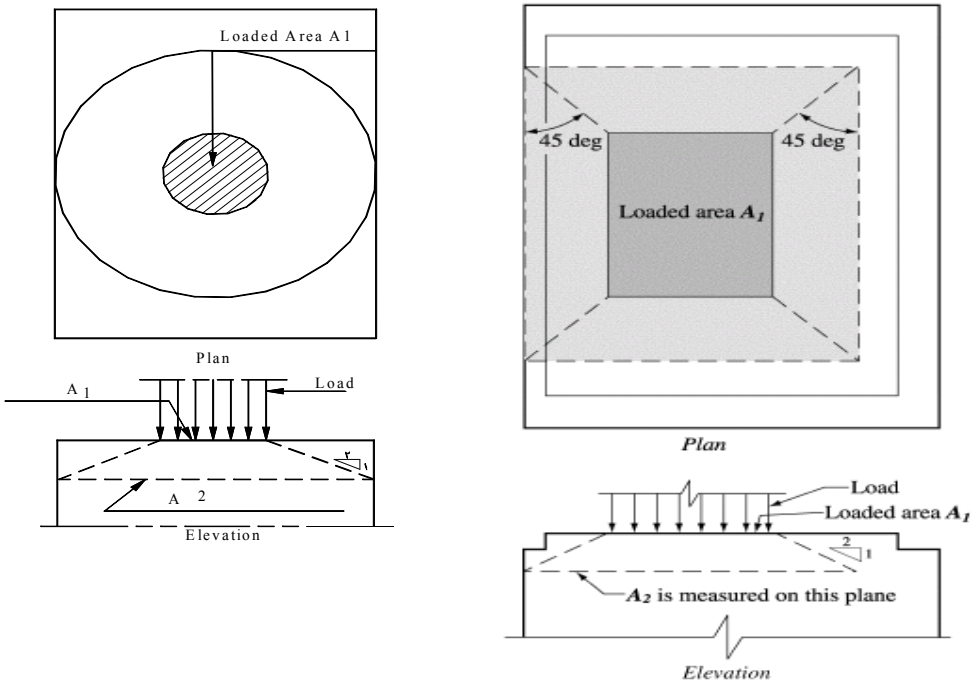


Figure (4-15) Determination of area A_2 in stepped or sloped supports

4-2-5 Development length, embedment length and splices of reinforcement

4-2-5-1 Development length

a - The calculated tension or compression in reinforcement at each section of reinforced concrete members shall be developed on each side of that section by development length L_d proportional to the force in the reinforcing bar at that section. The development length shall be measured from the critical sections of maximum tension or compression for bent-up bars or for bars that are no longer required as well as at locations of splices.

b - The development length L_d of reinforcing bars subject to tension or compression f_y/γ_s shall be computed from the following equation:

$$L_d = \frac{\alpha . \beta . \eta \left(\frac{f_y}{\gamma_s} \right)}{4 f_{bu}} . \phi \qquad (4\text{-}56)$$

Where:

f_{bu} = ultimate bond strength of concrete with reinforcing steel, and can be determined from the following relation:

$$f_{bu} = 0.30 \sqrt{\frac{f_{cu}}{\gamma_c}} \qquad N/mm^2 \qquad (4\text{-}57)$$

ϕ = Nominal bar diameter

α = Correction coefficient depending on end bar shape (Table 4-7)

β = Correction coefficient depending on bar surface condition (Table 4-8)

η = factor depending on bar location with respect to casting surface. It is equal 1.3 for tension reinforcement so placed that more than 300 mm of concrete is cast below the reinforcement. It equals 1.00 for all other cases.

c - The development length for reinforcing steel bars subject to tension or compression shall not be less than:

35 ϕ or 400 mm – whichever is bigger- for smooth bars with hooks
40 ϕ or 300 mm - whichever is bigger- for deformed bars

d - The distance between the bars and the concrete cover shall be taken in accordance to chapter (4) and chapter (7) with due consideration of the requirements of section (4-2-5-1-f) for the case of bundled bars.

e - The development length for individual bars may be from Table (4-9) with due consideration to the values of η.

f - The development length for bundled bars shall be calculated from equation (4-56) considering the bundle as an individual bar having an equivalent diameter ϕ_e . The equivalent diameter of a bundle consisting of bars of equal diameter shall be calculated as follows:

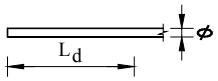
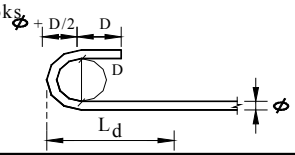
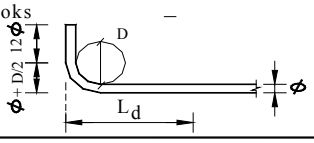
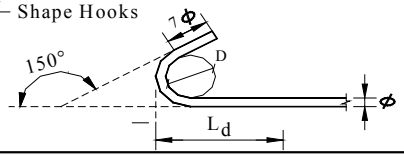
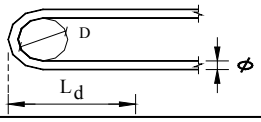
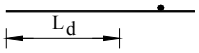
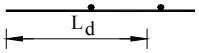
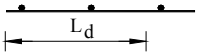
- In case of using a two-bar bundle $\phi_e = 1.4\ \phi$
- In case of using a three-bar bundle $\phi_e = 1.7\ \phi$

The equivalent bar diameter ϕ_e shall be used in the calculations related to the minimum concrete cover for the bundle according to Section (4-3-2-3-b) and Table (4-13) and for calculating the clear distance between adjacent bundles (Section 7-3-3-1) and for satisfying the limit state of cracking (Section 4-3-2-3-a). The thickness of the concrete cover of the bundle (c) and the clear distance between the bars of the adjacent bundles (a and b) shall be calculated according to the actual arrangement of the bars in the cross-section as shown in Fig. (7-2-b).

g - For elements subject to bending moments, it shall be permitted to reduce the development length if the existing area of reinforcing steel in the cross-section is in excess of that required by the analysis. The reduction factor shall be $\frac{A_{s,reqired}}{A_{s,provided}}$. Such a reduction shall not contradict other sections in this code such as the provisions related to the bottom reinforcement extended to the supports in solid slabs (Section 6-2-1-2-3), flat slabs (Section 7-5 and figure 7-4) and the provisions related to the termination of reinforcing bars used for negative bending moments (Section 4-2-5-3-2). It shall not also contradict the provisions related to beams used in moment resisting frames (Sections (6-8-2-2-1), (6-8-2-2-2) and (6-8-2-3-1)) and the provisions related to simply supported precast elements (section 6-9-

5-4). In all cases, the development length shall not be less than the minimum value given in section (4-2-5-1-c)

Table (4-7) Values of the correction factor α

α		End Condition of Bar	Type
Comp.	Tension		
1	1	1- Straight Ends 	Steel Reinforcement
1	0.75	1- U Shape Hooks 	
1	0.75	3- L Shape Hooks 	
1	0.75	4- ∠ Shape Hooks 	
1	0.75		
1	1	1- Straight Ends without cross bars within Ld 	Welded Steel Meshs
0.70	0.70	2- Straight Ends with one cross bars within Ld 	
0.50	0.50	3- Straight Ends with two cross bars within Ld 	

D = 4 ⌀ for steel 240 / 350
D = 6 ⌀ (or Φ) for 25 mm ≥ ⌀ (or Φ) > 6 mm
D = 8 ⌀ (or Φ) for ⌀ (or Φ) > 25 mm } for high grade steel

Table (4-8) Values of the correction factor β*

Surface Condition	In Tension	In Compression
Smooth	1.00	0.7
Deformed	0.75	0.45

* With due consideration of the requirements of section (4-2-5-1-C)

Table (4-9) Development length of individual bars Ld - multiplier of bar diameter (η = 1.0)

Charactristic Strength of Concrete N/mm^2	Type of Reinforcement			
	Normal Mild Steel Bars with kooks *** f_y=240 N/mm^2 or 280 N/mm^2		High Strength Bars** f_y=400 N/mm^2 or 360 N/mm^2	
	In Compression	In Tension	In Compression	In Tension
20	35	38	40	60
25	35	36	40	55
30	35	35	40	50
35	35	35	40	45
40	35	35	40	42
Greater Than or equal to 45	35	35	40	40

** In the case of High strength deformed bars with hooks multiply the values in the table by 0.75

*** It shall not be permitted to use smooth bars without end hooks

4-2-5-2 Anchorage of shear reinforcement

a - Bent-up bars shall be anchored using a length that shall be equal to the development length in tension or compression according to the location of the straight part following the inclined part of the bar. Such length shall be calculated in accordance to section (4-2-5-1-b).

b - Stirrups shall be placed in beams to surround reinforcing bars in tension as well as the compression zone and shall be anchored in the compression zone as shown in Fig. (4-16).

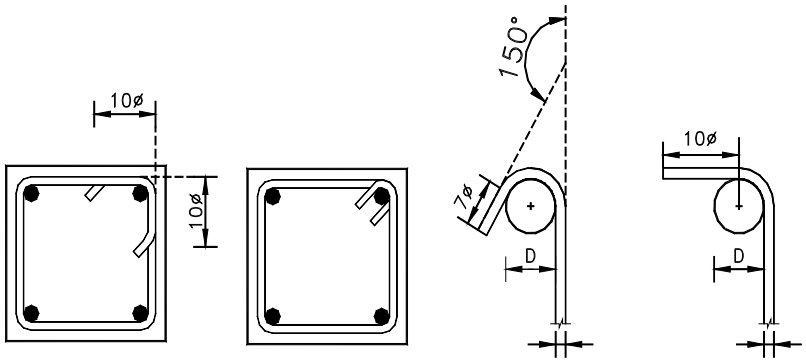


Figure (4-16) Anchorage of stirrups in beams

4-2-5-3 Development of flexural reinforcement

a - Reinforcement shall extend beyond the point at which it shall be no longer required to resist flexure for a distance that shall be not less than (L_d + 0.3d). The anchorage length, defined as the distance between the end of the bars and the section where these bars shall no longer be required to resist the bending moments, shall not be less than d or (L_d + 0.3d) whichever is bigger measured from the original bending moment diagram (Fig. 4-17).

b - Longitudinal bars shall not be terminated in a tension zone. In case of terminating flexural reinforcement in a tension zone, the following conditions shall be satisfied:

1 - The ultimate shear stress at the cutoff point does not exceed two-thirds of ultimate shear strength, including the shear strength of the provided web reinforcement.

$$q_u \le \frac{2}{3}\left(0.5\, q_{cu} + q_{su}\right) \qquad (4\text{-}58)$$

2 - Stirrup area at the section at which longitudinal bars are terminated are in excess of that required for shear and torsion by a value that shall not be less than $A_{st} = \dfrac{0.40 b.s}{f_y}$. The additional stirrups shall be provided along each terminated bar over a distance from the termination point equal to three-fourths of the effective depth of the member (Fig. 4-18). The spacing between these stirrups shall not exceed $\left(\dfrac{d}{8\beta}\right)$ where:

s = distance between stirrups

β = ratio of the area of terminated reinforcement to the total area of section reinforcement.

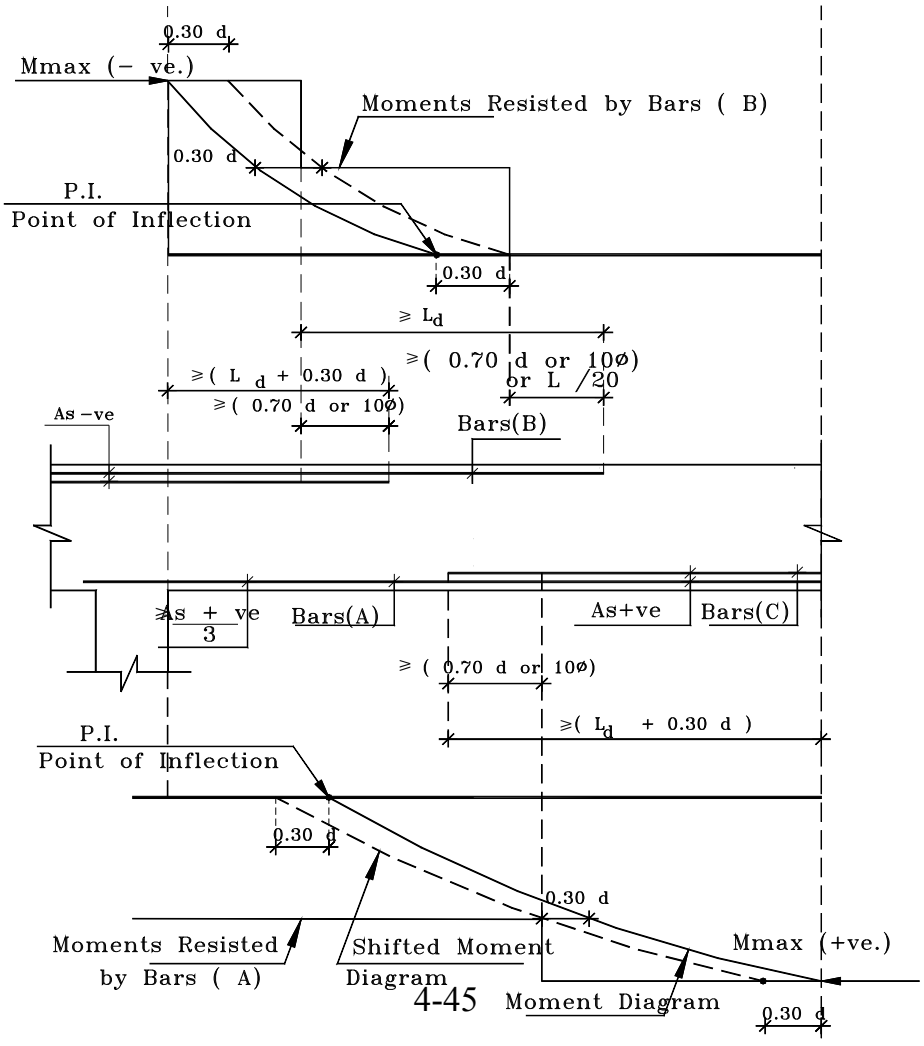


Figure (4-17) Development of reinforcing bars in elements subjected to bending moments

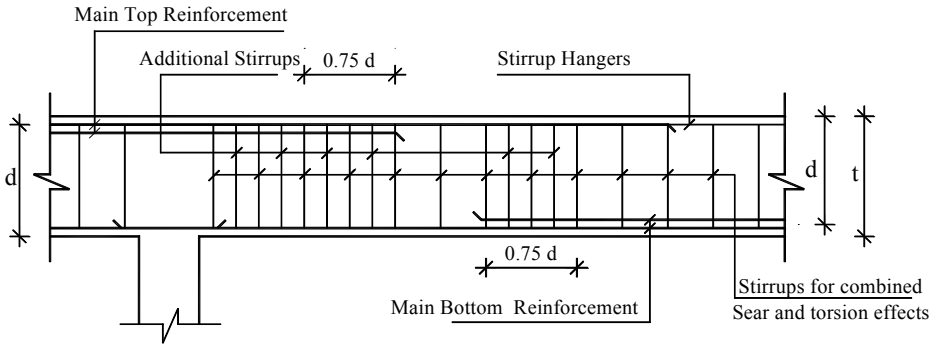


Figure (4-18) Development of bars in tension zone

4-2-5-3-1 Development of positive moment reinforcement

a - At least one-third of the positive moment reinforcement in simply supported members and continuous members shall extend into the support. In beams, such reinforcement shall extend into the support at least 150 mm. All the provisions required for satisfying the anchorage length shall be checked in accordance to Section (4-2-5-3-1-b).

b - At simple supports and at points of inflection in continuous members, the development length of tension reinforcement given in Section (4-2-5-1-b) shall satisfy the following relation (Fig. 4-19):

$$\alpha\left(\frac{M_u}{Q_u}\right) + L_a \geq L_d + 0.3\,d \qquad (4\text{-}59)$$

Where:

M_u = Maximum bending moment for the section reinforced with steel bars that are extended into the support and stressed to $\frac{f_y}{\gamma_s}$

Q_u = Maximum shear force calculated at the section considered

L_a = Length of bar beyond the centerline of the edge support or the length of the bar beyond the point of inflection with a limiting value of d or 12ϕ whichever is bigger.

α = 1.3 for simply supported edges when applied loads result in compression normal to the bottom edge of the support

α = 1.0 for all other cases

4-2-5-3-2 Development of negative moment reinforcement

a - One-third of the total tension reinforcement provided for negative moment at a support shall have an embedment length beyond the point of inflection of a value (0.3d + 10ϕ) or (0.3d + L/20) or d whichever is bigger, measured from the bending moment diagram (Fig. 4-17).

b - All the tension reinforcement provided for negative moment at an edge support shall extend into the support a distance not less than L_d measured from the interior face of the support.

c - When calculating the development length of earthquake resistant structures, the provisions of Section 6-8 shall apply.

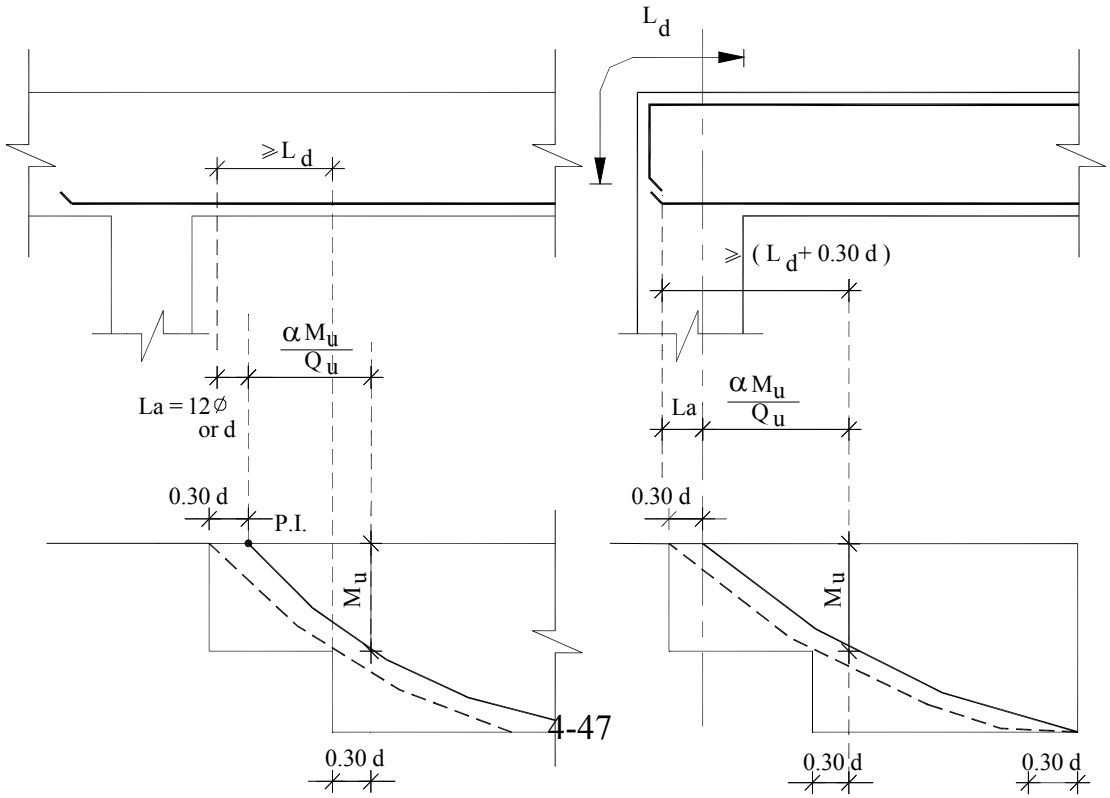


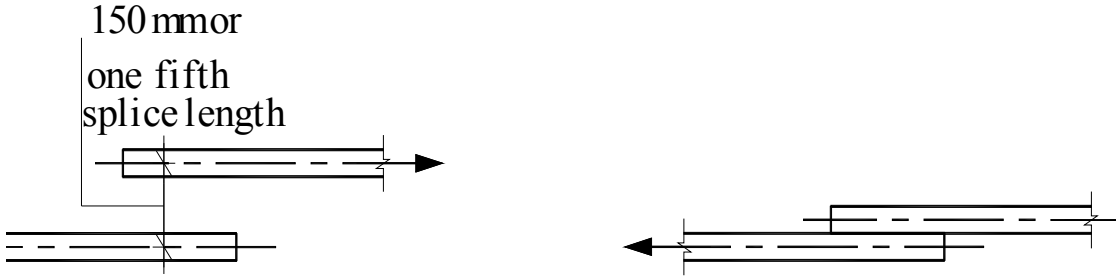
Figure (4-19) Termination of reinforcing bars at inflection points and at simple supports

4-2-5-4 Reinforcement splices

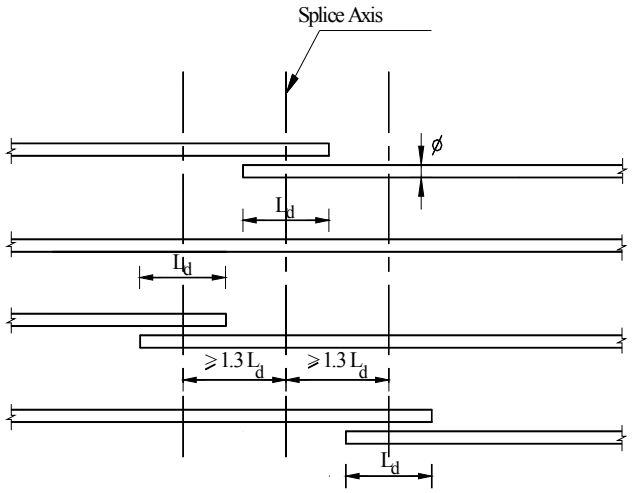
4-2-5-4-1 Splices of reinforcement shall be made only as required or permitted on design drawings or as authorized by the Engineer. They shall be made by lap splices, by welding if permitted according to the type of steel or by mechanical connections with due consideration of not splicing bars at locations of maximum stresses.

4-2-5-4-2 Lap splices

a - Bars can be spliced by contact lap splices (Fig. 4-20-a) or by non-contact lap splices (Fig. 4-20-b). Bars spliced by non-contact lap splices shall not be spaced transversely farther apart than 1/5 of the required lap splice length nor 150 mm.



b – non-contact bars **a – contact bars**



(c)

Figure (4-20) Lap splices

b - When satisfying the previous conditions a and b, the length of the lap splice for bars in tension shall be taken equal to the embedment length Ld provided that the area of the reinforcing bars in the section equal to or more than twice the required area and that area of the spliced bars is not more than 25% the total area of the bars at the section. In case the area of spliced bars is more than 25% of the total area of bars at the section or the area of bars at the section is less than twice the required area, the splice length shall be taken equal to 1.3 the development length L_d in tension.

c - It is permitted to splice all the reinforcing bars in compression at a section. The length of lap splice in compression shall be taken equal to the development length L_d in compression.

d - Lap splices shall not be permitted in tension tie members. Splices in such members shall be made with a full welded splice or a full mechanical connection and splices in adjacent bars shall be staggered by at least 750 mm. The provisions of Section (4-2-5-4-3) shall be satisfied.

e - When splicing bars having different diameters, splice length shall be computed based on the larger diameter.

f - Lap splices of bundled bars shall be based on the lap splice length required for individual bars within a bundle calculated in accordance

to Section (4-2-5-4-2-c), increased by 30%. It shall not be permitted to splice all the bars in the bundle at a certain section.

g - Lap splices shall not be used for bars having diameter more than 28 mm. For such diameters, welded splices or mechanical connections shall be used.

h - When splicing welded bars in tension the splice length shall not be less than the following values:
  1 - For deformed bars, the lap splice length shall be equal to 1.3 L_d but not less than 150 mm (Fig. 4-21).
  2 - For smooth bars, the lap splice length shall be equal to 1.5 L_d but not less than 200 mm (Fig. 4-22).

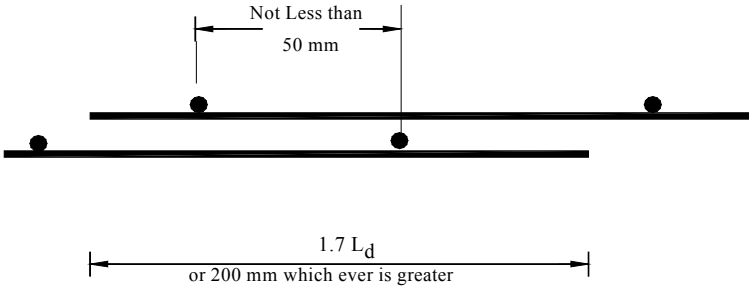


Figure (4-21) Lap splices of deformed fabric

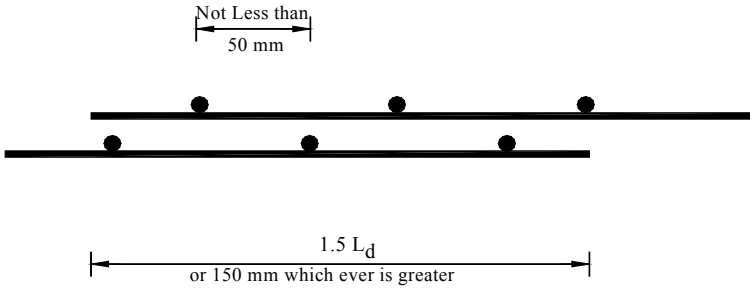


Figure (4-22) Lap splices of smooth fabric

4-2-5-4-3 Welded splices and mechanical connections

a - It shall be permitted to splice bars by welding according to the standard specifications of welding at the points where bars meet each other with due consideration of having the centerlines of the two bars lined-up.

b - A full welded splice or a full mechanical connection shall develop, in tension or compression, at least 125% of the specified yield strength of the spliced bars.

c - Welded splices or mechanical connections not meeting the requirements of Section (4-2-5-4-3-b) may be used provided that the distance between splices shall not be less than 600 mm and the splice strength in tension or in compression is not less than the yield strength.

d - Only electrical welding shall be permitted.

e - Welding shall not be permitted within a distance less than 100 mm from the point at which the bar is hooked provided that internal radius of the hook is not less than 12 times the bar diameter.

f - It shall not be permitted to splice cold-treated bars except after hot-treating the weld zone.

g - It shall not be permitted to splice bars by welding in structures subjected to dynamic loads.

4-3 Serviceability limit states

4-3-1 Deformation and deflection limit states

a - Reinforced concrete structural elements shall subjected to flexure shall be designed to have adequate stiffness to limit deflections or any deformations that adversely affect strength, serviceability and the nonstructural elements of the structure such as flooring and partitions.

b - Deformation and Deflection Limit States shall be satisfied through computing deflections in accordance to Section (4-3-1-1).

c - Cases that satisfy the provisions of Section (4-3-1-3) shall be waived from deflection calculations.

d - The minimum thickness of one-way solid slabs, two-way slabs and flat slabs shall not be less than the values given in Section (6-2-1-2), Section (6-2-1-3) and Section (6-2-5-2), respectively.

4-3-1-1 Calculation of deflections

4-3-1-1-1 Immediate deflections

a - Immediate deflection shall be computed by the theory of elasticity using the modulus of elasticity of concrete according to Equation (2-1) of Section (2-3-3-1) and calculating the effective moment of inertia of the section I_e according to Equation (4-60) with due consideration of the requirements of Section (4-3-1-1-1-b).

$$I_e = \left(\frac{M_{cr}}{M_a}\right)^3 I_g + \left[1 - \left(\frac{M_{cr}}{M_a}\right)^3\right] I_{cr} \qquad (4\text{-}60)$$

Where:

I_{cr} = Moment of inertia of cracked concrete section
I_g = Moment of inertia of gross concrete section about centroidal axis, neglecting reinforcement
M_a = Maximum value of bending moment in member at the stage of computing deflection.
M_c = Minimum moment resulting in concrete cracking and computed from:

$$M_{cr} = \frac{f_{ctr} \cdot I_g}{y_t} \qquad (4\text{-}61a)$$

Where:

y_t = Distance between extreme fiber in tension to neutral axis of gross section ignoring cracking and presence of reinforcement
f_{ctr} = Cracking-limit tensile stress of concrete subjected to tension resulting from bending, taken from experimental tests and can be calculated from:

$$f_{ctr} = 0.6\sqrt{f_{cu}} \qquad (4\text{-}61b)$$

b - For continuous spans, the effective moment of inertia shall be taken as the average of the values obtained from Equation (4-60) for the critical positive and negative moment sections.

4-3-1-1-2 Long-term deflections

Creep and shrinkage result in additional deflection of concrete elements subjected to bending moments. Such additional deflection

increases with time and its maximum value is affected by the amount of compression reinforcement in the section. It can be calculated by the value of the immediate deflection caused by sustained load by the factor α, which is taken equal to 2.0 for sections having no compression reinforcement and for other cases can be calculated from:

$$\alpha = 2 - 1.2\left(\frac{A'_s}{A_s}\right) \geq 0.6 \qquad (4\text{-}62)$$

With due consideration of has been mentioned in Section (4-2-1-2-d).

4-3-1-1-3 Total deflections

The total deflection shall be calculated as the summation of the immediate deflection calculated according to Section (4-3-1-1-1) and long-term deflection calculated according to Section (4-3-1-1-2).

4-3-1-2 Allowable limits of deflection for beams and slabs

a - The values of the total deflection of beams and slabs and cantilevers in ordinary structures under the effect of all loads taking into consideration the effect of temperature and time-dependent deflection resulting from shrinkage and creep according to Section 4-3-1-1-2) shall not exceed the following limits measured from the level of support provided the satisfaction of the requirements of section (4-3-1-1-1-a):

1 - Beams, one-way slabs and two-way slabs

$$\frac{L}{250} \qquad (4\text{ -}63\text{-a})$$

2 - cantilevers

$$\frac{L}{450} \qquad (4\text{-}63\text{-b})$$

b - For beams and slabs supporting nonstructural elements not likely to be damaged by deflection, immediate deflection resulting from live loads shall not be more than the following value:

$$L/360 \qquad (4\text{-}63\text{-c})$$

c - For beams and slabs supporting nonstructural elements likely to be damaged by deflection such as glass curtain walls, the part of the total additional deflection occurring after attachment of nonstructural elements and resulting from all loads including the effect of temperature and shrinkage and creep according to Section (4-3-1-1-2) shall not be more than the following value:

$$L/480 \qquad (4\text{-}63\text{-}d)$$

Where:

L = distance between of inflection in beams and slabs or cantilever length. It is calculated based on the short span of two-way slabs and on the long span in beamless flat slabs.

4-3-1-3 Clear span-to-thickness ratio unless deflections are computed

4-3-1-3-1 Beams, solid one-way slabs and cantilevers

a - In ordinary buildings, deflection calculations may be waived for beams with rectangular cross sections and spans less than 10.0 m, one-way slabs having spans less than 10.0 m and cantilevers lengths less than 2.0 m if the clear span-to-thickness ratios (L_n/t) will not exceed the values stipulated in Table (4-10).

Table (4-10) Clear span-to-thickness ratio (L_n/t) above which deflections must be computed for beams with rectangular cross sections and one-way slabs of spans less than 10.0m and cantilevers of lengths less than 2.0m

Cantilever	Both end continuous	One end Continuous	Simply supported	member
10	36	30	25	Solid Slab
8	28	25	20	Ribbed slabs and embedded beams
5	21	18	16	Beams

b - Values given in Table (4-10) shall be used directly for Grade 400/600 reinforcement. For other grades of steel, the values given in the table shall be divided by the factor given by Equation (4-64):

$$0.40 + \frac{f_y}{650} \qquad (4\text{-}64)$$

c - The values given in Table (4-10) shall not apply to beams and ribbed slabs supporting or attached to elements likely to be damaged by large deflections.

d - The values given in Table (4-10) shall not be applied to spans more than 10.0m and cantilevers of lengths more than 2.0m. They shall also not be applied for cases of heavy or non-uniform loads and for unordinary buildings. In such cases, deflections shall be computed and their values shall be verified in accordance with Section (4-3-1-2).

e - For T-beams, the values given in section (4-3-1-3-1-a) shall be modified by multiplying by the factor δ given in Fig. (4-23).

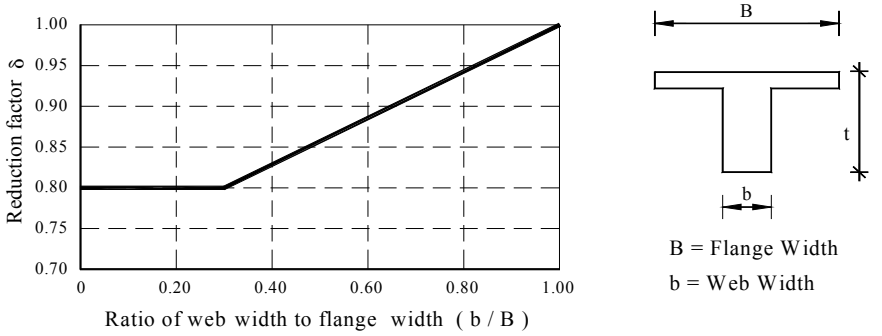


Figure (4-23) Modification of (L_n/t) Ratios for T-beams

4-3-1-3-2 Two-way slabs supported on rigid beams

Deflection calculations can be waived for two-way slabs located in ordinary buildings, not attached to non-structural elements likely to be damaged by deflection, subjected to uniform but not heavy loads and having spans less than 10.0m provided that the slab thickness is not less than 100 mm or the values given by Equation (4-65), whichever is larger.

$$t = \frac{a\left(0.85 + f_y/1600\right)}{15 + \frac{20}{b/a} + 10\beta_p} \qquad (4\text{-}65)$$

Where:

a = Short effective span of slab
b = Long effective span of slab

β_p = Ratio of continuous edges of the slab to its overall perimeter

f_y = Yield stress of reinforcing steel

4-3-2 limit states of cracking

4-3-2-1 For protecting concrete elements from defective cracking that might adversely affect the efficiency and the strength of the element against environmental factors, it is important to select the factors that affect the width of the cracks such as the concrete cover and the type, the distribution and the value of stresses in the reinforcing steel subjected to tension. The proper selection of such factors guarantees the satisfaction of the limit state of cracking according to this section.

4-3-2-2 For the satisfaction of the limit states of cracking, structural elements are categorized according to the state of exposure of their tensile surface to the environmental effects that adversely affects the performance of the structure as give in Table in (4-11).

4-3-2-3 Selection of the factors affecting the crack width

The provisions mentioned in this section shall be satisfied when evaluating the state of cracking of surface subjected to tension.

a-1 When designing reinforced concrete elements with cracking more or less normal to the direction of the reinforcing steel, the following relationship shall be satisfied:

$$w_k = \beta \cdot s_{rm} \cdot \varepsilon_{sm} \qquad (4\text{-}66)$$

Where :

$$s_{rm} = \left(50 + 0.25\, k_1\, k_2\, \frac{\phi}{\rho_r} \right) \qquad \text{mm}$$

$$\varepsilon_{sm} = \frac{f_s}{E_s} \left(1 - \beta_1\, \beta_2 \left(\frac{f_{sr}}{f_s} \right)^2 \right)$$

The values of w_k shall be less or equal to the maximum values $w_{k\,max}$ given in table (4-12)

Table (4-11) Classification of structural elements based on their tension surface exposure to environmental conditions

Class	Level of tension surface exposure to environmental conditions
First	**Elements with protected tension surfaces such as :** a. All internal protected elements in ordinary structures (buildings) b. Elements permanently submerged under water not containing injurious materials. Or elements permanently dry . c. Roofs well insulated against moisture or rain .
Second	**Elements with unprotected tension surfaces such as :** a. All structures in open air such as bridges and roofs not well insulated. b. Structures of first section above but adjacent to shores. c. Elements with exposed surfaces to moisture such as parking garages and open halls.
Third	**Elements with tension surfaces exposed to injurious conditions such as:** a. Elements exposed to high moisture percentages . b. Elements exposed to repeated moisture saturation . c. Water tanks . d. Structures subjected to injurious vapours , gases , and chemical materials .
Fourth	**Elements with tension surfaces exposed to oxidizing and injurious conditions causing rusting of reinforcement such as :** a. Elements exposed to injurious oxidizing conditions causing rusting of reinforcement including gases and vapors containing chemicals. b. Other tanks, sewers , and structures exposed to sea water .

Table (4-12) Values of the Coefficient W_{kmax} (mm)

Category	First	Second	Third	Fourth
W_{kmax}	0.30	0.20	0.15	0.10

Where:

ϕ = Bar diameter in mm. In case of using more than one bar diameter in the cross section, the average bar diameter shall be

used. When using bundled bars, the equivalent diameter of the bundle shall be used according to Section (4-3-7).

β = Coefficient that relates the average value to design value of the crack width and taken as follows:

1.7 for cracks induced due to loading

1.3 for cracks induced due to restraint in sections having width or depth (whichever is smaller) of less than or equal to 300mm.

1.7 for cracks induced due to restraint in sections having width or depth (whichever is smaller) of more than or equal to 800mm.

For sections having a width or a depth (whichever is smaller) between 300mm and 800mm, the value of β may be proportionally calculated.

β_1 = Coefficient reflects the effect of the bond characteristics of reinforcing steel on the average increase of steel strain relative to concrete around the steel. Its value shall be taken 0.8 for ribbed bars and 0.5 for smooth bars.

β_2 = Coefficient that takes into consideration the effect of the duration of loading on the average increase of steel strain relative to concrete around the steel. Its value shall be taken 0.1 for short term loading and 0.5 for permanent loads or cyclic loads.

k_1 = Coefficient that takes into account the effect of bond characteristics between reinforcing steel and concrete on the distance between the cracks. Its value shall be taken o.8 for ribbed bars and 1.6 for smooth bars.

- For the case of imposed deformation, the value of K1 is modified to Kk1 and shall be taken as follows:
- For case of tensile stresses resulting from restraint in general: k = 0.8. For rectangular sections, the value of k shall be taken as follows:

  Rectangular section with a height less than or equal to 300mm: k=0.8

  Rectangular section with a height more than or equal to 800mm: k=0.5

  For sections having a depth between 300mm and 800mm, the value of k may be proportionally calculated.

K_2 = Coefficient reflects the effect of the distribution of strains at a distance from the cracks. Its value shall be taken 0.5 for a section subjected to bending moment and 1.0 for a section

subjected to axial tension. For a section subjected to combined bending moment and axial force, k_2 shall be calculated from the following relationship:

$$k_2 = \frac{\varepsilon_1 + \varepsilon_2}{2\,\varepsilon_1} \qquad (4\text{-}67)$$

Where:

$\varepsilon_1, \varepsilon_2$ Maximum and minimum tensile strain, respectively, calculated at the cracking stage.

ρ_r Effective tension reinforcement ratio $\rho_r = \frac{A_s}{A_{cef}}$

A_s = Area of reinforcing steel in tension

A_{cef} = Effective concrete area in tension (determined according to Fig. 4-24) and equals to the width of the section multiplied by the depth t_{cef} where t_{cef} equals to 2.5 times the distance from the center of gravity of the tension reinforcement to the outermost fiber tensile fiber of the section, but not more than (t-c)/3 for slabs and with due consideration of the following definitions:

c = Distance from the extreme compression fiber to the neutral axis

t = Thickness of the structural element

f_s = Stress in reinforcing steel at the tension side of the section after cracking, calculated based on cracked section analysis under working loads provided that its value does not exceed that given in Table (5-1).

f_{sr} = Stress in reinforcing steel at the tension side of the section after cracking, calculated based on cracked section analysis under the effect of cracking loads

a-2 For cases in which the element is subjected to stresses resulting from intrinsic imposed deformation such as restraint shrinkage, f_s shall be taken equal to f_{sr}.

a-3 For walls subjected to shrinkage due to early thermal contraction, where the lower part of the wall is restraint in a previously cast foundation, the value s_{rm} in equation (4-66) shall be substituted by a value that is equal to the wall height in mm.

a-4 In elements having reinforcing steel arranged in directions x and y and having the angle of crack inclination to the reinforcing steel direction more than 150, equation (4-66) shall be satisfied by substituting the value s_{rm} by the value $\dfrac{1}{\dfrac{\cos\theta}{s_{rmx}}+\dfrac{\sin\theta}{s_{rmy}}}$

where:

θ = Inclination angle between reinforcing steel and principal tensile stresses

s_{rmx}, s_{rmy} = $\left(50+0.25\,k_1\,k_2\,\dfrac{\phi}{\rho_r}\right)$ calculated in directions x and y, respectively.

b- The concrete cover to the tension reinforcement shall not be less than the values given in table (4-13) nor shall it be less than the diameter of the largest bar utilized in the reinforcement. The concrete cover shall be increased for the cases mentioned in Section (9-7).

c- The minimum and maximum spacing for reinforcing steel shall be satisfied according to the provisions of chapters 6 and 7 of this code.

d- For elements in structures classified as category three and four, which should be liquid impermeable, the values of the tensile stresses calculated according to section (4-3-2-7) shall not exceed the values given in equation (4-69).

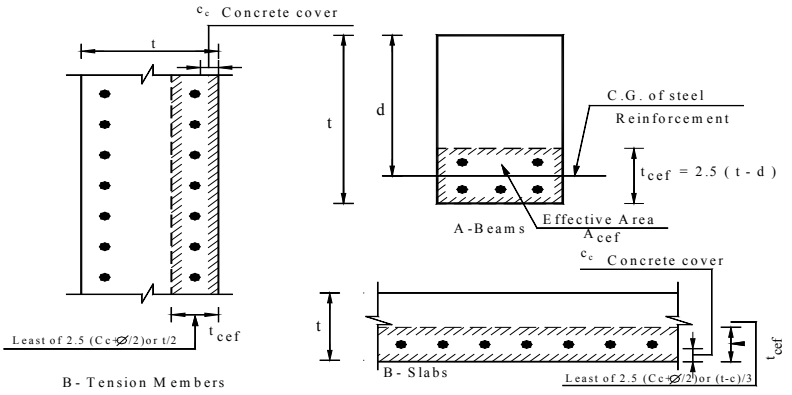


Figure (4-24) Area of Effective Concrete Section in Tension

Table (4-13) Minimum Limit of Thickness of Concrete Cover

Tension surface exposure section	Concrete Cover (mm)			
	For all elements except walls and solid slabs		For walls and solid slabs	
	$f_{cu} \leq 25$	$f_{cu} > 25$	$f_{cu} \leq 25$	$f_{cu} > 25$
First	25	20	20	20
Second	30	25	25	20
Third	35	30	30	25
Fourth	45	40	40	35

4-3-2-4 Cases for which the calculations of cracking limit state can be waived

The conditions of the cracking. The requirements of the cracking limit state (Section 4-3-2-3-a) can be considered to be fulfilled if one of the following conditions is satisfied:

A - For normal buildings included in categories one and two in which live load is not more than 5.0 kN/m^2:
- Solid slabs having thickness of not more than 160 mm.
- T- and L-shaped beams with flanges on the tension side subjected to the condition that the ratio of the flange width to the web width is not less than 3.0.

b - Elements subjected to bending moments combined with axial compression forces having values more than 0.2 f_{cu} A_c at service load level.

c - When the values of the tensile stresses in the reinforcing steel for sections subjected to bending moments or eccentric loads at the service load level are less than the values given in Tables (4-14) and (4-15). In these tables, the permitted values of tensile stresses are given for different values of bar diameter and for different types of structures according to the type environmental exposure of tension surfaces. The reinforcing steel ratio shall not be more than the values given in section (4-2-1-2-c).

d - When using the limit states design method to design sections subjected to bending moments or eccentric forces according to section (4-2-1), the requirements of cracking limit state for the stresses in the

reinforcing steel (Section 4-3-2-3-a) can be considered fulfilled if the value of the yield stress of steel f_y is multiplied by the coefficient β_{cr} given in Tables (4-14) and (4-15). Such a coefficient depends on the bar diameter and the type of exposure of tension surface (structural category). The reinforcing steel ratio shall not be more than the values given in section (4-2-1-2-c) according to the type of reinforcing steel and taking γ_c = 1.5 and γ_s = 1.15.

e - The requirements related to the tensile stresses in concrete mentioned in Section (4-3-2-6) for structures classified as category three or category four, according to Table (4-11) shall be satisfied.

4-3-2-5 For sections subjected to concentric tension force or eccentric tension force resulting in tension stresses acting on the whole section, calculation of stresses in the reinforcing steel to satisfy cracking limit shall be carried out according to Section (4-3-2-3-a). The previous requirement shall also apply when using smooth welded wire fabric.

Table (4-14) Working Stress of Steel and Coefficients of Reduction of Yield Stress of Steel (β_{cr}) that Satisfies Cracking Limit State for Smooth Bars

Steel working stress	β_{cr}	Tension surface of first section	Tension surface of second section	Tension surface of section three and four
N / mm^2		bar diameter (mm)	bar diameter (mm)	bar diameter (mm)
140	1.00	25	18	12
120	0.84	28	20	18
100	0.69	-	-	28

Table (4-15) Working Stress of Steel and Coefficients of Reduction of Yield Stress of Steel (β_{cr}) that Satisfies Cracking Limit State for Deformed Bars

Steel working stress	β_{cr}		Tension surface of first section	Tension surface of second section	Tension surface of section three and four
N / mm^2	Steel 360/520	Steel 400/600	bar diameter (mm)	bar diameter (mm)	bar diameter (mm)
220	1.00	0.92	18	12	8
200	0.93	0.83	22	16	10
180	0.85	0.75	25	20	12
160	0.75	0.67	32	22	18
140	0.65	0.58	.	25	22
120	0.56	0.50	.	.	28

4-3-2-6 For elements in structures classified as category three and four, which should be liquid impermeable, the values of the tensile stresses calculated according to section (4-3-2-7) shall not exceed the values given in equation (4-69).

4-3-2-7 Tensile stresses in concrete sections

a - When calculating the tensile stresses in concrete, the entire concrete section is considered effective under service loads. When taking the reinforcing steel into consideration, the elastic modulus of steel shall be taken as follows:

$$n = \frac{E_s}{E_c} = 10 \qquad (4\text{-}68)$$

b- The tensile stresses fct shall be calculated from the following equation:

$$f_{ct} = \left[f_{ct(N)} + f_{ct(M)} \right] \leq \frac{f_{ctr}}{\eta} \qquad (4\text{-}69)$$

Where:

fctr = Cracking-limit tensile stresses of concrete

fct(N) = Tensile stresses in concrete due to axial forces

fct (M) = tensile stresses due to bending moments

η = Coefficient determined from table (4-16) according to the virtual thickness of the section t_v given in the following equation:

$$t_v = t \left[1 + \left(\frac{f_{ct(N)}}{f_{ct(M)}} \right) \right] \qquad (4\text{-}70)$$

Where t is the thickness of the section.

Table (4-16) Values of the Coefficient η

Virtual thickness of section tv (mm)		Coefficient η
Less or equal	100	1.00
	200	1.30
	400	1.60
Greater or equal	600	1.70

c - For T- and L-shaped sections, it is preferable to take the flange width equals to one-half the value mentioned in Section (6-3-1-9).

d - For structures that are required to be liquid-tied, equation (4-69) shall be satisfied with due consideration of the values of the working stresses in steel according to Tables (4-14) and (4-15). As an alternative solution, it is permitted to calculate the amount of steel reinforcement using the limit states design method coupled with the use of the value of β_{cr} given in Tables (4-14) and (4-15).

use of the value of β_{cr} given in Tables (4-14) and (4-15).

CHAPTER 5
WORKING STRESS DESIGN METHOD

5-1 General considerations

This chapter provides minimum requirements for design of reinforced concrete sections, using the working stress design method (elastic method), under the effect of working loads and actions (section 3-2-1-1-a). To ensure safety requirements when using this method, the design shall satisfy the following:

a - The stresses in concrete and reinforcing steel, resulting from the action of service loads (without load factors) and computed by the straight-line theory, must not exceed the allowable stresses given in Table (5-1). This applies for sections subject to flexure, eccentric axial forces, shear forces, torsion, or shear forces combined with torsion.

b - Code provisions relating to deformation and deflection limit states (section 4-3-1) and cracking limit state (4-3-2), as well as the requirements of elastic stability (buckling) limit states (article 6-4) shall be satisfied for the stresses in both concrete and reinforcing steel.

c - Design of sections subject to flexure or eccentric axial forces shall follow the requirements of Section (5-3), while for sections subject to shear forces, the design shall follow Section (5-4) and for sections subject to torsion, the design shall follow Section (5-5). Bearing strength shall be determined according to Section (5-6) whereas bond is checked following Section (5-2-4).

5-2 Allowable working stresses

5-2-1 Table (5-1) gives the allowable working stresses for concrete with characteristic strength ranging from 20 to 30 N/mm^2 (MPa) and for different types of reinforcing steel, to be used with the requirements of Sections (5-1-a & b).

5-2-2 For sections subject to eccentric compression forces, the allowable working compressive stresses shall be calculated by the following relation:

Table (5-1) Allowable working stresses for concrete and reinforcing steel

Stress Type	Symbol	Allowable Working Stresses according to Concrete Characteristic Strength (N/mm^2)		
Concrete Characteristic Strength	f_{cu}	20	25	30
Axial compression ($e = e_{min}$)	f_{co} *	5	6	7
Flexure and axial Compression with big eccentricity	f_c **	8.0	9.5	10.5
Shear ***				
Concrete shear strength				
Without web reinforcement in slabs and foundations	q_c	0.8	0.9	0.9
Without web reinforcement in other elements	q_c	0.6	0.7	0.7
With web reinforcement in all elements (shear combined with torsion)	q_2	1.7	1.9	2.1
Punching shear	q_{cp}	0.8	0.9	1.0
Reinforcing Steel ****				
1- Milled Steel 240/350	f_s	140	140	140
2- Steel 280/450		160	160	160
3- Steel 360/520		200	200	200
4- Steel 400/600		220	220	220
5- Welded wire mesh 450/520 smooth		160	160	160
deformed		220	220	220

* This is the maximum allowable axial compressive stress under working loads.

** These allowable stresses are used for beams and for slabs with a thickness of more than 200mm and shall be reduced for thinner slabs by a value of 1.5, 2.0, 2.5, and 3.0 N/mm^2 for slab thickness of 200, 120, 100 and 80mm, respectively.

*** The considerations given in articles (5-4) and (5-5) shall also be satisfied.

**** Stresses in steel must be reduced to satisfy the cracking limit state according to article (4-3-2), if considered necessary.

$$\left(0.23+0.32\frac{e}{t}\right)f_{cu} \quad \text{where} \quad \left(\frac{e}{t}\geq 0.05\right) \qquad (5\text{-}1)$$

But shall not exceed the allowable working compressive stresses for flexure or compressive forces with big eccentricities, fc, listed in Table (5-1).

5-2-3 To satisfy the cracking limit state under working loads, for concrete elements with exposed tensioned faces in the third and fourth categories according to Table (4-11) and for other cases that call for this limit state, the allowable tensile stresses shall be determined in accordance with Sections (4-3-2-6& 7).

5-3 Sections subject to flexure or eccentric axial forces

5-3-1 Basic assumptions and general considerations

1 - Sections subject to flexure or eccentric axial forces shall be designed, using the working stress method, according to the following assumptions and general considerations.

2 - Strains vary linearly as the distance from the neutral axis. As a result, strains in both concrete and reinforcing steel are proportional to the distance from the neutral axis. This applies for all structural members except for deep beams, in which the strain distribution is nonlinear.

3 - For both concrete and reinforcing steel, the stress-strain relationship is a straight line under service loads within allowable service load stresses.

4 - In reinforced concrete members, concrete generally resists no tension while steel resists all tensile stresses.

5 - The modular ratio, n = Es /Ec, shall be taken as follows:

a - For sizing of members and calculation of stresses,

$$n=\frac{E_s}{E_c}=15 \qquad (5\text{-}2\text{-a})$$

b - For computing elastic deformations and analyzing statically-indeterminate structures as well as for determining tensile stresses in concrete for members required to be uncracked (Sections 4-3-2-6 & 7),

$$n = \frac{E_s}{E_c} = 10 \qquad (5\text{-}2\text{-}b)$$

The whole concrete section shall be considered effective in the later case.

6 - The allowable working stresses in steel shall be reduced to satisfy the cracking limit state conditions (Section 4-3-2).

7 - If it is proved- by tests in an accredited lab, that the yield strength of mild steel round bars exceeds 280N/mm2, the allowable steel stress, at service loads, shall be taken equal to ½fy but not more than160N/mm2.

8 - When the stresses resulting from actions like wind, shrinkage, earthquakes, temperature variation, friction at supports, or differential settlement is more than 15% of the stresses produced by the main loads, these actions must be considered in the design. In such cases, the allowable stresses may be increased by 15% for each action but the total stress increment shall not exceed 25% for all considered actions. However, the effects of wind and earthquakes shall not be added to the same combination.

9 - For rectangular sections subject to biaxial flexure, the compressive stress at the most stressed corner shall be permitted to exceed the allowable working stresses listed in Table (5-1) by up to 1N/mm2.

5-3-2 Sections subject to flexure

1 - Sections subject to single or double flexure shall be designed according to the basic assumptions and general considerations stated in Section (5-3-1). Stresses in concrete and reinforcing steel, resulting from service loads, must not exceed the allowable stresses given in Table (5-1). Requirements of Section (5-3-1-5) shall also be satisfied.

2 - The minimum reinforcement ratio for sections subject to flexure shall be in conformance with Section (4-2-1-2-h).

3 - The reinforcement ratio for rectangular sections with tension reinforcement only shall not exceed the limits given in Tables (4-1) and (4-2) according to the type of reinforcement.

4 - In statically indeterminate structures, it shall not be permitted to redistribute more than ±10% of the bending moments calculated by elastic theory. Conditions necessary for moment redistribution, stated

in Section (4-2-1-2-c), shall be fulfilled.

5 - Section flexural strength may be increased by using steel reinforcement in the compression side but the requirements of Section (4-2-1-2-d) must be fulfilled in this case.

6 - For T-sections, only two thirds of the values listed in Table (5-1) for the allowable concrete stresses for flexure and axial compression with big eccentricity shall be permitted.

5-3-3 Sections subject to flexure combined with axial forces

1 - Sections subject to eccentric axial forces shall be designed according to the main assumptions stated in Section (5-3-1) and the allowable stresses given in Table (5-1). It is preferable, however, to design these sections using the limit state method. In this case, the design of sections subject to eccentric axial compressive follows the requirements of Section (4-2-1-3) while the design of sections subject to eccentric axial tensile forces follows the requirements of Section (4-2-1-4).

2 - Sections subject to centric axial compressive forces combined with small bending moments ($<P.e_{min}$) shall be designed assuming a minimum eccentricity e_{min} where,

$$e_{min} = \frac{M}{P} = 0.05\,t \qquad (5\text{-}3)$$

Furthermore, the effect of eccentricity can be approximately accounted for by computing the allowable compressive axial force, at service load level, by Equation (5-4-a) for tied columns and by the smaller of Equations (5-4-b) and (5-4-c) for spiral columns.

$$P = f_{co}\,A_c + 0.44\,f_y\,A_{sc} \qquad (5\text{-}4\text{-}a)$$

$$P = 1.14\,f_{co}\,A_c + 0.51\,f_y\,A_{sc} \qquad (5\text{-}4\text{-}b)$$

$$P = f_{co}\,A_k + 0.44\,f_y\,A_{sc} + 0.92\,f_{yp}\,V_{sp} \qquad (5\text{-}4\text{-}c)$$

where A_c, A_{sc}, A_k, V_{sp}, and f_{yp} are as defined in Section (4-2-1-3-c-2). The ratio of volume of spiral reinforcement to total volume of core confined by the spiral (measured out-to-out of spirals) μ_{sp} shall not be taken less than the value given by Equation (4-12-e).

3 - Sections subject to uniaxial bending moment combined with centric axial compressive force that is less smaller than the value given by Equation (5-5) shall be designed for flexure only following article (5-3-2).

$$P \leq 0.026\, f_{cu}\, A_c \qquad (5\text{-}5)$$

5-4 Sections subject to shearing forces

5-4-1 Beams

5-4-1-1 The critical sections for shear shall be determined by article (4-2-2-1-1).

5-4-1-2 Calculation of the nominal shear stresses in beams

For beams and slabs with constant depth, the shear stress is calculated from Equation (5-6).

$$q = \frac{Q}{b.d} \qquad (5\text{-}6)$$

where:

Q = shear force

b = breadth of rectangular sections, or breadth of the web for T- and other sections.

For beams with variable depth , the shear force Q is replaced by the value Qr given by

$$Q_r = Q - \frac{(M.\tan\beta)}{d} \qquad (5\text{-}7)$$

where tanβ is the rate of change of beam depth with the distance along its centerline and tanβ≤0.33. The negative sign in Equation (5-7) assumes that the beam depth and the acting bending moment increase or decrease in the same sense (direction). For other cases, this sign shall be replaced by a plus sign.

5-4-1-3 For sections subject to shear forces, the shear stress, q, shall not be permitted to exceed the values of, q2, given in Table (5-1). Requirements of article (5-5-4) shall be considered for sections subject to shear forces combined with torsion,.

5-4-1-4 The concrete shear strength shall not be permitted to exceed the values of, q_c, given in Table (5-1). For sections subject to shear forces combined with axial tensile axial forces, q_c may be taken equal to zero.

5-4-1-5 Where the shear stress, q, exceeds the concrete shear strength, q_c, shear reinforcement consisting of one or more of the following types shall be used according to Section (5-4-1-7):

1 - Stirrups perpendicular to axis of member;

2 - Stirrups or bent bars making an angle of not less than 30° with axis of member, together with stirrups perpendicular to this axis.

5-4-1-6 The required strength of shear reinforcement is given by

$$q_s = q - 0.5\, q_c \qquad (5\text{-}8)$$

Figure (4-7) shows zones requiring shear reinforcement while Section (4-2-2-1-6) establishes the minimum shear reinforcement ratio that shall be provided in other parts.

5-4-1-7 Calculation of web reinforcement

a - The shear strength of stirrups perpendicular to axis of member shall be determined by Equation (5-9).

$$q_{st} = \frac{A_{st} \cdot f_s}{s \cdot b} \qquad (5\text{-}9)$$

b - Where Ast is the cross sectional area of stirrups and s is the spacing between stirrups.

When shear reinforcement in the form of inclined stirrups or bent-up longitudinal bars making an angle α with axis of the member, the web reinforcement shear strength shall be computed by Equation (5-10).

$$q_{sb} = \frac{A_{sb} \cdot f_s \cdot (\sin\alpha + \cos\alpha)}{s \cdot b} \qquad (5\text{-}10)$$

In which Asb is the cross sectional area of inclined stirrups or bent-up bars and,

$$q_s = q_{st} + q_{sb} \qquad (5\text{-}11)$$

In case the stirrup inclination or the bar bent-up angle α=45o, Equation (5-10) reduces to

$$q_{sb} = \frac{A_{sb} \cdot f_s \cdot \sqrt{2}}{s \cdot b} \qquad (5\text{-}12)$$

5-4-1-8 General requirements for selection and arrangement of web reinforcement

Requirements related to minimum web reinforcement ratio and detailing of the web reinforcement given in Section (4-2-2-1-6) shall be fulfilled.

5-4-2 Slabs and footings

Shear stresses in slabs and footings is governed by the more severe of the following two conditions:

1 - Beam action, in the longitudinal and transverse directions, according to Sections (5-4-1-2) to (5-4-1-4), but the nominal shear stress calculated from Equation (5-6) shall not exceed half the values given in Table (5-1) for qc. Two-way action (punching shear) according to Section (5-4-3).

5-4-3 Punching shear

5-4-3-1 The critical section for computing punching shear stresses in slabs and footings shall be taken at a distance of (d/2) outside the perimeter of the concentrated load area (Figure 4-9).

5-4-3-2 Punching shear stress shall be computed from the relation

$$q_p = \frac{Q_p}{b_o \cdot d} \qquad (5\text{-}13)$$

where bo is the perimeter of the critical section.

5-4-3-3 When calculating punching shear, the effect of the moments transmitted from flat slabs to columns shall be included according to Section (6-2-5-8).

5-4-3-4 Thickness of slabs and footings required to resist punching shear shall be determined on the basis that punching shear is resisted by concrete alone, without any contribution of the reinforcement. Furthermore, the nominal punching shear strength shall be taken equal to the smaller of the values computed by Equations (5-14-a) and (5-14-b).

$$q_p = 2.5\left(\frac{\alpha .d}{b_o} + 0.2\right) q_{cp} \quad \leq q_{cp} \qquad \text{(5-14-a)}$$

$$q_p = \left[0.5 + \left(\frac{a}{b}\right)\right] q_{cp} \quad \leq q_{cp} \qquad \text{(5-14-b)}$$

Values of qcp are given in Table (5-1) while a and b are the length and width of the rectangular loaded area. For loaded areas of other (non-rectangular) shapes, b shall be taken equal to the longest dimension of the effective loaded area while a shall be taken equal to the longest dimension perpendicular to b of the same area . For this purpose, the effective loaded area is the area having the least perimeter as shown in Figure (4-9-d) for an L-shaped loaded area. The factor α in Equation (5-14-a) shall be taken equal to 4, 3 and 2 for interior, edge and corner columns, respectively.

5-5 Sections subject to torsion

5-5-1 The critical section for torsion shall be taken according to Section (4-2-3-1).

5-5-2 Nominal shear stresses due to torsion

a - The nominal shear stress due to torsion for solid reinforced concrete sections shall be computed by the following equation:

$$q_t = \frac{M_t}{(2\,A_o \,.\, t_e)} \qquad (5\text{-}15)$$

where Ao snd te are as defined in Article (4-2-3-2).

b - For T- and L- sections, the effective part of slab may be neglected, hence, the section can be treated as a rectangular using Equation (5-15).

In case the effective part of slab is included, requirements of Section (4-2-3-2) and Figure (4-11-b) shall be followed.

c - Box sections shall be treated as is stated in Article (4-2-3-2-d).

5-5-3 Effect of torsional moments shall be neglected if the nominal shear stress due to torsion resulting is less than $0.04\sqrt{\frac{f_{cu}}{\gamma_c}}$ where f_{cu} is in N/mm^2.

5-5-4 Concrete dimensions of solid sections subject to shear forces combined with torsional moments and reinforced by web reinforcement as well as longitudinal bars shall satisfy the following relation:

For solid sections,

$$\sqrt{(q)^2 + (q_t)^2} \le q_2 \qquad N/mm^2 \qquad (5\text{-}16)$$

For box sections,

$$q + q_t \le q_2 \text{ N/mm}^2 \qquad (5\text{-}17)$$

In which, q is determined from Equation (5-6), qt is calculated using Equation (5-15) and q2 is as in Table (5-1).

5-5-5 Shear reinforcement for sections subject to shear forces combined with torsion

When the shear stress calculated according to Article (5-5-2) exceeds the value $0.04\sqrt{\frac{f_{cu}}{\gamma_c}}$ without violating the limit specified in Article (5-5-4), web as well as longitudinal reinforcement shall be used to resist stresses due to torsion. This reinforcement shall be added to other reinforcement required to resist shear forces and bending moments according to Table (5-2) as follows:

a - Area of transverse reinforcement, in the form of closed stirrups or welded wire mesh shall be determined by Equation (5-18-a) and Figure (4-12).

$$A_{str} = \frac{M_t \, . \, s}{2 \, A_o \, . \, f_s} \qquad (5\text{-}18\text{-}a)$$

For rectangular sections, Equation (5-18-a) reduces to:

$$A_{str} = \frac{M_t \, . \, s}{1.7 \, (x_1 . \, y_1) \, f_s} \qquad (5\text{-}18\text{-}b)$$

In these equations, Astr, s, x1 and y1 are as defined in Article (4-2-3-5). The cross sectional area of stirrups that resist shear forces and torsion shall not be less than the amount determined by Equation (4-52)- Article (4-2-3-5).

b - The area of additional longitudinal reinforcement, Asl, shall be computed as

$$A_{sl} = \left(\frac{A_{str} \cdot p_h}{s} \right) \left(\frac{f_{yst}}{f_y} \right) \qquad (5\text{-}19)$$

where ph is as defined in Section (4-2-3-2). This longitudinal reinforcement shall be distributed around the perimeter of the exterior closed stirrups. The area of additional longitudinal reinforcement shall not be taken less than that given in Section (4-53).

The following requirements shall be satisfied:

- The spacing of stirrups shall not exceed the smaller of ph /8 or 200mm.
- For sections with stirrups having more than two branches, exterior stirrups with two branches shall only be considered to resist torsion as shown in Figure (4-13).
- Additional longitudinal bars shall have a diameter at least 0.067 times the stirrup spacing, but not less than 12mm.
- The additional longitudinal reinforcement required for torsion shall be distributed around the perimeter of the exterior closed stirrups with a maximum spacing of 300mm. There shall be at least one longitudinal bar in each corner of the stirrups.
- The additional longitudinal reinforcement required for torsion shall be added to the longitudinal reinforcement required for resisting bending moments
- Both transverse and longitudinal reinforcement required for torsion shall be provided for a distance of at least half the perimeter of exterior stirrups beyond the point required by analysis.
- It shall not be allowed to redistribute the torsional moment in statically indeterminate structures when it is necessary for equilibrium (equilibrium torsion).

Table (5-2) Transverse reinforcement to resist torsion and shearing forces

	$q_t \le 0.04\sqrt{\frac{f_{cu}}{\gamma_c}}$ N/mm^2	$q_t > 0.04\sqrt{\frac{f_{cu}}{\gamma_c}}$ N/mm^2
$q < q_c$	The minimum percentage of shear reinforcement according to Article (4-2-2-1-6)	Reinforcement to resist q_t
$q > q_c$	Reinforcement to resist $(q - q_c/2)$	Reinforcement to resist both of: q_t and $(q - q_c/2)$

5-5-6 The torsional rigidity of concrete sections shall be computed according to Section (4-2-3-7).

5-6 Bearing loads

5-6-1 The bearing load shall not exceed 0.30fcuA1 where A1 is the area of bearing surface.

5-6-2 When the supporting surface is wider on all sides than the loaded area, then the maximum bearing load given by Section (5-6-1) shall be permitted to be multiplied by $\sqrt{\frac{A_2}{A_1}}$ but not more than 2.

Where, A2 =the largest area within the support base that is symmetrical-to and concentric-with the loaded area A_1(Figure 4-14). The thickness of the supporting surface shall be designed to resist the shear stresses stated in Section (4-2-2).

5-6-3 When the support area is stepped or has sloped sides, area A2 shall be taken equal to the largest frustum of a pyramid, cone, or tapered wedge contained wholly within the support and having for its upper base the loaded area, and having side slopes of 1 vertical to 2 horizontal (Figure 4-14).

CHAPTER 6
ANALYSIS OF STRUCTURAL ELEMENTS

6-1 General considerations

a- It shall be permitted to use any structural analysis method that provides full compliance with the requirements of the conditions of equilibrium and strain compatibility.

b - Members of the structure shall be designed to resist the maximum effects of all applicable loads

c- It shall be permitted to analyze ordinary building assuming that the spans of all elements of the building are subjected to full loads.

d- It shall be permitted when calculating of the reactions for continuous beams and slabs having approximately equal spans and loads to take the effects of continuity by increasing the magnitudes of the reaction and shear forces at the first interior support of the exterior spans by 10%, and 20%, respectively.

e- The effects of continuity of continuous slabs and beams having unequal span lengths with the longer of two adjacent spans is greater than the shorter by more than 20% shall be taken into consideration. For such cases the analysis shall be carried out assuming fully loaded slabs and beams.

f- The reactions at the external supports of cantilever slabs shall be evaluated with due consideration of the effects of the cantilever on the magnitude of reaction.

g- The effects of temperature and shrinkage on the structural response of ordinary buildings shall be ignored, except for the cases were it can be demonstrated that such effects are significant. Expansion joints for long buildings shall be in accordance with provisions 9-5-7 and 9-5-8 of this code.

h- The effects of time dependent stains on the internal forces and moments in ordinary buildings shall be ignored, except for the cases were it can be demonstrated that such effects are significant.

i- The design of concrete structures for resisting seismic loads shall be carried out with due consideration of the effects of inter-story drifts and the choice of the sizes of the seismic joints in accordance with provisions 6-8 and 9-5-9 of this code, as well as, the corresponding provisions of the Egyptian code for loads on the structures (ECP # 201)

6-2 Slabs

Provisions of this section shall apply to the following types of slabs:

1 - Solid slabs
2 - Hollow block slabs
3 - Waffle slabs
4 - Paneled beams
5 - Flat slabs

6-2-1 Solid slabs

6-2-1-1 General

6-2-1-1-1 Spans

a - Effective span of slabs shall be taken equal to the net span between supports, plus slab thickness, or 1.05 times the net span whichever is grater, but it shall not exceed the distance between support centerlines.

b - Continuous slabs monolithically cast with supports having width of support greater than 20% of net span may be considered as a slab with both ends fixed.

c - The effective lengths of cantilever slabs shall be taken equal to the least value of:

- The cantilever slab length measured from support centerlines in case of overhanging slabs.
- Clear length of cantilever slab plus the greatest thickness of the cantilever slab.

6-2-1-1-2 Supports

The slab support width shall not be less than three quarts of the slabs' thickness, or 100 mm whichever is greater with due consideration of section (4-2-3-6). These requirements shall not apply for pre-cast slabs. For slabs supported by brick walls, the minimum thickness of the wall shall not be less than 200. For slabs supported by beams the minimum thickness of the beams shall not be less than three times the thickness of slab, unless otherwise determined by structural analysis taking into consideration the stiffness of the supporting beams.

6-2-1-1-3 Rectangularity ratio

Rectangular slab supported on its four edges shall be considered unidirectional if the rectangularity ratio "r" of the portion of slab enclosed

between the lines of inflection in the span exceeds two. Shall be considered bidirectional if rectangularity ratio is less than or equals two. Accordingly, the rectangularity ratio "r" shall be determined using with equation (6-1a), and equation (6-1b);

$$r = \frac{m_b \cdot b}{m_a \cdot a} \qquad (6\text{-}1a)$$

and shall be used with table (6-1),

$$r = \frac{b}{a} \qquad (6\text{-}1b)$$

and shall be used with tables (6-2), (6-3).

Where:

a = short effective span.

b = long effective span.

m_a = ratio of length between lines of inflection in a loaded strip of the slab in direction of span a, to span length a.

m_b = ratio of length between lines of inflection in a loaded strip of the slab in direction of span b, to span length b.

Values of m_a and m_b are determined based on theory of elasticity. The following approximate values of both of m_a and m_b, may be used:

- m_a or m_b = 0.76 for spans continuous from both sides.
- m_a or m_b = 0.87 for spans continuous from one side only.
- m_a or m_b = 1.00 for simple spans

6-2-1-2 One way solid slabs

Definition:

1 - one way solid slabs refer to slabs where loads are transferred in one direction only to two supports along the opposite sides. Supports may be either walls or beams.

2 - Rectangular solid slab supported on four sides and having rectangularity ratio "r" exceeding 2, according to equation (6-1) shall be considered as one way slab.

One way solid slab may be calculated based on using a strip having unit width in the direction of short effective span between the opposite two supports.

6-2-1-2-1 Minimum thickness

1 - Slab minimum thickness shall be determined so that deflection limit in accordance with section (4-3) shall not exceed. Calculation of deflection shall not be required if slab thickness in ordinary buildings is not less than the values given in table (4-10).

2 - The minimum slab thickness shall not be less than the following:

- Simply supported slab $t_{min} = \frac{L}{30}$
- Continuous slab from one side $t_{min} = \frac{L}{35}$
- Continuous slab from two sides $t_{min} = \frac{L}{40}$
- Where L is the effective span of one way slabs

3 - Slab thickness, in ordinary buildings, shall not be less than the following values:

- 80mm for cast in place slabs, subjected to static loads.
- 120 mm for slab subjected to dynamic loads or moving loads.

4 - The preceding minimum thicknesses may be reduced for pre-cost slabs.

6-2-1-2-2 Bending moments

1 - Continuous slab may be analyzed beam theory as continuous beams supported by free rotating rigid supports, provided that special care shall be taken to ensure the exact placement of reinforcing steel for resisting negative bending moments.

2 - For slabs supported on walls or monotonically cast with supporting beams, the negative bending moment may be reduced according to parabolic curve, as shown in figure (6-1), where M_1 is the value of the difference between the moment at support centerline, and moment at support face.

3 - In continuous slabs design positive bending moment shall not be less than $wL^2/16$, with due consideration of minimum reinforcement ratio, according to section (6-2-1-2-3).

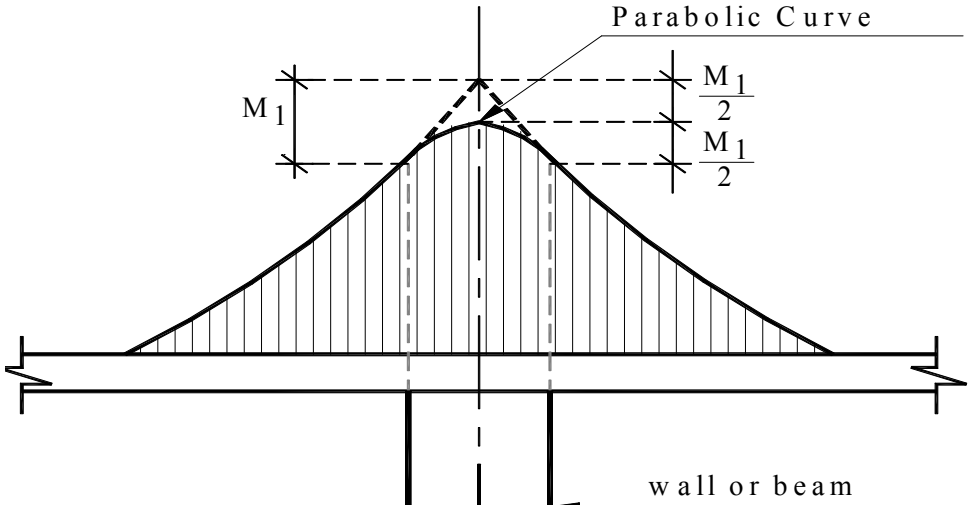


Figure (6-1) Reducing negative bending moment of continuous slabs

4 - Negative bending moment at external supports of slabs fixed in brick, stone or ordinary concrete walls, making partial fixation at slab edge shall not be less than

$$M = \frac{-w L^2}{16} \qquad (6\text{-}2)$$

Positive moment in outer spans shall be calculated neglecting the partial fixation at edges.

5 - Negative bending moment at outer supports of monolithically cast slab with supporting beams that result in partial fixation at slab edge shall not be less than

$$M = \frac{-w L^2}{24} \qquad (6\text{-}3)$$

Positive moment in outer spans shall be calculated neglecting partial fixation at edges.

6 - Slab shall be considered as fully fixed at its edges either when such edges are effectively connected to other parts of the structure such that rotation of slab edges under all loading conditions are totally prevented, or when requirements of section (6-2-1-1-1-b) are fulfilled.

7 - In cases of continuous slabs subject to equal uniformly distributed loads on all spans, having the magnitude of live load is equal or less than that of the dead load ($p \leq g$)and where spans are equal, or differences between spans do not exceed 20% of the greater span , following maximum values of bending moment may be assumed:

a - One span slab, the maximum positive bending moment

$$M = \frac{+ w L^2}{8} \qquad (6\text{-}4\text{-}a)$$

b - Continuous two span slab, the maximum positive bending moment

$$M = \frac{+ w L^2}{10} \qquad (6\text{-}4\text{-}b)$$

And the negative bending moment at middle support:

$$M = \frac{- w L^2}{8} \qquad (6\text{-}4\text{-}c)$$

c - Multi span continuous slabs, the maximum bending moment

$$M = \pm \frac{w L^2}{K} \qquad (6\text{-}4\text{-}d)$$

Where; the values of K are shown in figure (6-2). The values of the negative bending moment on any support may be taken equal to arithmetic mean of negative moment calculated on each side of the common support of t two adjacent spans

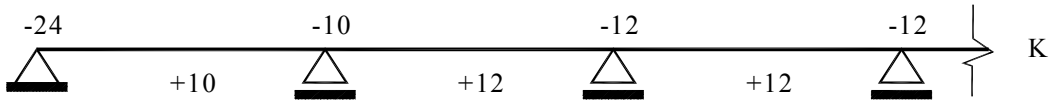


Figure (6-2) Bending moment in continuous slabs

8 - Negative moments shall be calculated at the mid spans when continuous slab are subjected to heavy live loads (p> 2g). In cases of monolithically cast slabs and beams, negative moments are permitted to be reduced at the span centers resulting from live loads only to its half value, due to the resistance of supporting beams to torsion, and negative moments may be taken at inner span centers, according to equation (6-5).

$$M_{min} = \frac{\left(g - \left(\frac{p}{2}\right)\right) L^2}{24} \qquad (6\text{-}5)$$

9 - In case of design by limit states method, (g_u , p_u , and w_u) shall replace g , p , w respectively.

6-2-1-2-3 Reinforcement

1- Reinforcement ratio in main direction shall not be less that $0.6/f_y$ of the area of effective concrete section , or 0.25 % of actual concrete section area in case of using mild reinforcement steel , and equivalent to 0.15 % in case of using high tensile steel.

2 - Reinforcement shall be arranged to cover the entire tension areas, and extend inside the support a distance equals the anchorage length according to section (4-2-5).

3 - In continuous slabs having equal span lengths or, span lengths that do not differ by more than 20 % of the longer span, and subjected to normal loading conditions and in cases that bars have not been arranged according to the bending moment curve, half of main reinforcement may be bent at a distance equals to 1/5 of clear span from face of interior supports, and extending in adjacent span to a distance equals to 1/4 of the longer of the two spans.

4 - The maximum spacing between main reinforcement bars in areas of maximum moments shall not exceed 200 mm.

5 - Cross-sectional area of bottom reinforcement bars extending to supports shall not be less than one third of the cross-section area of positive reinforcement used at the mid span.

6 - All requirements of the items shall also apply for the cases of using reinforcement mesh.

7 - Cross-sectional area of distributed bars perpendicular to main reinforcement shall not be less than one fifth of main reinforcement area of steel. The minimum number of distributed bars per meter width of slab shall not be less than four bars.

8 - Minimum diameter of main bars shall be 6 mm for straight bars, and 8 mm for bent-up bars. Bars of smaller diameters may be used in case of using mesh, or in pre-cast units.

9 - Top mesh reinforcement shall be used in slabs of thickness greater than 160 mm. The area of the steel mesh in every direction shall not be less than 20 % of that of main reinforcement with a minimum of 5ϕ 8/m, for mild steel or 5 ϕ 6/m for high grade steel.

6-2-1-3 Two- way rectangular solid slabs

6-2-1-3-1 General

1- Rectangular solid slab supported on four sides and having rectangularity ratio "r" less than 2, according to equation (6-1) shall be considered as two-way slab.

2 - Such slabs may be analyzed using theory of elasticity, provided that special care shall be taken to ensure the exact placement of reinforcing steel for resisting negative bending moments.

3 - The following methods of design shall be applicable only for the design of ordinary buildings. Accordingly, it shall not be permitted the use the method outlined in this section to the design of other types of structures such as bridges or liquid tanks … etc.

6-2-1-3-2 Minimum thickness

- Minimum thickness of two-way slabs shall be taken as follows:

  Simply supported slabs:

$$t_{min} = \frac{a}{35} \qquad (6\text{-}6\text{-a})$$

  Slabs continuous from one side:

$$t_{min} = \frac{a}{40} \qquad (6\text{-}6\text{-b})$$

  Slabs continuous from two sides:

$$t_{min} = \frac{a}{45} \qquad (6\text{-}6\text{-}c)$$

Where:

a- is the effective short span of the slab in accordance with items 3, 4 of section (6–2–1–2–1).

6-2-1-3-3 A simplified method for calculation of bending moments in two- way solid slabs subjected to uniformly distributed loads

In General, the analysis shall be carried out in accordance with section (6–2–1–3-1 clause2). The following simplified method may be used in calculating bending moment in monolithically cast rectangular slabs with beams, and supported on its four sides, provided that rectangularity ratio " r " does not exceed 2. Value of bending moments may be taken in two way slabs as follows:

- For simply supported spans:

$$M_a = +\frac{\alpha . w . a^2}{8} \quad \text{or} \quad M_b = +\frac{\beta . w . b^2}{8} \qquad (6\text{-}7\text{-}a)$$

-For continuous spans from one side only:

$$M_a = \pm\frac{\alpha . w . a^2}{10} \quad \text{or} \quad M_b = \pm\frac{\beta . w . b^2}{10} \qquad (6\text{-}7\text{-}b)$$

- For continuous span from both sides:

$$M_a = \pm\frac{\alpha . w . a^2}{12} \quad \text{or} \quad M_b = \pm\frac{\beta . w . b^2}{12} \qquad (6\text{-}7\text{-}c)$$

Table (6-1) gives the value of coefficients α , β used in calculating of bending moments of slabs in the directions a, b respectively corresponding to different values of “r", and for case of slabs subjected to live loads not exceeding 5 kN/m^2.

Table (6-1) Values of coefficients α , β corresponding to values of "r" for solid slabs monolithically cast with beams, subjected to uniform live load does not exceed 5 kN/m^2.

2.0	1.9	1.8	1.7	1.6	1.5	1.4	1.3	1.2	1.1	1.0	r
0.85	0.80	0.75	0.70	0.65	0.60	0.55	0.50	0.45	0.40	0.35	α
0.08	0.09	0.11	0.12	0.14	0.16	0.18	0.21	0.25	0.29	0.35	β

Where:

$$\alpha = 0.5r - 0.15 \quad \& \quad \beta = \frac{0.35}{r^2} \qquad (6\text{-}8)$$

But in case live loads greater than 5 kN/m^2, then values of α , β in table (6-3) may be used.

In case of diversity between bending moments on both sides of contact line between two slabs, contact moment M_C between them may be calculated using equation.

$$M_c = \frac{M_1 L_1 + M_2 L_2}{L_1 + L_2} \qquad (6\text{-}9)$$

Where M_1, L_1 are negative moment calculated for a slab, and span used in calculating such moment respectively.

And M_2, L_2 are negative moment calculated on the adjacent slab and span used in calculating such moment respectively.

6-2-1-3-4 Reinforcement of two way slabs

a - Maximum distance between main reinforcement bars shall not exceed at locations of maximum moments 200 mm. Cross-sectional area of reinforcement in secondary direction shall not be less than one quarter of main reinforcement cross-section, and the number of bars in areas of maximum moment shall not be less than five bars per meter. For other requirements of reinforcement refer to section (6-2-1-2-3).

b - Positive reinforcement adjacent and parallel to the slab continuous edges may be reduced when slab is continuous in a direction perpendicular on such edges. The reduction in area of reinforcement

shall not be less than one quarter, within a width of slab that does not exceed one quarter of the shortest dimension of the slab, and with due consideration of the preceding item , a .

6-2-1-3-5 Load distribution in slabs supported on masonry walls

Uniformly distributed loads in slabs supported on masonry walls shall be distributed according to table (6-2) in case that the live loads do not exceed 5 kN/m$^{2.}$ For live loads exceeding 5 kN/m^2, values of coefficients in table (6-3) may be used.

Table (6-2) Values of coefficients α , β corresponding to values of " r " for solid slab supported on wall, and (two way) ribbed slabs, with complete compression flange.

2.0	1.9	1.8	1.7	1.6	1.5	1.4	1.3	1.2	1.1	1.0	r
0.849	0.830	0.806	0.778	0.746	0.706	0.660	0.606	0.543	0.473	0.396	α
0.053	0.063	0.077	0.093	0.113	0.140	0.172	0.212	0.262	0.333	0.396	β

6-2-1-4 Design of slabs by yield line method

The yield line method may be used in design of slabs based on slab behavior when reaching failure limit. It is conditional when using such a method the fulfillment of the minimum slab thickness. It should be pointed out that such method shall not fulfill crack width requirement in slab tension zones when subjected to aggressive environmental conditions of third and fourth types specified in sections (4-3-2-4-e). Hence, it shall not be permitted to use such design method for such cases.

It should be noted that the ratio of section resistance of negative moments $M_u^{/}$ to section resistance of positive moments M_u, in the same direction, may average between 1.00 and 1.50.

$$\frac{M'_u}{M_u} = 1.00 \sim 1.5 \qquad (6\text{-}10)$$

6-2-1-5 Concentrated loads on slabs

Concentrated loads on slab may be in one of the following cases

1 - Separate (single) concentrated loads, figure (6-3-a), and figure (6-3-b).

2 - Linear concentrated loads (such as walls), figure (6-3-c), figure (6-3-d).

Slabs subjected to concentrated loads may be calculated in accordance with theories of elasticity, but rules shown in items (6-2-1-5-1), (6-2-1-5-2) may be followed.

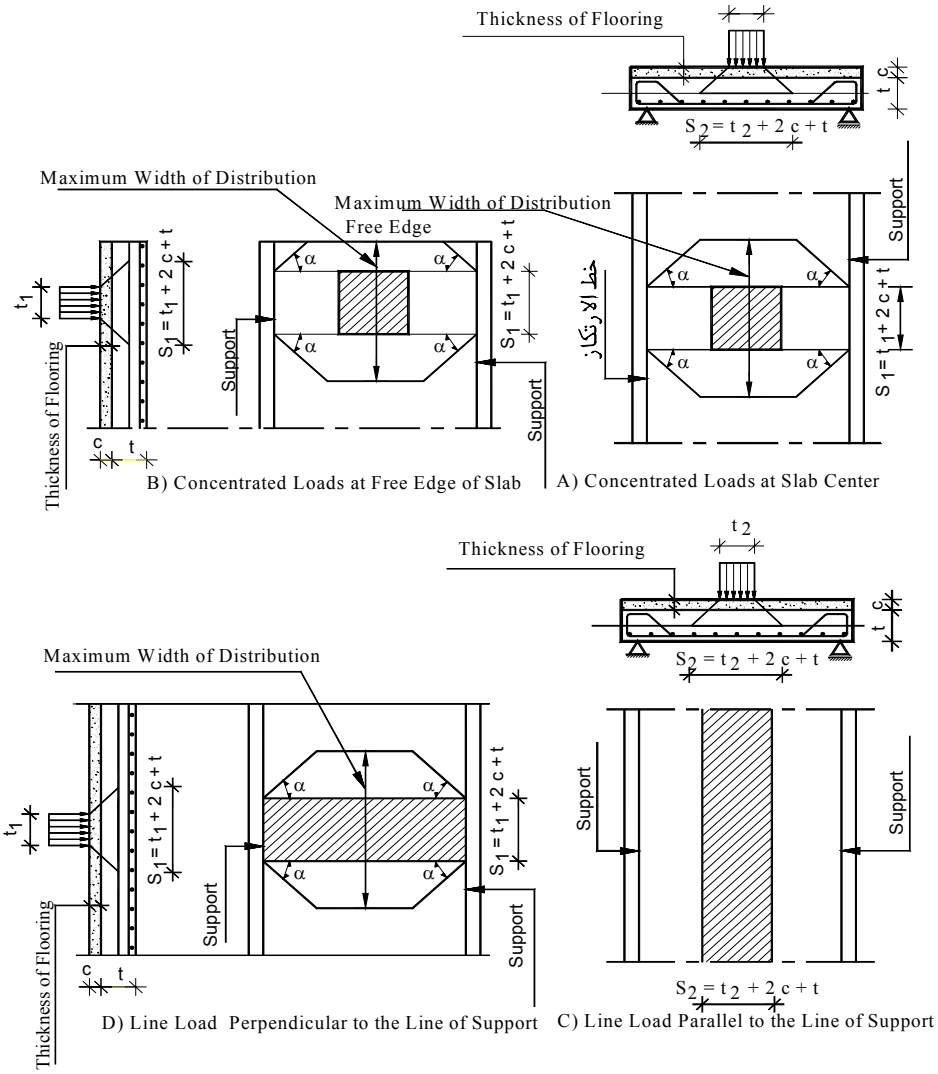


Figure (6-3) Distribution of separated and linear concentrated loads on one -way slab

6-2-1-5-1 One- way Slabs

1- Maximum distribution width of concentrated load

Primary width of concentrated load distribution on slab shall be defined according to equations (6-11) and figure (6-3).

$$S_1 = t_1 + 2c + t \qquad (6\text{-}11\text{-a})$$

$$S_2 = t_2 + 2c + t \qquad (6\text{-}11\text{-b})$$

Where:

t_1 = Load width in direction perpendicular to main reinforcement
t_2 = Load width in direction parallel to main reinforcement
c = Thickness of flooring cover
t = Slab thickness
S_1 = Load distribution width in direction perpendicular to main reinforcement at the support
S_2 = Load distribution width in direction parallel to main reinforcement

Distribution width would be equal S_1 at the support, gradually increasing until reaching maximum distribution width stipulated later on. Increase in width follows lines inclining by angle α to main reinforcement direction, as it shown in the plan.

Where:

$\tan \alpha = 1.00$ when calculating bending moments
$\tan \alpha = 0.50$ when calculating shear forces

Thus, maximum width of distribution in direction perpendicular to main reinforcement shall be equal to

$$S_1 + \left(\frac{A'_S}{A_S}\right) L \qquad (6\text{-}12)$$

Where L is the effective span in simply supported slab, or the distance between inflection lines in continuous slabs, provided that the ratio of secondary reinforcement $A_{S'}$ to that of the main reinforcement A_S in such equation shall not exceed 0.67, and maximum width shall not exceed the following:

a - For calculation of bending moments

- Maximum width given in equation (6- 12) shall not exceed (S1 + 2.0 meters), or slab length in direction perpendicular to main reinforcement, whichever is smaller.

- When concentrated load is close to the slab unsupported edge , or near the shorter side beams in slab, effective width of distribution , perpendicular to main reinforcement shall be taken equals to half the values stipulated previously plus the distance between load center and the unsupported edge or the slab shorter side beam edge (figure 6-3).

b - For calculation of shear forces

- Maximum width given in equation (6-12) shall not exceed (S_1 + L/3) or (S_1 + 1.00 meter) or slab length in direction perpendicular to main reinforcement, whichever is smaller.
- When concentrated load is close to line of support, the maximum allowed width of distribution when calculating shear forces between slab and carrying beam is (S_1 + 4t).
- When concentrated load is close to the beam throughout the slab short side, the maximum allowed width of distribution for calculating shear forces between slab and beam is (S_2 + 4t)

2- Bending moment and design

a - The additional torque resulting from concentrated load, shall be calculated with due consideration that the concentrated load is distributed on a length of the slab effective span equals S_2 and that the width affected by concentrated load in the direction perpendicular to main reinforcement is equal to that previously given .

b - Design bending moments of the slab within the maximum width of distribution shall be equal to the sum of bending moments resulting from slab dead and live loads, and additional bending moments as a result of the concentrated load.

c - Main reinforcement shall be calculated in accordance with bending moment previously given, additional secondary reinforcement of concentrated load shall extend for a length equals at least the width of distribution considered.

6-2-1-5-2 Two -way rectangular slabs

The following load distribution shall be used in the two directions for the cases where short and long suspended spans, a1, b_1 respectively conform to $b1/a_1 \leq 1.5$, otherwise the slab shall be considered as a one-way slab.

Distribution of isolated concentrated load in two directions

Isolated concentrated load distribution on slab in each of the two directions shall be taken as the reverse ratio to span length, as follows:

$$P_{a1} = P\left[\frac{b_1}{(a_1 + b_1)}\right] \qquad (6\text{-}13\text{-a})$$

$$P_{b1} = P\left[\frac{a_1}{(a_1 + b_1)}\right] \qquad (6\text{-}13\text{-b})$$

The maximum width of distribution in the short suspended span a_1 shall be:

$$S_2 + 0.4\, a_1 \qquad (6\text{-}14)$$

The maximum width of distribution in the long suspended span b_1 shall be:

$$S_1 + 0.4\, a_1\left[2 - \left(\frac{a_1}{b_1}\right)\right] \qquad (6\text{-}15)$$

Calculation of bending moments resulting from concentrated load in two directions

For calculating additional bending moment resulting from concentrated load in direction a_1, it shall be taken into consideration that load P_{a1}, distributed on length "a" of effective span shall be equal to the value given by equation (6-14), and that the width affected by concentrated load in the perpendicular direction to a_1 shall be equal to the value given by equation (6-15). Similarly, for calculating bending moment resulting from concentrated load in direction of b_1, it shall be taken into consideration that load P_{b1} distributed on length "b" of effective span shall be equal to the value given by equation (6-15), and that the width affected by concentrated load in the perpendicular direction to b_1, shall be equal the value given by equation (6-14).

Such additional moments shall be added to those resulting from permanent loads and live loads. The value of the total reinforcement shall

be calculated in each direction, and shall be placed within widthes affected by concentrated load.

6-2-2 Hollow block slabs

6-2-2-1 General

- When calculating hollow block slabs, such blocks shall be considered statically ineffective.
- The following condition regarding dimensions shall be fulfilled (figure 6-4).
  1 - Net distance between ribs "e" shall not exceed 700 mm.
  2 - Web width "b" shall not be less than 100 mm or one the third of depth "t", whichever is greater.
  3 - Compression slab thickness "t_s" shall not be less than 50 mm or one tenth of distance "e", whichever is greater
- The slab between ribs shall safely carry concentrated loads acting directly on it.

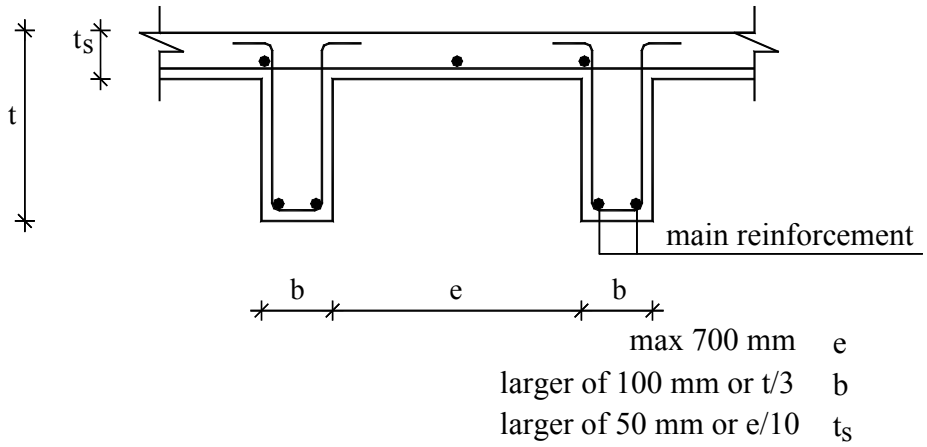


Figure (6-4) Hollow block slabs section and dimensions.

6-2-2-2 One- way hollow block slabs

- Cross sectional area of distributing bars perpendicular to ribs per meter shall not be less than the values given in section (6-3-1-10), and the minimum amount of distributing bars in slab (parallel to ribs) shall be 3ϕ 6 mm / m provided that a bar of 6 mm diameter shall be placed between every two ribs, and a bar at every rib, as shown in figure (6-4).
- If live load is less than or equals 3 kN/m^2 and spans are longer than 5.0 meters, the slab shall be provided with at least one cross rib at the span

center. Section dimension and bottom reinforcement of cross rib shall not be less than those of the main ribs, and the top reinforcement of the cross ribs shall be at least one half of the bottom reinforcement.

- If live load exceeds 3 kN / m ^{2}, and spans range between 4.0 and 7.0 meters, the slab shall be provided with one cross rib. However, for spans exceeding 7.0 meters, the slab shall be provided with three cross ribs; such cross ribs shall have the same dimensions and reinforcement as previously mentioned.

6-2-2-3 Two-way hollow block slabs

There are two cases for the beams supporting such slabs:

a - For the cases of beams having the same thickness as that of the slab (embedded or hidden beams), the slab shall be designed as a flat slab, or by the following the of item (b).

b - For the cases of rigid beams of having thicknesses greater than that of the hollow black slab thickness. Two types of such slabs are considered, as follows:

1 - The type where ribs have complete compression flanges: For such cases if the magnitude of the live load does not exceed 5 kN/m^2", the loads shall be distributed using the coefficients given in table (6-2). If live load exceeds 5kN/m^2, loads shall be distributed using coefficients given in table (6-3).

2 - The type where ribs have in-complete compression flange, (i.e. rib in the form of T section with limited compression flange width, or without compression flange); loads shall be distributed in both directions using coefficients given in table (6-3).

6-2-2-4 General notes

The following notes shall be applied to both one-way and two- way hollow block slabs.

- Shear forces in ribs shall be treated according to section (6-3-1-7). However, in case of design of the two- way hollow block slabs, as flat slabs, shear forces shall be treated according to section (6-2-5-8).
- Slab parts at supports shall be solid, to resist negative bending moments and shearing forces.

- Effective spans and bending moments in the slabs shall be determines according to sections (6-2-1-1-1) and (6-2-1-2-2)
- The minimum width of support on brick or stone walls shall be 200 mm.
- In case of simply supported hollow block slabs, the presence of hollow blocks over supports shall not be allowed; the slabs over support shall be solid.

Table (6-3). Values of coefficient α , β corresponding to values of "r" for hollow block slabs with in completed compression flange.

2.0	1.9	1.8	1.7	1.6	1.5	1.4	1.3	1.2	1.1	1.0	r
0.941	0.928	0.914	0.893	0.867	0.834	0.797	0.742	0.672	0.595	0.500	α
0.059	0.072	0.086	0.107	0.133	0.166	0.203	0.258	0.328	0.405	0.500	β

6-2-3 Waffle Slabs

Design waffle slabs are similar to that of flat slabs (figure 7-6) with due consideration of the following points:

1 - Distance between web (rib) centerlines (e + b) in figure (6-5) shall be increased up to 1.50 meter.

2 - Upper slab (flange) thickness t_s shall not be less than e/12 or 50 mm whichever is greater.

3 - Minimum web (rib) width "b" shall not be less than one quarter of slab thickness "t" or 100 mm whichever is greater. Concrete cover requirements, the distance between bars and fire requirements shall be satisfied.

4 - The requirements for punching shear resistance over columns shall be fulfilled.

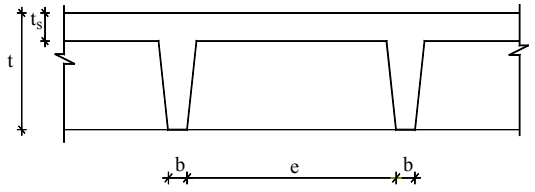


Figure (6-5) Waffle slab

6-2-4 Paneled Beams

a - When total dimensions of two- way slabs are relative big so that it is impractical to design the slab as solid slab, hollow block slab or waffle slab, a structural consisting of intersecting beams equal in depth and form a grid with monolithic slabs shall be more appropriate to use.

b - Intersected beams are normally arranged in two perpendicular directions forming rectangular grid. Beams may be also be arranged diagonally to form skew grid, arranged in three directions to form triangular grid, or arranged in four directions to form triangular grid.

c - It is structurally recommended to use beams parallel to the edges when rectangularity ratios of the floor vary between the values of 1.00 and 1.50. In the case that the rectangularity ratio is greater than 1.50 it may be suitable to use skew grid.

d - Internal forces shall be calculated, and slab between paneled beams shall be designed in accordance with section (6-2-1-3) and section (6-2-2).

e - Internal forces in paneled beams shall be analyzed using theory of elasticity. One of the simplified methods may also be used, provided that the design shall be in full complaisance with the actual behavior of the paneled beam system.

f - The design shall also satisfy the requirements of section (6-3)

6-2-5 Flat slabs

6-2-5-1 General

Flat slabs are reinforced concrete slabs with or without drop panels, supported on columns with or without column heads, as shown in figure (6-6). It includes solid slabs, slabs with ribs in two directions with or without hollow blocks.

Symbols:

L_1 = span length in the considered direction, measured center to center of support.

L_2 = span width in direction perpendicular to the direction under consideration, measured center to center of support.

L = longer panel length

L_x = shorter panel length measured from column centerlines

L_y = longer panel length measured from column centerlines

D = Diameter of the greater circle that can be drawn within the column section ,or column head, if any.

W = Total load per unit area of the panel
t = Slab thickness
d = Slab effective depth

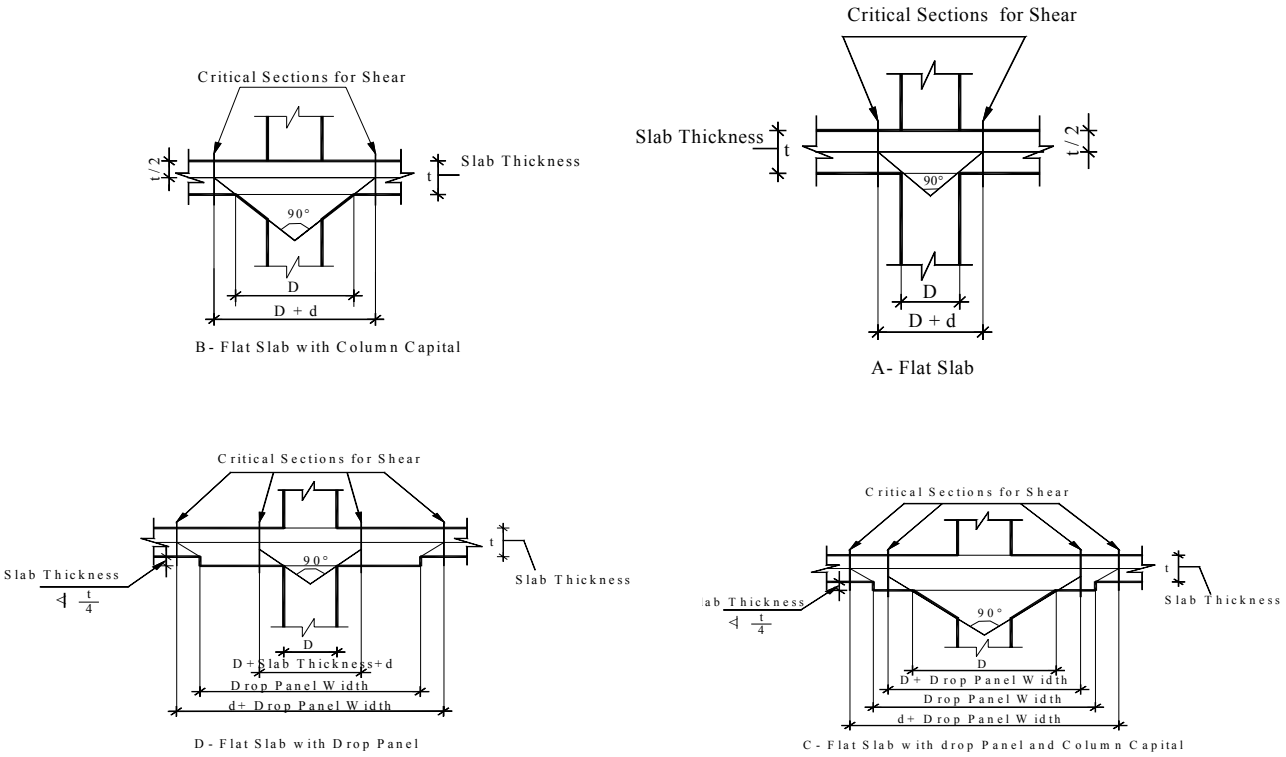


Figure (6 – 6) Critical sections for shear

6-2-5-2 Limits of concrete dimensions

a- Minimum slab thickness

Slab thickness shall not be less than the greatest of the following values:

1 - 150 mm.

2 - L/32 for external panels without drop.

3 - L/36 for continuous internal panels without drop, or external panel with drop.

4 - L/40 for continuous internal panels with drop.

b- Minimum dimension of columns

The diameter of circular column or the length of any of the sides of rectangular column shall not be less than the greatest of the following values:

1 - 1/20 of panel length in the considered direction.

2 - 1/15 of the total floor height.

3 - 300 mm.

The 300 mm limit may be reduced if the column and slab are designed to resist forces and moments transferred between them according to section (6-2-5-8-1).

c- Column heads

For columns provided with column heads, the following requirements of interior and exterior column heads shall be satisfied:

1 - The maximum head inclination to the vertical shall not exceed $45^{o.}$

2 - Effective diameter D considered in design shall not exceed one quarter of the smaller span of adjacent slabs. For noncircular column or column heed, the effective diameter D shall be considered as the diameter of the greatest circle that can be drawn within the column section or column head, (if any).

d- Drop panel

Drop panels are thickened slabs above the columns or their column heads for resisting negative bending moments or punching shear, and reducing reinforcement steel, the following shall be considered:

1 - Drop panel thickness below slab shall not be less than one quarter of the slab thickness.

2 - Drop panel shall extend to a distance not less than one sixth of shorter panel length in the same direction, measured from column centerlines, so as not to exceed one quarter of length of the panel of the shorter length.

e- Flat slab design strips

Flat slab panels are divided into the following design strips as shown in Figure (6-7):

- Column strip: a strip having a width equals to half of the shorter length, except in case of using drop panel the width shall be taken equal to drop panel width.

- Field strip: a strip having width equals to the difference between panel width and column strip width.

6-2-5-3 Structural analysis methods

a - Flat slabs may be analyzed according theory of elasticity. Yield lines method may be also used, provided achieving the ratio of negative moments to positive moments according to section (6-2-1-4). It is noted that the requirements of crack limit state in the tension surfaces of slabs subjected to environmental conditions of third and fourth sections according to section (4-3-2-4-e) shall not be fulfilled when using the yield line analysis method. Accordingly, it shall not be used for the analysis of such cases.

b - Flat slabs with columns on perpendicular straight axes in both directions may be analyzed according to one of the following two methods:

1 - As continuous frames, using the method conforming to section (6-2-5-4)

2 - The empirical method conforming to section (6-2-5-5)

Offset of column locations by not more than 10% of the average length of the two perpendicular panels shall be permitted.

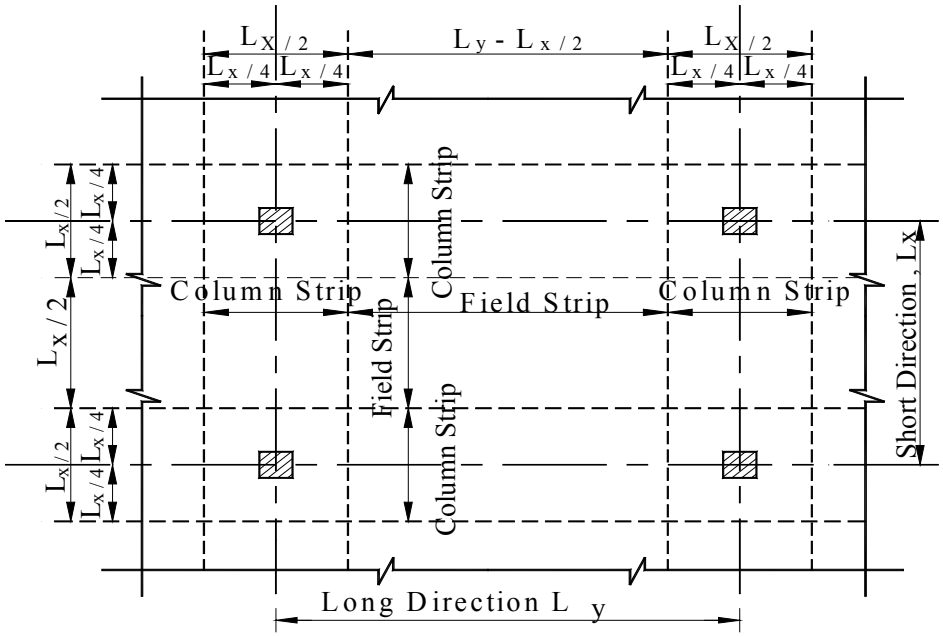
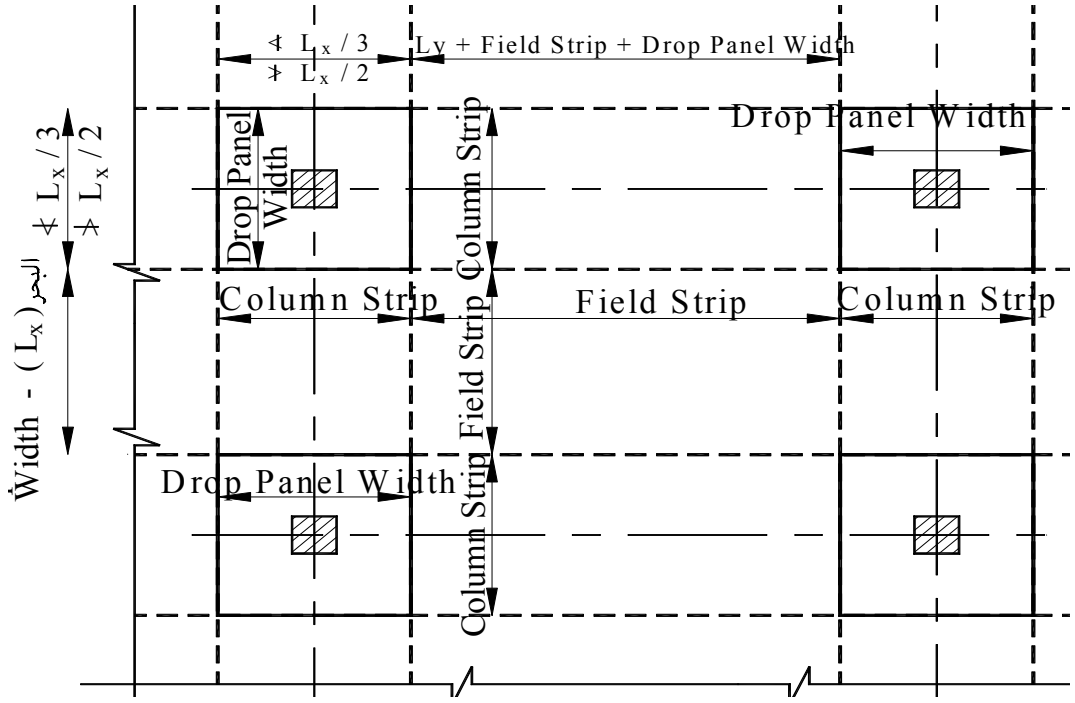


Figure (6-7-a) Dividing flat slab panels into column strips and field strips for slab without drop panel



Figure(6-7-b) Dividing flat slab panels into column strips and field strips for slab with drop panel

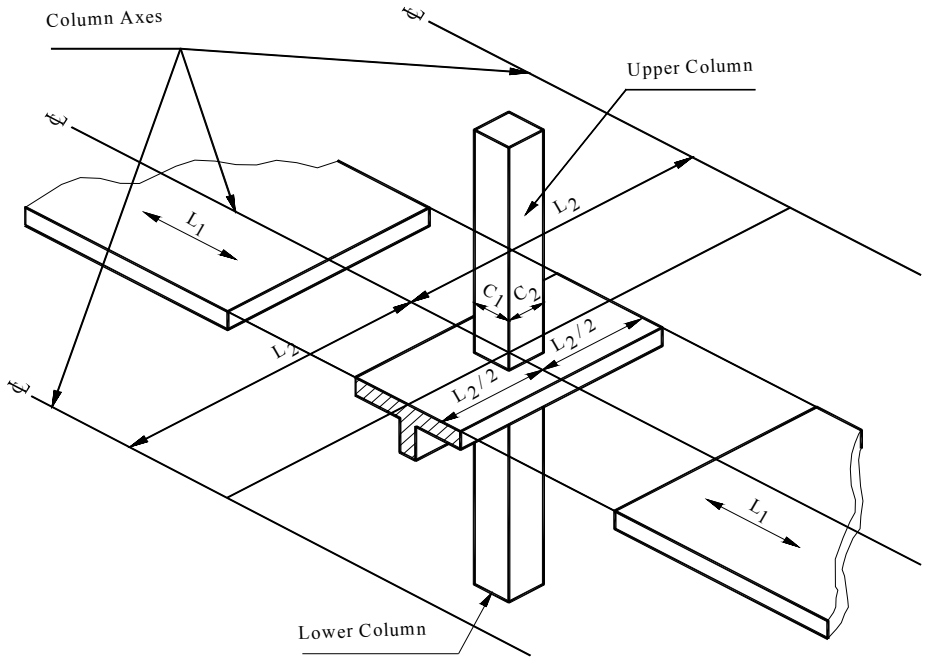


Fig (6-8) Equivalent column (columns and torsional elements)

6-2-5-4 Analysis of flat slabs as continous frames

As an alternate method of the precise structural analysis of flat slabs using theory of elasticity, the following analysis may be used:

A - Bending moments and shear forces shall be calculated by analyzing the structure as continuous frames subject to the following assumptions:

- The structure shall be divided longitudinally and transversally into frames onsisting of a row of columns and strips of slabs situated on both sides of the column row with width equals the distance between panels centerlines.
- Each continuous frame shall be analyzed as a seperate frame consisting of a strip of slabs, and row of columns above and below with ends totally fixed. The full magnitudes of both dead and live load shall be taken in each direction seperatly, with due consideration of placing live load at the locations that shall give the maximum values of internal stresses in the various members of the frame. Spans used in such analysis shall be taken equals to distances between column centerlines. Differences in rigidity frame elements shall be also taken into consideration.
- For vertical load analysis, the flat slab flexure stiffness shall be calculated using the total width slab, (i.e. distance between column centerlines).
- For lateral load analysis, effective width shall be taken for calculation of rigidity equals to column width plus three times the slab thickness including drop panel, (if any) on both sides of the column, provided that the effective width shall not exceed one third of the distance between column centerlines. Internal forces from lateral loads shall be applied to such effective width.
- When calculating flexure stiffness for equivalent columns, one of the following two methods may be used:

A-1 By considering the combined effects of both column flexural stiffness and torsional stiffness of the monolithically connected torsional elements to the column. Beams and the effective torsional parts of the slab in direction perpendicular to frame plane represent the torsional elements monolithically connected torsional element considering that for beamless torsional element comprises the column width c_1 plus three times the slab thickness according section (4-2-3-2) and figure

(4-11-b). The equivalent column flexural stiffness k_{ec} shall be calculated using the following relation and figure (6-8).

$$K_{ec} = \frac{\Sigma K_c}{\left[1 + \frac{\Sigma K_c}{K_t}\right]} \qquad (6\text{-}16\text{-a})$$

Where ΣK_C = sum of the flexural stiffness of the column above and below slab level, assuming that the columns are totally fixed at the upper and lower ends. Column flexural stiffness shall be given by the relation.

$$K_c = \left(\frac{4E_c I_g}{h}\right) \qquad (6\text{-}16\text{-b})$$

Where:

h = column height.

Ig = gross moment of inertia outside the connection of the concrete section of the column about the neutral axis, neglecting reinforcement steel and effect of cracks.

E_C= modulus of elasticity of concrete, conforming to section (2-3-3-1)

For slabs with drop panel or column heads or non-prismatic column sections, it is preferable to calculate column stiffness values K_C taking considering the actual stiffness for such cases.

K_t = torsional stiffness of the torsional elements of the equivalent column, calculated from following relation:

$$K_t = \Sigma\left[\frac{9E_c \,.\, C}{L_2 \,.\left(1 - \left(\frac{c_2}{L_2}\right)\right)^3}\right] \qquad (6\text{-}16\text{-c})$$

Where c_2, L_2 are column dimension and span length in the direction perpendicular to analysis direction as shown in figure (6-8). C is section constant, calculated by following relation:

$$C = \Sigma \left[\left(1 - 0.63 \left(\frac{b}{t} \right) \right) . \left(\frac{b^3 . t}{3} \right) \right] \qquad (6\text{-}16\text{-}d)$$

Where t, b are the longer and shorter dimensions of torsional element, respectively. The value of C for a T or L sections shall be equal to the sum of the values of C obtained for the various rectangular sections that make up the T or L sections

A–2 Calculation of equivalent moment of inertia for column I_{ec} using the following relation:

$$I_{ec} = \psi . I_g \qquad (6\text{-}17\text{-}a)$$

Where Ψ, is a coefficient conforming to following relation:

$$\psi = \left[0.6 + 0.4 \left(\frac{\alpha . L_{2a}}{L_{1a}} \right) \right] \left(\frac{L_{2a}}{L_{1a}} \right)^2 \quad \text{For external columns} \quad (6\text{-}17\text{-}b)$$

$$\psi = \left[0.3 + 0.7 \left(\frac{\alpha . L_{2a}}{L_{1a}} \right) \right] \left(\frac{L_{2a}}{L_{1a}} \right)^2 \quad \text{For internal columns} \quad (6\text{-}17\text{-}c)$$

For exterior columns $\psi = \left[0.6 + 0.4 \left(\frac{\alpha . L_{2a}}{L_{1a}} \right) \right] \left(\frac{L_{2a}}{L_{1a}} \right)^2$ (6-17-b)

For interior columns $\psi = \left[0.3 + 0.7 \left(\frac{\alpha . L_{2a}}{L_{1a}} \right) \right] \left(\frac{L_{2a}}{L_{1a}} \right)^2$ (6-17-c)

Provided that , $0.30 < \Psi < 1.00$, and the ratio $\frac{\alpha L_{2a}}{L_{1a}}$ shall not exceed 1.00, where:

α = The ratio of the moment of inertia of the torsion resistant beam (if any) to the moment of inertia of the slab strip.

L_{1a} = Average of the two span lengths on both column sides in analysis direction.

L_{2a} = Average of the two span lengths on both column sides in direction perpendicular to the analysis direction.

b - Slab shall be designed at any section, for bending moments calculated as previously outlined. It shall not be required to consider negative bending moment values greater than those at the face of the column. Bending moments calculated by the previous method shall be divided between both column and field strips by ratios given in table (6-4).

c - When column strip shall be taken equal to drop panel width, and field strip width shall be increased to a value greater than one half of span width, accordingly moments that field strip resists shall exceed the values given in table (6-4) in proportion to the increase in column strip width. For such cases moments that column strip resists shall be reduced to values lower than those given in table (6-4) such that there shall be no reduction in total values of positive moments and the total negative moments resisted by the column and field strips .

Table (6-4) Distribution of bending moments subject to vertical loads, between column strips and field strip for flat slab panels designed as continuous frames

Type of moment	Distribution of bending moments between column strips and field strips as a percentage of total positive and negative bending moments	
	Column strip	Field strip
Negative moments in internal span	75	25
Negative moments in external span	80	20
Positive moments	55	45

6-2-5-5 Empirical analysis for flat slabs subject to uniformly distributed loads

a- Limits of using the method

Such method may be used subject to fulfilling the following conditions:

1 - Flat slabs shall have a number of rectangular panels of almost constant thickness arranged in at least three rows in two perpendicular

directions, provided that the ratio of span length to its width shall not exceed 1.3.

2 - Length and width of any adjacent panels shall not differ in any group by more than 10% of the greatest length or width, provided that the separate spans shall not differ from each other in the group by more than 20% of the greatest span. End spans may be shorter than interior spans, and shall not be longer than them. In case of different length adjacent spans, the greatest span length shall always be used in calculation of bending moments.

3 - Live load shall not exceed double the slab permanent load.

b- Critical sections of bending moments in flat slabs

In continuous interior panels, critical sections of bending moments shall be as follows:

1 - For positive moments, critical sections shall be along panel centerline.
2 - For negative moments, critical sections are at panel edges throughout the line connecting column centers, and around the perimeter of column heads.

c- Bending moments in flat slab panels

Bending moment M shall be calculated in both directions of the panel according to the following equation:

$$M = \left(\frac{w\, L_2}{8} \right) \left[L_1 - \left(\frac{2D}{3} \right) \right]^2 \qquad (6\text{-}18)$$

Where L_1 is the length in considered direction, and L_2 is the length in the perpendicular direction, and w is the density of total load per square meter of the slab.

Value of M shall be divided between field strip and column strip in the considered direction according to the ratios given in table (6-5) and figure (6-9) the requirement of section (6-2-5-4-c) shall be satisfied.

Table (6-5) Distribution of bending moments due to vertical loads in flat slab panels as a percentage of M

Strip	Type of end support*	Exterior panel			Interior panel	
		Exterior negative moments	Positive moments	Interior negative moments	Negative moments	Positive moments
Column strip	a	25	30	50	45	25
	b	20	30			
Field strip	a	5	20	20	15	15
	b	10	20			

* Types of end support

a- No beams.

b- Beams with total depth equal or greater than three times slab thickness t.

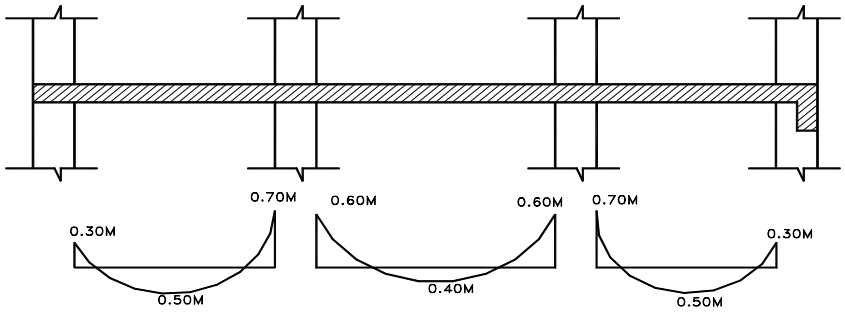


Figure (6-9) Total moments in panels for column and field strips in a flat slab supported on concrete columns

d- Negative bending moments in mid spans in case of heavy live loads

In case of heavy live loads (p > 1.5 g), negative bending moments in interior mid span shall not be less than the following values:

$$M_{-ve} = \left[g - \left(\frac{2p}{3} \right) \right] \left(\frac{L_2}{40} \right) \left[L_1 - \left(\frac{2}{3} \right) D \right]^2 \qquad (6\text{-}19\text{-a})$$

for column strip in direction L_1

$$M_{-ve} = \left[g - \left(\frac{2p}{3} \right) \right] \left(\frac{L_2}{100} \right) \left[L_1 - \left(\frac{2}{3} \right) D \right]^2 \qquad (6\text{-}19\text{-}b)$$

for field strip in direction L_1

Where p, g shall be the uniform live and permanent (dead) load on unit area respectively.

e- Bending moments in columns

1 - Internal and external columns shall be designed to resist bending moments equal 50 % and 90 %, respectively of the negative moment in column strip according to table (6-5). Such moments shall be divided between upper and lower columns by the ratio of its stiffness. In internal columns direct load, acting along with the moment, may be reduced, considering that the span on one side is free from live load.

2 - In case of external columns carrying parts of the slab and walls as cantilever loads, bending moments in columns may be reduced as determined in the pervious clause, by as much as the resulting moment of dead load on the cantilever part.

6-2-5-6 Bending moments in spans with or without marginal beams

a - For slab supported on marginal beam with a total depth equals or exceeds three times slab thickness, bending moments acting on the half column strip parallel to beam shall be equal to one quarter of the values given in table (6-4) or table (6-5).

b - In normal cases, where no marginal beams are present, bending moments acting on the half column strip shall be equal to half the values given in table (6-4) or table (6-5).

6-2-5-7 Design loads acting on marginal beam

1 - Total load that marginal beam carries shall include direct loads on the beam, in addition to distributed load equals the load acting on one quarter of total panel, and the torsional moments transferred from the monolithically connected slab.

2 - Marginal beam resistance to torsion shall be calculated according to equation (4-54) shown in figure (6-10), with moment redistribution according to figure (6-11).

6-2-5-8 Negative moments transferred from slab to columns

6-2-5-8-1 Total negative moments M_f in external spans (figure 6-12-a) or the difference of negative moment M_f in internal spans (figure 6-12-b) shall be transferred to columns according to the following distribution:

a - A part transfers directly to columns by bending moments ($\gamma_f M_f$) and γ_f shall be taken according to the following equation:

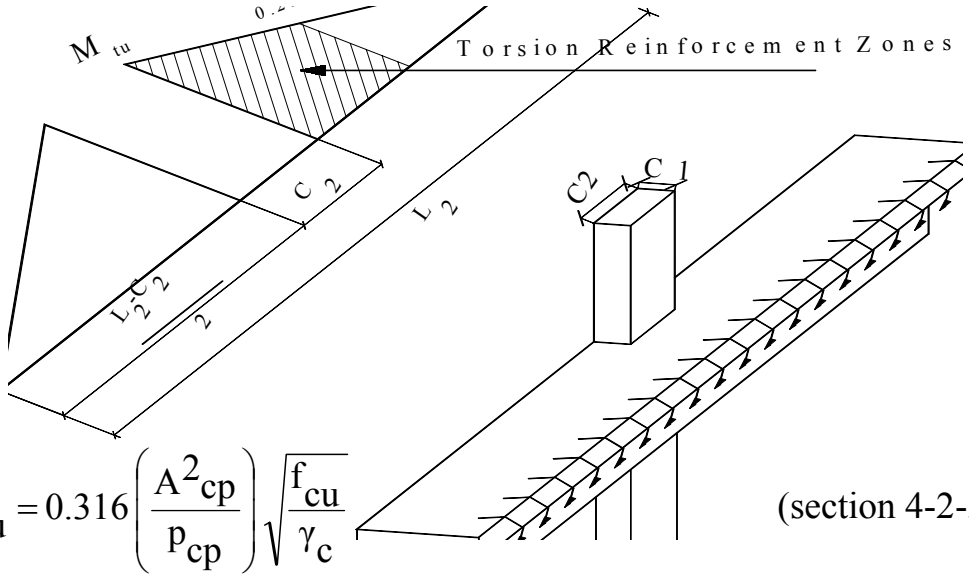
$$\gamma_f = \frac{1}{1+\left(\frac{2}{3}\right)\sqrt{\frac{b_1}{b_2}}} \qquad (6\text{-}20)$$

Where:

γ_f = Coefficient of moments transferred by bending

b_1 = Length of critical section in punching shear, measured in analysis direction

b_2 = Length of critical section in punching shear, measured in the direction perpendicular to b_1.



$$M_{tu} = 0.316\left(\frac{A^2_{cp}}{p_{cp}}\right)\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad \text{(section 4-2-3-6)}$$

Figure (6-10) Torsion moments acting on marginal beam

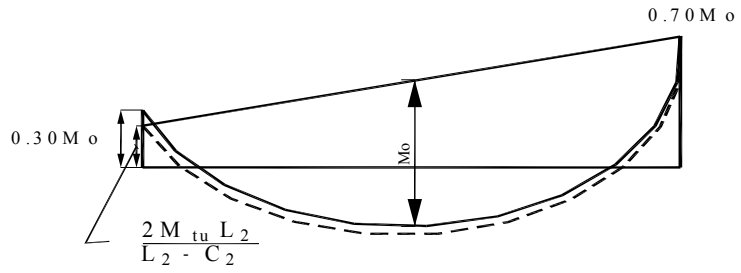


Figure (6-11) Redistribution of moments in external panel as a result of drop in marginal beam section torsional rigidity due to cracking

Steel reinforcement required to resist such moments in effective width (b_e) shall be concentrated as shown in figure (6-13).

b – That part of the moment transferred to columns by torsion moments ($\gamma_q M_f$), and γ_q shall be taken according to the following equation:

$$\gamma_q = 1 - \gamma_f \qquad (6\text{-}21)$$

Where:

γ_q = Coefficient of moments transferred by torsion causing punching shear on critical section, shown in figure (6-14) and figure (6-15) and are calculated in both directions according to following equations:

Punching shear stress q_x resulting from moment M_{fx}, considering γ_{qx} is coefficient of moments transferred by torsion

$$q_x = \frac{M_{fx} \cdot \gamma_{qx} \cdot C_{CB}}{J_{cx}} \qquad (6\text{-}22\text{-a})$$

Punching shear stress q_y resulting from moment M_{fy}, considering γ_{qy} is coefficient of moments transferred by torsion

$$q_y = \frac{M_{fy} \cdot \gamma_{qy} \cdot C_{AB}}{J_{cy}} \qquad (6\text{-}22\text{-b})$$

Such stresses shall be added to punching shear stress resulting from vertical loads according to equation (4-31) section (4-2-2-3) for design by limit state method, or section (5-4-3) for design by working stress method.

Where:

J_{cx} , J_{cy} = constant for critical moment in shear similar to polar moment of inertia around x, y coordinates respectively. Figures (6- 14), (6-15) show shear stresses resulting from moment M_y Values of J_{cy} shall be determined as follows:

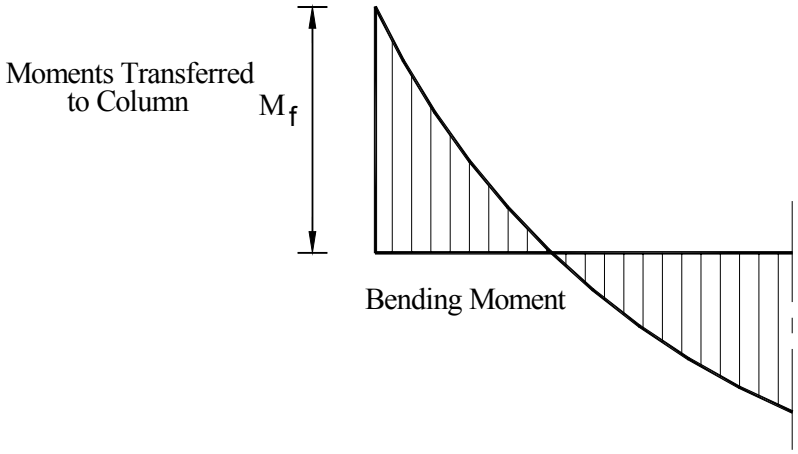


Figure (a) Bending moments in case of external edge column

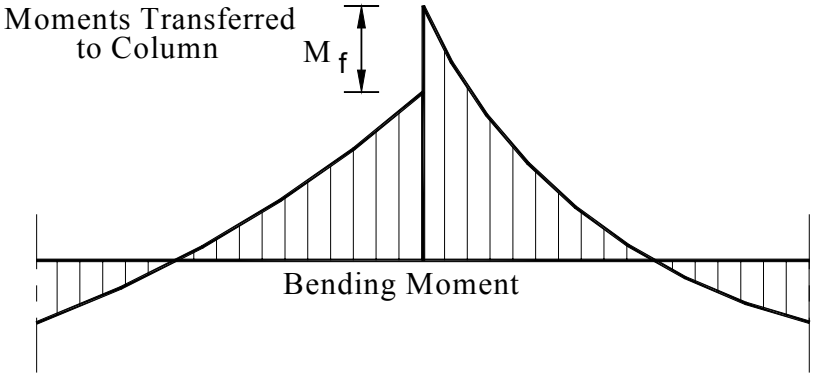


Figure (b) Bending moments in case of internal column.

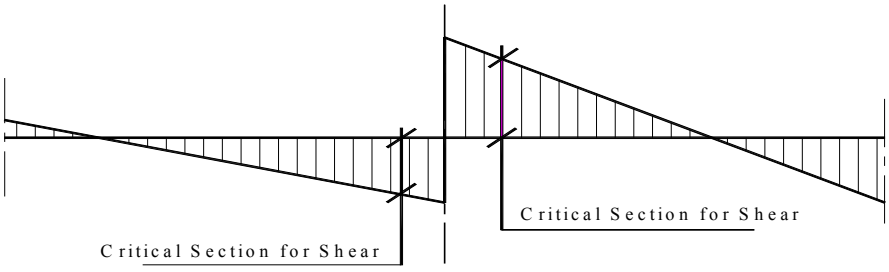


Figure (c) Shear forces in case of internal column

Figure (6-12) Bending moments and shear forces transferred to columns

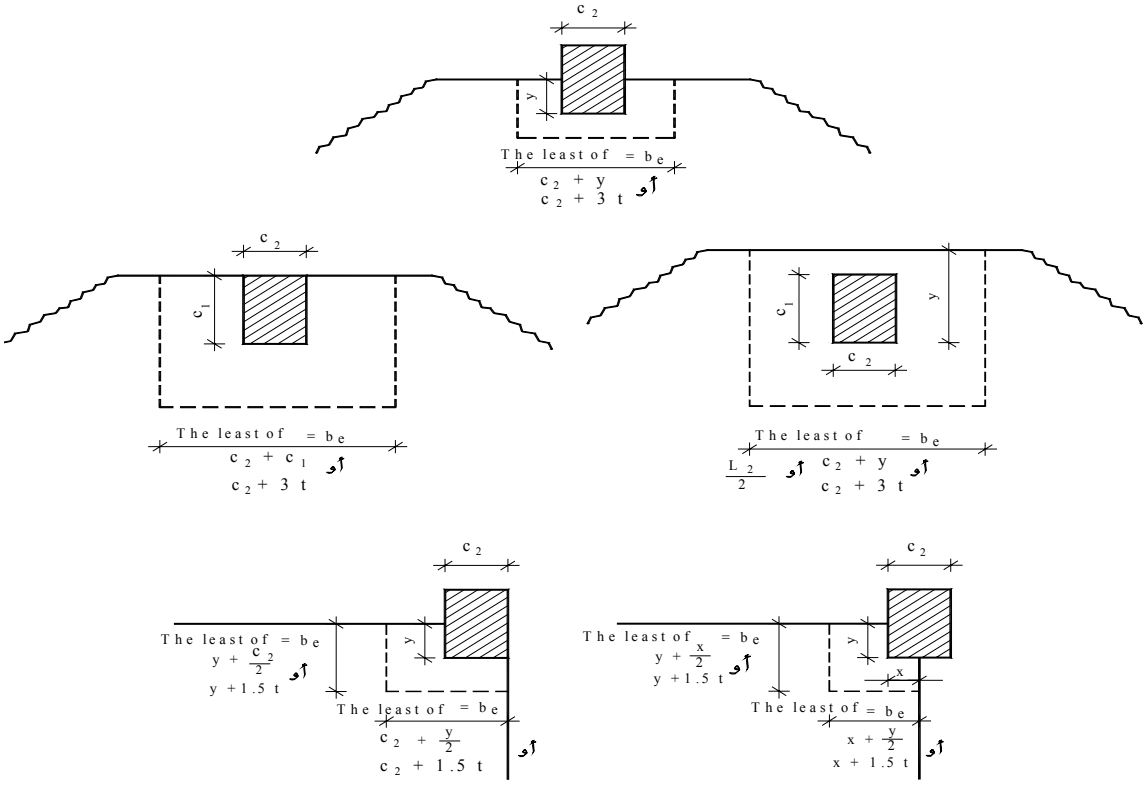


Figure (6-13) Width of strip (b_e) transferring bending moments in different cases

1 - In case of internal columns, figure (6-14), J_{cy} value shall be as follow:

$$J_{cy} = d\left(\frac{(c_1+d)^3}{6}\right) + d^3\left(\frac{c_1+d}{6}\right) + \frac{d\left((c_1+d)^2\ (c_2+d)\right)}{2} \qquad (6\text{-}23)$$

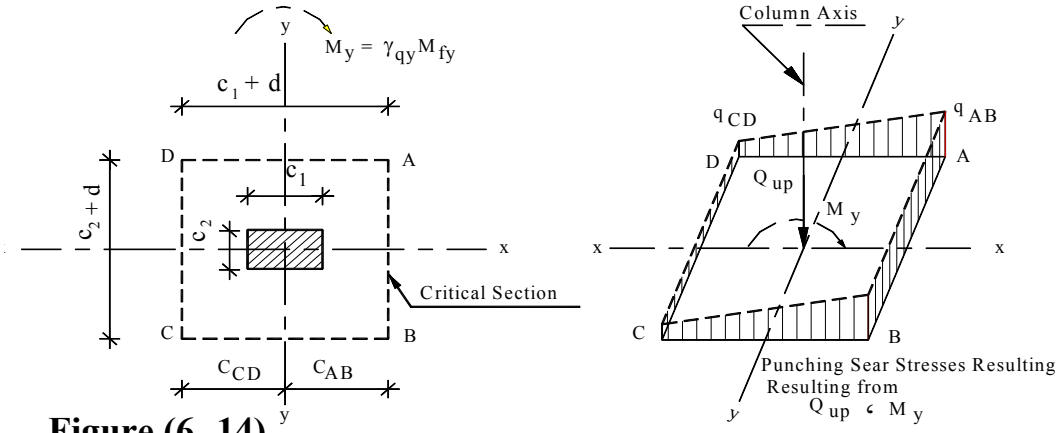


Figure (6 -14)
Distribution of punching shear stresses (internal column)

2 - In case of external columns, figure (6-15), J_{cy} shall be calculated by equation:

$$J_{cy} = d\,(c_2 + d)\left(C_{AB}\right)^2 + \left(\frac{2}{3}\right) d\left(C_{CD}\right)^3 + \left(\frac{2}{3}\right) d\left(C_{AB}\right)^3 + \left(\frac{1}{6}\right)\left(c_1 + 0.5d\right) d^3 \qquad (6\text{-}24\text{-a})$$

Where :

$$C_{AB} = \frac{(c_1 + 0.5d)^2}{[(c_2 + d) + 2\,(c_1 + 0.5d)]} \qquad (6\text{-}24\text{-b})$$

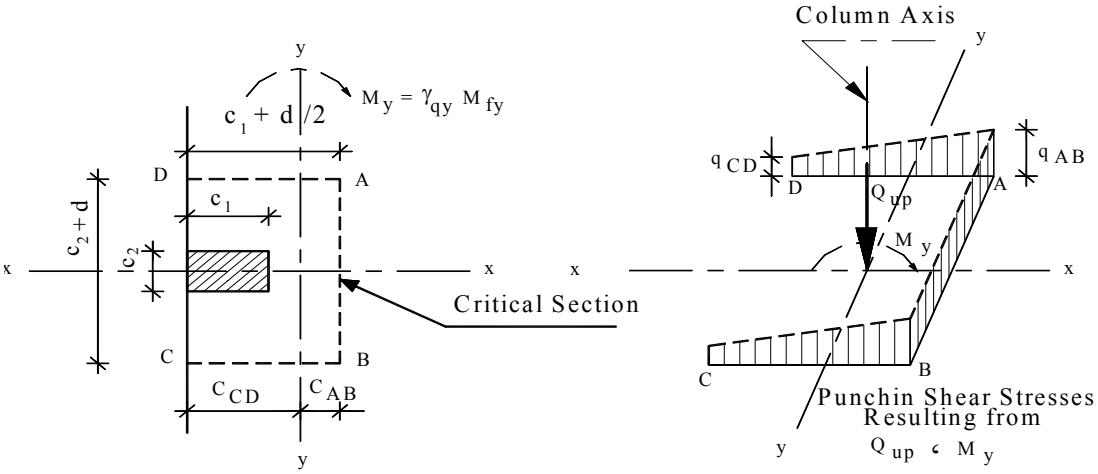


Figure (6 – 15) Distribution of punching shear stresses (external column)

6-2-5-8-2 Conditions of item (6-2-5-8-1), of transferring negative moments from slabs to columns, may be neglected in the following cases:

a- For internal columns in case of availability of both conditions:

1 – Live loads shall not exceed 4 kN/ m^2.

2 – Equal adjacent span or difference by a ratio not exceeding 20%.

B – For external columns in case of availability of either condition:

1 – Presence of rigid marginal beam of a depth not less than three times the slab thickness.

2 – Presence of cantilever slab outside the columns by a distance not less than one quarter of span length measured from column outer face, and loaded with the same slab load.

6-2-5-8-3 Total shear stresses (including stresses resulting from the effect of bending moments transferred between flat slab and columns), under effect of vertical loads may be calculated using the following simplified method:

$$q = \frac{Q \cdot \beta}{b_o \cdot d} \qquad (6\text{-}25)$$

Where:

Q = Design shear forces transferred to the column when loading the spans surrounding it with the entire design load.

d = slab effective depth.

b_o = Length of the critical perimeter section in punching shear according to section (4-2-2-3) and figures (6 – 14), (6 – 15).

β = A coefficient depends on effect of the eccentricity of shear forces, and taken as follows:

β = 1.15 in case of internal columns

β = 1.30 in case of external columns

β = 1.50 in case of corner columns

6-2-5-9 Arrangement of reinforcement in flat slabs

Flat slabs shall be reinforced, according previous methods, in two directions, and as shown in figure (7 – 4), so that each strip shall be reinforced through its entire width, considering items in section (7 – 5), and requirements of section (6 – 8 – 2 – 2) regarding earthquake design.

6-2-5-10 Reinforcement of column heads

Regarding the requirements of distances between bars, column heads shall be reinforced as shown in figure (6-16) with bars (1), (2) which are anchored by reinforcement steel (3) (stirrups) as shown in figure (6-16) that is sufficient to resist maximum bending moments as in section (6-2-5-4) clause (a), and section (6-2-5-5) clause (e). Total area of such

reinforcement shall not be less, in every direction (1), (2) than the following:

1 - When column head is rectangular, reinforcement steel area shall not be less, in each direction than (0.04) of the negative reinforcement steel area per meter for the column strip of the slab in the considered direction multiplied by the length of perpendicular panel to this reinforcement.

2 - When column head section in circular, the sum of reinforcement steel (1), (2) obtained as above shall be distributed as shown in figure (6-16), along the perimeter of the column head.

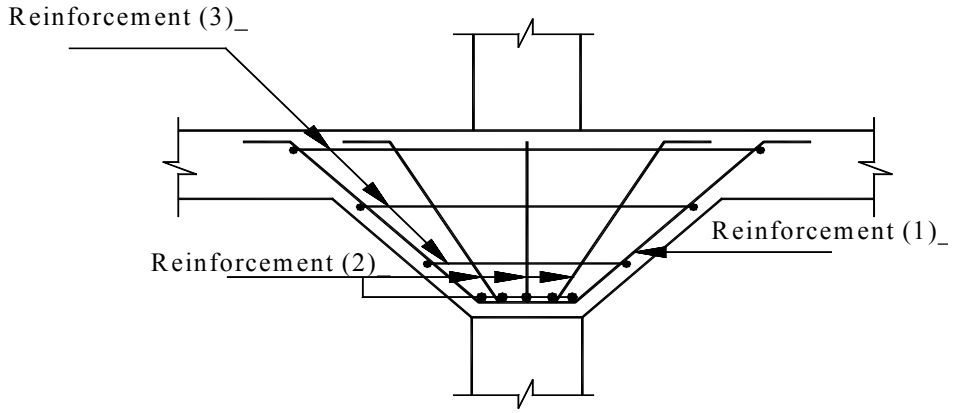


Figure (6-16) Reinforcement of column heads for flat slabs

6-2-5-11 Opening in flat slabs

According to figure (6-17) and figure (6-18):

a - Preferably openings within column heads shall be avoided.

b - It shall be permitted to make openings in intersecting areas between field strips of area A, figure (6-17), provided the fulfillment of the following:

1 - Greatest dimension of opening shall not exceed 0.40 of span length in direction parallel to axis.

2 - Positive and negative total design bending moments shall be redistributed on the rest of the structure based on the change due to opening presence.

c - It shall be permitted to make openings in the intersecting area between column strip and field strip of area B, figure (6-17), provided the fulfillment of the following:

1 - Total opening length or width shall not exceed one quarter of strip width in either directions.

2 - Sections of the two strips in the opening area can resist the design moments.

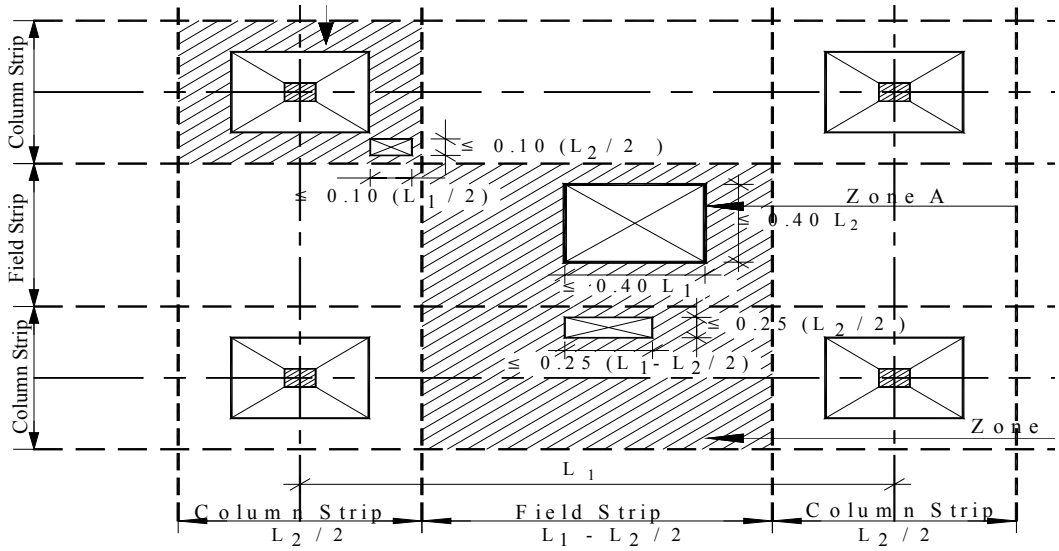


Figure (6-17) Allowed locations and dimensions of openings in flat slabs

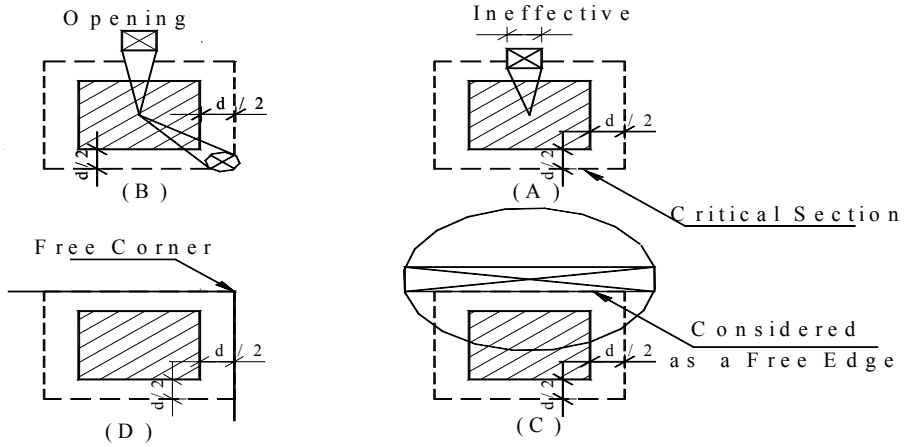


Figure (6-18) Effect of openings in flat slabs on critical section of punching shear

d - It shall be allowed to make openings in the intersecting area between two column strips area C, figure (6-17), provided the fulfillment of the following:

1 - Total opening length or width shall not exceed 0.10 of column strip width in either direction.

2 - Sections of the two strips, in the opening area can resist the design moments.

3 - Effect of openings in flat slabs on critical section of punching shear shall be taken according to figure (6-18).

e - In case of the opening dimensions in flat slabs exceeding values in items a, b, c, d, accurate structural calculations shall be made, fulfilling conditions of resistance and serviceability limit states.

6-3 Beams

This part includes the following beams:

1 - Ordinary beams

2 - Deep beams

6-3-1 Ordinary beams

6-3-1-1 General considerations

a - This section addresses the ordinary beams having effective span to depth ratio greater than 4.

b - The deep beams are characterized as beams effective span to depth ratios greater than 1.25 for the simple beams and 2.5 for the continuous beams. It is preferable to design deep beams using the strut and tie design method conforming to sections (4-2-2-6-2) and (6-3-2-3). Provisions of this section may also be applied to the design of deep beams.

6-3-1-2 Effective span

1- The effective span of simply supported beams

The effective span of the simply supported beams shall be taken equal to the least value of:

a - The distance between the supports axes.

b - The clear span between the supports plus the beam depth.

c - 1.05 of the clear span.

2- The effective span of continuous beams

a - Monolithically cast beams with supports:
The effective span of the continuous beams shall be taken equal to the distance between supports centerlines or 1.05of the clear span, whichever is less.

b - Beams supported on masonry supports:
The effective span shall be taken equal to the distance between supports centerlines or the clear span plus the beam depth, whichever is less.

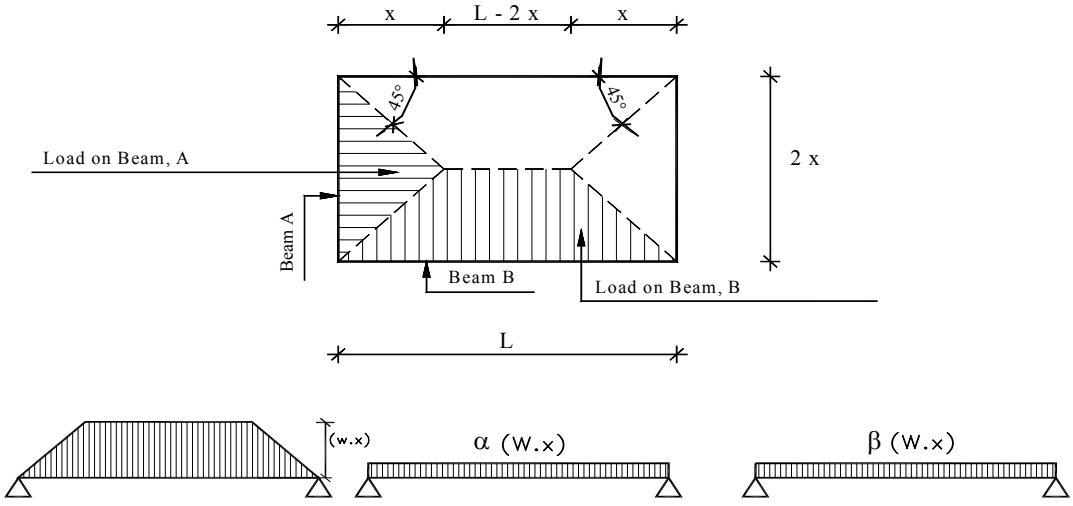
3 - The effective span of the cantilever

The effective span of the cantilever shall be taken equal to the least value of:

- Cantilever length measured from the support centerline.
- The net length plus the greater depth of the cantilever.

6-3-1-3 Load distribution on beams

a - Distribution of loads on slabs transmitted to the beams using the areas formed by angle bisectors lines at the corners of any panel as shown in figure (6-19).



The equivalent uniform load for shear forces calculation

The equivalent uniform load for bending moments calculations

The effective load on beam (B)

Figure (6-19) The effective slabs loads on beams and the equivalent uniform loads

b- The distribution given in, (a) shall be subject to fulfilling the following conditions:
- Greatest intensity of the load is located at mid span.
- Load distribution covers the entire span of the beam.
- Load distribution is symmetrical around the beam mid span.

Loads satisfying the preceding conditions shall be considered uniformly distributed along the beams span- except for the cantilever beams, as follows:

Assuming that: w = uniformly distributed slab load per unit area.
L = beam span length between supports centerlines.
Then: α w.x = The uniform equivalent load to original transmitted loads for calculating bending moments of beams as shown in figure (6-19).
β w.x = The uniform equivalent load to original transmitted loads for calculating shear forces and reactions of beams as shown in figure (6-19).
α and β values are to be taken from table (6-6).

Table (6-6) α and β coefficients values for estimating equivalent uniform loads to original loads acting on beams

L/2X	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
α	0.667	0.725	0.769	0.803	0.830	0.853	0.870	0.885	0.897	0.908	0.917
β	0.500	0.554	0.582	0.615	0.642	0.667	0.688	0.706	0.722	0.737	0.750

6-3-1-4 Structural analysis method

It shall be permitted to calculate the internal forces, actions and moments by any structural analysis methods that satisfy the applicable conditions of equilibrium.

Linear elastic analysis method shall be used to compute the internal forces, actions and moments in beams for both working stress and limit state design methods. Moments may be redistributed according to section (4-2-1-2-c).

6-3-1-5 Flexural rigidity

It is generally acceptable to compute the relative flexural rigidity of concrete members using un-reinforced gross concrete section i.e. $E_c I_{g,,}$, and the modulus of elasticity of E_c confirming to section (2-3-3-1).

Other assumptions that consider the effect of cracking may also be used in which the values of E_cIg and 1/2 E_cI_g are considered for columns and beams, respectively. In all cases, one base shall be used for estimating the rigidity of all elements of the structure. In the cases of T or L sections the flange width shall be equal to half of the flange width specified in section (6-3-1-9).

6-3-1-6 Bending moments and shearing forces of continuous beams

Bending moments of continuous beams can be calculated assuming that beams are supported on rigid knife edge supports; in the case of continuous beams of equal depths and spans that are subjected to uniformly distributed loads or beams of varying spans or loads where the larger of two adjacent spans not greater than the shorter span by more than with a maximum of 20%, the following values may be assumed for bending moments, provided that moment redistribution is not allowed.

a - Two-span beams

The maximum bending moment: ($M= wL^2/K_m$); K_m values are to be taken as shown in figure (6-20-a), where L is the effective span value.

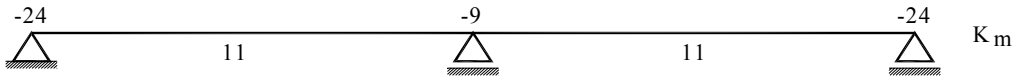


Figure (6-20-a) Bending moments coefficients K_m for two-span beams

The maximum shear force: ($Q= K_q$ wL); K_q values are to be taken as shown in figure (6-20-b).

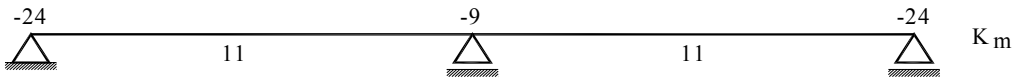


Figure (6-20-b) Shearing force coefficients K_q for two-span beams

b - Beams with more than two spans

The maximum bending moment: ($M=wL^2/K_m$); K_m values are to be taken as shown in figure (6-20-c), where L is the effective span value.

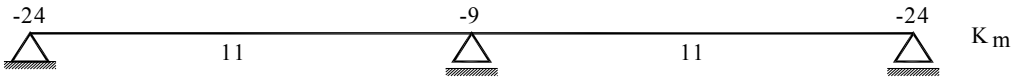


Figure (6-20-c) Bending moments coefficients K_m for Beams with more than two spans

when calculating negative bending moments over any support, arithmetic mean (average) values for the two adjacent spans and the two loads on either sides of this support are to be taken.

The maximum shear force: ($Q = K_q$ wL); K_q values are to be taken as shown in figure (6-20-d).

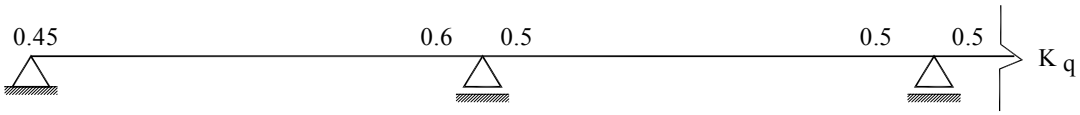


Figure (6-20-d) Shearing force coefficients K_q for beams with more than two spans

- Positive bending moments shall not be less than $wL^2/16$ and $wL^2/24$ for exterior and interior spans, respectively.
- Negative bending moments of continuous beams on rigid knife supports shall be calculated at the mid span when continuous beams are subjected to heavy live loads (p>1.5 g) using the beam theory, provided that negative moments are allowed to be reduced for live loads only at the mid span to two thirds of its value as a result of columns rigidity or monolithically cast supporting girders. In the case of equal spans – or spans that do not differ by more than 20% and for beams that are under the effects of heavy live loads (p>1.5 g), negative bending moments can be calculated at the mid span, as follows:

$$M = \left(g - \left(\frac{2}{3} \right) p \right) \left(\frac{L^2}{24} \right) \qquad (6-26)$$

Where:

L = the greater length of the two adjacent spans.
P = the live uniformly distributed load per unit length

G = the permanent (dead) uniformly distributed load per unit length

6-3-1-7 The critical sections for bending moments and shearing forces

The critical sections in monolithically cast beams shall be taken at the face of the interior support and at section of the zero shear force, for the negative moments and the positive moments, respectively.

1 - The critical section for the shearing forces shall be taken at the supports face (figure 6-21), except for the cases given in section (6-3-1-7-3).

2 - The critical section for shearing forces shall be taken at a distance d/2 from the support face in the cases when reaction produces compression in this distance, as shown in figure (6-22).

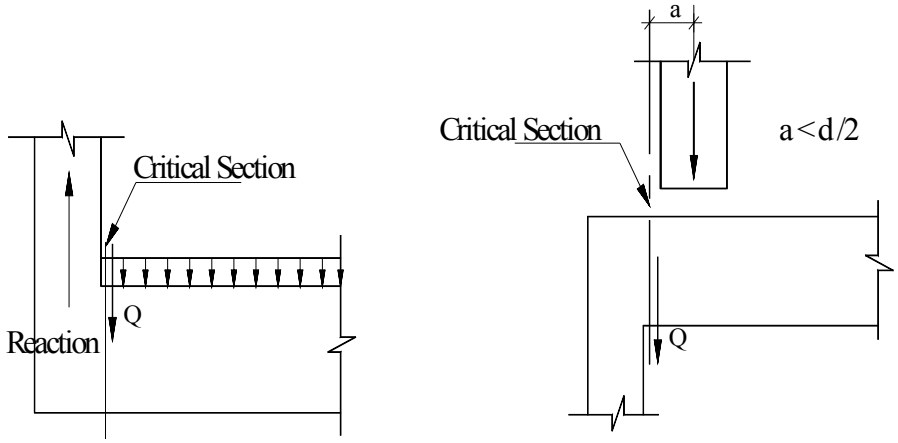


Figure (6-21) Critical section for shear at the support face

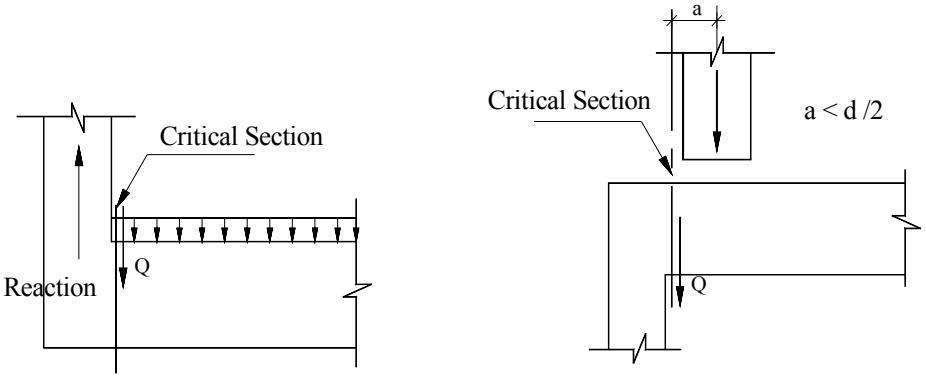


Figure (6-22) Critical section for shear at a distance of d/2 from the Support face

6-3-1-8 Slenderness limit

The laterally unsupported lengths between points of inflection in the lateral direction of the beam shall not exceed the following values:

a - For simply-supported or continuous beams: $40b_c$ or $200b_c^2/d$, whichever is less.

b - For cantilever beams laterally braced only at the support: $20b_c$ or $80b_c^2/d$, whichever is less. Where:

b_c = the beam width at the face subjected to compression.

d = the effective depth

6-3-1-9 Effective flange width for T or L sections

When determining the maximum resistance of the beams having T or L, shapes, the effective flange width shall be taken as the smaller value of the following:

$(16t_{s+} b)$ or $(\frac{L_2}{5} + b)$ for beams with T section (6-27-a)

$(6t_{s+} b)$ or $(\frac{L_2}{10} + b)$ for beams with L section (6-27-b)

Where L_2 is the distance between the points of inflection and shall be taken equal to 0.70 of the effective span in the continuous beams having both ends continuous and 0.80 of the effective span for continuous beams from one end only. The effective width of flange shall not exceed the width of the web, b plus one half the distances between the two adjacent beams from both sides. When concrete slabs are monolithically connected to the beams, the slab thickness shall not be less than 80 mm.

6-3-1-10 General considerations

- In order for the beam to be considered in the design as T or L section, the slab has to be cast monolithically with the beam or they have to be effectively connected together.
- In order to guarantee the monolithic action between flange and web, the top reinforcement of the flange in the direction perpendicular to the web shall not be less than 0.30% of slab section area. Reinforcement shall be continued at the full width of the flange given in section (6-3-1-9) and the distance between the bars of this reinforcement shall not be greater than 200 mm.
- Stirrups shall extend from the web to the ultimate surface of the flange in order to guarantee the monolithic action between flange and web.

- When T section is used for the isolated beams to provide the section with additional area for compression, the flange thickness shall not be less than one half of the web width, and the flange effective width shall not be greater than six times of slab thickness plus web width.
- Beams with depth greater than 600 mm, regardless the slab thickness, shall be provided with side shrinkage bars. The area of these bars shall not less than 8% of the tension reinforcement area and the distance between side bars shall not greater than 300 mm.

6-3-1-11 The minimum ratio of the main reinforcement

The reinforcement ratio shall not be less than the values given in section (4-2-1-2-h).

6-3-2 Deep beams

6-3-2-1 General considerations

a - This section is concerned with beams having effective span to depth ratio conforming to:

$$L/d \leq 4.0 \qquad (6\text{-}28)$$

Where: d = the effective depth of the section
L = the effective span of the beam

b - Deep beams subject to loads on the top surface or beams subject to loads on compression surfaces shall be designed using either the empirical design method given in sections (4-2-2-6-1), and (6-3-2-2) or by using the strut and tie method given in sections (4-2-2-6-2), (6-3-2-3), and (6-11) .

C - Nonlinear solution methods that consider cracks effect when designing deep beams can be used.

d - In cases that deep beam loads result in tension on the loading surface, the beams can be designed by either method specified in item, b provided that the requirements of section (4-2-2-6-3) are satisfied .

6-3-2-2 The empirical design method of deep beams

a - This method of design can be applied, for the following cases

$1.25 \leq L/d$ for simply supported beams (6-29-a)
$2.50 \leq L/d$ for continuous beams (6-29-b)

b - The moment arm y_{ct} for deep beams satisfying the requirements of item, a, shall be estimated as follows, provided that the moment arm does not exceed 0.87 of the effective depth d.

1 - For simply supported beams

$$yct = 0.86\ L \qquad (6\text{-}30\text{-}a)$$

2 - For continuous beams

a- At the mid span

$$yct = 0.43\ L \qquad (6\text{-}30\text{-}b)$$

b - At the interior support

$$yct = 0.37\ L \qquad (6\text{-}30\text{-}c)$$

c - The maximum resistance of shear for the deep beams that fulfill the requirements of section (6-3-2-2-a) shall be calculated according to section (4-2-2-6-1). The ratio of web shear reinforcement shall not be less than value given in section (4-2-2-6-1-k).

6-3-2-3 Design by using Strut and tie model

a - Strut and Tie model may be used to design deep beams defined in section (6-3-2-1-a) and in accordance with section (6-11).

b - For beams loaded with concentrated loads, the ratio of web shear reinforcement shall not be less than the values given in section (4-2-2-6-1-k) or section (4-2-2-6-2-b) according to the beam effective span to depth ratio and the ratio of shear span to the beam depth.

6-3-2-4 Minimum reinforcement ratio for deep beams

a - The ratio of main longitudinal steel reinforcement in deep beams shall not be less than the value given in section (4-2-1-2-h).

b - The main reinforcement shall be totally extended to the supports and adequately anchored either by providing the necessary bond length as given in section (4-2-5) or by using mechanical anchors.

6-4 Columns

6-4-1 Definitions

a - Columns are the compression members having length or height in the direction of the compression force exceeds five times the smallest dimension of the section. Columns have various shapes of cross-sections such as circular, polygonal, rectangular or sections comprising a number of rectangular sections for which the length to width ratio for each rectangular portion does not exceed 4. For members having cross -sections that do not satisfy the preceding conditions shall be designed as reinforced concrete walls in accordance with section (6-5).

b - Columns are designed in braced and unbraced buildings according to section (4-2-1-3) and section (5-3-3), respectively, considering the moments affecting the column in accordance with section (6-4-5) or moments resulted from the minimum eccentricity value of loads according to section (6-4-3), whichever is greater.

6-4-2 Laterally braced and unbraced buildings

a - The building shall be considered braced if it will be provided with supporting elements taking the form of continuous concrete walls having the same height as that of the building, symmetrically distributed in the horizontal projection of the building and fulfilling the following conditions:

- In case of buildings having four floors or more:

$$\alpha = H_b \sqrt{\frac{N}{\sum EI}} \quad < \quad 0.6 \qquad (6\text{-}31\text{-a})$$

- In case of buildings having less than four floors:

$$\alpha = H_b \sqrt{\frac{N}{\sum EI}} \quad < \quad 0.2 + 0.1\,n \qquad (6\text{-}31\text{-b})$$

Where:

H_b= the total height of building over the top surface of the foundation.

N= the sum of working loads at the foundation level resulting from all vertical elements of the building

$\sum EI$ = Sum of flexural rigidities of the vertical concrete walls of the building in the direction under consideration

n = number of floors in the building.

b - Concrete walls referred to in equation (6-31) of item, a, shall be monolithically connected to the foundation. The connection shall be capable of safely resisting horizontal forces and moments resulting from the wall.

6-4-3 Minimum eccentricity

The minimum eccentricity of sections subject to compressive forces shall not be less than the greater value of the following:

a - 0.05 of the cross section dimension of the column in the direction under consideration

b - 20 mm

6-4-4 Short columns

a - Columns shall be considered as short if the slenderness ratio λ of the column section is less than the values given in table (6-7). For rectangular columns the slenderness ratios, ($\lambda_b = H_e/b$) and ($\lambda_t = H_e/t$) shall be calculated in the two directions. For circular columns the slenderness ratio shall be expressed by ($\lambda_D = H_e/D$). In general, slenderness ratio can also be evaluated in the form ($\lambda_i = H_e/i$).

Where:

i = Radius of gyration of column section shall be taken according to the following:

$i = (0.30\ b)$ or $(0.30\ t)$ for rectangular columns (6-32-a)

$i = 0.25\ D$ for circular columns (6-32-b)

and

H_e= Effective height of the column in the direction under consideration

b& t = dimensions of rectangular column cross-section.

d = diameter of circular column

b - In unbraced buildings, design moments for short columns shall to be taken according to section (6-4-5-3-a).

Table (6-7) Maximum limits of slenderness ratio for short columns

Building Condition	Rectangular columns slenderness ratio λ_t or λ_b	Circular columns slenderness ratio λ_D	Slenderness coefficient λ_i
Braced	15	12	50
Unbraced	10	8	35

6-4-5 Slender columns

Slender columns are those columns having slenderness ratio λ exceeding the values specified in table (6-7), provided that slenderness ratio λ for any column shall not exceed the values given in table (6-8).

Table (6-7) Maximum limits of slenderness ratio for slender columns

Building Condition	Rectangular columns slenderness ratio λ_t or λ_b	Circular columns slenderness ratio λ_D	Slenderness Coefficient λ_i
Braced	30	25	100
Unbraced	23	18	70

6-4-5-1 Buckling length

1 - In the case of laterally braced buildings, the buckling length of column H_e shall be taken equal to the least value of the following:

$$H_e = H_o\,[0.7 + 0.05\,(\alpha_1 + \alpha_2)] \le H_o \qquad (6\text{-}33\text{-a})$$

or,

$$H_e = H_o\,[0.85 + 0.05\,(\alpha_{min})] \le H_o \qquad (6\text{-}33\text{-b})$$

In the case of laterally unbraced buildings, the buckling length of column H_e shall be taken equal to the least value of the following:

$$H_e = H_o\,[1.0 + 0.15\,(\alpha_1 + \alpha_2)] \ge H_o \qquad (6\text{-}34\text{-a})$$

or

$$H_e = H_o\,[2.0 + 0.3\,(\alpha_{min})] \geq H_o \qquad (6\text{-}34\text{-}b)$$

α, shall conform to the following :

$$\alpha = \frac{\sum \frac{E_c\, I_c}{H_o}}{\sum \frac{E_c\, I_b}{L_b}} \qquad (6\text{-}35)$$

Where:

H_o is the clear height of the column,

α_{min} is the least value of α_1 and α_2 at the lower and the upper ends of the column, respectively, considering that the maximum value of α shall be ten for hinged ends and the minimum value of α shall be one for totally fixed ends.

2 - EI value shall be calculated according to section (6-3-1-5) considering that the two ends of column shall be monolithically connected to other structural elements. The following simplified assumptions can also be used for the following cases:

a - For flat slabs, EI shall be calculated using an equivalent beam having a width and thickness equal to the width and thickness of column strips in the direction of analysis.

b - α shall be taken equal to 10 at the connection between columns and base not designed to resist moments

3 - Values given in tables (6-9) and (6-10) may be used for columns in braced and unbraced buildings, respectively for the following end conditions.

Condition (1): End of column or wall is cast monolithically with beams or slabs having depths not less than the column dimension in the direction of analysis. This case shall also be applicable for the cases of column to foundation connections where connection between the column and foundation is designed to resist bending moments.

Condition (2): End of column or wall is cast monolithically with beams or slabs of having depths smaller than the dimension of the cross section of the column or wall in the direction of analysis.

Condition (3): End of column or wall is connected with parts that are not designed to prevent rotation but is capable of providing marginal resistance.

Condition (4): Column is totally un-braced and is not capable of preventing horizontal resistance to movement or rotation like in the case of cantilever columns.

Table (6-9) The ratio of H_e/H_o for columns in braced buildings

Upper End Condition	Lower End Condition		
	1	2	3
1	0.75	0.80	0.90
2	0.80	0.85	0.95
3	0.90	0.90	1.00

Table (6-10) The ratio of H_e/H_o for columns in unbraced buildings

Upper End Condition	Lower End Condition		
	1	2	3
1	1.20	1.30	1.60
2	1.30	1.50	1.80
3	1.60	1.80	---
4	2.20	---	---

6-4-5-2 Slender columns in laterally braced buildings

- **Additional moments resulted from buckling (M_{add})**

Buckling effect in slender columns shall be taken into consideration by designing the column for additional moments shown in figure (6-23). The magnitude of the additional moment shall be computed by the following equation:

$$M_{add} = P.\delta \qquad (6\text{-}36)$$

Where , δ shall be taken as follows:

- In the case of rectangular columns in the direction (t) of the column

$$\delta = \frac{(\lambda_t)^2 \cdot t}{2000} \qquad (6\text{-}37\text{-a})$$

- In the case of rectangular columns in the direction (b) of the column

$$\delta = \frac{(\lambda_b)^2 . b}{2000} \qquad (6\text{-}37\text{-b})$$

- In the case of circular columns with diameter (D)

$$\delta = \frac{(\lambda_D)^2 . D}{2000} \qquad (6\text{-}37\text{-c})$$

- In the general case

$$\delta = \frac{(\lambda_i)^2 . t'}{30000} \qquad (6\text{-}37\text{-d})$$

Where t' = the side length in the direction of buckling and the dimensions are measured in millimeters.

• Moments for uniaxially loaded slender columns

a - For the uniaxial loaded columns, additional moments ،M_{add} about the main or secondary axes shall be combined with the moments resulting from the analysis of structure for the cases where both additional moments and initial moments have the same signs . Their combined values are shown in Figure 6-23. The column shall be designed for the greatest value obtained from the following moment combinations:

1- M2	2- Mi + Madd	
3- M1 + (Madd /2)	4- P. emin	(6-38)

Where, M_i is the initial moment to be estimated at a critical section near the mid height of the column and shall be obtained from the following relation:

$$Mi = 0.4\ M1 + 0.6\ M2 \geq 0.4\ M_2 \qquad (6\text{-}39)$$

M_i shall be taken with negative sign in equation (6-39) in the case of biaxially loaded columns.

b - In the case of columns subjected to bending moments about the main axis only, the columns shall be designed as biaxially loaded column having initial moments obtained according to section (6-4-6), and considering that the initial moment M_i around the secondary axis is equal zero.

c - In the case of building comprising beams and columns, where the columns shall not be subjected to moments resulting from side sway, columns bending moments shall be computed as follows:

1- Bending moments M_1, M_2 shall be considered equal zero in the case of interior columns that connected to set of beams having almost symmetrical configurations and loadings. In the case of flat slab structure, the bending moments for the interior columns are to be calculated according section (6-2-5-4) or section (6-2-5-5). In all cases, design moment shall be taken according to the equation (6-38).

2 - Moments in exterior columns shall be estimated according to the values given in table (6-11).

Table (6-11): Moments for exterior columns

Position of moments in columns	Moments in case of frames with one panel	Moments in case of frames with two panels or more
Moment at the bottom of the upper column	$\frac{K_u.M_f}{K_1+K_u+0.50K_b}$	$\frac{K_u.M_f}{K_1+K_u+K_b}$
Moment at the top of the lower column	$\frac{K_1.M_f}{K_1+K_u+0.50K_b}$	$\frac{K_1.M_f}{K_1+K_u+K_b}$

Where M_f is the exterior connecting bending moment of the beam that form a frame with the column, assuming that it is totally fixed at its ends.

Equations shown in table (6-11), which give the moments at the top end of the lower columns, can also be used to compute the moments at the top end of the upper floor columns by considering that K_u equals zero.

Where

K_u= stiffness of the upper column $K_u = 4EI_u / h_u$
K_l= stiffness of the lower column $K_l = 4EI_l / h_l$
K_b= beam stiffness $K_b = 4EI_b / L_b$
h_u, h_l = the height of the upper and lower columns, respectively.
L_b= beam length
I_u, I_l, I_b=moment of inertia for the upper and lower columns and the beam, respectively.

Other assumptions may also be used that take into account the effect of cracking on rigidity by using EI_g for columns, and 0.50 EI_g for beams.

The approximate values of Table 6-11 were based on the following assumptions:

a -Constant moment of inertia for all members.

b -Connection points are prevented from vertical or horizontal movements.

c -All members have the same degree of fixation at far ends.

d -Points of zero bending moment are be considered to be located at one third the height of columns from the point of total fixation, and at one fourth the height from the point of partial fixation.

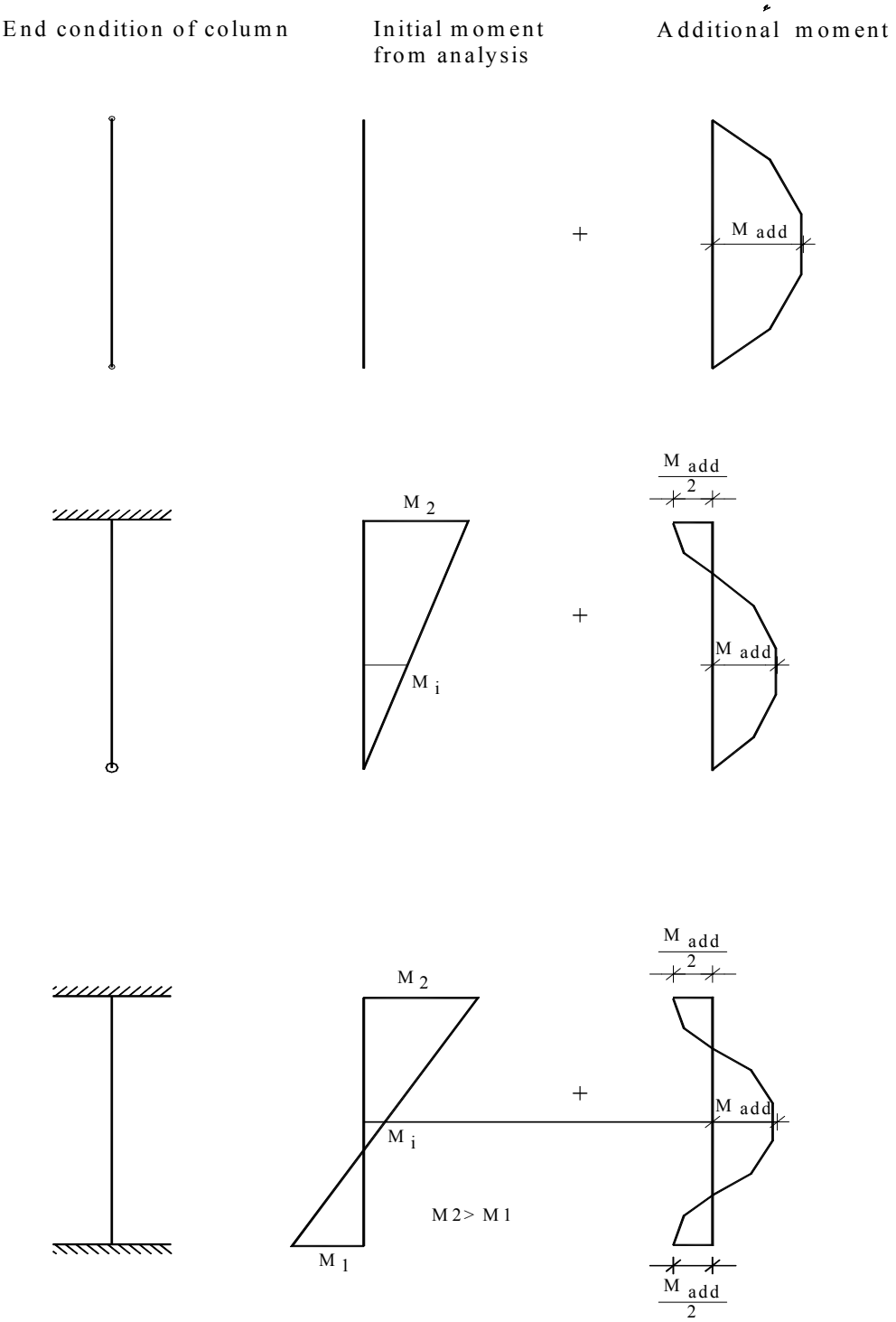


Figure (6-23) Moments of slender columns in laterally braced buildings

6-4-5-3 Slender columns in laterally unbraced buildings

a- Additional moments resulted from buckling

In the case of floors in which side sway values for all columns are almost equal, the effect of buckling can be taken into consideration by designing the column for additional moments the magnitude of which shall be computed by the following equation:

$$M_{add} = P.\delta_{av} \qquad (6\text{-}40)$$

Where

$$\delta_{av} = \frac{\Sigma\delta}{n} \qquad (6\text{-}41)$$

Where: n is the number of columns in a floor and, δ shall be calculated by using the equation (6-37). When calculating δ_{av}, δ values that exceed twice the value of δ_{av} shall be neglected, provided that these moments M_{add} shall be taken into account when designing beams or slabs connected monolithically with columns.

b - Design moments for uniaxially loaded columns (figure 6-24) shall be equal to the greater value of:

$P.e_{min}$ or M_2+M_{add}

With the additional moments acting at the column ends.

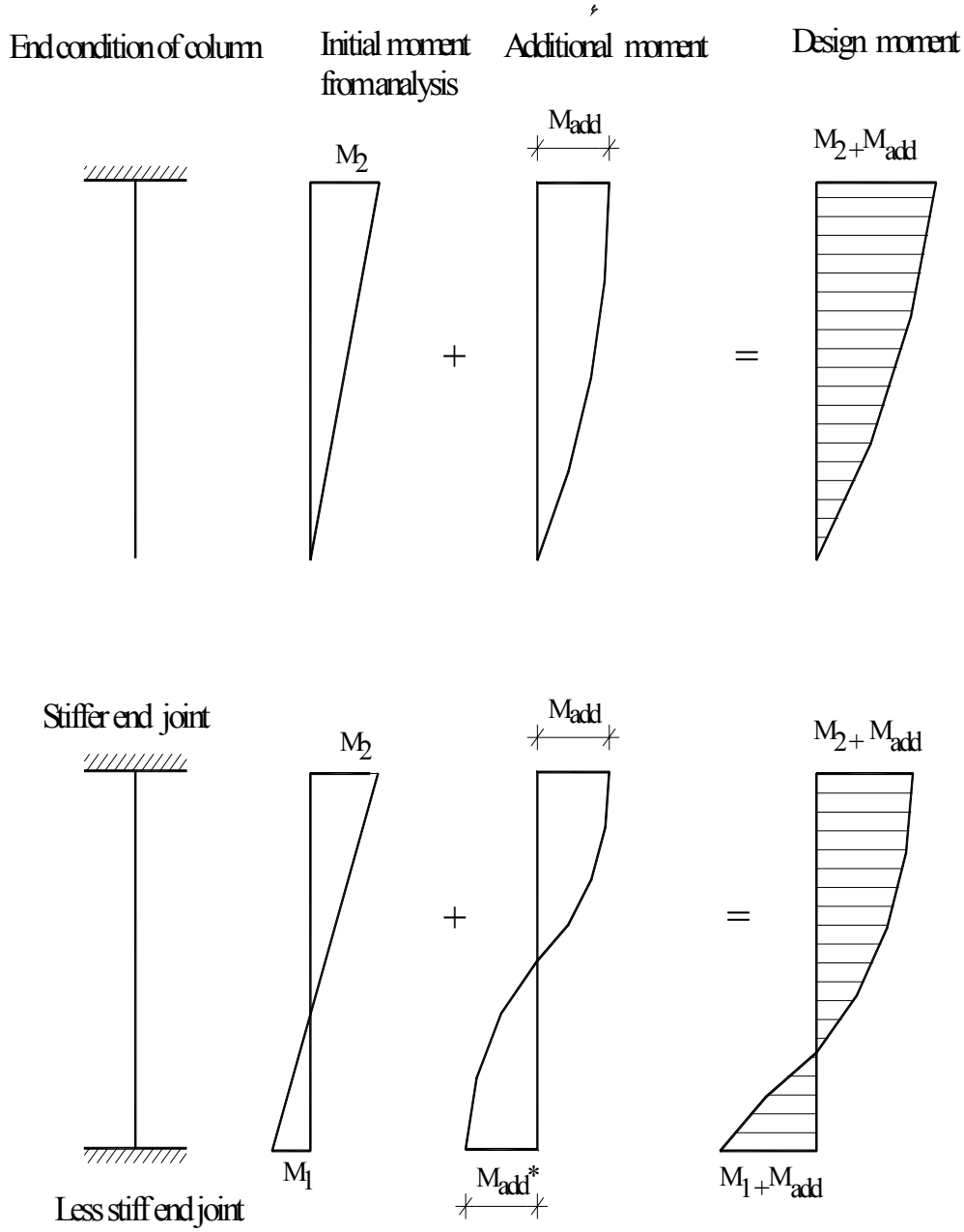


Figure (6-24) Design moments for slender columns in laterally unbraced buildings

6-4-6 Biaxially loaded columns

1 - Columns shall be designed to resist axial forces combined with biaxial moments calculated around the main and secondary axes of the cross-section according to section (4-2-1-3) and section (6-4-4-b) for short columns and section (6-4-5-2) and section (6-4-5-3) for slender columns.

2 - Either moment affecting the column can be ignored if the eccentricity resulted from this moment is less than the minimum value given in section (6-4-3).

3 - In the case of rectangular sections that are equally reinforced in all faces (figure 6-25-a), equivalent moment can be taken approximately around one axis, as follows:

a - In the case that ($M_y/b' \geq M_x/a'$)

Design moment M'_y around the axis (y) can to be taken according to the following equation:

$$M'_y = M_y + \beta\left(\frac{b'}{a'}\right)M_x \qquad (6\text{-}42)$$

b - In the case that ($M_y/b' < M_x/a'$)

Design moment M'_x around the axis (x) can to be taken according to the following equation:

$$M'_x = M_x + \beta\left(\frac{a'}{b'}\right)M_y \qquad (6\text{-}43)$$

Where a', b' are the effective depths of both M_x, M_y respectively, and β values are to be determined according to table (6-12-a) or from figure (6-25-b).

Table (6-12-a) Values of β

$R_b = \frac{P_u}{f_{cu}.b.a}$	≤0.2	0.3	0.4	0.5	≥0.6
β	0.80	0.75	0.70	0.65	0.60

C - Four bars shall be placed in the column corners; the remaining area of reinforcement steel shall be equally distributed on the four faces.

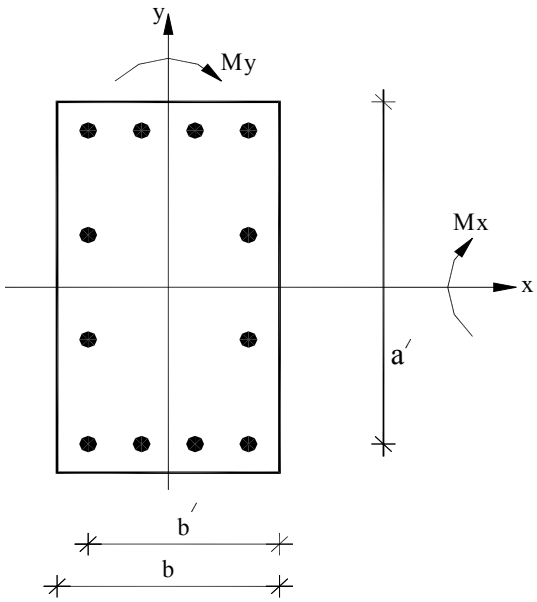
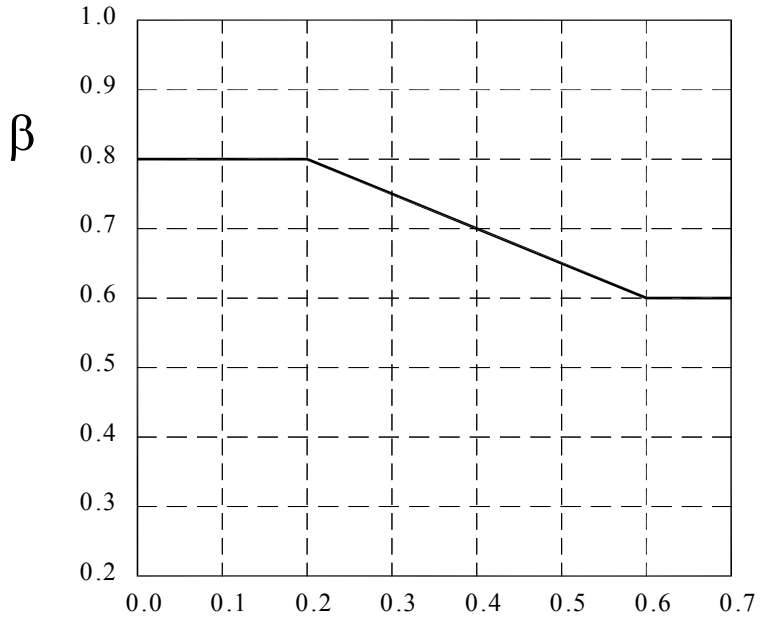


Figure (6-25-a) Columns biaxially loaded and equally reinforced in all faces



$$R_b = \frac{P_u}{f_{cu}.b.a}$$

Figure (6-25-b) β values

4 - In the case of rectangular sections with equally reinforced steel on two opposite faces in the column section (figure 6-26), provided that the

value of P_u / (f_{cu} .b.a) is to be less than or equal to 0.50, the column can be designed simply to resist the axial force P_u accompanied by each of the following bending moments separately:

$$M'_x = M_x \cdot \alpha_b \qquad (6\text{-}44\text{-a})$$

$$M'_y = M_y \cdot \alpha_b \qquad (6\text{-}44\text{-b})$$

Where α_b value shall be determined from table (6-12-b).

Table (6-12-b) α_b Value

$(M_x/a')/(M_y/b')$ $R_b = P_u/(f_{cu}\ b.a)$	∞	3	2	1	0.5	0.33	0
$R_b \leq 0.1$	1	1.20	1.25	1.30	1.25	1.20	1
$R_b = 0.2$	1	1.35	1.50	1.75	1.50	1.35	1
$R_b = 0.3$	1	1.25	1.35	1.40	1.35	1.25	1
$R_b = 0.4$	1	0.95	0.95	0.95	0.95	0.95	1
$R_b \geq 0.5$	1	0.65	0.70	0.75	0.70	0.65	1

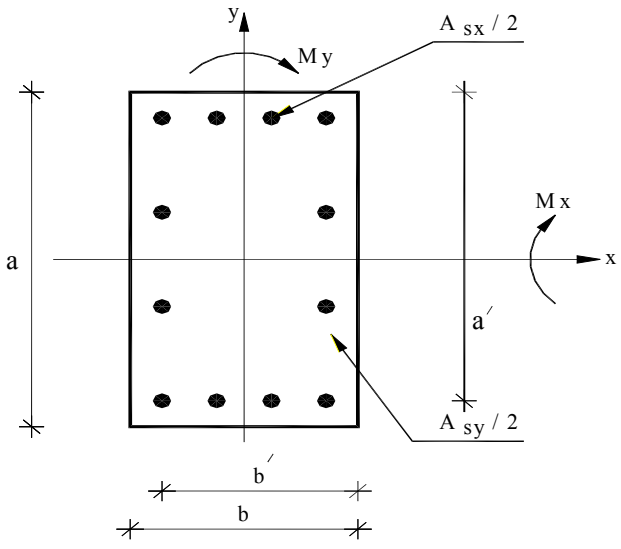


Figure (6-26) Biaxially loaded columns with equal reinforcement on the two opposite faces ($R_b \leq 0.5$)

6-4-7 Details and notes

a - The minimum percentage of longitudinal reinforcement

1 - In columns with tie reinforcement, the minimum percentage of longitudinal reinforcement shall be 0.80% of the required concrete section area (arithmetically), provided that the minimum shall not less than 0.60% of the actual section area, if the slenderness ratio λ_b or slenderness coefficient λ_i do not exceed the values given in table (6-7) and section (6-4-4-a). If the slenderness ratio and slenderness coefficient exceed that limit, the minimum percentage of reinforcement shall be:

$$0.25 + 0.015\,\lambda_i \qquad (6\text{-}45)$$

And for columns with rectangular sections:

$$0.25 + 0.052\,\lambda_b \qquad (6\text{-}46)$$

2 - In columns with spiral reinforcement , the minimum longitudinal reinforcement shall be 1% of the total section area or 1.20% of the core area defined by spiral stirrups, whichever is greater.

b - The maximum percentage of longitudinal reinforcement in columns shall not exceed the following percentages of the concrete column cross -sectional area:

4% for interior columns
5% for exterior (edge) columns
6% for corner columns

Provided that the reinforcement ratio shall not exceed 8% at the overlapping joint area.

c - The column shall contain a longitudinal bar at each corner.

d - The minimum diameter of the longitudinal bars shall be 12 mm.

e - The minimum side length for columns with rectangular section or the minimum diameter of the circular column shall not be less than 200 mm.

f - The maximum side length for columns having corner bars only shall not be more than 300 mm, or otherwise side bars shall be placed at a maximum distance of 250 mm. These bars shall be tied if the distance between the untied and tied bars exceeds 150 mm (figure 7-7-a). Also,

for columns having circular cross-sections the minimum number of longitudinal bars shall be 6 bars.

g - The distance between stirrups(ties) in the longitudinal direction of the column shall not exceed 15 times the diameter of the smallest longitudinal bar but shall not exceed 200 mm.

h - The minimum diameter of stirrups is 1⁄4 the diameter of the greatest longitudinal bar, provided that it shall not be less than 8 mm, and that the least volume of the stirrups shall be 0.25% of the concrete volume.

i - Ordinary and spiral reinforcement shall continue inside the joint zones between columns and beams.

j - The maximum pitch of the spiral reinforcement shall be 80 mm while the minimum pitch shall not be less than 30 mm. It is recommended that the pitch be kept constant for the entire length of the column. Also, spiral reinforcements shall have three turns at each end with a pitch equals half that of the ordinary pitch, along with bending the spiral reinforcement end inside the column cross-section a distance that shall not be less than 100 mm or 10 times the diameter of the spiral reinforcement.

k - The least diameter of the spiral reinforcement shall not be less than 8 mm.

l - In the case that the characteristic strength of concrete used in the columns concrete strength is higher by more than 140% of that of the concrete used in the floor, the following conditions shall be fulfilled:

1 - Floor parts shall be cast around the columns using concrete having the same characteristic strength as that of the column. Such parts shall extend at least 600 mm from columns faces. Care shall be taken to guarantee that concrete used in these parts and concrete used in the floor shall be well bonded.

2 - The ultimate capacity of columns shall be calculated using the lower characteristic strength value of the concrete used along with using vertical dowels and spiral reinforcement if needed, that may contribute to the increase of the ultimate capacity of columns

3 - For columns laterally surrounded on four sides by beams with approximately equal depths or slabs, the ultimate capacity of columns shall be calculated using assumed values for compressive strength of the concrete at the joint between the

column and the floor that equals to the sum of 75% of the columns concrete strength and 35% of the floor concrete strength , provided that the ratio between the concrete strength of column and that of the floor does not exceed 2.5.

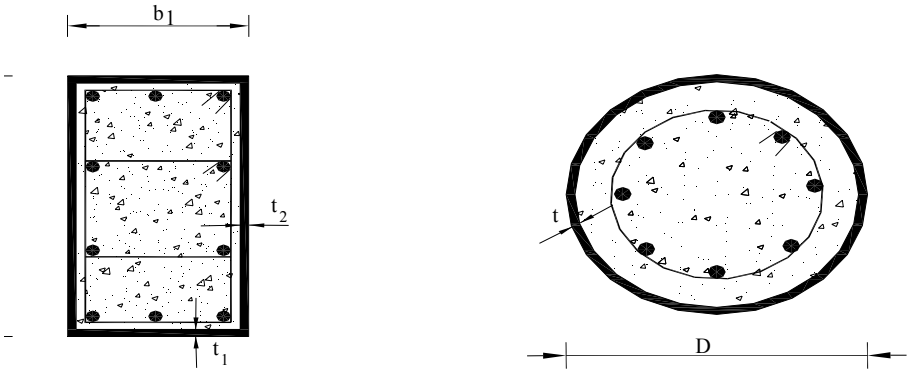
m - All forces and moments at the column base shall be transmitted to the foundation by means of direct bearing on the concrete, as well as by reinforcement steel dowels having splices according to section 7-3-2. For the cases that the loads transmitted from the column to the foundation are tensile, such tensile forces shall be resisted by the reinforcement steel only with due consideration of the requirements of limit state of cracking. Also, the values of bearing stresses resulting from the column at the foundation shall satisfy the ultimate strength bearing requirements given in section (4-2-4). The longitudinal steel reinforcement steel, dowels and splices shall be capable of providing the required strength in excess to that of the bearing strength for both the foundation and the column, provided that such steel reinforcement shall not be less than the column reinforcement. In the case of having lateral forces acting at the interface between the column and the foundation, such forces shall to be adequately transmitted by the shear friction according to section (4-2-2-4) or by any other suitable means.

6-4-8 Composite columns

6-4-8-1 General

1 - Composite columns are reinforced concrete columns having I longitudinally reinforcement in addition to structural steel sections such as I, pipes or tubes. Figure (6-27) shows typical types of composite columns .

2 - Forces and loads resisted by the reinforced concrete column of the composite column shall be resisted by the concrete column through direct bearing, with due consideration of the bearing strength requirements according to section (4-2-4-1) or section (5-6). The remaining forces and loads shall be resisted by the structural steel sections through bearing by means of properly designed steel joints capable of resisting such loads



Structural steel tube with reinforced concrete column having rectangular section

Structural steel pipe with reinforced concrete column having circular section

(a) Composite sections having structural steel sections surrounding reinforced concrete column



(b) Composite sections having structural steel sections inside reinforced concrete columns

Figure (6-27) Types of composite columns

3 - The ultimate strength of the composite columns sections subject to eccentric compression loads shall be calculated using the same way as that of the reinforced concrete columns according to section (4-2-1-3) with due consideration of the value of longitudinal reinforcement yield strength for steel sections according to sections (6-4-8-2) and (6-4-8-3).

4 - Yield stress value used in calculations for the steel sections shall not exceed 350 N/mm^2.

5 - In the case of using spiral reinforcement, the values of the pitch and diameters of the spiral reinforcements shall be the same as those used for reinforced concrete columns.

6 - The longitudinal reinforcements shall be taken as the sum of areas of the structural steel section and the longitudinal reinforcement of the concrete column according to the relation, $A_t = A_{sc} + A_{ss}$

7 - The ratio of longitudinal reinforcement, A_t shall not be less than 1% and not more than 6% of the net concrete section area ($A_g - A_t$).
Where:

A_g = section's total area
A_t = total steel section area
A_{sc} = area of longitudinal reinforcement
A_{ss} = steel section area

8 - The contribution of longitudinal reinforcement in the calculation of the moment of inertia of the section shall include the sum of the contributions of both steel reinforcements and the structural steel section according to the relation $I_t = I_{sc} + I_{ss}$

Where:

I_t= total steel section moment of inertia
I_{sc}= longitudinal reinforcement moment of inertia
I_{ss}= steel section moment of inertia around the neutral axis

9 - In order to calculate the slenderness ratio for the composite section, the radius of gyration, i of the cross-section shall be taken according to the following equation:

$$i = 0.8\sqrt{\frac{(E_c I_g/5) + E_s I_t}{(E_c A_g/5) + E_s A_t}} \qquad (6\text{-}47)$$

Where:

E_c = concrete modulus of elasticity according to the equation (2-1)
E_s = steel section modulus of elasticity
I_g = the gross moment of inertia for the entire concrete section neglecting the reinforcement

6-4-8-2 Composite sections having structural steel sections surrounding concrete columns

1 - The thickness of the steel covering the concrete core shall not be less than:

a - Steel cover with rectangular section

$$t_{min} \geq b\sqrt{f_y / 3E_s} \qquad (6\text{-}48)$$

It is to be calculated for each face separately as shown in figure (6-27-a).

b - Steel cover with circular section.

$$t_{min} \geq D\sqrt{f_y / 8E_s} \qquad (6\text{-}49)$$

Figure (6-27) shows some patterns of these sections.

2 - The maximum resistance of the axially loaded sections shall be calculated, in addition to simple moments with values less than $P_u.e_{min}$ according to the following equations (6-50) and (6-51):

a - In the case of columns with rectangular sections

$$P_u = 0.35f_{cu}A_c + 0.67f_{yss}A_{ss} + 0.67f_{ysc}A_{sc} \qquad (6\text{-}50)$$

Where:

f_{yss}= yield stress of the steel section

f_{ysc}= yield stress of the reinforcement steel

b - In the case of columns with circular sections and spiral stirrups

$$P_u = 0.4f_{cu}A_c + 0.67f_{yss}A_{ss} + 0.76f_{ysc}A_{sc} \qquad (6\text{-}51)$$

Considering what was mentioned in section (4-2-1-3-c-2)

6-4-8-3 Composite sections having structural steel sections inside reinforced concrete columns

1 - The ultimate compressive strength of the composite sections subject to axially loaded forces in addition to simple moments having values less than $P_u.e_{min}$, shall be calculated according to the following:

a - In case of using tied stirrups:
Ultimate compressive strength of the composite sections shall be calculated according to the equation (6-50) with due consideration of the following:

- Tied stirrups having minimum diameter of 8 mm shall be extended around steel sections
- The stirrups diameter shall not be less than 1⁄50 of the greatest dimension of the composite section, but shall not exceed 16 mm.
- Distances between stirrups in the longitudinal direction shall not exceed 16 times the longitudinal bar diameter. A vertical bar shall be placed at each corner of the section along with other bars at distances that shall not exceed 1⁄2 the smallest dimension of the concrete section according to the conditions given in section (6-4-7).

b - The case of using spiral reinforcement:
The ultimate compressive strength of the composite sections shall be calculated according to the following equation:

$$P_u = 0.35 f_{cu} A_k + 0.67 f_{yss} A_{ss} + 0.67 f_{ysc} A_{sc} + 1.38 f_{yp} V_{sp} \quad (6\text{-}52)$$

With due consideration of section (4-2-1-3-2)

Where:

f_{yp} = yield stress of the spiral reinforcement

V_{sp} = Ratio of spiral reinforcement steel volume for the single turn of the stirrups according to the equation (4-12-d).

6-5 Walls

6-5-1 General

1- Walls are flat elements, usually vertical, having length of cross-section greater than five times the width. Thickness of wall shall not be less than 120 mm.

2- Reinforced walls are divided into:-

a - Bearing walls: subjected mainly to compression forces accompanied or unaccompanied by lateral forces.

b - Stiffening walls: to support bearing walls against buckling as well as bearing walls.

c - Non bearing walls: subjected to lateral forces in addition to their own weight.

3- Walls shall be considered laterally braced if the building is laterally braced according to section (6-4-2).

4- Walls used as part of the earthquake resistant structural system shall meet the requirements of section (6-7-3).

6-5-2 Reinforced concrete walls

a. Vertical walls contributing to building bracing shall be constructed and connected rigidly to the bearing walls. The total resistance of multi-storey building having higher than 4 storeys shall not depend on walls laterally unbraced.

b. Reinforced concrete walls subjected to axial forces with or without bending moments shall be designed according to section (6-5-2-1).

c. Walls shall be designed to resist shear forces according to section (4-2-2-1) or section (5-4-1). The horizontal reinforcement ratio shall not be less than that specified in section (6-5-2-2-2).

d - The effective depth, d for wall section may be considered equal to 0.8 times that of wall length for calculating wall shear strength.

6-5-2-1 Design of reinforced concrete walls

Reinforced concrete walls may be designed by any of the two methods described in sections (6-5-2-1-1) and (6-5-2-1-2)

6-5-2-1-1 Design of walls as column section subject to Bending Moments accompanied by axial compressive forces.

a. Reinforced concrete wall section subject to concentric or eccentric compressive force may be designed as a column section according to sections (6-4-2) to (6-4-6). The wall slenderness shall be determined according to sections (6-5-2-1-1-b) and (6-5-2-1-1-c). Reinforcement ratio of wall shall be determined according to section (6-5-2-2).

b. For walls without lateral stiffeners, effective length and slenderness ratios shall be determined according to sections (6-4-4), and (6-4-5).

c. For walls with lateral stiffeners shown in figure (6-28), the reinforced wall shall be considered slender if the slenderness ratio ($\lambda t = He/t$) of the wall is equal to or greater than the values in table (6-13-a), where, t is the wall thickness. The slenderness ratio shall not exceed the values given in table (6-13-b).

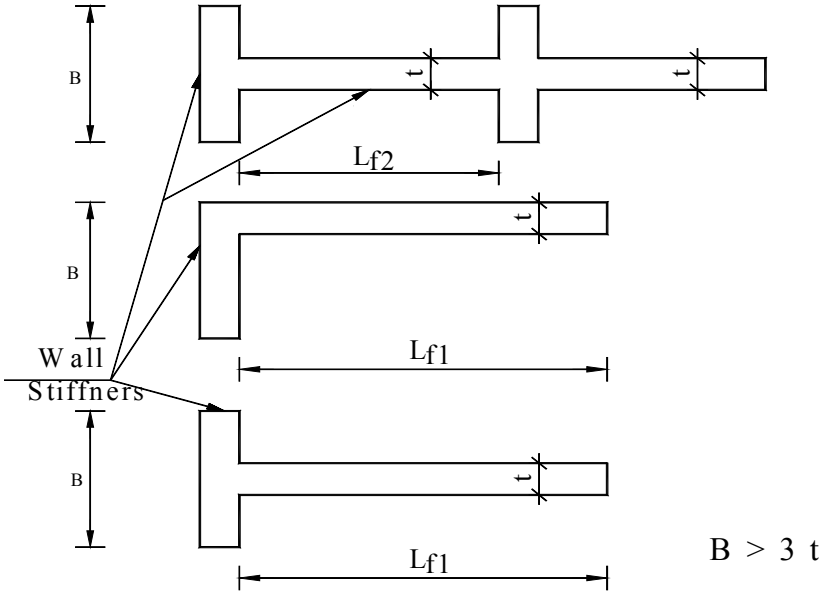


Fig. 6-28 Walls with lateral stiffeners

Table(6-13-A) Slenderness Ratio for Short Walls

λt	Wall Condition
15 10	Braced Unbraced

Table(6-13-A) Slenderness Ratio for Slender Walls

λt	Wall Condition
40 30	Braced Unbraced

The effective height (He = KH) shall be determined as follows:

1. For walls with more than one lateral stiffener, the value of, k shall be taken as follows:

$$k = 1.0 \qquad \frac{H}{L_{f2}} < 0.5 \qquad (6\text{-}53\text{-a})$$

$$k = 1.5 - \frac{H}{L_{f2}} \qquad 0.5 \le \frac{H}{L_{f2}} \le 1.0 \qquad (6\text{-}53\text{-b})$$

$$k = \frac{1}{1 + \left(\frac{H}{L_{f2}}\right)^2} \qquad \frac{H}{L_{f2}} > 1.0 \qquad (6\text{-}53\text{-c})$$

Where:

H = clear height of wall

Lf_2 = the average horizontal distance between lateral stiffeners.

2. For walls with one stiffener, the value of, k is taken as follows:

$$k = 1.0 \qquad \frac{H}{L_{f1}} < 1.0 \qquad (6\text{-}54\text{-a})$$

$$k = 1.0 - 0.423 \frac{H}{L_{f1}} \quad 1 \qquad 1.0 \le \frac{H}{L_{f1}} \le 2.0 \qquad \text{(6-54-b)}$$

$$k = \frac{1}{\sqrt{1 + 0.5 \left(\frac{H}{L_{f1}}\right)^2}} \qquad \frac{H}{L_{f1}} > 2.0 \qquad \text{(6-54-c)}$$

Where :

H = clear height of wall

Lf_1 = the horizontal distance between lateral stiffener and free edge of wall.

6-5-2-1-2 Simplified design method for design of reinforced concrete walls with solid rectangular sections

The following simplified method may be used for design of solid rectangular section of reinforced concrete walls if all following requirements are satisfied:-

a. The resultant of all ultimate loads including effect of lateral forces shall not be outside middle third of rectangular section.

b. Reinforcement ratio of wall shall not be less than that specified in section (6-5-2-2).

c. Wall thickness shall not be less than 0.04 of effective wall height or wall length whichever is shorter. In any case, wall thickness shall not be less than 120 mm.

Ultimate load of section, in this case, shall be evaluated from following equation

$$P_u = 0.8 \left[0.35 f_{cu} A_c \left(1 - \left(\frac{k.H}{32 t} \right)^2 \right) \right] \qquad \text{(6- 55)}$$

Where :

A_c = Concrete wall sectional area

H = Clear height of wall between stiffeners

K = Coefficient of effective height of braced wall against lateral movement at top and bottom of wall.
= 0.80 for wall prevented from rotation at any end or both (top and / or bottom)
= 1.00 for wall free to rotate at both top and bottom ends
= 2.00 for wall free to move laterally perpendicular to wall plane.
t = wall thickness

6-5-2-2 Minimum and maximum reinforcement ratios

Steel reinforcement consisting of two meshes at the two wall faces shall be placed in the wall. Vertical and horizontal reinforcement ratios shall be determined according to sections (6-5-2-2-1) and (6-5-2-2-2)>

6-5-2-2-1 Vertical reinforcement:

- Total vertical reinforcement ratio is used for the control cracks. Table (6-14) specifies minimum reinforcement ratios. Reinforcement ratio shall not be less than 0.5 % of concrete cross section required from design (Ac_{req}) and shall not be greater than 4 % of actual concrete cross section. Bar diameter shall not be less than 10 mm, and the distance between bars shall not exceed 250 mm. If welded wire fabric in used, bar diameter shall not be less than 5 mm.
- When all cross section is subjected to tensile stresses, minimum total vertical reinforcement ratio μ shall not be less than 0.8 % for normal mild steel and 0.45 % for high grade steel.
- When all cross section is subjected to compressive stresses, minimum total vertical reinforcement ratio μ shall not be less than 0.4 %.
- For sections subjected to bending moments, minimum main reinforcement ratio in tension side is 0.25 % for normal mild steel and 0.15 % for high grade steel, while total vertical reinforcement ratio shall not be less than 0.4 %.

Minimum Percentage of Steel Reinforcement		Condition
$f_y = 400$ N/mm^2	$f_y = 240$ N/mm^2	
0.45	0.80	Entire cross - section is subjected to Tension
0.40	0.40	Entire cross - section is subjected to Compression
0.15	0.25	Section subjected to flexure

Table 6-14 Minimum Percentage of Vertical Steel Reinforcement

6-5-2-2-2 Horizontal reinforcement

For walls subjected to compression, horizontal reinforcement encloses vertical steel and the minimum area of total horizontal shall not be less than the following:

- 0.3% of actual area of concrete section in case of steel with yield stress ($f_y = 240$ / N/mm^2) .
- 0.25% of actual area of concrete section in case of steel with yield stress ($f_y = 400$ / N/mm^2).
- Diameter of horizontal rebar shall not be less than 0.25 of vertical rebar and shall not be less than 8mm except for the case of using wire mesh reinforcement where minimum diameter shall not be less than 5mm.
- When area of vertical reinforcement exceeds 1% of cross sectional area, additional single closed stirrups (with minimum diameter 6 mm or $\frac{1}{4}$ of vertical steel diameter whichever is larger) shall be provided to tie vertical and horizontal reinforcement with each other across wall thickness at 4 points in meter square at least .
- The distance between horizontal reinforcement shall not exceed 15 times vertical steel diameter or 200 mm whichever is less.

6-5-2-3 Horizontal displacement of walls

When height of wall exceeds 12 times its length, the horizontal displacement under service loads shall not exceed (1/500) of wall height.

6-5-2-4 Concrete cover of steel reinforcement

Minimum concrete cover of steel reinforcement shall be specified according to section (2-4) and section (4-3-2-3-b) .

6-5-2-5 Calculation of effect of forces on lateral stiffeners

Horizontal stiffeners shall be able to transfer all following horizontal forces to foundations:

a. Static reaction of the sum of all ultimate horizontal forces at the locations of lateral stiffeners.

b. 1% of summation of ultimate vertical forces at stiffener

6-5-2-6 Concentrated loads on walls

In calculating bearing strength under concentrated loads, the effective horizontal dimension shall not exceed the distance between points of application of loads or width of bearing plus four times of wall thickness whichever is less. Additional reinforcement shall be placed equally at two wall faces in vertical distance under concentrated load not exceeding double wall thickness, as shown in figure (6-29).

6-5-3 Concrete walls considered as unreinforced.

Concrete walls with reinforcement percentages that do not satisfy requirements of previous sections of this chapter shall be considered in the design as unreinforced walls. However, reinforcement percentage of these walls shall not be less than those of section (6-5-3-7) and this thickness shall not be less than 120mm.

6-5-3-1 Design

- For design of walls considered as unreinforced, no tensile stresses on concrete section, or shear stresses exceeding allowable working stresses given in table (5-1) for concrete section without shear reinforcement under any case of loading shall be permitted.
- Walls considered as unreinforced may be designed using simplified method in section (6-5-2-1-2). However, ultimate strength of wall section shall be reduced by 20% of that calculated by equation (6-55).

6-5-3-2 Slenderness limits

In all cases, maximum slenderness of concrete wall considered as un-reinforced ($\lambda_t = He/t$) shall not exceed 30, where t is the smaller dimension of wall horizontal cross section, and He is the wall effective height according to section (6-4-5-1).

6-5-3-3 Minimum eccentricity of loads

Eccentricity of loads not less than 0.05 t or 20mm whichever is greater shall be considered in design.

6-5-3-4 Eccentricity of loads from slabs and floors

For walls connected to slab from one side only, it may be assumed that loads are applied at $\frac{1}{3}$ of wall thickness measured from face of wall connected to slab.

6-5-3-5 Load eccentricity in wall plane

This eccentricity shall be calculated using principles of statics.

6-5-3-6 Shear strength

For walls considered as un-reinforced, shear strength may not be calculated if one of the following two conditions shall be satisfied:

a. If design horizontal shear force is less than 0.25 of design axial force.

b. If average working shear stress is less than 0.4 N/mm^2.

6-5-3-7 Minimum reinforcement ratio in concrete walls considered as un-reinforced

Internal or external concrete walls considered as un-reinforced shall be supplied with reinforcement to control cracking due to flexure, shrinkage and temperature gradient. The total steel reinforcement area in both vertical and horizontal directions shall not be less than 0.3% of concrete section for mild steel, and 0.2% of concrete section for high grade steel or mesh reinforcement. However, the concrete cover shall not be less than the values specified in section (4-3-2-3-b).

For walls with openings, steel reinforcement at each opening side shall not be less than half of steel cut by opening in that direction.

However, this steel shall not be less than 2 bars 16mm diameter for mild steel and 2 bars 12mm for high grade steel. This steel shall be placed on the two wall faces if thickness exceeds 150mm .

6 – 6 Monolithic beam-column connections (Joints)

6-6-1 Types of beam-column connections

Monolithic beam-column connections (joints) shall be classified, depending on the nature of applied loads, by the following two types:

Joints Type (I): These are monolithic beam-column connections that transfer bending moments and shear forces produced by vertical loads and lateral forces caused by wind or any other loads excluding earthquakes. The design of connections Type (I) shall conform to section (6-6-2) or by the strut-and-tie method given in section (6-11).

Joints Type (II): These are monolithic beam-column connections that transfer bending moments and shear forces produced by vertical loads and lateral forces caused by earthquakes. The design of connections Type (II) shall conform to sections (6-6-2) and (6-8-2-3-3).

6-6-2 Design of connections

1 - Forces acting on connections are those produced by various load combinations and causing the largest stresses at column faces as shown in Figure (6-30).

2 - Strength of beam-column connections (joints) shall be determined using the appropriate strength reduction factors given in section (3-2-1-2).

3 - Longitudinal reinforcement, terminated in a column, shall be extended beyond the column centerline a distance equal to the full development length according to section (4-2-5-1).

4 - Ultimate design shear force acting on the joint (Q_{ju}) shall be calculated assuming that moments of opposite sign shall be formed at opposite joint faces (i.e. ends of columns and beams) as shown in Figure (6-30).

5 - Ultimate design shear force acting on a beam-column joint (Q_{ju}) shall satisfy the following relation:

$$Q_{ju} \le k_j A_j \sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (6\text{-}56\text{-a})$$

Value of the shear force (Q_{ju}) shall be calculated from:

$$Q_{ju} = \frac{A_{su} \lambda f_y}{\gamma_s} + \frac{0.67\, b\, f_{cu}\, a_t}{\gamma_c} + A'_{su}\, f_{st} - Q_{ucol} \qquad (6\text{-}56\text{-b})$$

Where:

A_j = the effective cross-sectional area within the joint. It shall be equal to the shear-resisting area for the loading in the direction under consideration. The joint depth shall be equal to the overall depth of the column while the effective joint width shall be taken equal to the smaller value of the following ,as shown in Figure (6-31),:

- beam width plus joint depth (b+c2), or
- twice the smaller perpendicular distance from the longitudinal axis of beam to column side (b+2x).

But the effective joint width shall not be taken greater than the overall width of the column.

k_j = the joint confinement parameter depending on the condition of beams connected to the joint as given in Table (6-15). A beam shall be assumed to confine the joint if at least three-quarters of the face of the joint shall be covered by that beam, as shown in Figure (6-31).

f_{st} = stress in compression steel.

λ = 1.00 for joints Type (I)

= 1.25 for joints Type (II).

6 - Effects of shear forces on beam-column joints shall be determined for each direction separately (Figure 6-31).

7 - Column stirrups shall continue inside the beam-column joints with a stirrup cross- sectional area of not less than the larger of the following two values:

$$A_{st} = 0.313 \left(\frac{s.y_1 (f_{cu}/\gamma_c)}{(f_{yst}/\gamma_s)} \right) \left[\left(\frac{A_g}{A_k} \right) - 1 \right] \qquad (6\text{-}57\text{-a})$$

$$A_{st} = 0.1\left(\frac{s.y_1(f_{cu}/\gamma_c)}{(f_{yst}/\gamma_s)}\right) \qquad (6\text{-}57\text{-b})$$

where:

A_j = element gross cross-sectional area.

A_k= element cross-sectional area enclosed by the exterior stirrup.

f_{st}= yield strength of stirrup steel.

s = stirrup spacing along the column longitudinal axis.

y_1= cross-sectional dimension of column core, measured center-to-center of outer legs of stirrups, perpendicular to the considered direction.

A_{st}= total area of stirrups' cross-section including cross ties within a distance s and perpendicular to the distance y_1.

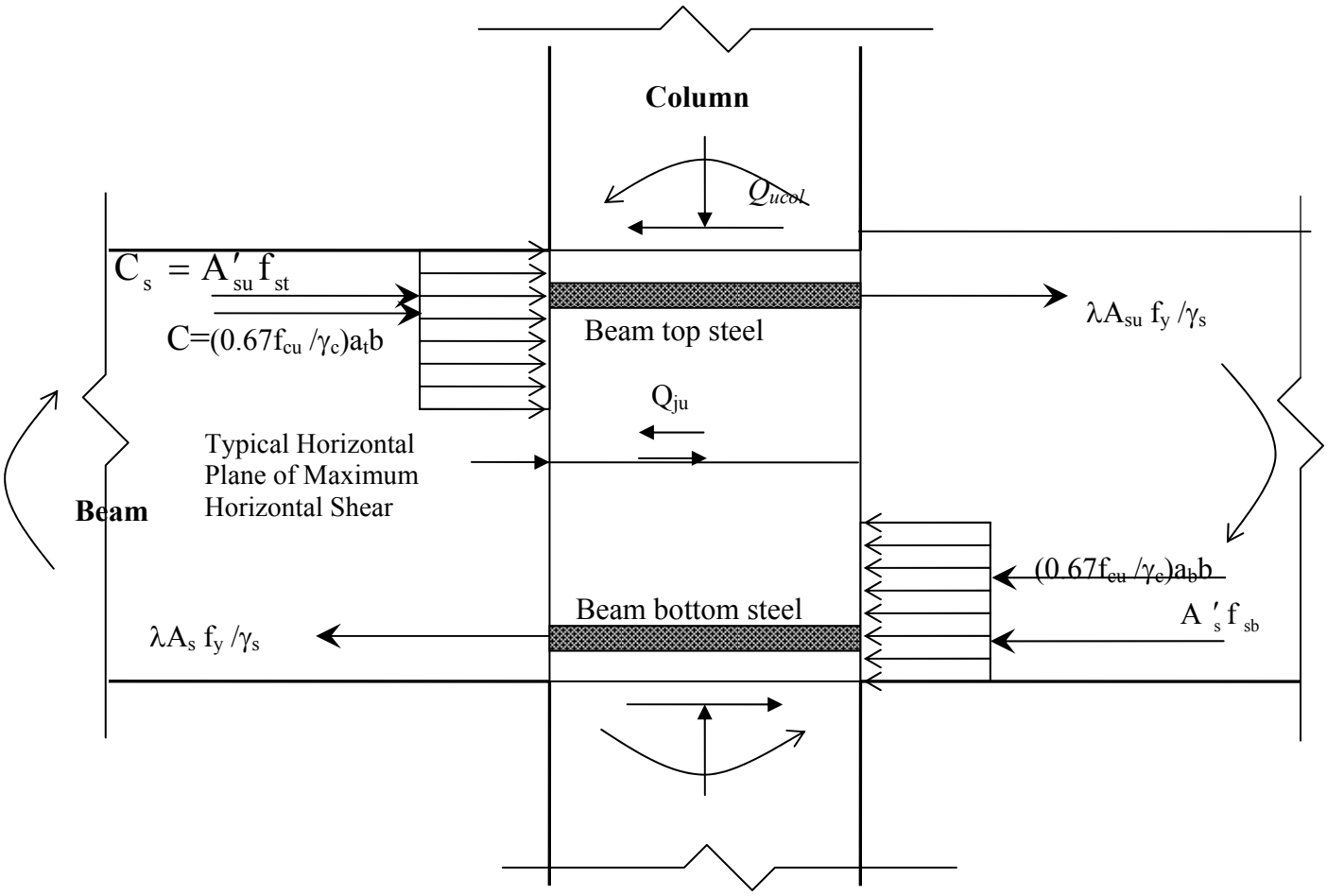


Fig. (6-30) Forces acting on beam-column joints

Table (6-15): Parameter of joint confinement

Types of Connections with Surrounding Structural Elements	Type of Joint	
	(I)	(II)
Joints of Continuing Columns (at intermediate floors)		
1- Joints confined on four faces	2.0	1.6
2- Joints confined on three faces	1.6	1.2
3- All other types of joint	1.2	0.9
Joints of Terminating Columns (at Roof floors)		
1- Joints confined on four faces	1.6	1.2
2- Joints confined on three faces	1.2	0.9
3- All other types of joint	0.9	0.6

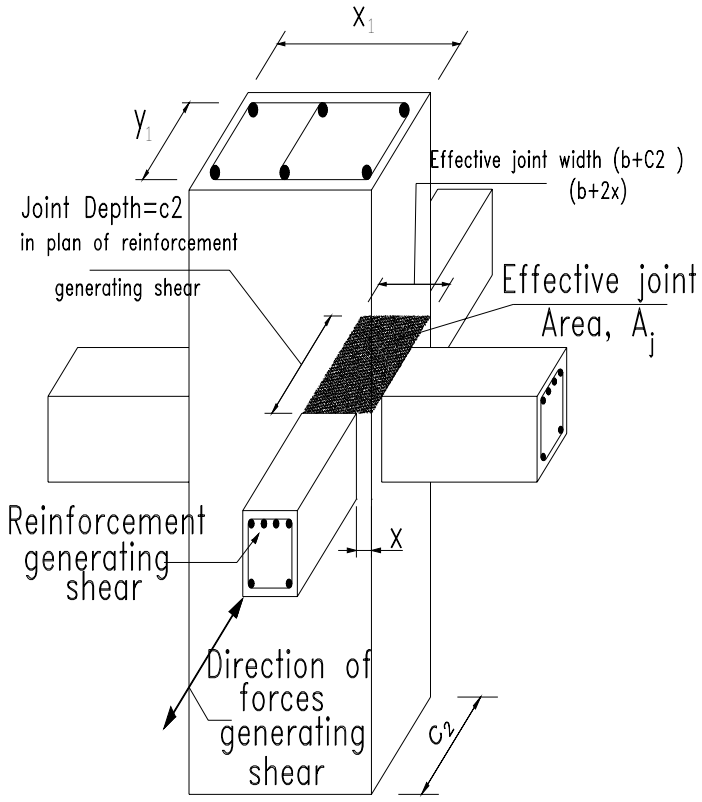


Fig. (6-31) Effective area of beam-column joints, A_j

6-7 Foundations

- Base area of footings or number and arrangement of piles shall be determined using the working loads. The design shall ensure that permissible soil pressure or permissible pile capacity, as well as the effects of differential settlement calculated according to the Egyptian Code for Soil Mechanics and Foundations, ECP 202, shall not be exceeded.
- The area of main tension reinforcement in foundations shall not be less than 0.25% of the gross concrete cross section area for milled steel reinforcement with f_y=240 MPa. When a high-grade steel reinforcement is used, this value shall be reduced in proportion to the ratio between the two yield stress values but in no case shall a value of less than 0.15% be used.
- The minimum amount of the area of shrinkage and temperature reinforcement (which is perpendicular to the tension reinforcement) is 20% of the area of the main reinforcement.
- Moments and shear forces in pile foundations shall be computed assuming that the reaction from each pile is concentrated at the pile center.
- The thickness of reinforced concrete footings shall not be taken less than 300mm for footings on soil or 400mm for footings on piles, but it shall not be taken less than the smaller dimension of the column cross section. In addition, this thickness shall satisfy the provisions of shear and punching shear strengths of sections (4-2-2-1) and (4-2-2-3) respectively.

6-7-1 Isolated footings and pile caps

6-7-1-1 General

It shall be permitted to assume uniform distribution for the soil bearing pressure for shallow and pile foundations when the vertical load acts at the foundation center. For eccentric loads, a linear distribution shall be permitted for both the soil bearing pressure and the loads on piles.

6-7-1-2 Design of footings and pile caps for flexure

6-7-1-2-1 The design for flexure of foundation sections shall follow the requirements of either the limit states design method of section (4-2-1) or the working stress design method of section (5-3-2).

6-7-1-2-2 For bending moments in foundations, the critical sections shall be determined by passing a vertical plane through the footing as follows:

- At the face of the reinforced concrete column or wall that is monolithic with the footing, as shown in Figure (6-32-a).
- Halfway between the column face and the edge of the steel base plate beneath the column, as shown in Figure (6-32-b).
- Halfway between the middle and edge of a masonry wall, as shown in Figure (6-32-c).

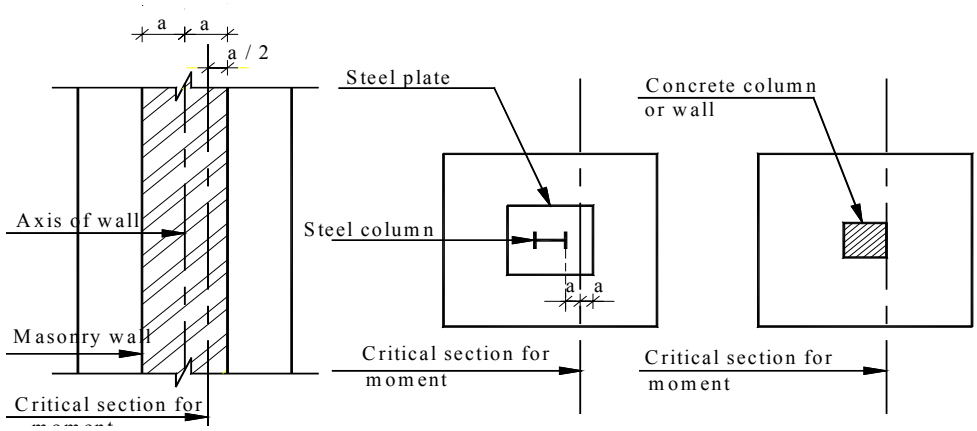


Fig. (6-32) Critical sections for bending moments in foundations

6-7-1-2-3 Bending moment on a critical section shall be determined by computing the moment of all forces acting on one side of the critical section.

6-7-1-2-4 In square footings, the reinforcement shall be distributed uniformly across entire width of footing in both directions; the reinforcement may also be distributed according to the bending moment diagram.

6-7-1-2-5 In rectangular footings, the reinforcement shall be distributed according to the bending moment diagram; the reinforcement may also be distributed following Figure (6-33) as follows:

- Reinforcement in long direction shall be distributed uniformly across entire width of footing.
- For reinforcement in short direction, a portion of the total reinforcement, A_{sm}, shall be closely distributed within a width centered with the column equals to the larger of:
  (a) Footing short side or,

(b) Length of column side parallel to footing long side plus thickness of footing, as shown in Figure (6-33). The ratio of the closely distributed reinforcement, A_{sm}, to the total reinforcement required in short direction, A_s, shall be determined from the following equation:

$$\frac{A_{sm}}{A_s} = \frac{2}{\left[\left(\frac{A}{B'}\right)+1\right]} = 2B' / (A+B') \qquad (6\text{-}58)$$

where,

A = length of long side of footing.

B = length of short side of footing.

B' = larger of the length of short side of footing, or the length of column side parallel to footing long side plus thickness of footing.

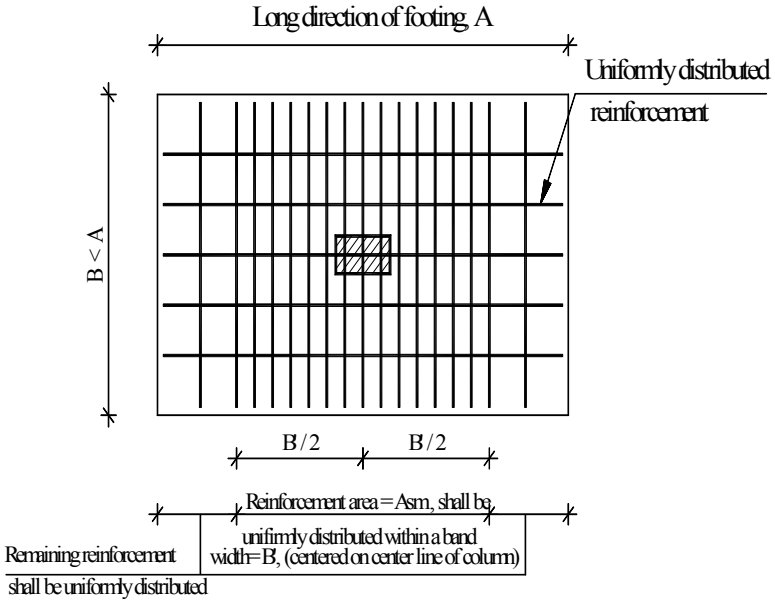


Fig. (6-33) Reinforcement distribution in rectangular foundations

6-7-1-3 Design of Footings and Pile Caps for Shear and Punching Shear

6-7-1-3-1 Shear strength and punching shear strength of footings shall be determined in accordance with section (4-2-2-2) and section(4-2-2-3), respectively.

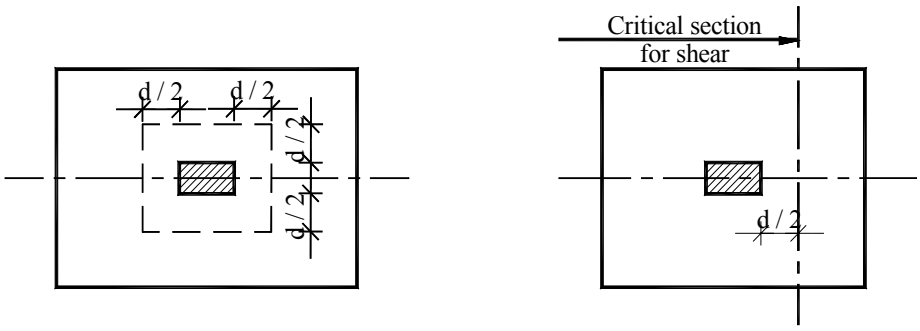
6-7-1-3-2 Location of critical section for shear shall be taken in accordance with section (4-2-2-1-1) and Figure (6-34-a).

6-7-1-3-3 Location of critical section for punching shear shall be taken in accordance with section (4-2-2-3) and Figure (6-34-b). When a steel base plate is used to fasten a column to a reinforced concrete base, the

location of critical section for punching shear shall be at a distance of (d/2) outside the critical section for flexure determined by section (6-7-1-2-2).

6-7-1-3-4 Computation of shear on pile caps shall be in accordance with the following:

- For any pile whose center is located a distance of pile radius or more outside the critical section, the full pile reaction shall be considered as producing shear on that section- case (a) in Figure (6-35).
- For any pile whose center is located a distance of pile radius or more inside the critical section, the pile reaction shall be considered as producing no shear on that section- case (b) in Figure (6-35).
- For intermediate positions of pile center, the portion of the pile reaction to be considered as producing shear on the section shall be based on straight-line interpolation between the above-mentioned two cases- case (c) in Figure (6-35).



a) Critical section for shear in shallow foundations

b) Critical section for punching shear

Fig. (6-34) Critical section for shear and punching shear

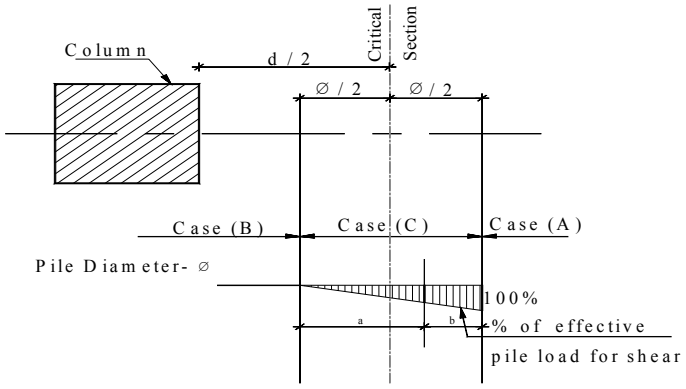


Fig. (6-35) Effective pile load in computation of shear in pile caps cases (a), (b), and (c)- Article (6-7-1-3)

6-7-1-4 Space-truss method for design of pile caps (strut-tie model)

6-7-1-4-1 For design of pile caps, it shall be permitted to use the plane-truss or space-truss method. In this case, compression members shall extend from the center of load bearing area to points of intersection of piles' axes with tension members which are formed by the reinforcement located in directions of the lines connecting piles' centers.

6-7-1-4-2 When pile spacing is larger than triple the pile diameter, the main reinforcement of ties shall be distributed in a band width equal to 1.5 pile diameter and centered on piles' axes.

6-7-1-4-3 It shall be permitted to use the strut-and-tie model section (6-11) for design of footings and pile caps.

6-7-1-5 Development of reinforcement

Development length, anchorage length, and splice length of reinforcement in footings shall be determined in accordance with section (4-2-5). Critical sections for development of reinforcement shall be assumed at the same locations of critical sections for flexure as defined in section (6-7-1-2-2).

6-7-1-6 Column loads shall be transferred to footings in accordance with section (6-4-7-n).

6-7-1-7 Any pile cap that is supported by a single pile shall be connected to other foundations by grade beams in at least two directions. Besides, pile caps supported by two or more piles lying in one vertical plane shall be connected to other foundations by grade beams in the direction perpendicular to this vertical plane. In addition, design of grade beams shall consider the largest probable offsets of pile centers and actual loads on piles. Furthermore, design of grade beams shall be rechecked using the as-built offsets in pile centers after pile construction.

6-7-2 Combined footings and raft foundations

6-7-2-1 combined footings and raft foundations shall be designed considering the relative stiffness between foundation and soil. It shall not be permitted to analyze combined footings and raft foundations using the methods recommended for solid slabs (Article 6-2) and beams (Article 6-3).

6-7-2-2 Combined footings and raft foundations may be considered as rigid and, thus, having linearly-variable soil bearing pressure distribution when either of the following two conditions is satisfied:

a - The relative stiffness, Kr, given by Equation (6-59-a) is larger than 0.5.

b - The average distance between any column and its adjacent columns in both directions is less than $\frac{1.75}{\beta}$ where β is a parameter determined by Equation (6-59-b).

$$K_r = \frac{E_c \cdot I_B}{E_{soil} \cdot b^3} \qquad (6\text{-}59\text{-a})$$

$$\beta = \sqrt[4]{\frac{K \cdot b}{4E_c \cdot I}} \qquad (6\text{-}59\text{-b})$$

where,

E_c= elastic modulus of concrete.

I_B= moment of inertia for foundation (or foundation together with frames and/or shear walls) per unit strip width.

E_s= elastic modulus of soil.

b= foundation strip width.

K= Winkler's modulus of soil subgrade reaction.

E_c.I= flexural rigidity of foundation strip.

6-7-2-3 Combined footings and raft foundations that do not satisfy the conditions stated in section (6-7-2-2) shall be treated as flexible and shall be analyzed as an elastic slab on a Winkler foundation or on a semi-infinite elastic medium. This analysis shall be based on actual soil properties determined by field or laboratory tests and shall satisfy equilibrium and compatibility conditions.

6-7-3 Concrete slabs on grade

6-7-3-1 General

- Slabs on grade are defined as rigid concrete slabs which are continuously supported by well-compacted soil, these slabs transfer to the soil vertical and/or lateral loads that act on the slabs either directly or through other structural members.
- Concrete slabs on grade can be classified as follows:
  a - Plain concrete isolated slabs.
  b -Concrete slabs, isolated or continuous, such as those in item "a." but provided with reinforcement to resist tensile stresses due to shrinkage and temperature only.

c - Structural reinforced concrete slabs, isolated or continuous, provided with effective reinforcement to resist bending moments and tensile forces produced by the loads acting on the slabs.

d - Reinforced concrete slabs provided with effective continuous reinforcement to resist bending moments and tensile forces produced by the loads acting on the slabs as well as the tensile stresses caused by shrinkage and temperature.

- In isolated slabs on grade, joints shall be arranged in accordance with the requirements of section (9-5-7).
- In all cases, the strength and serviceability requirements shall be satisfied including surface protection against cracks that may affect its use, if required.

6-7-3-2 Plain Concrete Slabs - Type "a"

The thickness of these plain concrete slabs, Type"a" shall be determined to ensure that the tensile stresses produced by loads that act on the slabs either directly or through other structural members do not exceed the cracking strength limit in accordance with section (4-3-2-7).

6-7-3-3 Concrete slabs provided with shrinkage and temperature reinforcement only, Type "b"

- The thickness of these concrete slabs Type "b" shall be determined to ensure that the tensile stresses produced by loads that act on the slabs either directly or through other structural members do not exceed the cracking strength limit in accordance with section (4-3-2-7).
- To resist the tensile stresses due to shrinkage and temperature in slabs of Type "b", these slabs shall be provided with distributed steel reinforcement- transverse and longitudinal- placed at slab midplane level or near its top surface. The required reinforcement ratio is determined following the subgrade drag method as:

$$\mu = \mu_f \cdot \omega \cdot L/2f_s \qquad (6\text{-}60a)$$

where:

L = distance between joints.

f_s = permissible working stress for steel reinforcement.

μ_f = coefficient of friction between concrete and soil according to type of soil (ranges from 1.5 to 2.5).

ω = unit weight of slab concrete

μ = ratio of area of required steel reinforcement to gross concrete area, A_s/A_c, and shall not be taken less than 0.15% for high-

grade steel or 0.25% for normal milled steel, with a minimum amount of 5 φ 10mm per meter width of slab in both directions.

6-7-3-4 Reinforced Concrete Slabs -Types "c" and "d"

- These slabs shall be designed assuming sections to be cracked under the effect of loads acting on the slabs in accordance with section (4-3-2-7), steel reinforcement shall be provided on the tension side to resist the tensile stresses produced by bending moments and axial tensile forces.
- The design shall follow the limit states design method and satisfy the serviceability limit states. The reinforcement ratio shall not be less than:

$$\mu = 0.3\frac{f_{ctr}}{f_y} \qquad (6\text{-}60b)$$

where:

μ = ratio of area of required steel reinforcement to gross concrete area, A_s/A_c.

$f_{ct\,r}$ = cracking strength limit state for concrete in tension in accordance with Equation (4-61b).

f_y = yield or proof stress for steel reinforcement.

- The ratio of steel reinforcement, $\mu=A_s/A_c$, shall not be taken less than 0.15% for high-grade steel or 0.25% for normal milled steel, with a minimum amount of 5 φ 10mm per meter width of slab in both directions.
- Requirements related to casting and treatment of slab-on-grade concrete as well as proper arrangement of joints shall be fulfilled in all cases.

6-7-4 Foundations subject to seismic loads

6-7-4-1 Footings, raft foundations and pile caps

6-7-4-1-1 Longitudinal reinforcement of reinforced concrete columns and walls shall be extended inside footings, raft foundations, or pile caps for a distance of not less than the development length of tension bars below the contact surface between foundation and column or wall. The reinforcement shall also be extended till the foundation bottom steel reinforcement and shall be provided with a right-angle leg

6-7-4-1-2 Longitudinal reinforcement of piles shall be extended inside pile caps for a distance of not less than the development length of tension bars above the contact surface between piles and pile cap.

6-7-4-1-3 Footings and pile caps supporting columns that are subject to axial tensile forces due to seismic loads shall be provided with top foundation reinforcement, sufficient to resist the bending moments produced by these column tensile forces.

6-7-4-2 Grade beams and slabs on grade

6-7-4-2-1 Grade beams shall be designed as part of the seismic, lateral-load resisting-system in accordance with section (6-8). They shall be design to act as horizontal ties between pile caps or footings and shall have continuous longitudinal reinforcement along the entire length of grade beams. The grade beam longitudinal reinforcement shall have a full developed length beyond the centerline of supported column. The structural design drawings shall clearly state that grade beams are parts of the seismic lateral-load resisting system.

6-7-4-2-2 Requirements of section (6.7.4.2.1) shall also apply for slabs on grade when such slabs are considered as part of the seismic, lateral-load resisting-system.

6-7-4-2-3 The smallest cross-sectional dimension of grade beams shall not be taken less than $\frac{1}{20}$ of the clear spacing between connected columns, but need not be greater than 450mm, provided that it will satisfy the slenderness limit stated in section (6-3-1-8).

6-7-4-3 Piles

6-7-4-3-1 Piles shall be provided with adequate longitudinal and transverse reinforcement to resist the moments and forces resulting from earthquakes. These moments and forces shall be computed considering the actual soil properties. Requirements related to area of stirrups and stirrups spacing shall be satisfied.

6-7-4-3-2 Piles stirrups shall be increased in accordance with section (6-8) at the following locations:

a -At the top of the pile for a length of at least 5 times the pile diameter, but shall not be less than 2m below the bottom surface of the pile cap.

b - For the portion of piles in soil that shall not be capable of providing lateral support and for portions with abrupt changes in soil properties.

The ratio of spiral stirrups shall not be taken less than that required by sections (4-2-1-3) and (6-4-7).

6-8 Special provisions for seismic design

6-8-1 General

This section contains special requirements for design of structural members for which the seismic design forces have been determined in accordance with the Egyptian Code for Loads on Structures (ECP 201).

6-8-1-1 Definition of structural members

- Beam: a structural member that shall be mainly subjected to bending moments accompanied by axial compressive force having a value not exceeding $0.04A_gf_{cu}$.
- Column: a structural member that shall be subjected to axial compression combined with bending moments and for which the axial compressive force shall be more than $0.04A_gf_{cu}$.
- Frame: a spatial structure composed of beams, columns, and joints which shall be capable of resisting bending moments, axial forces, and shear forces. The frame shall be classified as either ductile; having adequate ductility or ordinary; having limited ductility in accordance with section (6-8-2).
- Shear Walls: structural members having cross-section length to thickness ratio more than five. Walls resisting the forces induced by earthquakes shall be classified as follows:
  - Ductile shear walls: reinforced concrete structural walls resisting forces induced by earthquakes having a height to length ratio, h_w/L_w, equal to or more than 2 and fixed at the foundation level. The walls shall be capable of dissipating energy through the formation of a plastic hinge at location of maximum bending moment.
  - Low-rise shear walls: reinforced concrete structural walls resisting forces induced by earthquakes having a height to length ratio, h_w/L_w, of less than 2, and fixed at the foundation level, but shall be incapable of dissipating energy as they have limited inelastic deformation capacity . Their deformations are mainly caused by sliding shear.
  - Coupled shear walls: reinforced concrete structural walls consisting of two or more single walls connected together by ductile beams in a regular pattern and in which the beams shall be capable of reducing the sum of the base bending moments of the individual walls if working separately by at least 25%.

6-8-1-2 Seismic-load resisting structural systems

- Wall system: is a structural system in which the primary structural members that resist lateral and vertical loads shall be reinforced concrete walls- either coupled or uncoupled- and whose shear strength at building base shall be more than two thirds of the total building shear strength.
- Frame system: is a structural system in which the primary structural members that resist lateral and vertical loads shall be spatial frames whose shear strength, at building base shall be more than two thirds of the total building shear strength.
- Dual system: is a structural system in which the primary structural members that resist vertical loads shall be spatial frames while spatial frames and walls shall act together as a combined system for resisting lateral loads. Dual system shall be classified as one of the following two types:
  - Frame-equivalent dual system: dual system in which the shear resistance of the frame system at the building base shall be higher than 50% of the total shear resistance of the whole structural system.
  - Wall-equivalent dual system: dual system in which the shear resistance of the walls at the building base shall be higher than 50% of the total shear resistance of the whole structural system.

6-8-1-3 Design concepts

a - The design of earthquake resistant concrete buildings shall provide an adequate energy dissipation capacity to the structure without substantial reduction of its overall resistance against horizontal and vertical loading in conformance with the requirements of sections (6-8-2) and (6-8-3).

b - Concrete structures shall be analyzed, designed and detailed in accordance with the requirements of Chapters 3 to 7 and sections (6-8-2) and (6-8-3). The design shall be based on energy dissipation and shall ensure structural ductility so that the ductile flexural failure mode precedes the brittle shear failure mode.

c - Based on the structure capability to dissipate energy when subjected to reversed loading, two ductility classes are defined: limited ductility and adequate ductility. Each ductility class corresponds to structures designed and detailed according to specific provisions, enabling the structure to develop stable mechanisms associated with large

dissipation of hysteretic energy under repeated reversed loading, without suffering brittle failures.

d - A ductility class having either limited or adequate ductility shall be assigned to each structural member in accordance with the requirements of sections (6-8-2) and (6-8-3) for frames and walls, respectively.

e - Adequate ductility shall be ensured for the structure by implementing the structural details described in this chapter which permits non-linear deformations in the critical regions defined in the following item.

f - Critical region or energy dissipative zone: it is a region, in a main seismic-load-resisting member, where the dissipative capabilities shall be mainly located and where plastic hinges may be formed due to inelastic deformations caused by bending moments when subjected to the worst load combination (i.e.bending moments, axial forces, shear forces, and/or torsion). The length of critical region for each one of the main seismic-load resisting members shall be determined as follows:

- For beams: it is a distance equal to twice the beam depth measured from face of support as shown in Figure (6-36).
- For columns: it is a distance, L_o, measured from the column-beam intersection surface at each column end (Figures 6-36 and 7-6-b), where L_o is equal to the larger of:
  - One sixth of the column clear height
  - The longer side of column cross section
  - 500 mm
- For ductile shear walls: it is a distance of not less than the larger of:
  - One sixth of the entire wall height
  - The length of wall cross section
- But it shall not be taken more than twice the wall length (Figure 6-36).

g- The requirements of the Egyptian Code for Loads on Structures (ECP 201), related to seismic joints and interstory drift values and the requirements of section (9-5-9) of this Code shall be fulfilled.

h - Special requirements for the reinforcing steel and concrete of seismic-load resisting members:

- It shall not be allowed to use smooth bars for the longitudinal reinforcement. Only deformed bars shall be permitted for both milled and high-grade steel reinforcement.
- The ratio of ultimate strength to yield strength of reinforcing steel shall not be less than 1.25.
- The concrete characteristic strength shall not be taken less than 25N/mm^2.

i - For seismic-load resisting members, the effective moment of inertia shall be taken as follows:

- $I_{eff} = 0.70\ I_g$ for columns
- $I_{eff} = 0.35\ I_g$ for shear walls
- $I_{eff} = 0.50\ I_g$ for beams (including slab contribution)
- $I_{eff} = 0.25\ I_g$ for flat slabs

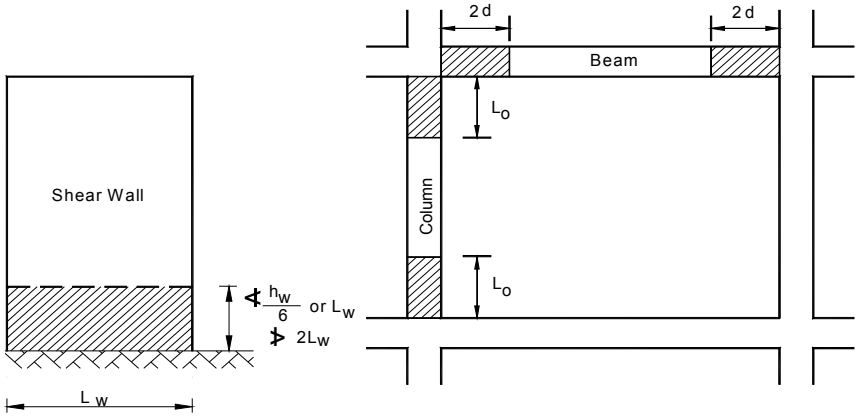


Figure (6-36) Critical regions in beams, columns, and walls

6-8-2 Requirements for frames resisting earthquake-induced forces

6-8-2-1 General

- Structural members of frames resisting earthquake-induced forces shall satisfy the following:

  a - Distance between column center and beam centerline shall not be greater than 25% of the column dimension perpendicular to the beam centerline (Figure 6-37).

b - Beam width shall not be taken greater than the smaller of:
- Column dimension (parallel to beam width) plus beam depth
- Twice the column dimension parallel to beam width
- Frames resisting earthquake-induced forces shall be classified as being either ductile having adequate ductility or ordinary having limited ductility in accordance with the reinforcement details used. Each frame class shall be assigned an appropriate value of the response modification factor, R, in accordance with the Egyptian Code for Loads on Structures (ECP 201).

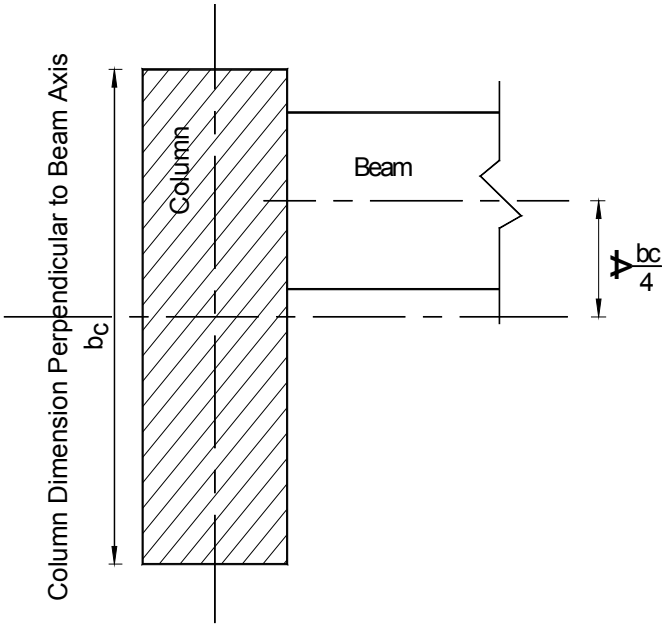


Figure (6-37) In-plan relationship between beam axis and column axis

6-8-2-2 Requirements for ordinary frames having limited ductility

This section sets up the requirements for flat slabs, frame beams, and frame columns.

6-8-2-2-1 Flat slabs

a - Bending moments related to earthquake effect transferred from slab to column shall be resisted by the column strip alone.

b - The negative bending moments, $\gamma_f M_f$, determined by section (6-2-5-8) shall be resisted by the effective slab width, c_2+3t, where t is the slab thickness (Figure 6-38).

c - Not less than 50% of the column strip total reinforcement shall be placed in the effective slab width.

d - Not less than 25% of the total top reinforcement in column strip shall be extended over the entire span length (Figure 6-39).

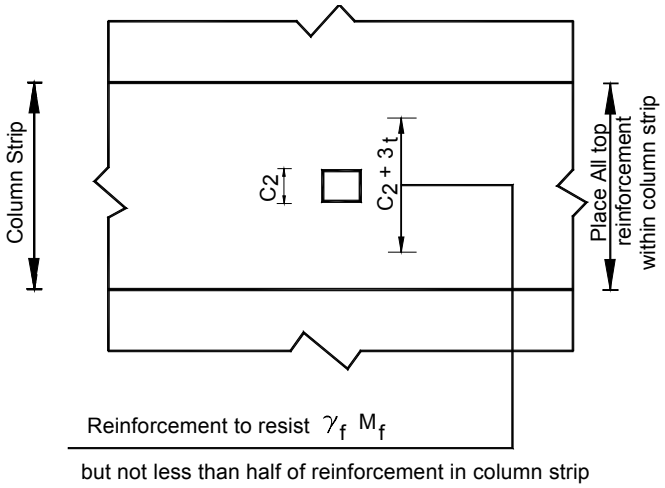


Figure (6-38) Effective width in flat slab constructions

e - Not less than 50% of the total bottom reinforcement in column and field strips shall be extended over the entire span length; the bottom reinforcement shall be adequately developed at support regions in accordance with section (4-2-5-3).

f - The amount of continuous bottom reinforcement in column strip shall not be taken less than 33% of the column strip top reinforcement at support regions.

g - At slab discontinuous edges, both its top and bottom reinforcement shall be adequately developed inside the edge support region in accordance with section (4-2-5-3).

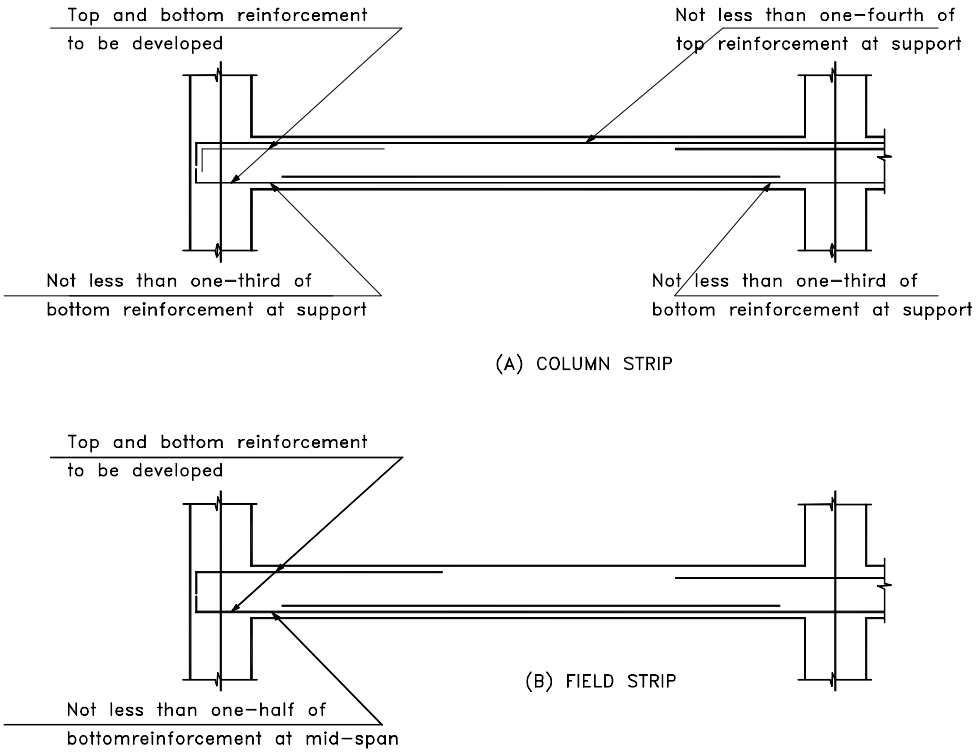


Figure (6-39) Reinforcement Arrangement in Flat Slabs

6-8-2-2-2 Beams in ordinary frames having limited ductility

a - The beam positive moment strength at the face of the support shall not be less than one-third the negative moment strength provided at that face of the support.

b - Neither the negative nor the positive moment strength at any section along the length of the beam shall be less than one-fifth the maximum moment strength provided at the face of either column.

c - Stirrups shall be provided over the critical regions such that the first stirrup shall be located at not more than 50 mm from the face of the support. Stirrup spacing shall not exceed the smallest of:

- One quarter of beam depth;
- Eight times the diameter of the smallest longitudinal bar enclosed;
- 24 times the stirrup diameter;
- 200 mm.

d - Stirrups throughout the rest of beam length shall be placed at not more than one half of beam depth or 200mm, whichever is smaller.

6-8-2-2-3 Columns in ordinary frames having limited ductility

a - Stirrups shall be provided over the critical regions with spacing, s_o, that shall not exceed the smallest of:

- Eight times the diameter of the smallest longitudinal bar enclosed;
- 24 times the stirrup diameter;
- One-half of the smallest column cross-sectional dimension;
- 150 mm.

b - The first stirrup shall be located at not more than 50 mm from the column-beam intersection surface. The stirrups spacing, throughout the rest of column length, shall not be more than $2s_o$ or 200mm, whichever is smaller.

6-8-2-3 Requirements for ductile frames having adequate ductility

This section outlines the requirements for frame beams and columns. It shall not be permitted to consider flat slabs as parts of ductile frames.

6-8-2-3-1 Beams in ductile frames having adequate ductility

Beams shall be designed in accordance with section (6-8-2-2-2) in addition to the following requirements:

a - Beam width shall not be less than the larger of 30% of its depth or 250 mm.

b - Positive moment strength at support face shall be not less than one-half of the negative moment strength provided at that face of the support. Neither the negative nor the positive moment strength at any section along beam length shall be less than one-fourth the maximum moment strength provided at face of either support.

b - All lap splices shall be designed as tension lap splices. The following requirements shall also be fulfilled:

- Transverse reinforcement provided over the lap length shall be composed of closed stirrups or spiral reinforcement.
- Spacing of the transverse reinforcement enclosing the lapped bars shall not exceed the smaller of d/4 and 100 mm.
- Lap splices shall not be used in critical regions and within the beam column joint.

c - The ultimate design shear force shall be computed assuming that moments of opposite sign, corresponding to probable flexural moment strength, act at the support faces (Figure 6-40) and that the beam shall be loaded with the factored tributary gravity load along its span.

Transverse reinforcement in critical regions shall be proportioned to resist the ultimate design shear force ignoring the concrete shear strength.

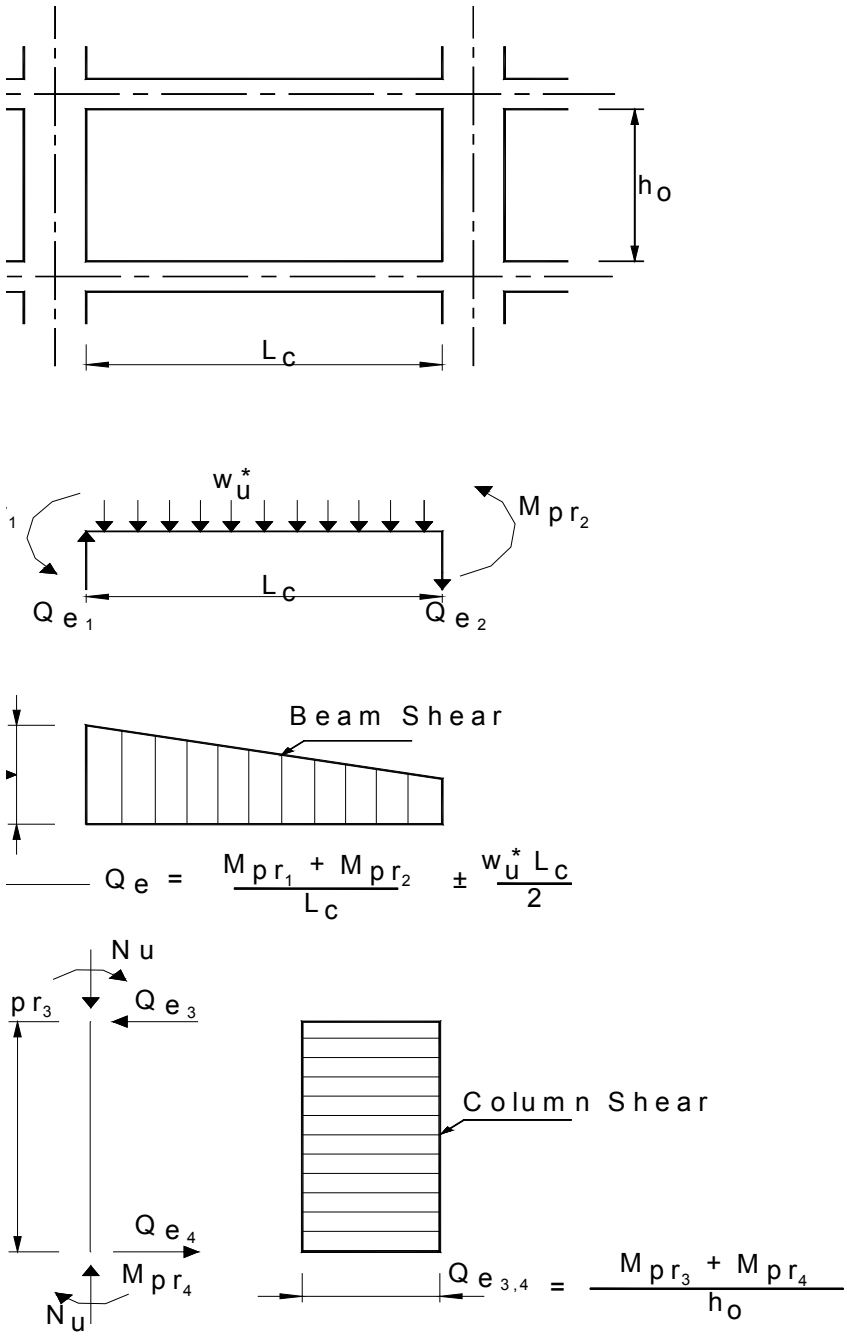


Figure (6-40) Calculation of ultimate design shears for girders and columns

6-8-2-3-2 Columns in ductile frames (having adequate ductility)

Reinforced concrete columns shall be designed in accordance with section (6-8-2-2-3) in addition to the following requirements:

a - The shortest dimension of column cross-section shall not be less than 300 mm; the ratio of column shortest cross-sectional dimension to the perpendicular dimension shall preferably be not less than 0.4.

b - The column dimension parallel to the beam reinforcement shall not be less than 20 times the diameter of the largest longitudinal beam bar.

c - For columns that are connected to frame beams and resist axial compressive forces exceeding $0.04A_g f_{cu}$, the ultimate flexural strength shall satisfy the following relation:

$$\sum M_c \geq 1.2 \sum M_g \qquad (6-61)$$

$\sum M_c$ = sum of flexural strengths of columns framing into the joint in the analysis plane, evaluated at the faces of the beam. Column flexural strength shall be calculated for the factored axial force, consistent with the direction of the lateral forces considered, resulting in the lowest flexural strength.

$\sum M_g$ = sum of flexural strengths of the beams framing into the joint in the analysis plane, evaluated at the faces of the column. In T-beam construction, where the slab is in tension under moments at the face of the support, slab reinforcement within three slab thickness in each side of the beam shall be assumed to contribute to M_g if the slab reinforcement is developed at the critical section for flexure.

For computing $\sum M_c$ and $\sum M_g$, flexural strengths shall be summed such that the column moments oppose the beam moments.

Lateral strength and stiffness of columns not satisfying Equation (6-61) shall be ignored when determining the calculated strength and stiffness of the structure, but such columns shall conform to section (6-8-2-2-3).

Equation (6-61) shall be satisfied for beam moments acting in both directions in the vertical plane of the frame considered.

d - The ratio of longitudinal reinforcement area to gross concrete cross-section area shall not be less than 0.01.

e - Lap splices shall be permitted only within the center half of column length (Figure 7-6-b).

f - All lap splices shall be designed as a tension lap splice.

g - Mechanical and welded splices shall be permitted outside the critical regions provided that the requirements of section (4-2-5-4-3) shall be fulfilled and that welded splices conform to the standard specifications.

h - Column stirrups shall continue through the beam-column joint. The cross-section area of these stirrups shall be determined in accordance with Equations (6-57a) and (6-57b).

i - The ultimate design shear force shall be computed assuming that moments of opposite sign, corresponding to probable flexural moment strength, act at column ends (Figure 6-40).

6-8-2-3-3 Beam to column connection

Forces in the beam longitudinal reinforcement at faces of supports shall be determined using a tensile stress of at least $1.25f_y$. Requirements of section (6-6-2) shall also be fulfilled.

6-8-3 Requirements for shear walls

6-8-3-1 Scope

a - Requirements of section (6-8-3) shall apply to ductile structural or shear walls used as part of the seismic load resisting system.

b - Requirements of section (6-5) shall not be permitted except within the limits stated in section (6-8-3).

6-8-3-2 Concrete dimensions

Concrete dimensions of shear walls shall be determined in accordance with Article (6-5-2-1-1), but the wall thickness in critical regions shall not be less than one tenth of the story clear height as shown in Figures (6-41a&b)

a) Wall with uniform thickness;

b) Wall strengthened by a boundary element

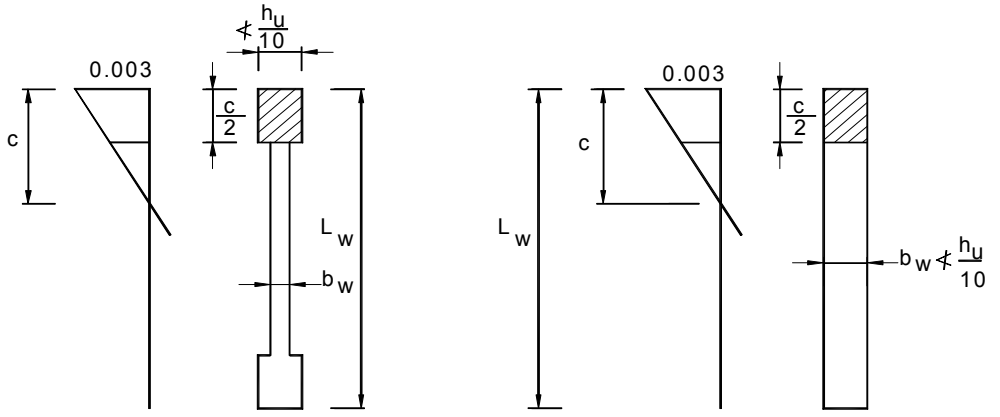


Figure (6-41) Minimum wall thickness in critical regions (plan section)

6-8-3-3 Reinforcement of ductile shear walls

Reinforcement of ductile shear walls shall consist of two meshes, one on each face. Ratios of distributed vertical and horizontal reinforcement shall conform to sections (6-8-3-3-1) and (6-8-3-3-2), respectively. In addition, concentrated vertical reinforcement shall be determined in accordance with section (6-8-3-3-3).

6-8-3-3-1 Distributed vertical reinforcement

a - The ratio of total, distributed, vertical reinforcement shall not be less than 0.25%.
b - The diameter of selected reinforcement bars shall not be less than 10 mm and the bar spacing shall not exceed 200 mm.

6-8-3-3-2 Distributed horizontal reinforcement

a - The ratio of total, distributed, horizontal reinforcement shall not be less than 0.25%.
b - The diameter of selected reinforcement bars shall not be less than 10 mm and the bar spacing shall not exceed 200 mm.
c - When the ratio of total distributed vertical reinforcement is more than 1%, additional crossties in the form of closed stirrups, that links the reinforcement meshes on both sides together and penetrate through the thickness shall be used. The diameter of these closed stirrups shall not be less than the larger of one quarter of the diameter of vertical reinforcement bars or 8 mm. Arrangement of closed stirrups shall provide least 4 stirrups per each square meter of wall.

6-8-3-3-3 Concentrated vertical reinforcement

a - Concentrated vertical reinforcement shall be used at wall edges, at wall corners and at intersections of walls as shown in Figure (6-42).
b - The diameter of concentrated reinforcement bars shall not be less than 12 mm.
c - The ratio of concentrated vertical reinforcement area to the total area of wall concrete section, outsides the critical regions, shall not be less than 0.1%. With the critical regions, this ratio shall not be less than 0.2%.
d - Stirrups, satisfying the requirements of section (6-8-2-3-2), shall be used to contain the concentrated vertical reinforcement. Concentrated vertical reinforcement shall preferably be used at all locations where the concrete compressive strain exceeds 0.0015.

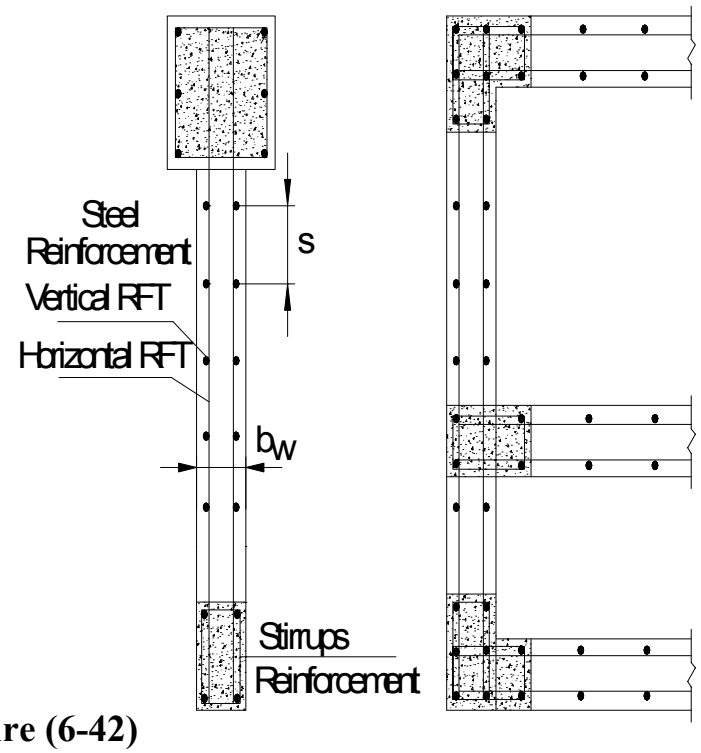


Figure (6-42) Distributed as well as concentrated vertical reinforcement in shear walls (plan section)

6-8-3-4 Flexural strength of shear walls

a - Flexural strength of shear walls shall be determined using the ultimate strength limit state method in accordance with section (4-2-1).

b - Both the distributed and the concentrated vertical reinforcement types shall be considered in determining the flexural strength of shear walls.

c - The ultimate flexural strength of shear walls shall not be less than the larger of the cracking strength of wall cross-section and the factored bending moment acting on the wall.

6-8-3-5 Shear strength of shear walls

a - The ultimate shear strength of shear walls shall be determined using the following equation:

$$q_{umax} = \left(0.9\alpha_c \sqrt{\frac{f_{cu}}{\gamma_c}} + \mu_{st} \frac{f_y}{\gamma_s} \right) = \frac{Q_u}{b_w \times L_w} \qquad (6-62)$$

where,

μ_{st} = ratio of web reinforcement as given by Equation (4-22)

α_c = 0.25 for $\frac{h_w}{L_w} \le 1.5$

= 0.17 for $\frac{h_w}{L_w} \ge 2$

and varies linearly from 0.25 to 0.17 for $1.5 \le \frac{h_w}{L_w} \le 2$.

b - For ductile shear walls, design shear forces shall be determined using a tensile stress of at least $1.25f_y$ in all reinforcing steel. Consequently, the design shear forces shall be equal to 125% of the shear forces produced by the seismic loads.

c - In wall critical regions, the concrete shear strength shall be ignored and transverse reinforcement shall be proportioned to resist the total ultimate design shear forces.

6-8-3-6 Structural members not-designated as part of the seismic-load resisting system

a - Structural members not-designated as part of the seismic-load resisting system and subject to the same deformation as the resisting system shall possess sufficient ductility to be able to support gravity loads while subjected to the design displacement.

b - Structural members not-designated as part of the seismic-load resisting system shall be designed to resist the ultimate bending moments that result from the lateral displacements (i.e. drifts) caused by the seismic loads, or they shall be designed and detailed in a manner that shall permit the formation of plastic hinges at their critical sections.

c - Stirrups used in structural members not-designated as part of the seismic-load resisting system shall conform to section (6-8-2-2-2) or section (6-8-2-2-3) and shall permit the formation of plastic hinges.

6-8-3-7 Coupling beams

a - Coupling beams with clear-span to depth ratios of not less than 4 shall conform to the requirements of Article (6-8-2-3-1).

b - Coupling beams with clear-span to depth ratios of not more than 2 shall be reinforced with two intersecting groups of diagonally placed bars symmetrical about the mid-span in accordance with Figure (6-43).

c - Coupling beams with clear-span to depth ratios that are larger than 2 but smaller than 4 shall either be detailed in accordance with the requirements of section (6-8-2-3-1) or shall be reinforced with two intersecting groups of diagonally placed bars symmetrical about the mid-span as shown in Figure (6-43).

d - Coupling beams reinforced with two intersecting groups of diagonally placed bars symmetrical about the mid-span shall satisfy the following:
- Each group of diagonally placed bars shall consist of a minimum of four bars;
- Each group of diagonally placed bars shall be enclosed in transverse reinforcement satisfying section (6-8-2-2-3);
- The diagonally placed bars shall be developed for tension in the wall.
- The ultimate shear strength of coupling beams shall be determined using the following equation:

$$q_u = \frac{2A_{sd}}{bd}\frac{f_y}{\gamma_s}\sin\alpha \quad \le 0.7\sqrt{\frac{f_{cu}}{\gamma_c}} \qquad (6\text{-}63)$$

where:

A_{sd} = area of one group diagonally placed bars

α = angle between diagonally placed bars and the beam longitudinal axis

e - The contribution of diagonally placed bars shall be considered in the determination of the ultimate flexural strength of coupling beams.

f - Coupling beams shall be provided with longitudinal and transverse reinforcement conforming to the minimum requirements of sections (4-2-1-2-h) and (4-2-1-2-6), respectively.

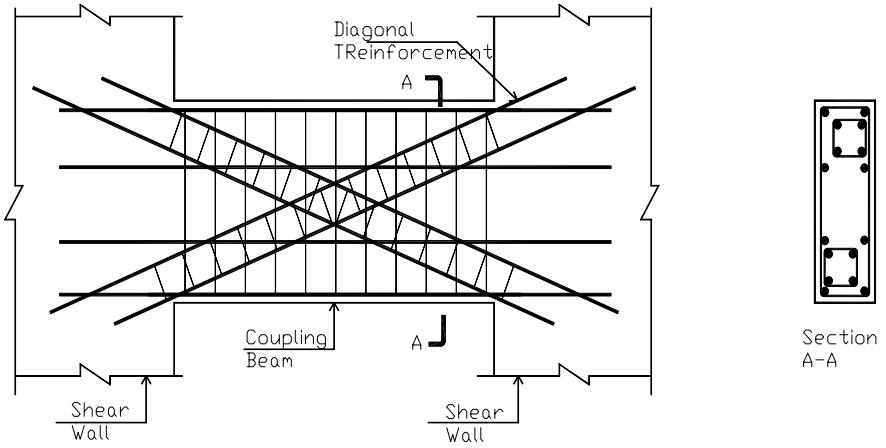


Figure (6-43) Reinforcement details for coupling beams

6-9 precast concrete:

Precast concrete structural members shall be designed according to provisions of this section (6-9). All provisions of this code, not in the conflict with the provisions of section (6-9), shall apply to analysis and design of precast concrete structural members. The requirements of this section shall not sufficient for safety requirements for earthquake resistance.

6-9-1 General

1- Precast elements, connections and joints shall be designed to resist all external loads affecting member during fabrications, storage, transportation, erection, construction and usage. In addition, stresses due to end restraint shall be considered.

2 - In analysis of precast structure, connection structural behavior assumptions shall represent actual behavior.

3 - Design and details shall take into consideration special requirements for erection and tolerances specified in section (9-8-3) in addition to stresses resulting from erection.

4 - In addition to requirements of details specified in section (7-2), the following shall be included in tender or workshop drawings:-

- a - Reinforcement details, connections, concrete cover, inserts and lifting devices required to resist temporary loads during construction stages.
- b - Characteristics strength of concrete during different construction stages.
- c - Surface finishing.
- d - Special tolerance (i.e. non-standard) required for element or structure.
- e - Locations of ties and joints with applied forces.
- f - Requirements and recommendations for erection and construction.

6-9-2 Distribution of forces among members

1 - Forces perpendicular to the plane of members shall be distributed based on structural analysis or tests

2 - In-plane forces shall be transferred between the members of a precast floor or wall system according to the following requirements:

- a - In-plane forces paths shall be continuous through both connections and members.

b - Continuous path of steel shall be provided to resist a resulting tension force

c - Connections, ties, and bearing areas shall be designed to resist all transferable forces including any special forces due to tolerance, elastic deformations, creep, shrinkage and temperature.

6-9-3 Reinforcement of precast elements.

- Elements shall be reinforced according to the requirements of this section. All code items not in conflict with this section shall be part of these requirements.
- Horizontal and vertical reinforcement in walls shall not be less than 0.25% of gross concrete section area.
- Slab reinforcement in each direction shall not be less than 0.15% of gross section area.

6-9-4 Structural integrity

6-9-4-1 For precast concrete structure with height not more than two stories, the following requirements shall be satisfied.

1 - Longitudinal, transverse, vertical, and around the perimeter of building ties shall be used to assure integrity of precast elements with the lateral force resisting structural system.

2 - For floors consisting of precast elements acting as rigid horizontal diaphragms, the connections between diaphragms and the vertical members resisting lateral forces and laterally supported on diaphragms shall have a nominal tensile strength capable of resisting at least $4.5\ KN/m$.

3 - Vertical ties shall be used in all vertical structural elements. This shall be satisfied by providing connections at horizontal joints according to the following:

a - Precast columns shall have a nominal ultimate tensile force not less than 1.4 Ag, (N), when Ag shall be the required concrete cross sectional area in mm^2. for columns with a larger cross section than required by consideration of loading, a reduced effective area (Ag'), based on cross section required but not less than one-half the actual area, shall be used.

b - Precast wall panels shall be provided with two ties per panel as a minimum with a nominal tensile strength not less than 45 KN per tie. These ties shall be symmetric about the vertical axis of the wall and exist in the external quarter of the wall where possible.

c - When no tension at the base exists due to design forces, the ties required by item (b) above, shall be appropriately anchored into a reinforced concrete floor slab on grade.

4 - Friction caused by gravity loads shall not be relied on when designing and detailing of connections.

6-9-4-2 For precast concrete bearing wall structures three or more stories high, the following provisions shall be applied as a minimum (figure 6-44):-

1 - Floor structural system shall be provided with longitudinal and transverse ties assuring nominal strength of 22 KN/m of width or length. These ties shall be positioned over interior wall supports and between members and exterior walls. These ties shall be spaced within 0.6 m of the level of the floor or roof system.

2 - Longitudinal ties parallel to floor spans shall be spaced not more than 3.0 ms. Enough precautions shall be taken for transferring of forces around openings.

3 - Transverse ties perpendicular to floor spans shall be spaced at distances not exceeding bearing wall spacing.

4 - Ties around the perimeter of each floor, shall be provided within 1.2 m of the floor edge, and shall assure a nominal strength in tension not less than 70 KN.

5 - Vertical ties shall be provided in all walls and shall be continuous over the height of the building. They shall provide a nominal tensile strength not less than 40 KN per horizontal meter of wall. At least, two ties shall be provided for each wall.

6-9-5 Design of connections and bearing zones

6-9-5-1 Grouted joints, shear keys, mechanical connectors, reinforcing steel connections, reinforced topping, or a combination of these means allow transfer of forces between members. Mechanical connectors with grouted joints or shear keys are preferred to be used for structure with three or more stories.

6-9-5-2 Analysis or testing shall be used to determine the adequacy of connections to transfer forces between members.. Where shear is the primary effective loading, the provisions of section (4-2-2-4) shall be satisfied.

6-9-5-3 For designing connections with materials having different structural properties, their relative stiffness, strengths, and ductility shall be taken into consideration.

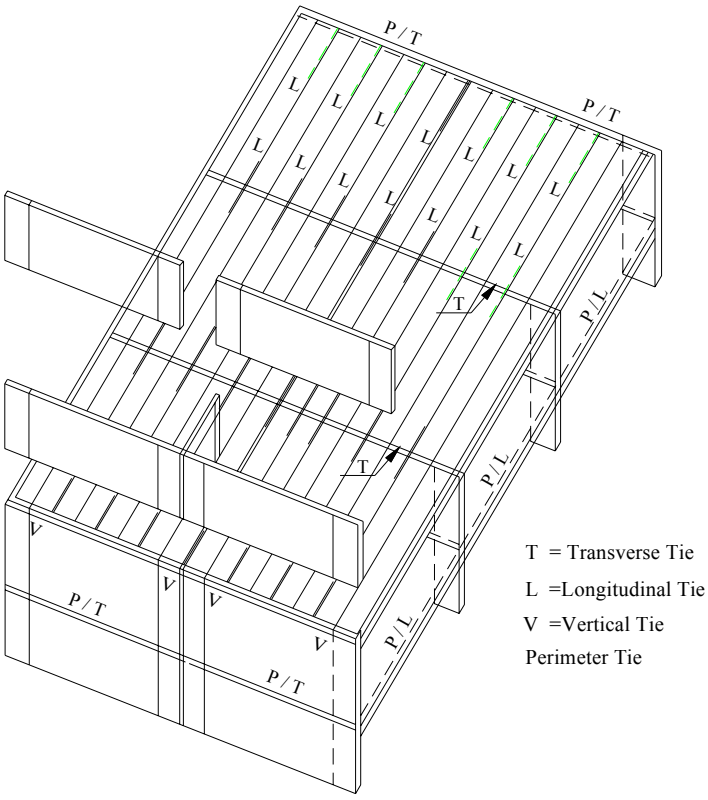


Figure (6-44) Typical linear distribution for ties in precast systems

6-9-5-4 Precast floor members bearing on simple supports shall satisfy the following:-

1 - The allowable bearing stresses at the contact surface between supported and supporting members shall not exceed the bearing strength for either surface or the bearing element. Concrete bearing strength shall be according to section (4-2-4) or (5-6).

2 - Unless shown by structural analysis or test that performance of connection or bearing zones of precast elements shall not be impaired, the following requirements shall be satisfied:-

a -The design dimensions for each member and its supporting elements with due consideration of the allowable tolerances shall satisfy the condition that the distance between support edge and end of precast element supported on it shall not be less than 1/180 of the clear span of the member but not less than 50 mm for slabs and 75 mm for beams as shown in figure (6-45).

b -Bearing pads for unarmored edges shall be provided at a distance not less than 15 mm from the face of the support, or at least the chamfer dimension at chamfered edges.

3 - The requirements of (4-2-5-3) shall not apply to the positive bending moment reinforcement for statically determinate precast members, but at least one-third of such reinforcement shall extend to the center of the bearing length.

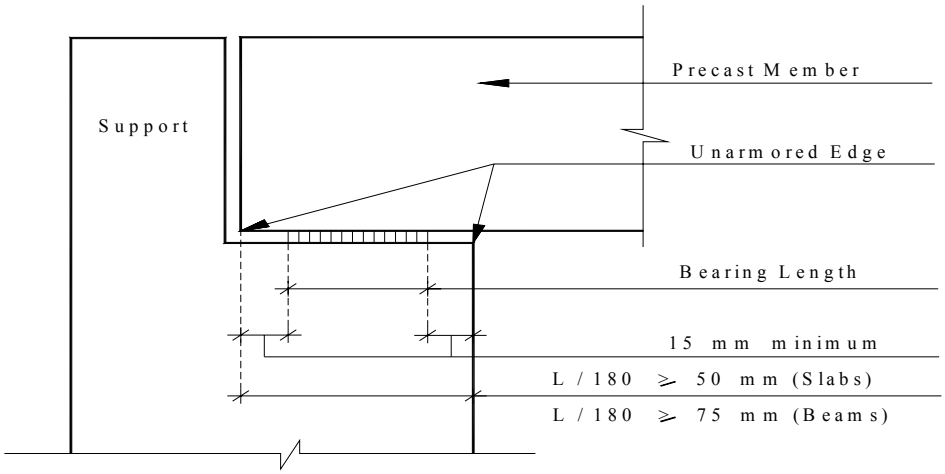


Figure (6-45) Bearing length for precast elements

6-9-6 Items embedded after concrete casting

Embedded items, such as dowels or inserts, that either protrude from the concrete or remain exposed for inspection may be embedded while the concrete is in plastic state provided that the following requirements are satisfied:

1 - Embedded items shall not have hook ends or tied to reinforcement inside concrete.
2 - Embedded items shall be positioned correctly while concrete in plastic stage.
3 - The concrete shall be well compacted around the embedded items.

6-9-7 Marking and Identification

1 - Each precast member shall be marked to identify its location and orientation in the structure and also date of manufacture.

2 - Identification marks shall be identical to the erection drawings.

6-9-8 Handling

1 - For design of precast elements, all forces and distortion resulting during curing, stripping, storage, transportation and erection shall be considered.

2 - Precast members shall be adequately supported during erection to ensure stability until permanent connections are cast.

6-9-9 Strength evaluation of precast members

1 - Precast members used with cast-in-place concrete shall be permitted to be tested in flexure as a precast element alone according to the following:

a - Test loads shall not be applied except when calculations indicate that the precast element alone will not be critical in compression or buckling.

b - The test load is that load which, when applied to the precast member alone, produces the same total force in the tension reinforcement as that produced by loading the composite member with the test load according to section (8-7-7).

2 - The precast member shall be considered acceptable if the requirements of section (8-9-6) shall be satisfied.

6-9-10 Horizontal shear strength of composite members

1 - This item apply to composite members defined as precast concrete part and in-situ concrete part cast in later stage but so inter-connected to act as one member responding to loads as a unit.

2 - Full transfer of horizontal shear forces shall be ensured at contact surfaces of interconnected elements.

3 - For design of horizontal shear forces, the following shall be satisfied:-

$$Q_u \le Q_{uhr} \qquad (6\text{-}64)$$

Where Q_u the factored is shearing force at the section under consideration and Q_{uhr} is the nominal horizontal shear strength.

3-a when contact surfaces are clean, free of laitance, and intentionally roughened with 5mm minimum full amplitude horizontal shear strength shall not be taken greater than $0.4\ b_v d$ (Newton) where b_v

(mm) is the contact width between precast and cast-in-situ parts, and d (mm) is the depth of the composite member.

3-b when minimum vertical ties between composite member parts are provided in accordance with section (4-2-2-1-6) , and contact surfaces are clear and free of laitance , but not intentionally roughened (as per 3-a) , horizontal shear strength shall not be taken greater than $0.45\ N/mm^2$

3-c when minimum vertical ties are provided between composite member parts in accordance with (4-2-2-1-6), and contact surfaces are clean, free of laitance, and intentionally roughened according to (3-a) shear strength shall be taken equal to $(1.35 + 0.5\mu_v \frac{fy}{\wp_s})$ in N but not greater than 3.0 N/mm^2 where μ_v in the reinforcement ratio for vertical ties.

3-d when factored horizontal shear strength exceeds 3.0 N/mm^2 , design for horizontal shear shall be in accordance with section (4-2-2-4).

4 - As an alternative to item 3 above, horizontal shear shall be permitted to be determined by computing the actual change in compressive or tensile force in any segment, and provisions shall be made to transfer that force as horizontal shear to supporting elements. The factored horizontal shear force shall not exceed horizontal shear strength as given in (3-a), (3-b), (3-c) and (3-d) where area of contact surface A_c shall be substituted for $b_v d$.

- When vertical ties are provided to resist horizontal shear, the distribution of these ties shall approximately reflect the distribution of shear forces in the member.

5 - When tension exists across the contact surface between interconnected elements, horizontal shear forces shall be transferred by vertical ties with area exceeding minimum area as specified in (3-b) above and in accordance with (4-2-2-1-6).

6-10 Mathematical modeling and validation of computer-aided structural models

6-10-1 Requirements of the mathematical model

It shall be permitted to use numerical methods, such as the finite element method, for the determination of internal forces in structures provided that these methods shall satisfy the equilibrium and strain-compatibility conditions, in addition to the following requirements:

6-10-1-1 Geometry requirements

1 - The mathematical model shall simulate the true behavior of the structure with respect to geometry, supports (size and stiffness of supporting elements), loads and end restraint/constraint conditions.
2 - When the finite element method is used, it shall be assured that the selected sizes and aspect ratios of these elements shall not have any adverse effects on solution accuracy.
3 - When numerical methods are used in the analysis, the mesh gridlines shall preferably pass by column locations. This may require the use of elements with variable sizes.
4 - Structure discretization at stress-concentration zones (e.g. at locations of columns, structural walls, openings, sudden changes of element sizes, and concentrated loads) shall be made in a manner that shall satisfy the requirements of section (6-10-2).

6-10-1-2 Structural requirements

1 - The transfer of various loads from one structure element to the other (load paths), starting from their place of application to the foundations, shall be checked.
2 - Secondary load effects, if any, shall be appropriately included in the model.
3 - When the floor beams are modeled as a two-way grid system, all produced internal forces,(i.e. bending moments, axial forces, shear forces and torsion moments), shall be considered in the design of these beams.
4 - Various, possible load patterns that may act on the structure shall be considered in the analysis to ensure that critical values for the internal forces are checked at every section.
5 - Generally, in the analysis of structures- and in particular for floor slabs, the flexural stiffness of columns in two planes shall be included in the mathematical model.
6 - The effects of section cracking may be accounted for in the structural analysis. However, for cases where including the effects of cracking shall be difficult, it shall be permitted to redistribute the internal forces obtained from analysis considering the anticipated effects of cracking on the structure behavior and the determination of steel reinforcement amounts and directions.

6-10-2 Review of input data and output results

6-10-2-1 Review of input data

1 - The input related to the structure geometry shall be checked with respect to:
   - Boundaries of the structure.
   - Section properties for different elements.
   - Axes (or planes) of symmetry.
   - Locations of supports.
   - Openings.

2 - Restraint and constraint conditions shall conform to the actual behavior of the structure. The structure model shall have sufficient restraints and/or constraints to satisfy its statical stability.

3 - Load data shall be verified with respect to load values, directions, points of application, and units of measure.

4 - Material properties shall be checked including the numerical values of Young's modulus, Poisson's ratio, and material strength, as well as interface (i.e. bond) strength between different materials.

6-10-2-2 Review of output results

1 - The general equilibrium of the structure shall be verified by comparing the total sum of applied loads with the corresponding support reactions.

2 - The structure deformed shape shall be checked in terms of overall mode of deformation and deflection direction.

3 - Deflection values at support locations shall be confirmed and program-calculated deflection values shall either be compared with values obtained using closed-form solutions or with simplified methods for few structure elements.

4 - It shall be ensured that the computer program adopted for analysis shall give results that shall practically be consistent to a reasonable degree with the results of traditional methods of analysis.

6-10-3 Slabs

In addition to the requirements of sections (6-10-1) and (6-10-2), the following provisions shall be considered:

1 - Slabs may be represented by general shell elements or plate-bending elements.

2 - In modeling of floors containing slabs and beams using shell or plate-bending elements for slabs while using frame elements for beams, the model shall account for the vertical shift between the levels of slab and beam centers of gravity. It shall be permitted to indirectly account for this shift by taking the beam cross section as a T- or L-shaped in the analysis; the flange protrusion (overhangs) shall be taken equal to half that given in section (6-3-1-9).

3 - When the steel reinforcement is uniformly-distributed and is placed in two perpendicular strips, the bending moment acting in each strip shall be given by:

$$\overline{m}_x = |m_x| + |m_{xy}| \quad (6\text{-}65)$$

$$\overline{m}_y = |m_y| + |m_{xy}| \quad (6\text{-}66)$$

where, $\overline{m}_x$ and $\overline{m}_y$ are the maximum bending moments, per unit strip width, acting in the x- and y- directions, respectively. Values of $\overline{m}_x$ and $\overline{m}_y$ need not exceed 1.5 times the mean bending moment in the strip. In Equations (6-65) and (6-66), $|m_x|$ and $|m_x|$ shall be the absolute values of bending moments, per unit strip width, while $|m_{xy}|$ shall be the absolute value of torsion moments, per unit strip width.

4 - The column in-plan restraining effects may be represented by a point at its center or over its full cross-section area. The bending moments for floor elements shall be taken as those acting at column faces in both cases.

5- It shall be permitted to position the main steel reinforcement in direction of the principal tensile stresses with a deviation tolerance of ±15o or, alternatively, the steel reinforcement may be placed in two perpendicular directions as stated above.

6-10-4 Rafts

In addition to the requirements of sections (6-10-1) and (6-10-2), the following provisions shall be considered:

1 - Column loads transferred to the raft may be distributed- in plan- over the column full cross-section area. The bending moments for raft shall be taken as those acting at column faces.

2 - The model adopted for soil modeling as well as the numerical values of its coefficients shall be conforming to the Egyptian Code for Soil Mechanics and Foundation (ECP202). Requirements concerning the soil bearing pressures and allowable settlements and differential settlements shall be satisfied.
3 - Rafts satisfying the conditions of section (6-7-2-2) shall be considered as rigid.
4 - In the analysis of soil-raft systems, the tensile stresses between raft and soil shall not be permitted. The bearing pressure shall not exceed the allowable bearing capacity of the soil.

6-10-5 Beams, columns, and frames

1 - In the analysis of columns, it shall be permitted to include the second order (P-Δ) effects but in no case shall the column design bending moments be taken less than that given in section (6-4).
2 - For structures analyzed as space frames, the design of sections shall be based on all internal forces produced at the section resulting from the same load conditions.

6-10-6 Deep beams, short cantilevers, and structural walls

It shall be permitted to use numerical methods to evaluate the stresses and strains in deep beams, short cantilevers, and structural walls provided that the obtained results shall not less than those given by other sections of this code.

6-11 Strut-and tie model

6-11-1 Introduction

Strut-and-Tie Model shall be used in designing D-regions of elements of concrete structures. Such regions results from concentrated loads, abrupt changes in dimensions or both as outlined in Section (6-11-2).

Strut-and-Tie Model follows the load paths in structural elements to model the internal forces as a truss in equilibrium under the effect of the external forces. Such a model contains elements in compression (struts), elements in tension (Ties) and connecting nodes (Nodes), as shown in Fig. (6-46). The designed shall be based

on the satisfaction of conditions of equilibrium and compatibility of strains given in Section (6-11-3).

6-11-2 Definitions

- Strut-and-Tie Model: Truss model consisting of a system of struts and ties connected together at regions called nodes to simulate the flow of forces at Bernouli regions (B-regions) and discontinuity regions (D-Regions), as shown in Fig. (6-46).
- B-Region: A part of a structural element in which Bernouli hypothesis (i.e. plane sections before bending remain plane after bending) can be applied. In general any part of the element located outside the discontinuity region shall be considered a Bernouli region.
- Discontinuity region (D-region): a part of the element in which an abrupt change in dimensions or loads occur. Such a region shall be determined based on St. Venant principle by a distance equals to the thickness of the element measured from the region of discontinuity (Fig. 6-47)
- Strut: It is the compression element in the Strut-and-Tie Model and modeled as the resultant of prismatic or a bottle-shaped field (Fig. 4-48). The strut can be either reinforced or un-reinforced.
- Tie: It is the tension element in the Strut-and-Tie Model and it models the tension field.
- Node: it is the point in the Strut-and-Tie Model at which the axes of the struts and ties meet (Fig. 6-53). The Nodal Zone is the concrete mass around the node where the forces of the Model equilibrate (Fig. 6-54).

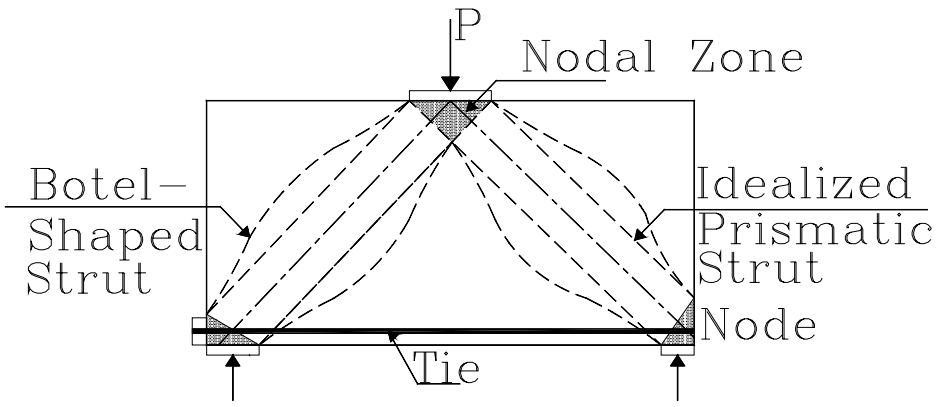


Fig. (6-46) Description of the strut-and-tie model

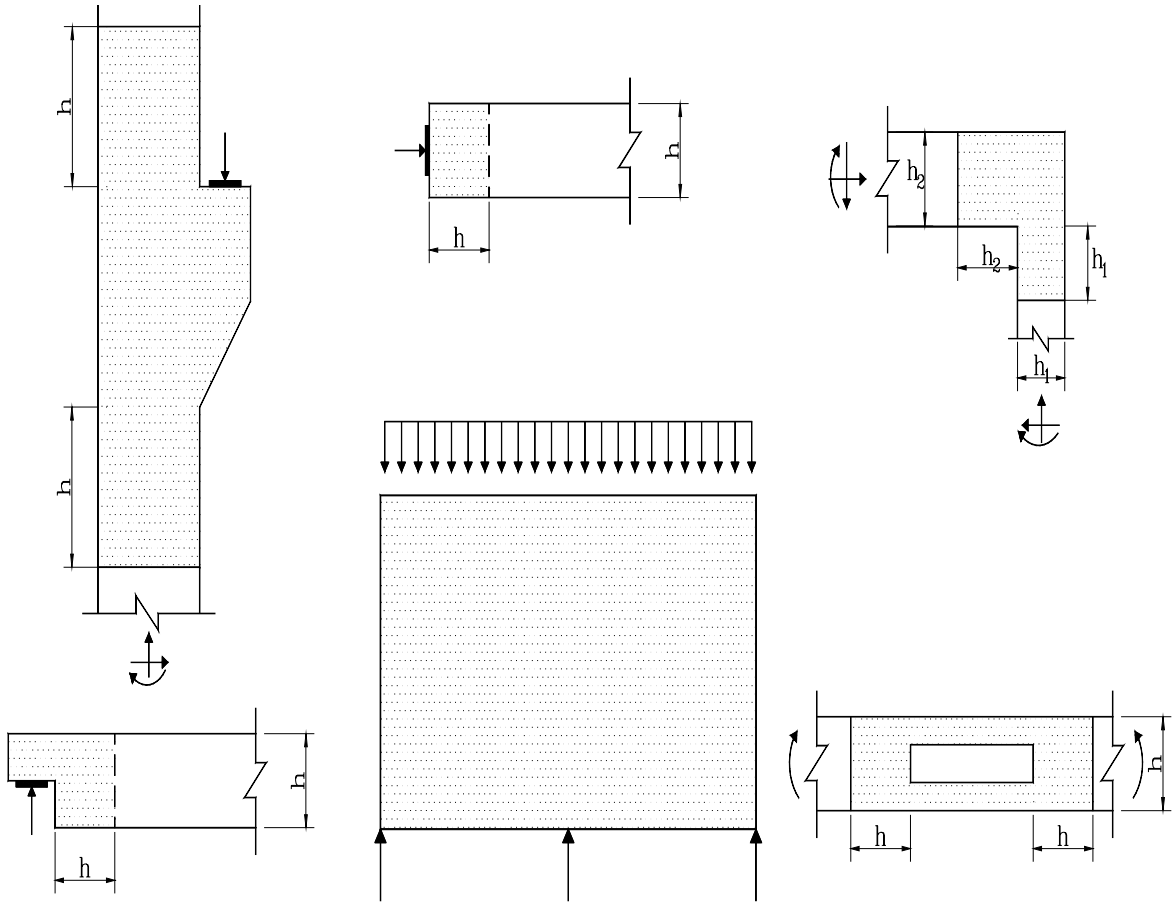


Fig. (6-47) Examples of D-regions

6-11-3 Design of the elements of the strut-and-tie model

6-11-3-1 General

- This section includes the design of the elements of the Strut-and-Tie Model according to the limit states design method.
- The elements of the Strut-and-Tie Model shall be designed to resist the acting ultimate forces and loads with due consideration of the requirements of the serviceability limit state.

6-11-3-2 Design of strut

6-11-3-2-1 Types of stress fields in struts

The value of the concrete strength in struts depends on the state of the surrounding stresses and their intersection with the cracks or the reinforcing steel according to Section (6-11-3-2-2).

According to the stress fields, the types of struts can be classified as follows;

Prismatic strut

Prismatic strut models a parallel path of stresses (Fig. 6-48-a) as for the case of the compression zone in beams subjected to bending moments. In such case, the thickness of the prismatic strut can be considered equal to the depth of the compression stress block (a) according to sections (4-2-1-10 and (4-2-1-2).

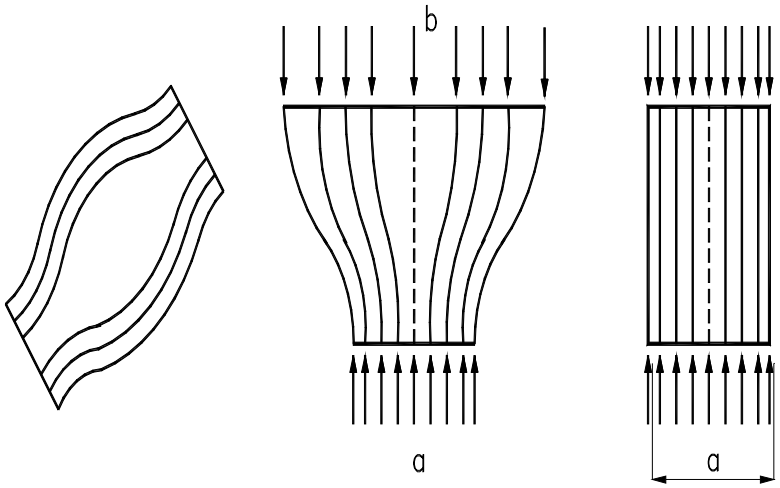


Fig. (6-48) Compression stress fields

Bottle strut

The bottle strut represents the general case of most of the struts in the Strut-and-Tie Model (Fig. 6-48-c). The internal lateral spread of compression forces leads to the development of lateral tensile stresses as shown in Fig. (6-49) . Such stresses shall be considered in design by providing reinforcement for controlling the developed of cracks, to increase the strength of the strut in the direction long its axis and to permit the redistribution of forces.

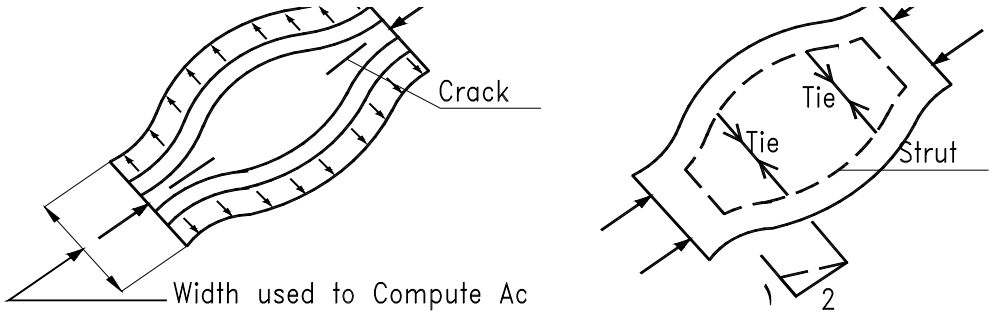


Fig. (6-49) Bottle-shaped strut

6-11-3-2-2 Ultimate strength of the strut

a - The ultimate strength of an un-reinforced strut at any end shall not be more than the following value:

$$Fc = fcd \cdot Ac \qquad (6\text{-}67)$$

Where:

F_{cd} = the smaller value of the effective concrete compression strength in the strut given by Equation (6-96) and the effective concrete compression strength in the node given by Equation 6-72).

A_c = Cross sectional area of the strut at one end. It is equal to (Ws. B), where;

b - is the beam width

Ws - is the strut width . It is equal to the smallest dimension normal to the axis of the strut at the node.

b - The ultimate strength of the strut can be increased through adding reinforcing steel in a direction parallel to the axis of the strut. Such reinforcing steel shall have sufficient development length and shall be tied with stirrups satisfying the requirements of Section (6-4-7). The ultimate strength of a reinforced strut shall be calculated according to the following equation:

$$F_c = f_{cd}\, A_c + A_s \frac{f_y}{\gamma_s} \qquad (6\text{-}68)$$

Where the value of γ_s shall be taken equal to 1.3.

c - The effective concrete compression strength shall be taken according to the following equation:

$$f_{cd} = 0.67\, \beta_S \frac{f_{cu}}{\gamma_c} \qquad (6\text{-}69)$$

Where the value of γ_c is taken equal to 1.6. The value of β_s shall be taken as follows:

β_s = 1.0 for prismatic strut as given in Section (6-11-3-2-1)

β_s = 0.70 for bottle-shape strut that is parallel to the cracks shown in Fig. (6-50-a). The value of β_s given in this section requires the use of reinforcing steel in a direction normal to the direction of the axis of the strut in order to resist the transverse tension force resulting from the spread of the compression

force in the strut with an inclination to axis of the strut of (2) longitudinal to (1) transversal.

β_s = 0.6 for bottle-shape strut that is inclined to the angle of the cracks as shown in Fig. (6-50-b).

β_s = 0.4 for struts in tension elements or elements with tension flanges.

β_s = 0.60 for all other cases.

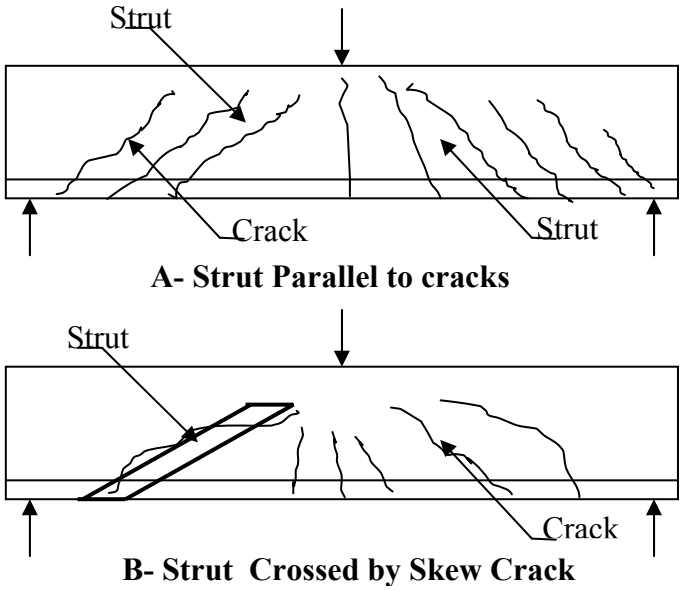


Fig. (6-50) Types of struts: (a) strut parallel to the cracks, (b) strut inclined to the cracks

d - In case of using confining stirrups, the concrete compression strength can be increased by taking the confining effect into consideration similar to the end zones of prestressed concrete members.

e - The angle of inclination of the compression strut to the axis of the structural element shall not be less than 26 deg.

6-11-3-3 Design of ties

- The ultimate strength of the tie shall be calculated from the following equation:

$$f_{cd} = 0.67\ \beta_S\ \frac{f_{cu}}{\gamma_c} \qquad (6\text{-}69)$$

Where:

T_{ud} =Design tension force for the case of ultimate limit state

A_s = cross sectional area of reinforcing bars.

The value of γ_s shall be taken equal to 1.5.

- The requirements of section (4-3-2) for the cracking limit for members subjected to axial tension forces shall be satisfied.
- At the ends of a tension tie, the satisfaction of the required development length (L_d) (Fig. 6-51) shall be verified. Other options are to use anchors or mechanical connections at the ends of Section (4-2-5).

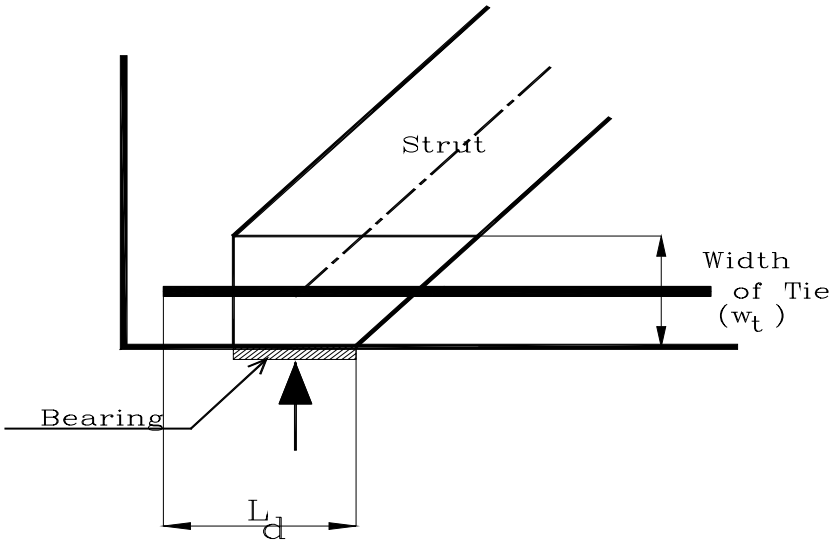


Fig. (6-51) Calculation of required development length at the nodal zone

6-11-3-4 Design of nodes

- The width of the reinforced tie (Wt) shall be designed to satisfy the safety requirements of compression stresses at the nodal zone for struts and ties meeting at the node (Section 6-11-3-4). An approximate tie thickness can be taken not more than 70% of the thickness of the largest strut connected to that tie at the nodal zone.
- The development length (L_d) shall be calculated according to section 4-2-5.

The node is defined as the region where the forces of the Strut-and-Tie Model meet. At least three forces shall be in equilibrium at the node.

6-11-3-4-1 Types of nodes

Nodes are classified as Singular and Smeared. Singular nodes are those having well-defined locations at the locations of concentrated loads such as nodes (I, III) in fig. (6-52). Smeared nodes are those having

locations determined according to the equilibrium of forces in the Strut-and-Tie Model, such as node (II) in Fig. (6-52). Calculation of stresses in smeared nodes can be neglected since their locations are hypothetical.

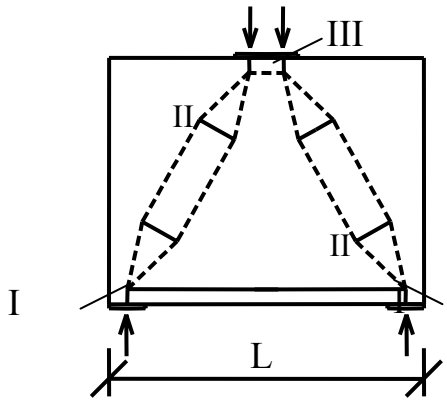


Fig. (6-52) Types of Nodes

6-11-3-4-2 Design of singular nodes

- Singular nodes shall be designed to satisfy the requirements of ultimate compressive strength of concrete, as well as satisfying the required development length of reinforcing steel in tension.
- Singular nodes are classified according to the type of forces in equilibrium at the node (Fig. 6-53) as follows:
  - Node C-C-C all members meeting at the node are struts.
  - Node C-C-T members meeting at the node are two struts and a tie
  - Node C-T-T members meeting at the node are two ties and a strut
  - Node T-T-T all members meeting at the node are ties.
- In order to satisfy the safety requirements for ultimate limit state, the compression strength of concrete at the node shall conform to the following equation:

$$F_{cn} = A_{cn} .f_{cd} \qquad (6\text{-}71)$$

Where:

A_{cn} = cross sectional area at the nodal zone for a section normal to the direction of the strut (Fig. 6-54).

f_{cd} = Effective compressive strength of concrete at the node calculated from the following equation:

$$f_{cd} = 0.67\ \beta_n \times \frac{f_{cu}}{\gamma_c} \qquad (6\text{-}72)$$

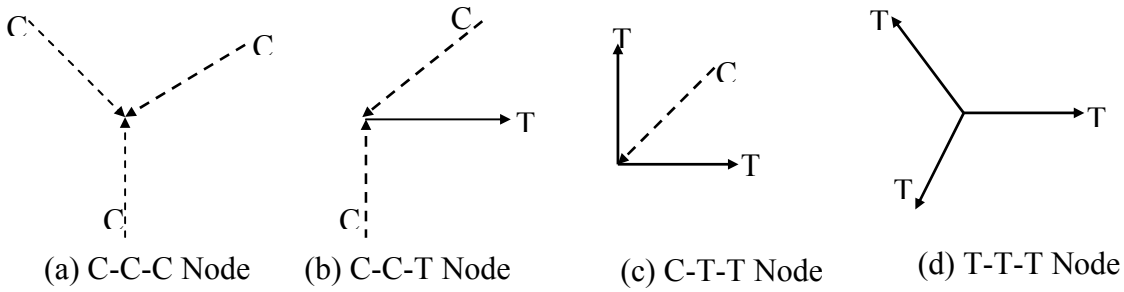
Where:

γ_c = Strength reduction factor of concrete and is taken equal to 1.6

f_{cu} = Characteristic compressive strength of concrete

β_n = A coefficient that takes into account the type of the forces acting at the node. It is taken as follows:

- β_n = 1.0 for nodes subjected to compression forces only (C-C-C), (Fig. 6-53-a). In such a case, the node region is subjected to compression acting in one- or two- or three-directions.
- β_n = 0.8 for (C-C-T) nodes, (Fig. 6-53-b). The use of β_n = 1.0 in this type of nodes shall be permitted if the tie shall extended through the node and mechanically connected as shown in Fig. (6-54).
- β_n = 0.6 for (C-T-T) or (T-T-T) nodes shown in Figs. (6-53-c and 6-53-d)



(a) C-C-C Node (b) C-C-T Node (c) C-T-T Node (d) T-T-T Node

Fig. (6-53) Types of singular nodes

- In case of singular nodes of type (C-C-T), the height Wt shall be calculated according to the distribution of reinforcing bars as follows (Fig. 6-54):
  - When using one row of reinforcing bars, and not achieving sufficient development length behind the node region (Fig.6-54-a):

$$W_t = 0 \qquad (6\text{-}73a)$$

- When using one row of reinforcing bars, and achieving sufficient development length behind the node region by a distance not less than 2c, where c is the concrete cover (Fig. 6-54-b):

$$W_t = \varphi_s + 2\,c \qquad (6\text{-}73b)$$

Where ϕ_s is the diameter of the used bars

- When using more than one row of reinforcing bars (Fig. 6-54-c):

$$W_t = \varphi_s + 2\,c + (n-1).s \qquad (6\text{-}73c)$$

Where, n is the number of rows and, s is the distance between the reinforcing bars with due consideration that the distance of the extended node ≥ (s/2)

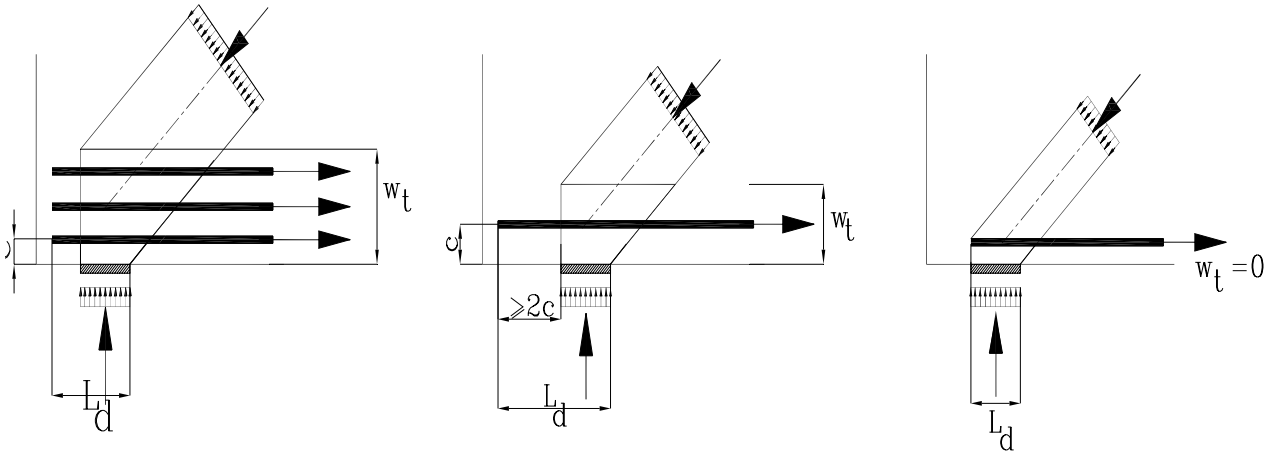


Fig. (6-54) Typical node of type (C-C-T) subjected to compression and tension at the support of a shallow or deep beam

CHAPTER 7

DETAILS OF REINFORCEMENT

7-1 General

Concrete workshop drawings shall be fully detailed and with complete dimensions. It shall also be prepared according to structural calculations and in such a way that makes the formworks and concrete casting as easy as possible.

7-2 Structural drawings and drawing specifications

Structural drawings shall be prepared according to the design made by specialist engineers and approved by the syndicate of engineers- either by the design office or by the contractor upon an assignment from the design office and approved by it - to include all the details necessary to carry out the project according to the guidelines explained in this chapter.

7-2-1 Scheme drawings

Scheme drawings are made according to the schematic drawings and requirements of the project. The purpose of these drawings is to show the column layout and to estimate approximate dimensions of structural members so that the architect can prepare the final drawings of the project. Scheme drawings are usually submitted in a scale of 1:100.

7-2-2 Tender and design drawings

Tender drawings are prepared in a suitable scale and all structural elements shall be shown clearly in a way that allows tendering contractors to estimate the quantities of concrete, formworks and reinforcing steel. Tender drawings shall include the following data:

7-2-2-1 Loads

Live loads and additional loads on each part of the building shall be shown as well as the dynamic effects of machines and apparatuses, if any. Flooring, cladding loads and allowable formwork loads shall also be shown. In case of using special forms, approval of design engineer shall be obtained.

In special structures like factories, power plants, water and sewage treatment plants, and storage silos … etc, where other loads exist, the

values of such loads shall be shown on drawings of concrete dimensions, or reference shall be made to mechanical drawings showing such loads.

7-2-2-2 Properties of materials

These data shall include characteristic strength of concrete for building members. Type of cement, and minimum cement content, shall also be specified. The type of admixtures and steel reinforcement as well as steel grade shall be specified. Different types of steel are as follows:

- Plain mild steel (Grade 240/350) indicated by ϕ.
- High tensile steel (Grade 360/520) indicated by $\Phi\!\!\!\!+$.
- High tensile steel (Grade 400/600) indicated by Φ.
- Welded wire mesh steel (Grade 450/520) indicated by # .
- Concrete cover of reinforcing steel, as determined by articles (4-3-2-3-b) and (9-7), shall be mentioned in drawings, for all elements.

7-2-2-3 Foundations data

Foundation drawings shall include foundation level, allowable stress on soil, and types of piles (if any) and their working loads. Locations and specifications of water insulation layers (if any) shall be shown. Number of floors for which the design is made shall also be indicated on the drawings.

7-2-2-4 Precast concrete

In case of using precast concrete the requirements of art (6-8) shall be fulfilled together with indicating the following data on drawings:

a) Minimum characteristic compressive strength of concrete before removal of forms and at time of transferring precast units from casting yards to storage or erection sites.

b) Positions of hanging precast units and details of additional reinforcement at these positions. Storage method shall also be determined to avoid impermissible stresses at any section due to lifting and storage.

c) Weight of each member to arrange for the suitable equipment for lifting, transfer, storage and erection.

d) Sufficient details, in a scale not less than 1:20, at all joints between precast units. Positions and methods of supporting these units till hardening of mortar or joint filling material shall also be determined.

7-2-3 Workshop drawings

Workshop drawings shall include the necessary details for construction of all structural members of the building. Drawings shall be made with a suitable scale, preferably not less than 1:50. Drawings shall include the following:

a - Concrete dimensions data

1) Plans and sufficient sections to show the concrete dimensions of all structural elements, spacing between axes, levels, slab thickness. Dimensions of beams, cantilevers and columns shall also be shown. In case of beams and cantilevers, width is mentioned first followed by total depth including thickness of slab.

2) Locations and details of openings, anchorage bolts and embedded parts required for sanitary works, air conditioning, machine installation … etc.

3) Locations and details of expansion, shrinkage, seismic and construction joints (if any), as well as camber of slabs, beams and cantilevers in structures with large spans.

b - Reinforcing steel data

Reinforcing steel data shall be correlated to concrete dimensions to make construction easy. The following shall also be considered:

1 - Showing the layer arrangement in case of using steel mesh like the one used in slab and wall reinforcement.

2 - Showing the bent and straight bars for slabs on plan with their true shape. Only one bar of each group of slab reinforcement may be drawn on plan with the number of bars either per unit length, or total number in each slab with spacing between bars. Where cantilevers exist, their cross section with the connected structural elements, with suitable scale, shall be drawn .

3 - When beam reinforcement details are required, they shall be drawn on elevations with a scale not less than 1: 50. Reinforcement shall be drawn in continuous lines giving sufficient cross-sections in each

beam with suitable scale. It is preferable to show steel detailing outside the beams and write the length of each bar.

4 - Sections of column showing reinforcement details shall be drawn to a suitable scale. An elevation shall be drawn when beam and column reinforcements are connected. In this case it is better to draw reinforcement, to a suitable scale, separately outside concrete limits in the elevation. The same shall be made for the case of changing column shape along its height. In all cases, any requirements for locations of splices, anchorage length, bending of lower bars at joints shall be shown on drawing in a way that permits continuity of steel reinforcement in its place along the total height of column.

7-2-4 Detail drawings

In some cases it is necessary to make detail drawings with a scale that suits the accuracy required in construction. For example:

1 - Some connections in concrete structures where reinforcement intensity is high. In such cases the intersections are drawn to explain bar arrangement and ensure the existence of spacing between bars sufficient for casting and compacting concrete.

2 - Making bar lists including bar details, lengths, and numbering to facilitate placing of bars in forms.

3 - In some cases, it is necessary to make detail drawings for wooden or steel forms and scaffolds to ensure the accuracy of construction. When forms and scaffolds are designed, its ability to resist applied loading and fresh concrete pressure at all casting stages shall be considered.

4 - In some special cases, it is necessary to determine the expected deflection of concrete members when forms and scaffolds are removed and to consider them in constructing the forms and scaffolds so that the operation of their removal and reconstruction becomes easier.

5 - In case of constructing mechanical and electrical equipment foundations where high accuracy is needed in determining the locations of anchorage bolts, detail drawings for methods of installing bolts in place, inside forms or with reinforcing steel shall be made.

7-2-5 Title and drawing information table

Title table shall be placed at a location where it appears on the face after folding the drawing paper. It shall include the following information:

- Project name, owner's name and address.
- Design office name and address including the name of engineer of record of the project.
- Name of party or authority responsible for reviewing project (if necessary).
- Contractor's name, if the drawing is prepared by him.
- Drawing scale(s).
- Date of issuance of drawing
- Drawing number and Title.
- Modifications, their dates and summary of modifications showing their locations on drawing. Project engineer shall keep a copy of each drawing before and after modification to refer to it when necessary.
- Issuance number of modified and coordinated drawings.

7-3 Special arrangement for reinforcing steel

Details of reinforcement are necessary and important to ensure perfect concrete works; taking into consideration that these details shall be sufficient to produce bending lists of reinforcement for all parts of the project.

7-3-1 Use of different types of reinforcement in the same structural element

a - It is preferred to avoid using different types or grades of steel in the same structural member to avoid mistakes in arranging reinforcement which may lead to structural risks.

b - It shall be allowed to use two different types of reinforcing steel in the same structural member if each of them resists stresses of different type or direction; such as for the cases when using one type of steel for main reinforcement and another type for secondary reinforcement of slabs when using one type of steel for longitudinal reinforcement in columns and beams and another type for stirrups.

7-3-2 Stopping of bar ends, development length and splices

The ends of reinforcing steel bars shall be anchored in concrete using one of the following methods:

a - Straight end bars.

b - Hooked ends like ⊂ or ⊃, or right angled like ⌋ as shown in Table (4-7) and section (4-2-5-1), or loops ⊃.

c - Using transverse bars or steel plates welded at ends of bars to be fixed in concrete. It shall not be permitted to end large ratio of bars at the same concrete section in order to avoid stress concentration in the section. It shall always be preferable to use large number of bars with smaller diameter in order to be able to ended bars at different locations..

d - Development length and splice and welding length shall be calculated according to section (4-2-5).

7-3-2-1 Lap splices

Location and number of bars in each splice, and spacing between bars shall be determined according to section (4-2-5-4-2).

7-3-2-2 Mechanical splices

a - Mechanical splices shall be used for bars with diameter not less than 16 mm and shall be made using steel couplers made from a steel with material properties at least equal to these of the coupled bars. The tensile strength of coupler section shall not be less than 125% of the strength the steel bars.

b - It shall not be allowed to have a slippage within the coupler more than 0.1 mm at working loads.

c - The couplers shall be made according to one of the following two methods:
I- By threading bar ends from outside and coupler from inside as shown in Fig (7-1-a).
II- By using couplers to be pressed along its outer surface area with bar ends inside using special jacks so that the stresses can be transferred between bars by friction between the internal surface of coupler and external surface of bars, as shown in Fig (7-1-b). This method shall be used for deformed steel bars

d - It is essential when using mechanical splices to make sufficient tests on specimens to verify the ability of the splice in resisting working stresses and the fulfillment of the preceding requirements and those given in section.(4-2-5-4-3). These tests shall be specified by the engineer of record of the project.

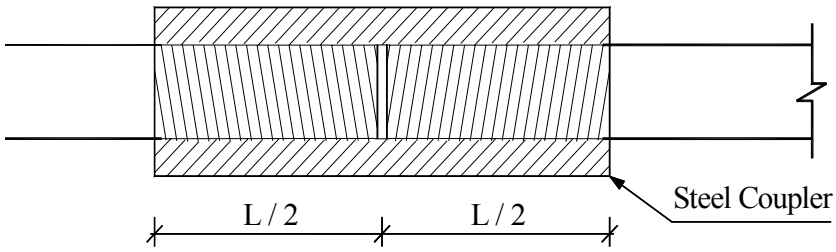


Figure (7-1-a) Mechanical splices using threaded coupler

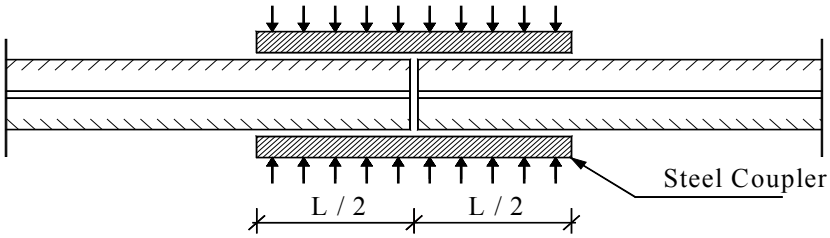


Figure (7-1-b) Mechanical splices for deformed bars

7-3-2-3 Welded splices

Welded splices shall be used for bars with diameter not less than 16 mm and for weldable steel as given in section (4-2-5) and according to section (4-2-5-4-3) and the details shown in Figs (7-1-c) and (7-1-d). Welding process shall be made according to the following:

1 - Only electric welding shall be used.

2 - Axes of welded bars shall be aligned in the same line.

3 - Welded splices shall be made staggered with not more than 25% of the total number of bars welded at the same location. The spacing of the welds of the rest of the bars shall be spaced not less than 20 times bar diameter.

4 - Length and thickness of weld shall be determined based on the ultimate tensile force of the welded bars.

5 - It is preferred not to make welded splices in the region of maximum bending moment.

6 - All welding shall be performed by certified technicians

7 - It shall be essential, when using welded splices, to make sufficient tests on specimens to ensure their ability to resist working stresses and fulfill the preceding requirements.

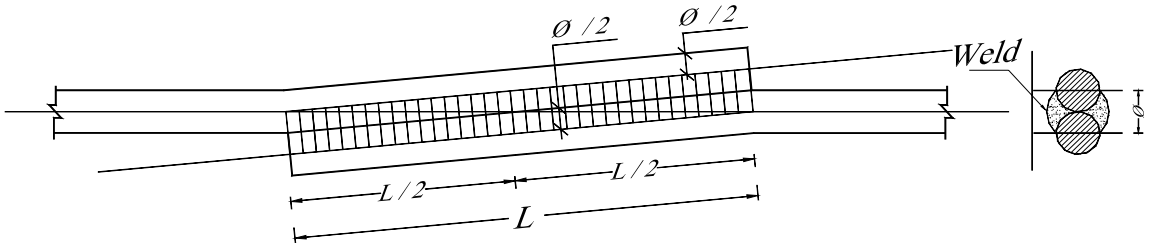


Figure (7-1-c) Details of welded lap splices

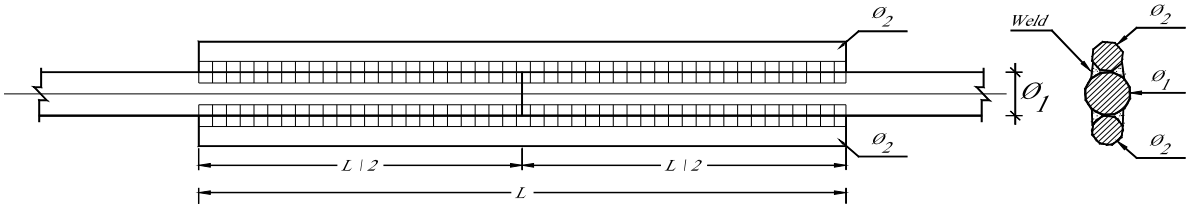


Figure (7-1-d) Details of welded splices using additional bars

7-3-3 Minimum and maximum bar spacing

7-3-3-1 Minimum bar spacing

The spacing between reinforcing bars shall be sufficient for casting and compacting concrete; using either manual or mechanical compaction. Figure (7-2-a) shows the minimum spacing between individual bars, and Fig (7-2-b) shows minimum spacing between bundled bars

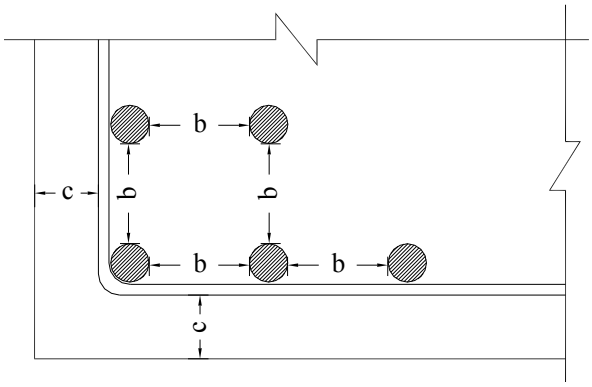


Figure (7-2-a) Minimum spacing between individual bars

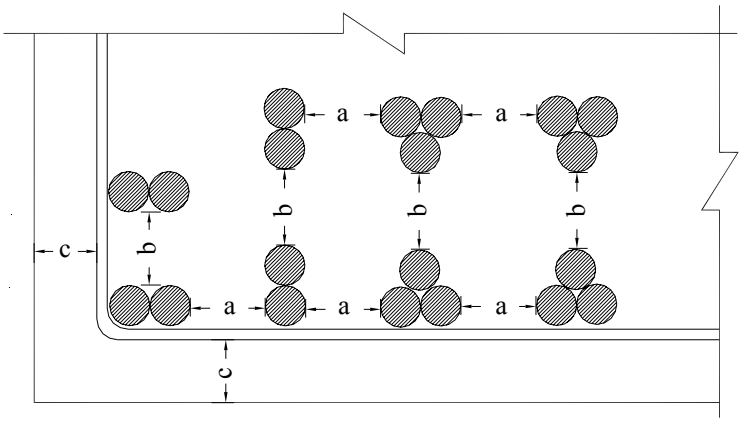


Figure (7-2-b) Minimum spacing between bundled bars

Where,

a = Concrete cover of bars determined according to the values given in Table (4-13) and section (4-3-2-3-b) with due consideration of section (9-7).

b = the largest bar diameter ϕ_{max} or 1.5 times the maximum nominal size of concrete aggregate, whichever is larger.

c = the largest bar diameter ϕ_{max} or 1.5 times the maximum nominal size of concrete aggregate or (maximum nominal aggregate size+15 mm) whichever is larger.

7-3-3-2 Maximum bar spacing

The following sections shall be referred to when determining the maximum bar spacing:

- Section (6-2-1-4) and section (6-2-7-8) for solid and flat slabs, respectively.
- Section (6-3-1-10) for beams.
- Section (6-4-7) for columns.
- Section (6-5-2-2) for reinforced concrete walls

7-3-4 Bundled bars

7-3-4-1 General

For structural elements that have high steel reinforcement ratio, steel bars can be gathered in bundles of two or three bars subject to fulfilling the following:

a - Bundled bars shall be allowed only when deformed bars are used.

b - Maximum bar diameter used in bundle shall be 28 mm.

c - Different bar diameters are allowed in the same bundle where the differences of bar diameters are not more than 4 mm.

c - Sufficient measures shall be made to keep contact between bars in the bundle during steel arrangement and concrete casting. This shall usually be done by using steel wires of suitable diameter at distances not more than 20 times the smallest diameter of the bars in the bundle.

7-3-4-2 Lap splices and stopping locations of bundled bars

a - Development length (L_d) and lengths of lap splices are calculated according to sections (4-2-5-1), (4-2-5-3) and (4-2-5-4). The correction factors shown in Table (4-8) for bundles shall be applied when calculating development and splice lengths.

b - It shall be allowed to end all bars in the bundle at the same section if the bundle equivalent diameter (ϕe) is not more than 28 mm.
Bundle equivalent diameter (ϕ_e) is calculated as follows:

- For two-bar bundle $\phi_e = 1.50\ \phi$
- For three-bar bundle $\phi_e = 1.75\ \phi$

Where ϕ is the largest diameter in the bundle?

c- In cases where equivalent bar diameter of the bundle is more than 28 mm, bundle bars shall be ended as shown in Fig (7-3-a) for the cases where no overlap occurs in the theoretical locations of influence region of the bundle, or as shown in Fig (7-3-b) for the cases where overlap occurs in the theoretical locations of influence region of the bundle which is indicated in figures by letter x.

d - In case of lap splices, where bars are staggered having additional bar as shown in Fig (7-3-c), the value of L_d shall be calculated according to section (4-2-5-4-2-g).

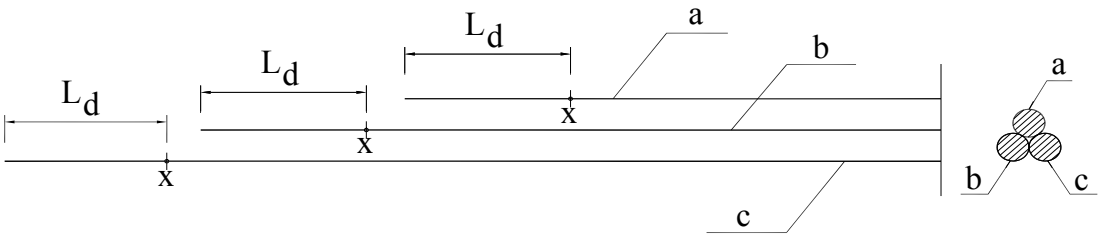


Figure (7-3-a): Arrangement of ending bundled bars (in case where no overlap occurs in the theoretical locations of Influence region of the bundle)

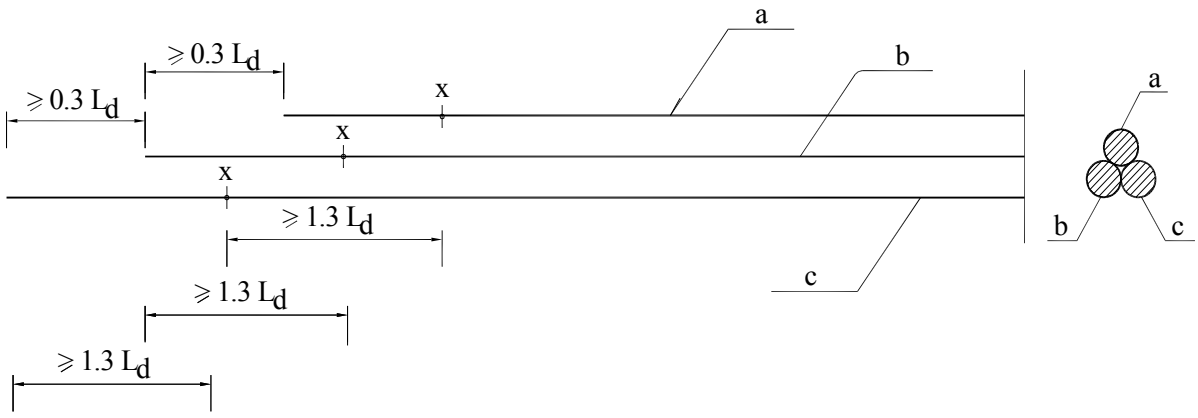


Figure (7-3-b) Arrangement of ending bundled bars (In case where overlap occurs in the theoretical locations of influence region of the bundle)

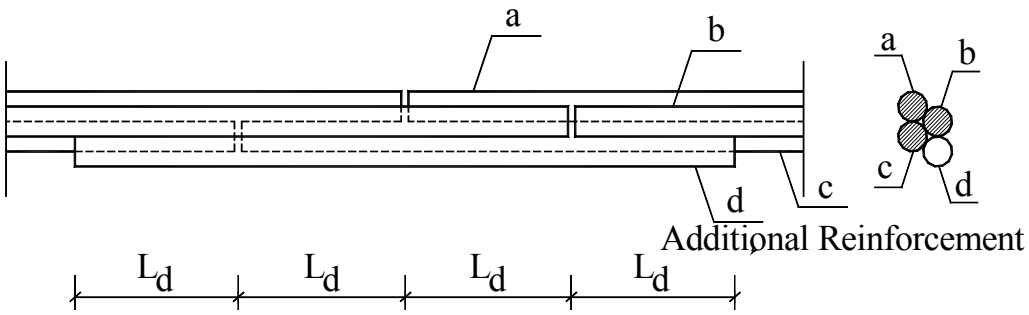


Figure (7-3-c) Arrangement of lap spliced bundled bars

7-4 Joints in concrete

7-4-1 Construction joints

These are the joints used for casting concrete on stages suitable to the capability of producing and casting concrete in the site. Locations of these joints shall be determined by the design engineer or contractor. Locations shall be chosen where minimum stresses, particularly shear stresses, exist. Measures stated in section (9-5-6) shall also be followed.

7-4-2 Shrinkage joints

These joints are made to avoid cracks due to concrete shrinkage in large areas such as walls, floors of water tanks and basements. In such cases, concrete shall be cast in distant parts or a shrinkage strip shall be left without concrete between the cast parts. It is preferable to provide keys in the sides of concrete. After hardening and curing of cast parts, the parts left between consecutive parts, (the shrinkage strips) shall be cast with die consideration of the measures specified for construction joints and given in section (7-4-1).

7-4-3 Movement joints

These joints are made to allow for any volumetric changes in concrete due to change in temperature or concrete shrinkage or vertical movement due to difference in values of loading in parts of the same building or due to change in foundation type.

These joints shall allow the movement of various parts of the structure and avoid unfavorable deformations or stresses resulting from restraining the movement.

Special care shall be taken when constructing these joints in order to avoid leakage of sub-soil water during the differential movement of parts of the structure. Locations of such joints are chosen by the design engineer of record and indicated on the detail drawings and specifications of the project. Measures stated in section (9-5-8) shall also be followed when making such joints.

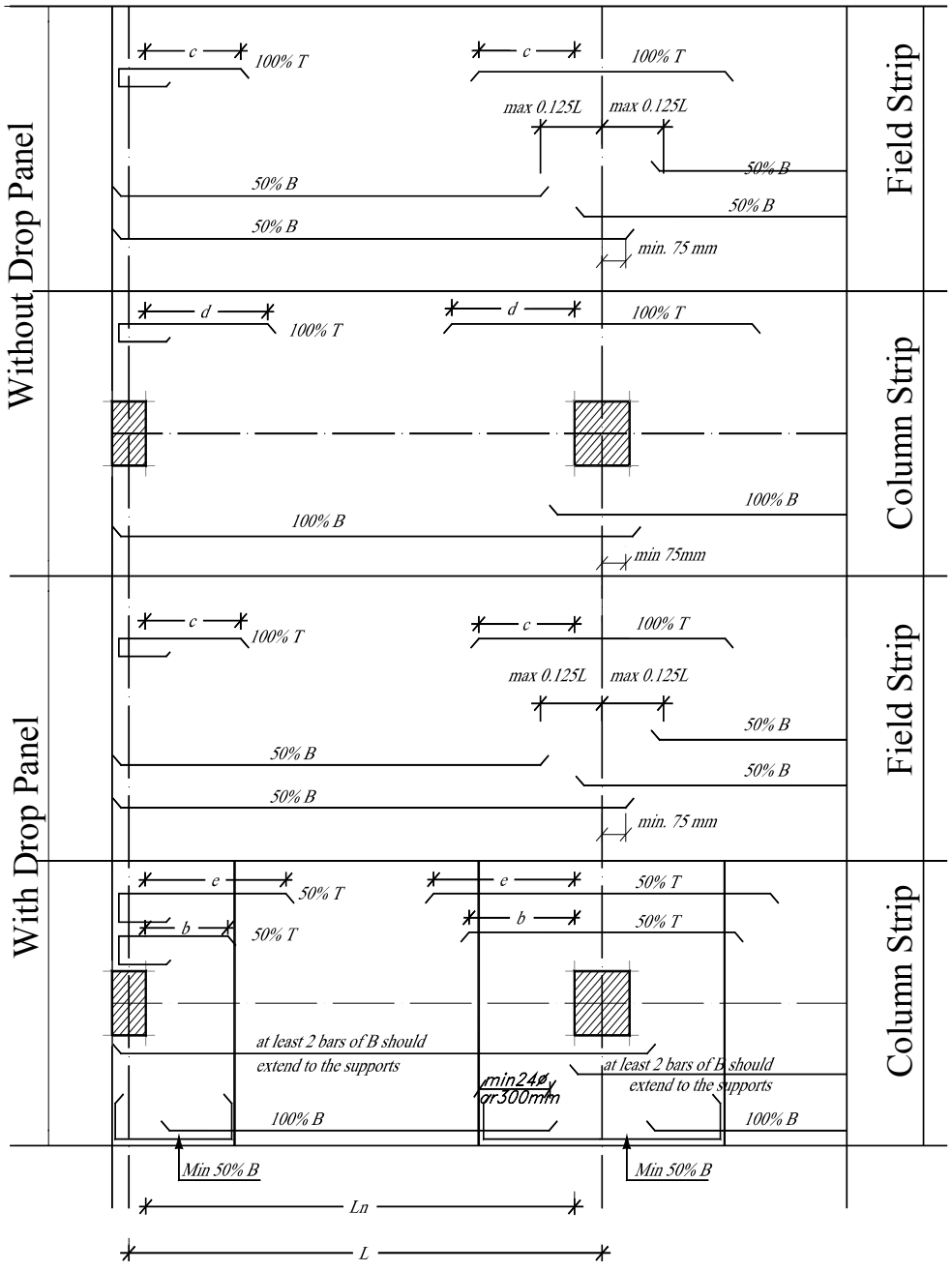
7-5 Typical details of reinforcement for structural members

Typical details of reinforcement for structural elements are presented, as follows:

1 - Typical details of reinforcement of flat slabs are shown in Fig (7-4). Figure (6-29) shall be referred to for the case of flat slabs resisting seismic loads.

2 - Typical details of reinforcement of beam and slab joints are shown in Fig (7-5)

3 - Typical details of column reinforcement are shown in Fig (7-6)



Minimum Distances			
b	c	d	e
0.20 L_n	0.22 L_n	0.30 L_n	0.33 L_n

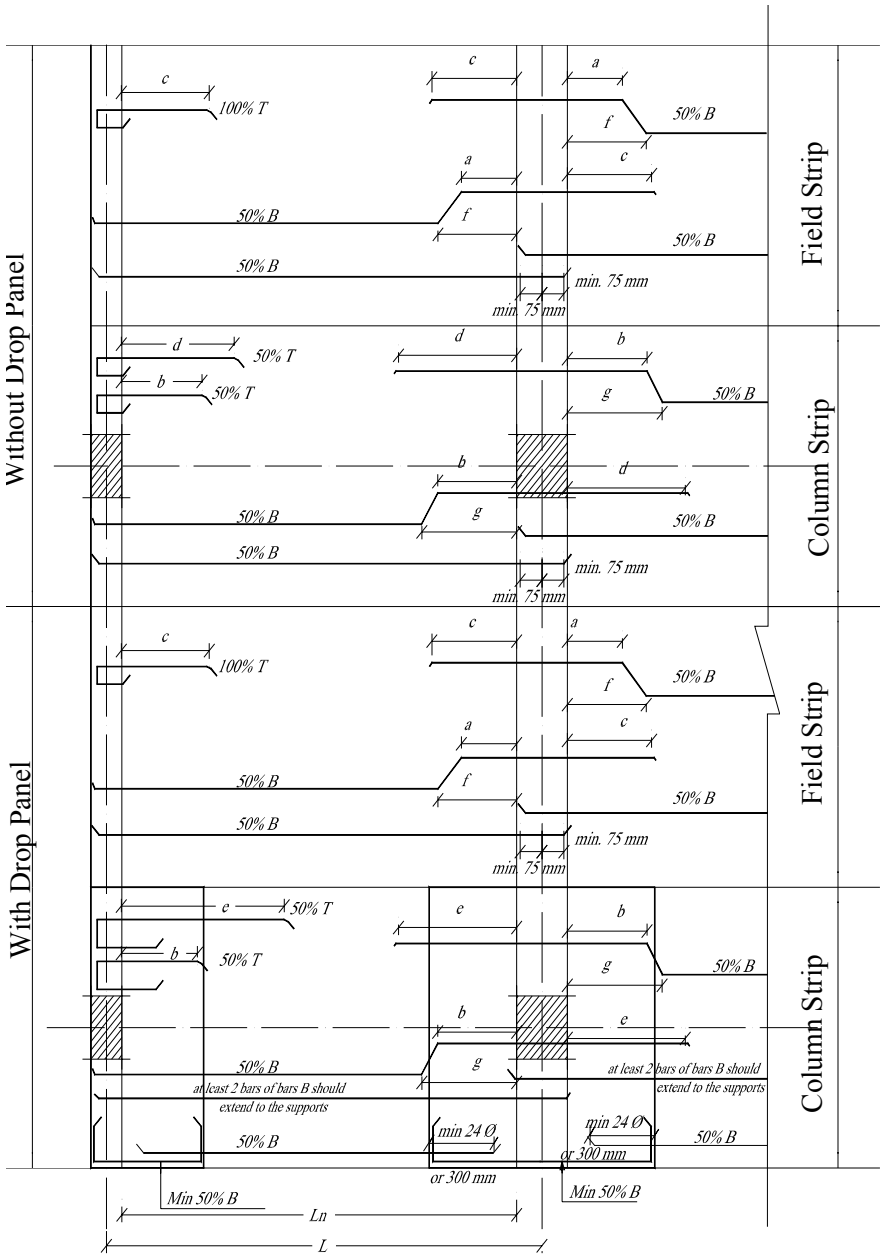
B: Bottom Reinforcement

T: Top Reinforcement

L: Distance between Axes of Columns

Ln : Clear Distance between Axes of Columns

Figure (7-4-a) Typical details of reinforcement of flat slabs

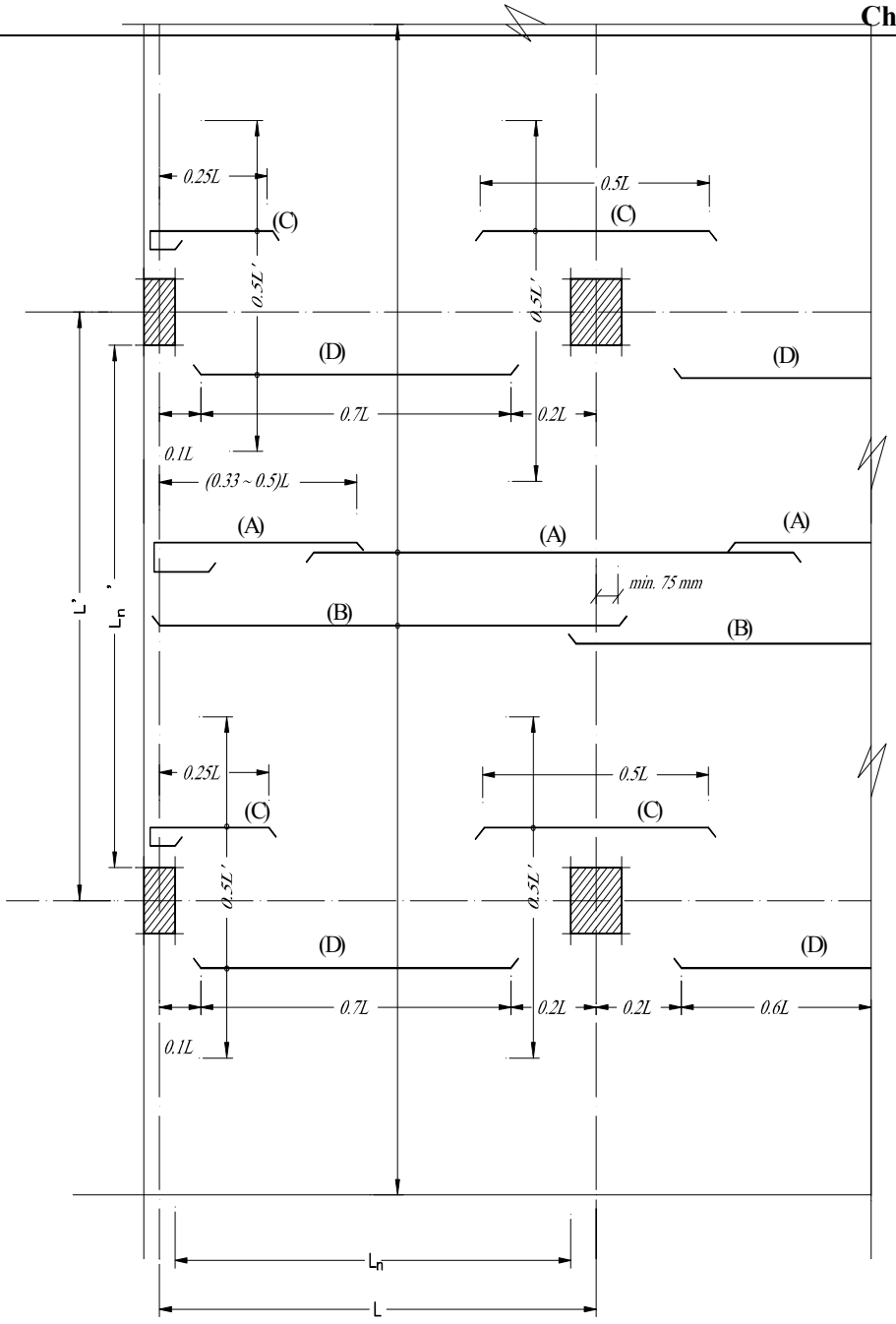


	Length of Bars From Face of Column						
	Minimum Length					Maximum Length	
الرمز	a	b	c	d	e	f	g
الطول	0.14 L_n	0.20 L_n	0.22 L_n	0.30 L_n	0.33 L_n	0.20 L_n	0.24 L_n

* Minimum Length of Reinforcement for Flat Slabs

* Top and Bottom Reinforcement of End Panels Shall be Extended to

Figure (7-4-b) Alternative reinforcement details of flat slabs using Bent-up Steel Reinforcement

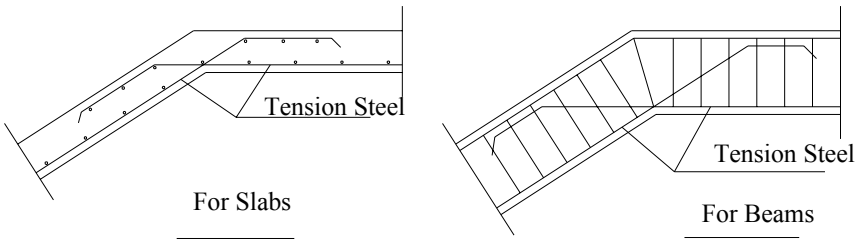


A	Top Reinforcement Mesh
B	Bottom Reinforcement Mesh
C	Additional Top Column Strip RFT
D	Additional Bottom Col. Strip RFT

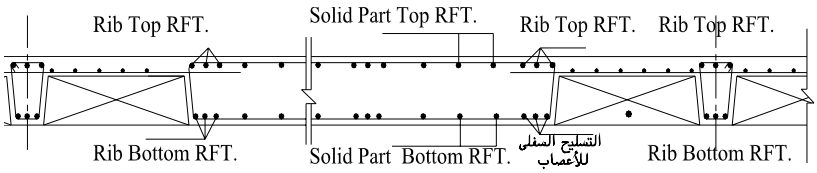
L or L':Distance Between Axes of Columns

Ln or Ln': Clear Distance Between Axes of Columns

Figure (7-4-c) Alternative reinforcement details of flat slabs using steel meshes combined with additional reinforcing bars

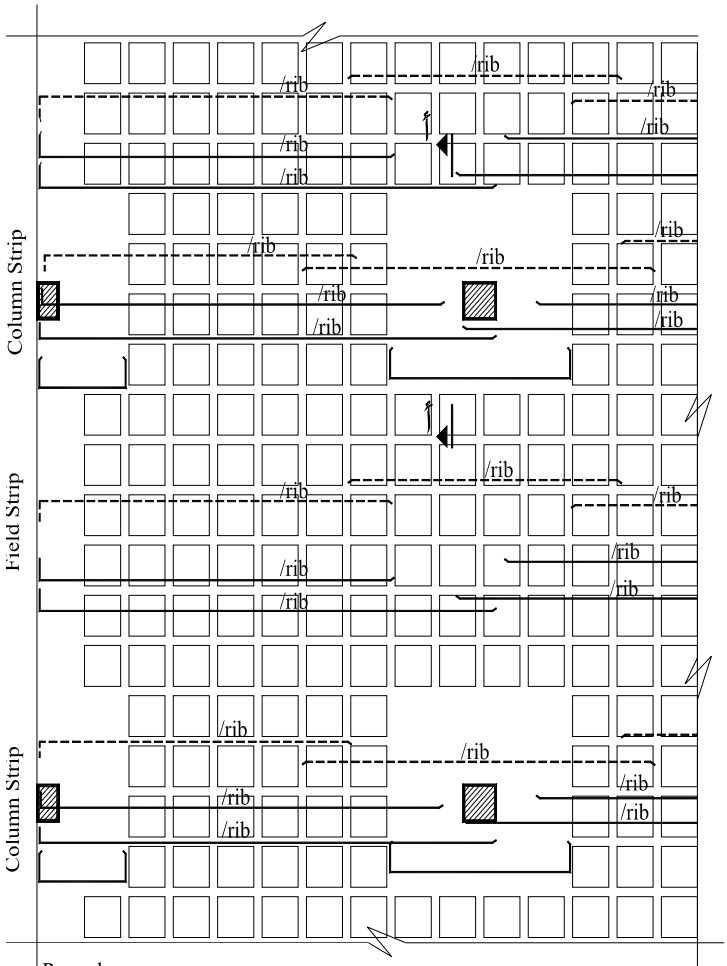


Figure(7-5) Details of Reinforcement of Beam and Slab Corner



Typical Details of Reinforcement of Waffle Slabs

Section (A-A)



Remarks:

Details of Reinforcement shall follow those of Flat Slabs

Solid, and Dashed Lines Denote Bottom and Top Reinforcement, Respictively

Figure (7-6) Details Of Reinforcement Of Waffle Slabs

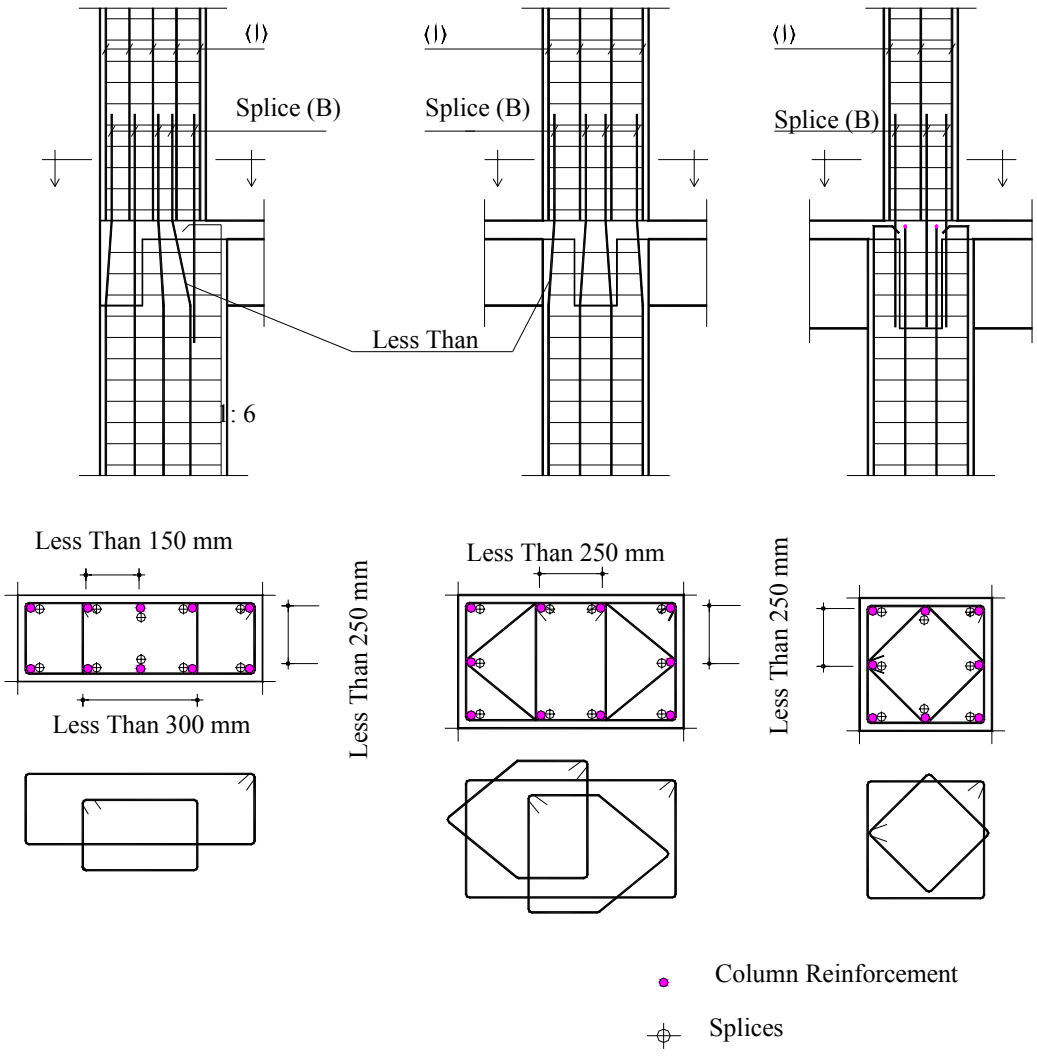


Figure (7-7-a) Typical details of reinforcement of columns for limited ductility cases

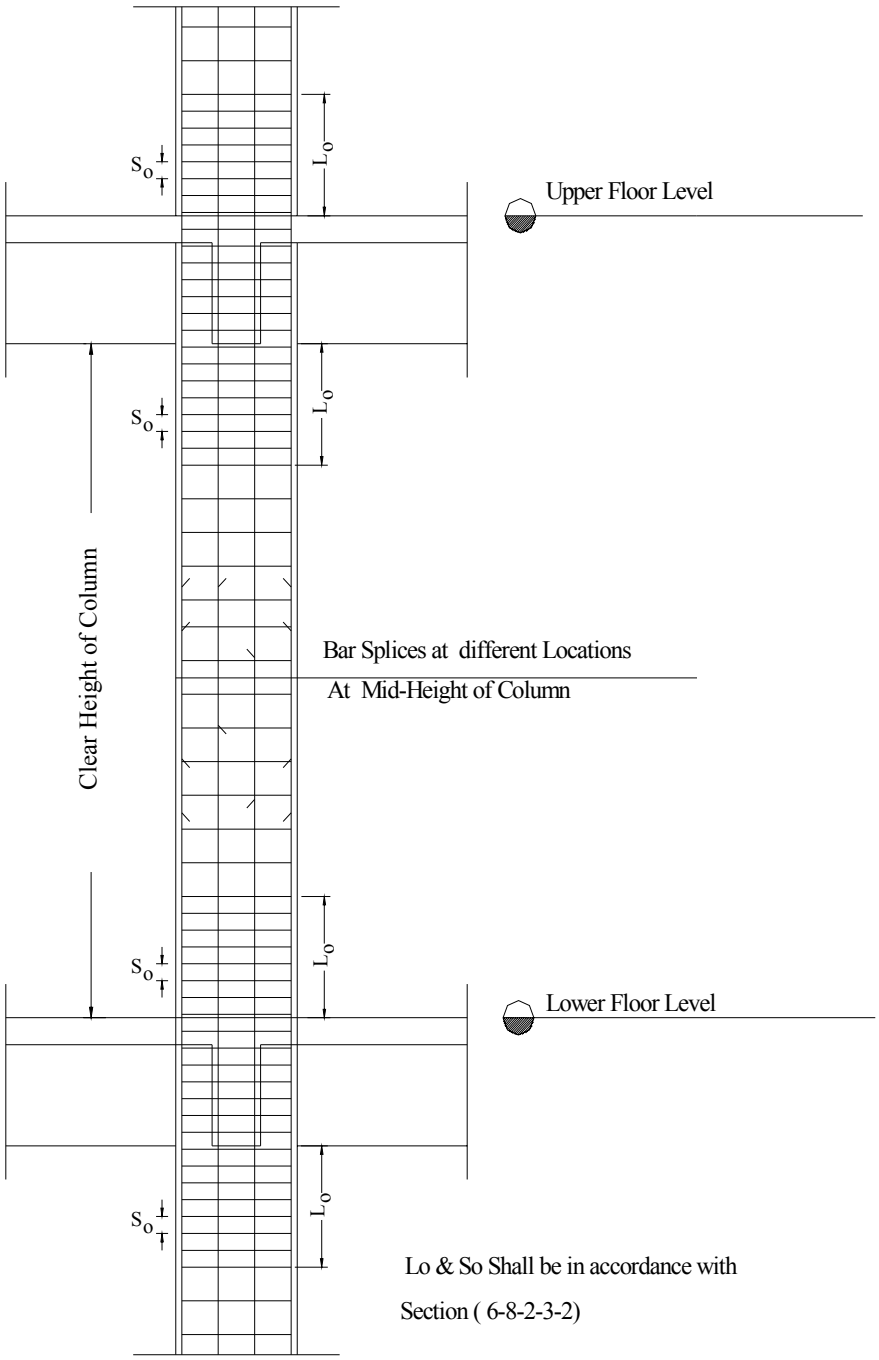


Figure (7-7-b) Reinforcement of columns having adequate ductility and subjected to large horizontal forces

CHAPTER 8
QUALITY CONTROL AND QUALITY ASSURANCE OF REINFORCED AND PRE-STRESSED CONCRETE WORKS

8-1 General considerations

This chapter considers the quality control procedures for reinforced and pre-stressed concrete works. These procedures guarantee the good quality and utilization of the materials as well as the attainment of the requirements and specifications of design codes, construction requirements and practices in order to meet the necessary targeted level of performance.

The quality control of the project could be achieved through the following:

- The internal quality control measures within the executing institute.
- The external quality control measures dictated by owner.

For pre-stressed concrete, in particular; the additional provisions according to section (10.6) shall be adopted.

8-2 Definitions

8-2-1 Quality target

It is the quality that achieves the targeted serviceability for which the building has been designed and constructed.

8-2-2 Quality assurance

It is the management tool which guarantee the satisfaction of the owner and users. It is composed of the systems, plans and programs necessary to assure the compliance of the constructed building with its targeted services as well as the compliance of all constructions works with the requirements of specifications and contract documents.

8-2-3 Quality control

It may be defined as the production tool composed of the group of adopted procedures/tests that assure the compliance of materials (concrete constituents) properties and concrete manufacturing to standard specifications and project requirements.

8-2-4 Quality manual

It is the document that elucidates the policy of all the participating parties, and explains the quality system.

8-2-5 Quality plan

It is a specific plan, approved by owner or consultant engineer, prepared for a specific project. The plan comprises the quality objectives and requirements as well as a detailed explanation for the work procedures and the organizing relationships between the different parties involved in the project.

8-2-6 Quality system

It is the organizational structure of the participating parties in the project. It comprises the responsibilities, procedures, operations and required resources to achieve quality objectives.

The owner is at the top of the quality system where the system comprises the internal policy of the owner, contracting procedures, quality plan (for the specific project) and quality manual attested by the project working team.

8-2-7 Elements and requirements of a quality system

Table (8-1) shows the elements of the quality system, while table (8-2) shows its requirements and the responsibilities during the phases of the project.

Table 8-1 Elements of Quality System

Document	Content	The party responsible for the document & its development
Quality plan	- Owner policy - Quality objectives - Scope of work - Organizational relationships - Responsibilities & role of all involved parties	All participating parties identified by the owner are required to prepare a quality manual.
Quality manual	- Elements pertaining scope of work of participating parties. - Work programs - Action steps of work	

Table 8-2 : Requirements of quality control and quality assurance during all stages of the project

Ser. No.	Project stages	Requirements	The responsible for the quality system stage	Notes
1	Concept & feasibility studies for the project	To focus & define quality target	The owner * should check the quality plan and attest it	How far the project is needed. How the project is meeting the owner needs & identifying the performance requirements.
2	Design	Specify quality	Designer provides his quality manual to the owner to be attested	Technical solutions
3	Planning for construction	Quality assurance	The owner * should check quality plan and bid documents which guarantee achieving the quality requirements	Preparation of bid documents which should include the requirements of achieving the target quality
4	Accreditation of material resources	Checking the compliance of material and its sources to the project standard specifications	The owner attests lab (or accredited labs for project) quality manual	Identifying the standard specifications and the acceptance/rejection limits for the used materials
5	Construction	Production and quality control	The owner * attests the quality manual of contractors or the sub-contractors involved in the project	Planning of activities for construction and the follow-up
6	Delivery	Verification of Quality	The owner * confirms the compliance of the executed work to the requested quality	Quality of the building & quality of the building documentation
7	Operation & use	Maintain quality	The owner * attests the periodical maintenance plan of the project	Periodical inspection and maintenance

* The owner or project director or the engineer whom he assigns

8-2-8 Quality assurance system

It is a management control system that organizes obligations, policies, responsibilities and owner requirements specified by the Quality Assurance Plan and which is included in Quality Assurance Program. It presents

quality control processes affecting the pre-mentioned activities and requirements.

8-2-9 Quality assurance plan

It is a project plan prepared and specified by the owner and his consultant or a quality control engineer. It comprise the owner's policies and objectives of quality objectives as well as a detailed description for the work plan and organizational relationships which guarantee the owner the start of his project according to a planned system to which the participating parties will abide .

8-2-10 Quality assurance program

It is a document that determines policies, practices, and work procedures which agree with quality requirements and contracting documentations.

8-2-11 Internal quality control

Internal quality control is continuously implemented in order to guarantee the achievement of the required specifications for the concrete and its constituents. It should be carried out by knowledgeable specialists usually employed by the executing institute. In case of unavailability of experienced personal at the executing institute, help from experts not affiliated to the executing institute is sought to perform the internal quality control tasks.

8-2-12 External quality control

The external quality control is carried out by parties affiliated to the owner and do not have any contractual form with the executing institute. The external quality control tasks comprise design review, special tests on materials (if necessary) and unannounced periodical inspection during all stages of the project.

8-2-13 Quality control requirements

Quality control and quality assurance is an integrated process that starts with the early investigation of the projects feasibility and continues through the preliminary project, the design, construction and turn – in stages as well as the operational period of the project. Table (8-2) shows briefly the quality control and assurance requirements during the different stages of the project's life span.

8-3 Technical Inspection

8-3-1 General

Technical inspection is the design and preparation of a program that assures the fulfillment of materials and concrete structure to the requirements specified in the contract documentations. The technical inspection covers mainly the following items:

- Concrete stockage, sources and tests.
- Site, inventory, equipment, scaffolds.
- Concrete mix design, constituents proportions, control, acceptance and testing.
- External factors and operating environment.
- Technical team necessary to run the site.

8-3-2 Inspector

8-3-2-1 External technical Inspector

The external technical aud inspector is affiliated to the owner or the supervision consultant office or the certified bodies or governmental organizations responsible for quality control in the construction industry. He should not be affiliated to the contractor or the internal auditor in any form, hence , his fees are bared by the organization he is representing .

Subsequently the external technical auditor is chosen from among unbiased experts that posses independent opinions.

8-3-2-2 Internal technical Inspector

Internal auditing must be performed by qualified personnel other than those directly supervising the audited activities. Qualifications of the internal inspector shall be approved by the consultant of the project and must fulfill the requirements of the special organization that issue certificates in that regards.

8-3-3 Material technical inspection

8-3-3-1 Phases of technical inspection

A – Outside the site

For major projects of special nature, production sites or supply sources must be inspected where samples are to be taken and periodically tested within the scope of technical inspection. Such inspection and its tests are insufficient and must be complemented with the periodical material inspection at delivery to project site. Contracts for inspection outside the project site with the producer or supplier must consider testing of materials at the same form and conditions it is delivered to the project site.

B – Primary inspection

The primary inspection is conducted in order to evaluate the efficiency and appropriateness of equipment, site facilities and testing laboratory for internal quality control (Human resources – laboratory capabilities). Such evaluation is according to the requirements identified in the project specifications, standard specifications for materials and concrete code.

C – Periodic technical inspection

The periodic technical inspection is conducted in order to fulfill the production and delivery requirements as well as internal and external quality control requirements. Hence, the periodic inspection does not start unless the results of primary inspection are acceptable. The periodic inspection is implemented without any notification at time intervals in accordance with the nature and program schedule of the project. Also periodic inspection is conducted on materials at site or at specialized laboratories outside the site. In all cases, the internal inspection revisions or modifications should be promptly accepted by the internal quality control inspector.

D – Additional tests for technical inspection

Additional tests are conducted in any of the following cases:

1 - Incompliance of materials with specifications limits of routine test.

2 - Cease of use of materials or operation at site for durations that exceed allowable storage periods.

3 - The contractor's violation to storage specification and conditions for material preservation at site.

At each case, the technical inspector determines the nature and limits of additional tests according to the targeted objective.

8-3-3-2 Attesting of concrete materials

A – Accreditation of sources

The external technical inspector accredits the suggested sources for materials and its capacity for fulfilling the project requirements. Based upon such accreditation, the responsible contractor shall contract the suppliers and producers. The accreditation should be supported by other information of which the most important are producer certificates, results of material testing at specialized laboratories and supply conditions. However, the accreditation of sources in any form does not exempt the contractor from being responsible for delivery of materials to the site with specifications less than those necessary for the accreditation of source since the contractor is responsible for the compliance of supplied materials delivered from the accredit sources or from other sources that may need to be accredited.

B – Acceptance based on certificate of origin

Sometimes the delivered materials are from suppliers of known outstanding experience in producing these materials. In such cases, materials may be accepted based on the certificate of origin to which all necessary information for attesting must be appended. The necessary information comprises results of quality control tests at production site, test results in specialized labs and information regarding date , size of sales and utilization record .

The acceptance based on certificate of origin will not limit the periodic or additional testing at any stage according to the opinion of the technical inspector.

C – Rejection of materials

In cases where materials do not comply with requirements of standard specifications (referred to some of these specifications in appendix three of the manual of laboratory tests for concrete materials) and/or project specifications, the materials must not be used. Rejected batches of materials must be disposed from stockage areas or at least isolated. The site

engineer or quality inspector must provide the technical inspector with their affirmation of the reasons based upon which the material was rejected. Tests may be repeated where test results are doubtful and hence material was rejected. In such cases, the test must be conducted on two separate samples taken at the same time; both samples must pass the test separately. Eventually, the acceptance report must comprise the unaccepted results that rendered the material unacceptable and led to the retesting.

8-4 Test laboratory

Establishment of site laboratories depend on the size and nature of the project as well as on the degree of targeted quality. The consultant engineer will determine the level of lab facilities which should be specified in the project documents. It is also possible to conduct some tests at other specialized laboratories. In all cases the laboratory equipment must be calibrated otherwise we limit the testing to certified laboratories.

8-5 Structural design review

Construction works shall not be started until the compliance of the structural design with this code and other construction works (architectural, electromechanical …..etc) is verified and attested by the authorized party according to the applied relevant legislations and regulations. The design will not be modified without the approval of the designer or the consultant engineer and attesting party.

8-6 Quality control procedure

After the achievement of the technical inspection to structural design requirements, construction monitoring and quality control of execution shall be adopted to achieve necessary requirements regarding preparation and handling of samples and materials incorporated in concrete production at its three stages; before casting, at casting and after casting.

8-6-1 Preparation and handling of materials

A – Sampling requirements

Sampling of each material shall be in accordance with Egyptian Standards, ES, in order to completely represent stocks out of which the samples have been drawn.

B – Sampling sources

Samples are collected from the following sources according to its use, site status and the point of view of the responsible for the samples:

- Material batches at delivery to site.
- Material stocks at site.
- Suppliers storing locations.
- Producers storing locations.

C – Samples handling

In handling the samples, one shall consider the following :

1 - Adoption of all necessary provisions that secure the delivery of the samples to the laboratories without subjecting the samples to changes such as partial loss of sample , subjecting samples to abnormal climatic conditions , damage of sample containers , loss of cover , intermixture of different samples , seepage of fluid materials ….etc.

2 - Use of clear undoubtful marking of samples as well as obtaining the signature of technical inspector and either the responsible for quality control or site engineers (whoever represents either of them).

3 - Registering the samples in the relevant special record which shall comprise :

- Producer or construction site.
- Location from which samples were drawn.
- Inventory volume if possible.
- Number and/or size of samples.
- Distinguishing marks for material source (local or imported).
- Distinguishing mark or code by sample collector.
- Tests required to be conducted and name of laboratory.
- Location and date at which samples were collected and tested.
- Production date and/or expiration date.
- Other information regarded by sample collector.
- Signatures of all of the above.

8-6-2 Monitoring and quality control for concrete constituents materials

8-6-2-1 Cement

Site engineer is not allowed to receive on site and store cement batches until he confirms its compliance with the requirements of the project specification and the Egyptian Standards, ES. The engineer has the right to run tests on similar samples at the site laboratory or any specialized laboratory (tests are to be conducted according to relevant standard specification depending on the cement type). Cement shall be stored according to the Section (9-2-1) after making sure that the cement delivered first shall be used first and that torn, opened or hardened cement packages shall not be used with due consideration of Section (2-2-1)

8-6-2-2 Aggregates

Aggregates samples must be monitored and undergo the quality control procedures before storage and stockage at site. Aggregates samples are not approved until their compliance with the requirements of the project and Egyptian Standards are confirmed (type wise and quality wise)

For major projects, vists to aggregates supply sources and their validity is considered part of the inspection scheme.

During construction, aggregate batches are not to be unloaded before confirming the compliance of the batches with the certified samples through visual inspection and some lab tests conducted at site laboratory. In cases where doubts that delivered batches show acceptable differences from certified samples, such acceptable differences must be recorded and reported to the engineer responsible for concrete mix design so that he may change proportions of concrete constituents if necessary.

8-6-2-3 Water used in concrete manufacturing

Water used in concrete manufacturing must be tested for validation as stated in Section (2-2-3). In cases where non-potable water will be used, tests on both setting time and concrete strength using the unpotable water shall be conducted. These tests shall be conducted in accordance with Section (2-2-3) where each test is conducted twice (at the same time, using the same attested cement type for the project and under the same conditions). In one of these two tests the intended type of water shall be used while in the second test potable or distilled water is used.

8-6-2-4 Admixtures

The properties of the admixtures shall agree with the limits specified in its standard specifications or in the specifications agreed upon. The use of admixtures follows – to a great extent- the brochures issued by the producers. In addition, the concrete constituent proportions shall be investigated through testing confirmative mixes where the effectiveness of the admixtures on fresh and hardened concrete properties is confirmed. Also, Section (2-2-4) shall be considered.

8-6-2-5 Concrete curing materials

Concrete shall be cured using water previously identified in Section (8-6-2-3). Also, concrete may be cured by surface sealing materials which are considered one of the main factors controlling the retention of mixing water inside the concrete rather than evaporating through the concrete surface. However, before allowing the use of such materials, one must test it and confirm its compliance to the specified limits in the specifications.

8-6-2-6 Reinforcing steel bars

It is recommendable to check the quality of reinforcing steel bars and grids and their compliance with the relevant Egyptian Standards at the factory. The reinforcing steel bars and grides must be delivered to site showing the distinguishing marks and shall be accompanied by the batch information card issued by the factory or from supply storage locations or from the institutes supervising the testing process. For pre-stressed concrete steels, one shall refer to Section (10-6-3).

The site engineer shall check the delivered batches of reinforcing steel bars or grids and record any visible rust, oil, grease and damage inflicted during loading and unloading as well as necessary corrective actions before use…. etc. Samples are taken for testing from batches supplied to the site at the rates indicated in table (8-4-A). The samples are collected and tested according to the testing guide appended to this code, and its modification.

The site engineer shall also consider that quality control for reinforcing steel bars and grids is not only based on compliance with limits in the Egyptian Standards, but also on necessary provisions during handling at the stages of storage , cleaning , cutting , forming , distinguishing , collecting , forming of frames and welding if applicable . These stages must be carefully planned and executed according to the requirements of detailing and shop drawing or its appendices shown in Section (9-6).

8-6-3 Monitoring and quality control before concrete casting

The internal inspector for quality control shall not allow concrete casting until he confirms the fulfillment of the requirements of the preparation phase which include the following:

- Survey works
- Efficiency & appropriateness of equipment
- Approval of materials and their sources
- Stocking
- Concrete mix design
- Fill & excavation works
- Foundation works
- Form works & scaffolding
- Reinforcement
- Joints
- Openings and embedded items
- Cleaning of form works surface just before casting
- Determination of tests dictated by monitoring and quality control of materials.

8-6-4 Monitoring and quality control during concrete casting

Monitoring and quality control during casting of concrete comprises:

- Proportioning of concrete mix constituents
- Taking necessary provisions for unusual conditions such as casting in hot or cold weather, casting below water surface, pumping concrete
- Homogeneity of concrete mixes
- Handling and casting of concrete
- Concrete compaction
- Concrete finish
- Preparation and casting of fresh and hard concrete test samples at site.

- Monitoring and recording the working conditions at site, equipment performance, atmospheric conditions as well as unusual conditions that led to stoppage of work.

8-6-5 Monitoring and quality control after concrete casting

Monitoring and quality control after concrete casting comprises:

- Concrete curing and taking necessary protection measures
- Dismantling of forms and scaffolding at determined time
- Visual inspection of concrete structure after dismantling of forms and scaffoldings.

8-6-6 Levels of quality control

The level of quality control is determined based upon the coefficient of variation (v) as shown in table (8-3)

Table (8-3) : The value of coefficient of variation corresponding to the level of Quality control

Level of quality control	Excellent	Good	Acceptable	Poor
Coefficient of variation (V) %	Less than 10	10-15	15-20	Greater than 20

8-7 Traceability and Non-Conformity

8-7-1 Traceability

In determining the main reason for a problem at construction, one shall determine the range and elements affected by the problem. Hence, a traceability system must exist through which one traces each of the following:

1 - The use of materials in concrete manufacture.

2 - Individuals responsible for work, inspection and quality control.

3 - Equipment used in a specific task.

4 - Determination of the method used in a specific task.

8-7-2 Controlling Non-Conforming cases

The control of non-conformity is divided into:

8-7-2-1 Determination of non-conformity as well as isolation and distinction of non conforming materials

This is applied to materials used in concrete manufacture or any other activity not fulfilling project requirements or its specification. The contractor shall isolate non conforming materials and distinguish them from conforming materials.

8-7-2-2 Determination of the required corrective actions

Upon the approval of the project supervisor or the owner representative, the corrective action is one of the following:

- Repair of defected element to an acceptable status (this status might not be of the same project conditions and requirements).
- Rework which is the resumption of the main requirements through re-operation.
- Acceptance as is, in a way that does not affect the performance requirements or fitness for use or safety.
- Down grading which is the use in a different item possessing lower level of requirements as compared to the original item.
- Rejection which is an unacceptable non-conformity which does not fulfill project requirements even after repair or rework.

8-7-2-3 Determination of the possible reasons for non-conformity

Under the supervision of the owner or his representative, the contractor shall investigate the reasons that led to the non-conformity and take the necessary provisions to limit or discontinue future non – conformity.

8-7-2-4 Re-inspection

Repaired elements shall be re-inspected according to its new status. New rejection/acceptance criteria are determined by the party conducting the retesting where the original requirements were deviated from. Reworked elements shall be re-inspected according to the original requirements where no deviations are allowed.

8-8 Records

The quality control system for concrete works shall comprise methods of recording documentation, its preservation as well as the kept period. The contractor shall guarantee the endorsements and signatures by the responsible parties of all records or reports or documents. All kept items must be indexed to facilitate its future access.

The following represents the list of documents which shall be kept by the contractor within the quality control system for concrete works:

8-8-1 General documents

- Contract documents
- Purchase documents
- Issued instructions regarding quality (quality plans, provisions, …etc)
- Detailed drawings
- Design drawings
- As Built drawings
- Project specifications
- Applications and requests for changes

8-8-2 Documents regarding quality control and assurance

- Laboratory test reports
- Reports on repair of equipment and machinery
- Forms for inspection and approval of works
- Codes and standard specifications used in the project
- Documents for inspection of project materials
- Certificates of origin
- Non-conformity cases and methods of handling.
- Records for training and qualifications of quality control team.
- Reports on concrete mix designs.
- Results of statistical evaluation.
- Photographs for the important phases of the project.
- Calibration reports for equipment and machinery.

8-9 Concrete tests

8-9-1 Test bases

Fresh concrete samples are drawn form mixes at arrival to site or at the location of site mixers. Each sample shall be composed of a mix of different portions drawn during unloading). Fresh concrete tests shall be conducted according to the specified fresh concrete requirements in the project's specifications. In cases where it is possible to run tests other than those specified in project's specifications, the equivalent properties correlations shall be considered. The hardened concrete specimens for compressive strength are prepared according to the Egyptian standards number 1658/1991 and its modifications. The hardened specimens shall be prepared just after testing the fresh concrete and assuring its compliance to the requirements indicated in project specifications.

In cases where moulds of different shape and/or size than those specified in the specifications are used, results must be adjusted to the standard specimens. In such cases, correction factors of section (2-3-2) shall be used as guide.

In all cases, specimens must be prepared according to provisions specified in the Egyptian standards number1658/1991 at all stages; mould filling, number of layers of filling, concrete vibration and compaction concrete surfacing, preservation of moulds in first stages of hardening, concrete curing and transportation of moulds to laboratory for testing.

8-9-2 Primary tests on concrete

Before construction of concrete works, concrete, whether mixed at site or central, shall undergo primary testing at its fresh and hardened forms. Construction works shall not be permitted before confirming the compliance of concrete with mix requirements of section (2-6-3-2). If the concrete will not fulfill the requirements, the mix designer shall be notified of the test results so that he modifies the mix design. This cycle shall be repeated until the confirmative mix after its latest modification fulfill the requirements of both the fresh and hardened forms.

8-9-3 Concrete tests during construction

Fulfillment of the concrete for the requirements specified in the project specifications must be confirmed. The site engineer shall inspect every mix before casting by conducting tests on fresh concrete samples. Also, the site engineer shall prepare hardened concrete specimens

according to the rate specified in the project specifications or the minimum rates indicated in the tables 8-4-A, B and C or if necessary, which ever is higher.

Table (8-4-A) shows the repeatability of quality control tests of concrete constituents and reinforcing bars and grids. The table provides the minimum required limit of tests for each material as well as the standard specifications for the necessary tests.

Table (8-4-B) shows the repeatability of quality control tests for ready mixed concrete or concrete mixed at site for fresh concrete according to the test plan indicated in the table which shows the type of test, the standard specification for conducting the test as well as the minimum limit for the repeatability and acceptance limits.

Table (8-4-C) shows the repeatability of quality control tests for ready mixed concrete or concrete mixed at site for hardened concrete according to the test plan indicated in the table which shows the type of test, the standard specification for conducting the test as well as the minimum limit for the repeatability and acceptance limits.

The tests are part of technical inspection of section (8-3). The concrete will be considered in compliance with the Characteristic Strength (f_{cu}) during construction if the test results fulfill the hardened concrete evaluation requirements of section (2-6-5-2).

8-9-4 Non destructive tests

As guidance, one may revert to non-destructive tests such as rebound hammer, ultrasonic or any other non-destructive test. Where the compressive test results do not fulfill the strength requirements or where concrete strength in an element for which test results are not available or will be doubtful. The use of calibration equipment shall be considered in addition to fulfilling the requirements of all provisions related to the use equipment as well as section (8-2) and (8-3) of the manual of Laboratory Tests for Concrete Materials.

8-9-5 Concrete core test

Concrete Cores may be taken in cases where the compressive strength do not fulfill the strength requirements or where concrete strength in an element for which test results are not available will be doubtful. Cores shall be taken, prepared, tested and test results shall be evaluated according to Egyptian Standards number 1658/1995 with due consideration of test (8-1) of the

manual for Laboratory Tests for Concrete Materials. The core results shall be considered acceptable if the calculated average strength for three cores will be greater than or equal to 75% of the required strength and if the calculated strength of any of the three cores will not less than 65% of the required strength.

Table 8-4-A : Repeatability of Q/C tests of concrete reinforced concrete

Material	Test	The number in the Guide for Laboratory Tests for Concrete Materials Issued 2003 Third appendix for code	Minimum limit of repeating the test	
Cement	Physical & mechanical properties		At the beginning of delivery, whenever the source is changed, every one month stockage & whenever necessary	
	Setting time	1-6		
	Compressive strength of cement mortar	1-16		
	Soundness	1-7		
Aggregate	Wear resistance	2-17	At accreditation of the source, at the beginning of delivery, and when source is changed	
	Crushing coefficient	2-16		
	Impact factor	2-19		
	Alkali activity of carbonate rocks	2-26	Whenever necessary	
	Visual inspection		Every batch	
	Grain size distribution	2-2	Every 100 m^3 delivered	
	Clay and fine material	2-11, 2-12	Every 100 m^3 delivered	
	Organic impurities for fine aggregate	2-14	At the beginning of delivery and every 500 m^3 of each batch.	
	Sulphate content SO_3	2-22-2		
	Chlorides content Cl	2-22-1		
	Soundness	2-24		
Mixing water	Determination of suspended materials	3-6	At the beginning of use for the first time (except the potable water) and when changing the source.	
	Chlorides CI	3-2		
	Sulphate SO_3	3-3		
	Total dissolved salts	3-1		
Additives	Homogeneity requirements	4-1	Before contracting & at Delivery of each batch	
	Performance requirements	4-2		
Reinforcing steel and grids			Number of samples for a batch of 50 tons & less	Number of samples for a batch more than 50 tons
	Dimensions & weights	5-1	2	3
	Tension	5-2	2	3
	Cold bend	5-3	2	3
	Special tests		1	1

For carbonate aggregate, a certificate shall be provided by supplier. The certificate shall show that the above tests had been carried out on aggregate at quarries in addition to dispersed X-Ray analysis and petrographic analysis. Also the certificate shall report the percentage of dolomite in the lime-rock.

Table # 8-4-B : Repeatability of quality control tests for fresh concrete

Test	Test Procedure	Min. limit of repeating the test		Limit of acceptance and rejection
		Ready-mix concrete	Concrete mixed at site	
Confirmative mix	Second Chapter of this Code	Before delivering to site for every grade	Before execution for every grade	Check the validity of the mix
Slump Evaluation	E.S.S 1658/1989	When taking strength samples		Required slump less than 50 mm and tolerance + or – 10 mm
				Required slump 50-100 mm and tolerance + or -20 mm
				Slump greater than 100 mm, tolerance + or -30 mm.
Trapped air	Tests guide, The two test 6-6, 6-7	When conducting the confirmative mix, once every month, and when changing the type of additives	When conducting the confirmative mix, once every month in case of using additives, when changing the type of additives and once every three months in case where additives are not used	Item 2 3%, 2%
Concrete density	Test guide. Test 6-9	When conducting the confirmative mix and once daily	When conducting the confirmative mix	
Temperature	-----	Temperature is measured for every sample on which the slump test is conducted		Not more than 35 degree centigrade
Special tests	As per project specification	According to the project specifications		Achieving the requirements of project specifications

Note : For ready mixed concrete, the sample is drawn from the middle or after discharging 15% of batch, while for concrete mixed at side the sample is drawn from the middle third of the mixed quantity at site.

Table 8-4-C : Quality control tests of hardened concrete.

Test	Specification	Minimum Limit for repetition of test		Limits of acceptance & rejection
		Ready mix concrete	Concrete Mixed at site	
Compressive Strength	E.S.S 1658/1989 Half the number of samples tested at the age of one week, and the other half after 28 days. Test after 56 or 90 days can be conducted according to the wish of the consultant engineer	At different concrete grade or structural elements (foundations, walls, columns, beams, slabs). Six cubes are drawn from the first 50 m^3 and six cubes for every extra 100m^3 on the same casting day		Fulfilling the requirements of article 2-6-5-2
Special tests	Specifications followed in the project	As specified in project specification		Achievement of requirements of project specification.

Note : For ready mixed concrete, the sample is drawn from the middle or after discharging 15% of batch, while for concrete mixed at side the sample is drawn from the middle third of the mixed quantity at site.

8-9-6 Load tests of concrete structures and elements thereof

The test shall be conducted on beams, slabs and ceilings of reinforced concrete structures. The test shall also be conducted on the structure after completion if it is specified in the project specifications or in cases of doubtful structural integrity. In all cases, the test shall not be conducted before the elapse of six weeks from the last concrete casting date. The vertical deflection shall be recorded before and after loading of the tested part of the structure with an equivalent load equal to 0.85 (1.4 permanent Load + 1.6 Life Load).

The equivalent load shall be applied on four almost equal increments where impact is completely prohibited during application of load.

The permanent load comprises the weight of the element and any other permanent load such as flooring and partitions. Hence, the actual existing permanent load at testing is deducted from the equivalent load.

Sufficient vertical supports shall be provided to sustain the total load. It shall be arranged to allow ample space for the expected deflections of the tested elements.

The tested structural elements and the neighbouring elements shall be loaded in such a way as to provide the most critical loading case. Spaces shall also be provided between the materials providing the load.

The deflections as well as crack widths shall be recorded after 24 hours of total load application. The loads shall then removed and after 24 hours the deflections as well as crack widths shall once again recorded.

The structure shall be considered safe if the following will be fulfilled:

a - If the maximum deflection δ max in the test element will be less than or equal to

$$\delta_{max} \leq \frac{L_t^{\,2}}{2000t} \qquad (8\text{-}1)$$

where:

L_t : is the span of the tested element in mm. The span in the case of Flat Slab or two way slabs is the smaller span. As for cantilevers the span is twice the distance from the support faces to the end of the cantilever.

T : the element thickness in mm.

b - If the max deflection of the tested element exceeds the value given by the Equation. 8-1, then the retrieved part of max deflection after 24 hrs from the removal of the load shall not be less than 75% of the max deflection. Cracks width shall also be within the allowable range.

c – If 75% - at least – of the maximum deflection that was recorded during 24 hrs of loading was not retrieved during 24 hrs after the removal of the load equivalent to the live load, the test shall be repeated as previously.

Part of the structure shall be considered unacceptable if at least 75% - at least – of the maximum deflection that was recorded during the second test was not retrieved or if the cracks width will be wider than allowable. The load test shall not be repeated before 72 hrs elapse from the moment of

removing the load of first test. The test may be conducted on pre-stressed concrete elements.

If part of the tested structure shows during testing or after removal of load any undesirable reaction or construction defects, the designer shall adopt one of the following solutions:

- Use additional supports if possible.
- Possible reduction of the live loads, improve the load distribution, and rearrange the concentrated loads.
- Possible reduction of the dead loads.
- Possible reduction of the dynamic effect if applicable

The structure shall be considered unacceptable for its purpose of use if all of the preceding measures could not be implemented

Load tests shall not be conducted on elements not subjected mainly to bending. The safety of such elements shall be determined through structural analysis.

CHAPTER 9

CONSTRUCTION REQUIREMENTS

This chapter is concerned with the construction of concrete structures including the production of good quality concrete satisfying the design requirements of the project in accordance with the code requirements.

9-1 Handing over and preparation of project site

Site preparation and organization shall be followed according to the subsequent steps for handing over the project site:

9-1-1 Confirm the acquisition of all permits and approvals before the commencement of work, also the geological suitability of the site shall be verified. Special precautions shall be taken in case of the presence of faults, collapsible areas, or flood pathways, especially in new cities and areas.

9-1-2 Specify the project site according to the general project layout which indicates the location, dimensions, axes for each structure and its relationship to other structures. The site shall be cleared from obstacles, including buildings, trees and foundations which might obstruct the construction of the structures. The type and quantities of these obstacles shall be specified. In case of the existence of underground utilities, the site engineer shall contact appropriate authorities for proper action.

9-1-3 Prepare the leveling grid of the site to determine natural land elevations, compute cut and fill quantities, and leveling operations. A starting reference point for surveying shall be specified and maintained intact and clear during the project construction.

9-1-4 Take security precautions and follow the instructions of industrial safety.

9-1-5 Site planning and specifying locations of structures, storage areas, and knowledge of surrounding areas to prepare pathways which shall facilitate arrival of supplies, equipment, and materials. Identify and secure site entrances and exits. Supply the site with electricity, water, necessary maintenance workshops, communication facilities (wired and/or wireless), fences, closed and open storage facilities and offices for engineers and workers.

9-1-6 After establishing the structures' locations in site, boring holes shall be performed and soil specimens extracted from different depths according to the Egyptian code for soil mechanics and foundations, ECP202 and the project specifications and requirements to confirm the foundation level and soil stress stated in the structural drawings. The ground water table elevation and movement and different soil layers shall be evaluated in order to determine the necessary precautions for dewatering during construction. Special precautions shall be taken to maintain the safety of neighboring structures during foundation construction. Proper design shall be made for the lateral earth support systems before foundation work starts.

9-1-7 The locations of experiments performed before work commencement shall be located. These experiments include; pumping experiments to test ground water reservoirs and the appropriate means of ground water disposal, by constructing a pipe network to divert the ground water away from the equipment pathways and storage areas sensitive to humidity. Another type of experiments includes load tests on non-working piles outside of the specified construction area.

9-2 Materials storage

Materials shall be stored on-site in the storage places in the specified sites in such a way as to guarantee its safety and to avoid any probable damages. All materials shall be subjected to quality control measures upon arrival on-site according to the frequency specified in Table (8-4-a) for quality control to ensure there conformity to the Egyptian Standards, ES.

9-2-1 Cement

1 - Cement shall be delivered on-site either in tight bags or closed containers that shall be stored in away to protect cement from moisture and direct sunlight. Different types of cement shall be stored separately.

2 - In case of storing bagged cement, the bags shall not be in direct contact with the ground. Bags shall be staked to permit continuous ventilation. The maximum number of stacked bags shall not exceed 10. The production date shall be written on each stack allowing the use of earlier produced cement first in accordance with section (8-6-2-1).

3 - In case of delivering cement in containers, the cement shall not be used until its temperature is lower than 75°C.

4 - Cement shall be tested upon delivery to the site before use, and if stored at site for a period exceeding one month, even if properly stored, to ensure that its properties conforms with the Egyptian Standards, ES according to Table (8-4-a).

9-2-2 Aggregate

Small and large aggregates shall be stored separately in a manner to avoid its pollution and mixing with other materials and according to the gradation pre-specified in the project mixtures design for projects that require special or high grade concrete, hard well-drained flooring shall be manufactured to store the aggregate according to the different sizes and in agreement with the required grading.

The aggregate shall be visually inspected before storage. The acceptance certificate issued from the quarry for use in concrete works must be revised. The aggregate shall be insured to be free from organic materials such as grass, plants, and roots. Also, the aggregate shall be checked not to be mixed with foreign materials or silt blocks ether big or small. It shall also be verified that the aggregate surface shall not be covered by fine layer of silt. The suitability of bottom layers of stored aggregate and accumulation of fine materials shall be checked.

9-2-3 Reinforcing steel

The reinforcing steel shall be stored such that it is protected from exposure to corrosion by covering it to prevent its exposure to humidity or water. The reinforcing steel shall not be in direct contact with the floor so it shall not be exposed to any materials that affect its bond with the concrete. It is preferable to provide the reinforcing steel directly before use.

The integrity of the reinforcing steel shall be visually corroborated before storage on-site. The surface of the reinforcing steel shall be free from oil or fats or organic materials as well as corrosion.

9-2-4 Admixtures

The admixtures shall be stored in their original containers labeled with all admixtures information. Storage shall be in accordance with the conditions listed in the product datasheet, taking special precautions for the maximum storage temperature. Admixtures shall not be stored in open air, taking into consideration items stated in section (8-6-2-4) .

Any special usage instructions or necessary safety precautions (e.g. if additives are caustic, poisonous or corrosive), shall be stated on the label.

Any recommendations or steps to be followed before use shall be reported especially in the case of long storage periods (e.g. stirring or rolling the barrels…ect.).

9-2-5 Water

Potable water shall be used in concrete mixes. In case there is no continuous source of water on-site, water can be stored in closed containers that do not permit the water to be polluted with harmful materials such as oil, acids, organic materials, or any materials that could have a deteriorative effect on concrete components or reinforcing steel.

9 - 3 Materials measurements

The degree of accuracy of measurement devices for concrete materials depends on several factors according to the project size, production rate, and the concrete specifications. These devices are periodically calibrated. The allowable tolerances shall be considered as reported in section (9-8).

9-3-1 Cement

Cement calibration by volume shall not be permitted. It is preferable that concrete mixing container accommodate whole number of cement bags. In case of using loose cement it shall be measured by weight using calibrated accurate scales.

9-3-2 Aggregate

Aggregate shall be measured by weight as it yields more accurate results. It shall be permitted to measure aggregate by volume using measuring boxes with specific capacity. These boxes shall be filled without compaction. The top and bottom of the surface of the aggregate (inside the box) shall be leveled with the box sides.

9-3-3 Water

Water shall be accurately measured according to required values. The amount of aggregate moisture shall be considered.

9-3-4 Admixtures

Additives shall be mixed according to the quantities specified in the design mix. Additives shall be measured accurately.

9 – 4 Scaffolds and forms

The following requirements shall be fulfilled when executing scaffolding and form works:

a - Knowledge of the types of forms and scaffolds used by both the designer and the contractor.
b - Providing sufficient safety for all the concrete structure components during preparation, lying reinforcing steel, pouring, and hardening period till the time of scaffolds removal.
c - In case of openings in ceilings, beams, and walls for air conditioning pathways or pipes or otherwise, these openings shall be accounted for in the scaffolds before laying the reinforcing steel or pouring the concrete.
d - Following instructions of industrial safety for all the workers and supervisors during the execution with the availability of inspection and monitoring by easely and safely.

9-4-1 Design, preparation and setup of forms and scaffolds

All forms and scaffolds shall be designed and prepared to satisfy the following:

9-4-1-1 Scaffolds, supports, and ties shall be stable to maintain the position of the concrete components in their proper place

9-4-1-2 Forms shall be rugged, and tight to prevent the leakage of cement and water mixture (Laitance) from the concrete during different work stages.

9-4-1-3 If forms were exposed to sun and weathering conditions for prolonged periods before concrete is poured, it shall be checked to ensure there are no distortions or changes in its dimensions.

9-4-1-4 Tying the supports especially the vertical ones so they shall not be affected by horizontal shocks caused by the movement of workers or equipment or the thrust force resulting from pumping the concrete or wind load and vibrations from the equipment used at work.

9-4-1-5 Vertical supports rest on stable ground having a bearing capacity proportional to the applied load.

9-4-1-6 In case of using scaffolds or forms of a special nature, they have to be implemented according to design drawings and specifications for scaffolds type. They shall also be inspected before laying the reinforcing steel.

9-4-1-7 Cambering the bottom surface of beams and slabs forms according to the information in the project documents. In case this information is not available, the forms shall be cambered for spans equal or greater than 8 meter by (1/300) to (1/500) of the span length. For cantilevers with an unsupported length that exceeds 1.5 meter, the camber is approximately (1/150) of the cantilever length.

9-4-1-8 The tolerances for the inside dimensions of the forms –i.e. dimensions of the concrete cross sections - shall not exceed the values stated in section (9-8-3).

9-4-1-9 Forms shall be carefully cleaned from inside –i.e. surface in contact with the concrete- before the reinforcing steel is laid and directly before pouring concrete. Cleaning shall be performed by removing the dust and garbage using water or compressed air. In case of columns, walls, and deep beams, openings shall be made at the lowest level in the forms to make cleaning easier. These openings shall be closed after finishing cleaning process and directly before pouring concrete.

9-4-1-10 In case of wooden forms, surfaces in contact with concrete shall be sprayed with water before pouring to prevent the wood from absorbing water from the concrete mix.

9-4-1-11 It is preferable to paint or spray the surface of the forms that are in contact with concrete using special materials that prevent the concrete from sticking to the forms. This shall take place before laying reinforcing steel. This makes dismantling the form easier and protects the concrete surface from sticking to the form.

9-4-1-12 Pathways shall be made for the workers such that their movement does not affect the dimensions and shapes of the reinforcing steel.

9-4-2 Dismantling scaffolds and forms

The time between pouring concrete and dismantling forms and scaffolds is influenced by: temperature, span length, type of cement used, grade of concrete, type of curing, and the load applied to the structure after dismantling. Before dismantling, it shall be confirmed that the concrete strength reached a level that provides sufficient safety with condition that dismantling shall not cause unacceptable instability or deflection or cracks. If there were no results from testing concrete cubes before dismantling and no structural computations regarding deflections and cracks, the forms shall not be dismantled till a minimum period of time from pouring has passed according to the following rules:

1 -When using ordinary Portland cement

- It shall not be permitted to dismantle the form sides that only work as a cover for the concrete before 48 hours from pouring beams, columns and walls. For special cases such as tunnel or sliding forms, refer to the design engineer of record.
- It shall not be permitted to dismantle the forms and scaffolds before a waiting period (in days) equal to twice the span in meters plus 2 days. When computing the dismantling time for slabs, the span shall be taken as the shorter length, noting that the waiting period shall not be less than one week.
- In case of cantilevers, the waiting period before dismantling the form (in days) shall be equal to four times the cantilever length (in meters) plus two days, such that the period shall not be less than one week for cantilevers with an unsupported span of 1.50 meter.

2 -When using rapid hardening Portland cement

- The scaffolds and forms carrying beams and slabs can be dismantled after half the duration needed when using ordinary Portland cement but not less than three days. Upon dismantling, concrete shall be capable of withstanding the stress resulting from the actual acting loads. It is preferable to conduct compressive strength tests on cubes from the used concrete before dismantling the scaffolds to make sure that concrete reached the required strength.
- In case of temperature drop below $15C^{o}$ and especially when using rapid hardening Portland cement, caution shall be taken and dismantling

forms and scaffolds shall be delayed a suitable period of time, in addition to the periods mentioned before.

9-4-3 Special precautions for dismantling scaffolds and forms

9-4-3-1 When forms and scaffolds are carrying additional loads as the case of a floor carrying the weight of a newly poured floor, it shall not be permitted to dismantle the vertical supports before twenty eight days. All necessary precautions shall be taken to ensure that the vertical supports rest on ground that can safely withstand the loads after confirming that the strength of the concrete meets the project specifications. This duration may be reduced if the structural safety of all structural components holding forms was confirmed and after the approval of the design engineer of record.

In special cases such as inverted beams and slabs hanging by tension columns, the computed duration to dismantle the scaffolds shall start from the date of pouring the inverted beam or the slab carrying the hanging slab.

9-4-3-2 In all cases when dismantling forms, care shall be taken to ensure the stability of the structure and avoid the occurrence of any opposing stresses in its components.

9-4-4 Dismantling tunnel and half tunnel forms

For tunnel and half-tunnel forms, compressive strength tests shall be conducted before dismantling scaffolds and confirming the conditions presented in section (9-4-2) are met.

9-4-5 Concrete breaking after form removal

It is completely impermissible to break or make cavities in columns or make openings in beams or slabs after pouring, or cutting reinforcing steel for any reason without referring to the design engineer.

9 – 5 Production, manufacturing, and curing of concrete

9-5-1 Preparation for pouring

9-5-1-1 All mixing and transportation equipment shall be clean. All measuring equipment shall be calibrated before work commencement. Calibrations shall be repeated periodically as determined by the supervising engineer every two months, and after fixing the equipment, or according to the project quality plan.

9-5-1-2 The surface of all wooden forms shall be sprayed with water before pouring. Hollow blocks shall also be sprayed with water before pouring. In case of pouring concrete directly on the foundation layer, soil shall be well sprayed with water after compaction, ensuring there is no water accumulation.

9-5-1-3 Before pouring new concrete onto old one, it is necessary to remove all broken parts of the old concrete and all materials attached to it. Then the surface shall be treated to ensure bonding between the old and new concrete.

9-5-1-4 The reinforcement steel shall be clean of harmful materials and free from corrosion, and the following shall be considered:

- The reinforcement steel bars shall be laid on plastic spacer or pieces of mortar to maintain the concrete cover during pouring.
- It shall not be permitted to bend the reinforcement steel while pouring.
- Walking shall be completely forbidden on reinforcement steel bars after fabrication and placing.

9-5-1-5 Before pouring concrete, water shall be removed. In case of neighboring structures or foundations, consideration shall be taken to the proper engineering methods and design. If it is necessary to pour concrete under water level, underwater pouring shall be used after the approval of the consulting engineer and considering section (9-5-3-6).

9-5-1-6 Pouring, compaction, finishing and backup equipment shall be prepared. Labor specialized in pouring, leveling surface, compaction, and finishing concrete shall be arranged in numbers proportional with the rate of pouring to avoid the occurrence of pouring joints in locations that were not previously pre-determined.

9-5-2 Mixing concrete ingredients

9-5-2-1 Mechanical mixers shall be used in mixing concrete ingredients. Mixers capacity shall be proportional to pouring rate so that the ingredient distribution is homogeneous. The mixer shall be emptied completely before refilling. The concrete mix shall be transported from the mixer to the pouring location via conveying belt, or crane or sliding channel or concrete pump. It shall also be permitted to empty the mixture on a solid table till it is manually transported. No new concrete shall be poured on the table till the previous mix is completely moved.

9-5-2-2 If it is extremely necessary to manually mix the concrete, it shall be permitted with the approval of the project consulting engineer of record. In this case, mixing is performed by stirring the materials using the required ratios on a flat solid table. Cement shall be mixed with dry aggregate and shall be stirred a minimum of three times before the water is gradually added in the amount needed for the mix. Stirring and mixing shall continue till the mix is homogenous in color and consistency such that the design requirements are met.

9-5-2-3 When using pre-mixed, self compacting and hot weather concrete, it shall be necessary to refer to their respective technical specifications, and approve them from the project consulting engineer of record before usage.

9-5-2-4 The following information shall be written in the field notebook:

- The concrete grade, type and percentages of mix ingredients.
- Number and volume of mixes used in pouring different parts of the structure.
- Locations of pouring concrete.
- Time and date of mixing.
- Quality control procedures.

9-5-3 Pouring concrete

When pouring concrete, care shall be taken to maintain the stability of the form. The following precautions shall be taken:

9-5-3-1 The concrete shall be poured after thorough mixing avoiding segregation. The time between adding water and pouring concrete shall not exceed 30 minutes at a temperature not exceeding 30°C in the shade or 20 minutes in hot weather. If these times are to be exceeded, it shall be permitted to use suitable additives while mixing, as approved by the project consulting engineer of record. The additives shall be added using the predetermined dosage. These percentages shall be experimentally verified before pouring starts.

9-5-3-2 Concrete that has set or partially solidified or polluted with foreign materials shall not be used.

9-5-3-3 The locations of the construction joints (i.e. places of stopping pouring) shall be determined before pouring starts. Pouring shall continue

regularly until finishing the pre-determined section taking into consideration section (9-5-6).

9-5-3-4 In case of pouring concrete with large thicknesses, care shall be taken to pour concrete on layers with thicknesses between 300-500 mm. A mechanical vibrator shall be used to compact concrete. Care shall be taken so the duration between pouring different layers does not exceed 30 minutes in normal temperature or 20 minutes in hot temperature, so that the lower layer does not solidify when pouring the next layer. This duration can be exceeded if there is enough reinforcement to connect the consecutive layers, or using temperature lowering additives to decrease the heat generated from the hydration process. Care shall be taken to keep the amount of water minimal taking into consideration the requirements in sections (4-2-2-4) and (9-5-1-4).

9-5-3-5 For columns with heights exceeding 3.0 meters, it shall not be permitted to pour concrete through the full height. One side of the form shall be divided into parts with height not exceeding 3.0 meters, which shall be closed periodically so pouring continues. It shall be necessary to compact the concrete using mechanical vibrator.

9-5-3-6 If it is necessary to pour concrete underwater without removing the water, care shall be taken to have cement rich and high workability concrete mix with the minimum possible water content. The concrete shall be poured through a pipe with an approximate diameter of 200 mm that reaches the bottom where concrete is to be poured. The pipe shall be lifted while pouring by a rate that shall not allow the pipe moving out of the mix such that the water shall not leak inside.

9-5-3-7 If the air temperature exceeds 35°C in the shade during mixing and pouring concrete, refer to the technical specification with regards to pouring concrete in hot weather. The following precautions shall be considered:

- Using shades for aggregate (coarse and fine) storage areas. Coarse aggregate can be cooled using water sprinklers.
- For loose cement stored in silos, the silos shall be painted from outside with a sunlight reflective material. If the cement is in bags, it shall be arranged under a ventilated shade.
- Cooling water before usage in mixing concrete.
- Painting mixers with materials that reflect sunlight and/or covering the pan with a layer or more of burlap and spraying with water.

- Spraying the forms with water before pouring. In case of producing precast concrete elements, pouring shall take place in shaded areas.

9-5-4 Concrete compaction

Compaction and vibration processes shall take place to guarantee the mix is flowable around the reinforcement steel. Compaction shall continue till pouring ends.

Mechanical compaction shall be performed using submerged vibrators inside the mix or vibrators fixed on the forms and scaffolds. For special cases, manual compaction can be used with the approval of the consulting engineer. Mechanical compaction is to be performed by a specialist trained such that compaction is stopped after the appearance of air bubbles. The submerged vibrator shall be kept away from the reinforcement steel while vibrating. During compaction, care shall be taken to avoid disturbing previously poured concrete or displacing reinforcement steel or changing form sizes.

9-5-5 Concrete treatment and protection

9-5-5-1 Concrete shall be treated such that it shall be maintained completely wet for a minimum duration of seven days from the time the surface solidifies when using ordinary Portland cement. When using rapid hardening Portland cement or accelerating additives, the minimum treatment duration shall be four days from the time the surface solidifies. Treatment shall be performed by thoroughly spraying the surface with water free from salts or harmful materials or covering its surface with burlap or sand or hay or mats or any suitable coverage while keeping it wet by continuous spraying. If not using wet treatment, it shall be permitted to use certified treatment compounds that shall be homogeneously sprayed to guarantee the complete coverage of the concrete surface to protect it from losing mixing water. Steam or other treatment methods can be used. Treatment by wetting shall be continued to guarantee that the concrete reaches the required strength according to the project specifications.

9-5-5-2 Steam treatment shall be used for precast concrete elements after two hours from pouring time. This shall be accomplished by raising the temperature of the concrete elements to 60°C within duration of four to six hours according to the thickness and width of the concrete element followed by decrees to the normal temperature within three hours. Treatment by wetting

shall continue to guarantee reaching the required concrete strength after 28 days.

9-5-5-3 Newly poured concrete shall be protected from rain, quick drying from hot or dry weather or storms. This shall be achieved by covering the concrete using suitable covers from the time of pouring till the surface adequately hardens so it can be treated using different methods.

9-5-5-4 During curing, reinforced concrete shall not be exposed to water containing harmful salts that exceed the allowable values according to section (2-2-3).

9-5-5-5 Concrete shall not be subjected to any loads such as ground water pressure or earth fill especially saturated with water until the concrete compressive strength reaches the required strength by the project specifications.

9-5-5-6 If concrete will be subjected to loads from natural disasters such as earthquakes, and floods within seven days from pouring, the homogeneity integrity of the concrete and structural joints as well as the non-existence of cracks shall be confirmed.

9-5-6 Construction Joints

The following conditions and precautions shall be considered when preparing construction joints:

9-5-6-1 The joint shall be perpendicular to the internal effective forces.

9-5-6-2 The joints shall be located at bending moment inflection points for beams and slabs or at sections of minimum values of shear forces next to the supports. If necessary, it is preferable to have the joint location at the end of one third of span next to the supports.

9-5-6-3 The site engineer shall specify the pouring joints locations on the construction drawings. The reinforcement steel required to transfer the shearing forces, and main tensile forces shall be clarified according to section (4-2-2-4) provided the approval of the design engineer prior to construction.

9-5-6-4 The minimum distance between the location of the joint in the main beams and the support of secondary beams shall not be less than twice the width of the secondary beam.

9-5-6-5 The joints between deep or inverted beams and the slabs connected to it shall be constructed at connection locations. The sloping edge borders of the slabs (haunches) or below the drop level around columns (drop panels) if existing with the slabs.

9-5-6-6 When pouring the joints is resumed after the concrete hardens, the surface of the concrete is roughened well to show the coarse aggregate. The surface is cleaned to remove the residuals and loose materials using compressed air and washed with water, then a layer of a mixture of cement and water slurry or any other material certified to ensure the bonding between the old and new concrete, fulfilling specifications regarding the prevention of water permeability in case it is requested by the design engineer of record.

9-5-7 Shrinkage joints

9-5-7-1 In case of pouring large areas of unreinforced concrete slabs that necessitate using shrinkage joints to avoid the occurrence of cracks such as floorings of airports, factories, garages and others, these surfaces shall be divided into longitudinal strips with widths not exceeding 30 times the slab thickness with a maximum limit of 5 meters. The longest dimension shall not exceed 25 meters. Pouring of odd or even numbered strips shall start provided the rest of the strips are poured alternately. Vertical pouring joints shall be constructed between these longitudinal strips with a minimum thickness of 20 mm. After pouring, these joints shall be filled with mastic or any other similar material according to the instructions of the design engineer. Special precautions shall be taken to prevent the relative settlement between the strips.

9-5-7-2 The longitudinal strips shall be divided using secondary shrinkage joints located not more than 1.25 times the strip width. The minimum strip width is 20 mm with a minimum depth equal to one third the slab thickness. These joints shall be filled with mastic or any other similar material. These joints shall be executed using a mechanical saw cutter after the final setting time but not exceeding three days from the pouring date.

9-5-7-3 It shall be permitted to pour large surfaces and floorings at the same time on the condition of executing the joints in both directions after pouring, according to section (9-5-7-2).

9-5-7-4 The spacing between the shrinkage joints can be increased in case of using an upper reinforcement mesh in the concrete slab to resist stresses due to concrete shrinkage.

9-5-8 Expansion joints

For regular structures, the maximum spacing between expansion joints shall be:

- From 40 to 45 meter in normal weather regions.
- From 30 to 35 meters in hot weather regions.

The spacing can be increased if temperature differences and the influence of expansion shrinkage and creep shall be accounted for in the design.

In case of mass concrete works such as retaining walls and frames, the spacing between the joints shall be reduced. Adequate precautions shall be taken to prevent water leakage from these joints.

9-5-9 Seismic joints

When choosing the seismic joints, the following requirements must be satisfied:

- Horizontal and vertical uniformity of the structure and joint locations
- Relative displacements between structure floors.
- Compute the width of the seismic joint between the different parts of the structure and neighboring structures according to the requirements of the Egyptian Code for calculating loads and forces, ECP 201

9 – 6 Fabrication of steel reinforcement

9-6-1 All types of steel reinforcement shall be cold fabricated according to the reinforcement steel bar list.

9-6-2 In case that steel reinforcement is subjected to corrosion or is delivered to the site with the manufacturing scales, it shall be permitted to use this steel if it is possible to remove the surface corroded layer or manufacturing scales using wire brushes or sand blast. The loss in the weight of the reinforcement steel shall not exceed 2%. The decrease in bar diameter shall not exceed:

- 0. 20 mm for bar diameters up to 10 mm
- 0.30 mm for bar diameters between 10-mm and 20-mm.
- 0.50 mm for bars with diameters larger than 20-mm

9-6-3 The steel reinforcement shall be carefully placed in its locations according to the structural construction drawings. Steel bars shall be properly fixed so they shall not be dislocated during pouring and compaction. A distance shall be kept between the reinforcement steel and forms to be filled with concrete during pouring. It shall not be permitted to expose steel reinforcement on the surface of the concrete to avoid weathering conditions which will help to initiate corrosion.

9-6-4 The supervising engineer shall check the reinforcing steel after placement with respect to the structural drawings. All remarks shall be implemented before giving permission to pour concrete.

9-6-5 If there is a high percentage of steel reinforcement in the concrete sections, it shall be permitted to use bundles according to sections (4-2-5) and (7-3-4).

9 – 7 Minimum concrete cover for steel reinforcement

9-7-1 The minimum concrete cover for steel reinforcement shall not be less than the values given in section (4-3-2-3-B).

9-7-2 For buildings that may be subjected to fires; the dimensions of the concrete cover shall not less than the values in Tables (2-14-A) and (2-14-B).

9-8 Allowable tolerances in concrete works

This section specifies the allowable tolerances in concrete works after approving its components in the laboratory and calibrating the measuring devices with the rates of mixing and preparing of concrete. For special structures, the design can specify tolerances more stringent than those specified in this section.

9-8-1 Allowable tolerances in the measurement of quantities of concrete ingredients

When using bagged cement, the negative allowable tolerance in the weight of each bag of cement shall be 1% from the weight stated on the bag. If the average weight of the bags in any load delivered to the site computed from the weights of fifty randomly chosen bags was found to be less than the weight stated on the bag, it shall be permitted to reject the whole consignment or replace the difference in weights in case of using the cement in the mix.

Table (9-1) shows the allowable tolerances in the weights of the materials used in each mix.

Table (9-1) Allowable tolerances in concrete ingredients

Ingredients	Allowable tolerances
Aggregate	+/- 3%
Added Water[(1)]	+/- 1%
Cement	+/- 1%
Additives	Tolerance is not desirable

[(1)] Tolerance shown in table includes all the water added to the mix including the moisture of the aggregate

9-8-2 Tolerances in slump test measuring concrete consistency

1 - The allowable tolerances for the slump test shall be specified for the samples taken from the mix directly before pouring.
2 - For concrete consistency, refer to the project specifications with the maximum allowable tolerances in accordance with Table (8-4-B), on condition of achieving the required characteristic strength and the approval of the project consultant.

9-8-3 Allowable tolerances in dimensions

The tolerances stated in this article shall be the references to abide with when there shall be no special tolerances in the contract requirements or drawings. These tolerances are subjective and shall be used for validity and acceptability range rather than as a refusal limit.

These tolerances shall not be used to exceed the property limit or land dimensions or increase in permissible extensions or heights in accordance with the building laws and regulations.

1 - Maximum tolerances in horizontal dimensions (columns, beams and wall axes)
For any span or for every 6.0 meter in any direction +/- 5 mm
Total structure dimension +/- 25 mm

2 - Tolerances for vertical plumb

a - Columns and wall surfaces and intersection lines between surfaces

For each 6.0 meter in height	+/- 5 mm
Total structure dimension (30 meter maximum)	+/- 25 mm

b - Corner column surfaces and vertical expansion joints

For each 6.0 meter in height	+/- 5 mm
Total structure dimension (30 meter maximum)	+/- 25 mm

c - Walls and columns executed using sliding forms

For each 1.5 meter in height	3 mm
For each 15 meter in height	25 mm
Maximum for total structure dimension (180 meter)	75 mm

For buildings with heights exceeding the preceding maximum limit, the allowable tolerances shall be specified by the design engineer.

3 - Allowable tolerances in levels

The tolerances given in this article are limited compared to the information given in the contract documents and before dismantling the forms.

a - Beam and slab bottoms

For each 3.0 meter horizontal dimension	+/- 5 mm
For each span or 6.0 meter horizontal dimension	+/- 10 mm
Complete structure width or length	+/- 20 mm

b - Lintels, window sills, parapets and architectural cornices in facade

For each span or 6.0 meter horizontal dimension	+/- 5 mm
Complete structure width or length	+/- 15 mm

c - Points used for determining slabs and inclined beams levels

For each span of 6.0 meter length	+/- 10 mm
Complete structure width or length	+/- 20 mm

4 - Locations and sizes of connection bolts and openings

For locations of opening axes	+/- 15 mm
For opening sizes	+/- 5 mm

5 - Sizes of columns, beams, smells and thicknesses of slabs and walls

For sizes up to 400 mm	+ 10 mm or – 5 mm
For sizes larger than 400 mm	+ 15 mm or – 10mm

6 - Reinforced footings

Horizontal footing dimensions	+50 mm or -15 mm
Dimensions between axes	+/- 50 mm
Footing thickness	without maximum limit or -2%
Top footing level	+/- 15 mm or -5 mm

7 - Stairs

For a step	
Height	+/- 3 mm
Horizontal distance	+/- 6 mm
For each one flight or summation of flights for one floor	
Height	+/- 5 mm
Horizontal distance	+/- 10 mm

9-8-4 Allowable tolerances in the dimensions of ordinary and high strength steel reinforcement

1 - Allowable tolerances in forming the reinforcement steel shown in Figure (9-1) are given in Table (9-2) for bar diameters between 8 mm and 32mm.

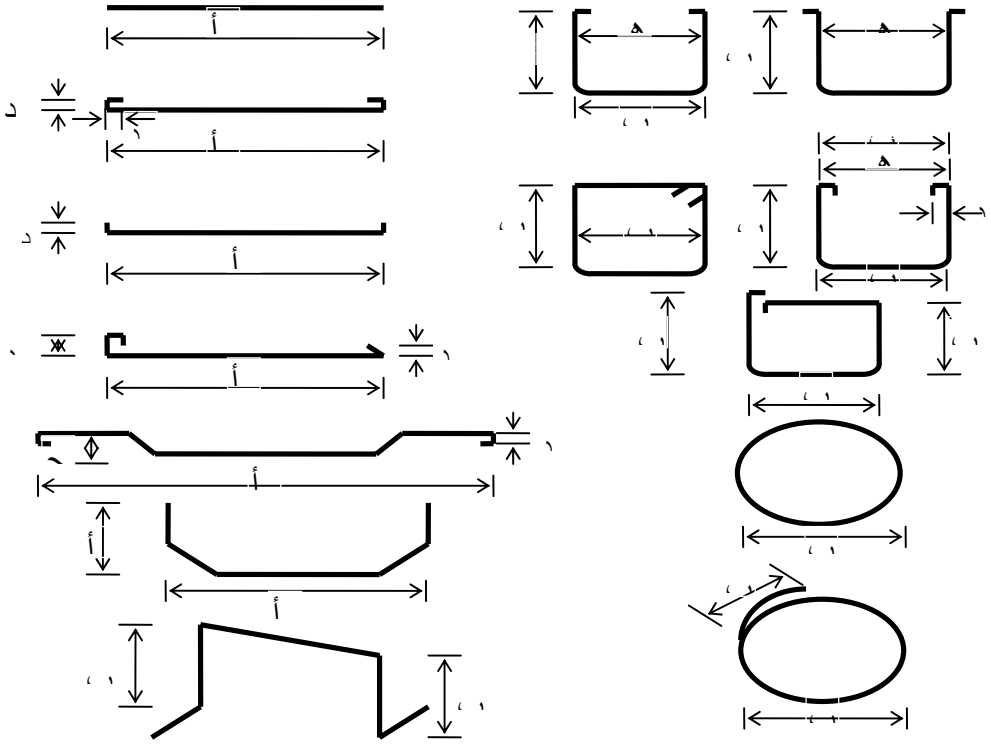


Figure (9-1) Dimensions and deforming of reinforcing bars

Table (9-2) Allowable tolerances in steel reinforcement

Dimension	For sections with thickness less than 250 mm	For sections with thickness more than 250 mm
a	+/- 15 mm	+/- 25 mm
b	+/- 10 mm	+/- 15 mm
c	+/- 8 mm	+/- 12 mm
d	+/- 8 mm	+/- 12 mm
e	It is permissible to have the tolerance for this dimension the same as the one across it with an additional tolerance of +/- 10 mm	

2 - Allowable tolerance in arrangement of reinforcing steel bars

a - Allowable tolerance in depth d

The depth d is the distance between the outside compressive surface and the center of steel reinforcement in tension

Depth d less than 250 mm +/- 10 mm

Depth d greater than 250 mm +/- 15 mm

b - Allowable tolerance to reduce the steel reinforcement concrete cover

Depth d less than 250 mm - 6 mm

Depth d greater than 250 mm - 8 mm

(These values shall not exceed one-third the concrete thickness specified on drawings).

c - Allowable tolerance in reducing the spacing between bars in beams

The tolerance in the spacing between bars in beams must not exceed - 5 mm

d - Allowable tolerance in the spacing between bars

Slabs and walls +/- 20 mm

Stirrups +/- 20 mm

Welded mesh +/- 5 mm

Total number of reinforcing bars per meter must not be less than those given in the construction drawings.

f - Allowable tolerances in bending locations and bar ends in the longitudinal direction

In continuous beams and slabs	+/- 25 mm
Bar ends in beams and slabs on outer edges	+/- 15 mm

g - Allowable tolerance in reducing the length of splices

Tolerance in the length of the splice does not exceed	- 25 mm

h - Allowable tolerance in reducing the length of reinforcement bond length inside the concrete

For bars of diameter from 10 mm to 32 mm	-25 mm
For bars of diameter greater than 32 mm	-50 mm

9-8-5 Allowable tolerance in precast concrete element dimensions

9-8-5-1 Tolerances in the horizontal element length dimensions

Element length up to 3.00 meter	+/- 3mm
Element length from 3.00 meter to 4.5 meter	+/- 5 mm
Element length from 4.5 meter to 6.00 meter	+/- 6 mm
Element length every additional 6.00 meter	+/- 6 mm
Element length exceeding 18.00 meter	+/- 20 mm

9-8-5-2 Tolerances in the dimensions of the element cross section

Element thickness up to 150 mm	+/- 3 mm
Element thickness from 150 mm to 450 mm	+/- 5 mm
Element thickness from 450 mm to 900 mm	+/- 6 mm
Element thickness greater than 900 mm	+/- 10 mm

9-8-5-3 Allowable tolerances in straightness relative to the element length

Element length up to 6.00 meter	+/- 3 mm
Element length from 6.00 meter to 12.00 meter	+/- 6 mm
Element length from 12.00 meter to 18.00 meter	+/- 10 mm
Element length greater than 18.00 meter	+/- 12 mm

9-8-5-4 Allowable tolerances in element convexity camber

Element length up to 3.00 meter	+/- 3 mm
Element length greater than 6.00 meter	+/- 6 mm

9 - 9 Project management

9-9-1 General

Project management based on knowledge and acquired experiences are considered main element in the project success and in fulfilling the purpose from establishing the project. Project management is highly needed as an independent element in projects involving various jobs or specializations required to execute the project.

One of the important functions of project management is fulfilling the project aims. This shall be achieved by studying and determining the most suitable means to reach these goals starting from tendering means, type of contract used, determination of the construction methods to attain the targeted quality; to match the site, project timetable and estimated cost; in order to coordinate these tasks to suit the cash flow to be determined with the project owner.

9-9-2 Project management tasks

The tasks of project management in its different stages are summarized as follows:

9-9-2-1 Design and tender documents preparation stage

At this stage, the project management unit shall be responsible for:

a - Revising architectural, structural, electromechanical and other designs in light of the preliminary project and design recommendations in order to ensure compatibility and suitability of the design for construction. In addition to follow up with completing and updating these designs, (if needed).

b - Revising quantities and specifications and ensuring their compatibility with the drawings.

c - Preparing timetable, cash inflow table according to the quantities and the method of construction proposed by the designer. Specifying work packages to be offered and the most suitable means for contracting according to this test (main contractor, or itemized work contract or fixed value contract, etc….)

d - Projects with special nature that demand contractors with special qualifications, the project management unit shall prepare a pre-qualification list for the contractors to determine a short list of contractors to be called for bidding.

e - Preparing the bidding instructions documents to assist the contractors for preparing the bid in an acceptable form and to simplify the job of the evaluation committees.

f - Preparation of the general and specific requirements documents that show the rights and obligations of both parties of the contract during construction, testing and delivery stages.

9-9-2-2 Bidding Stage

In this stage, the project management department shall participate in:

a - Specifying the bidding method (general or limited bidding).

b - Specifying bases and elements for evaluating companies proposed for bidding.

c - Offering the bid and responding to inquiries.

d - Bid evaluations and negotiating the contractors with the best bids to reach the bid that is most suitable technically and financially.

9-9-2-3 Construction stage :working method for project management

In this stage, the project management unit shall perform the following:

a - Setting the method, foundations, and models for quality assurance and ensuring its applications. This shall guarantee the control of the timetable, data and document flow with preparing the forms and specifying the procedures to be followed when giving instructions to the contractors and suppliers.

b - Revising organizational hierarchy of contractors participating in the project. Also revising the authority and responsibilities of the key personnel of the different parties to avoid an authority conflict or lack of responsibility or its vagueness.

c - Preparing the document cycle and determining the communication channels.

d - Revising the general timetable for execution presented by the contractor that shows the following:

  1 - Type of activity, its duration, dates of early starts and ends and the dates of late start and ends.

  2 - Curves showing the distribution of workers and main equipment in the project over the duration of execution.

e - Revising the timetable presented by the contractor regarding: workshop drawing preparation and their approval; endorsement programs;

equipment and material delivery to the project in accordance with the general timetable for the project execution.

f - Preparation/revision of the project general site planning drawing that shows the locations of the workshops, storage, temporary roads, tower winches, equipment movement, fences, management offices, security, etc.

g - Checking security and safety precautions according to section (9-10).

h - Studying modifications and changes requested by the owner or suggested by contractors or the design engineer or supervision team personnel. Also investigating the abilities to execute these changes and their impact on the different project contracts, its cost and timetable. Making change orders after discussions with the design engineer and getting the owner's approval.

i - Investigating the effect of change orders on the project duration and cost.

j - If there is a claim from the executing company, the project management department shall analyze it and respond to it in coordination with the owner. The project management department shall also be responsible for putting a follow up system for invoice that guarantees checking them in an accurate and efficient manner.

k - Arrange coordination meetings with the project parties to guarantee full coordination and solving problems (if any) as soon as possible and to follow up with all project correspondence between the different parties.

The report shall include the project financial status, important issues to be solved, and problems hindering the execution, ways of solution and how the consultant dealt with these issues. The report shall also include photographs illustrating the work progress, information regarding periodical meetings with all the contractor companies and unresolved issues that have not been resolved. The report shall include the percentage executed to date from the different project tasks compared to the timetable; approval of workshop drawings, material samples, technical publications for equipment and otherwise; equipment delivery program; and updating the anticipated cash flow. The following shall be appended with the report:

1 - Percentage executed from the different project tasks.
2 - Approval of workshop drawings.
3 - Equipment delivery program.
4 - Updating the cash flow.
5 - Anticipated tasks to be executed.
6 - Tables for checking documents.

Remarks:

a - From practical experience, the execution of large projects is probably accompanied with modifications in the planned timetable at the start of work. This is attributed to many reasons including delays in the delivery of some equipment or because the actual rates of execution do not agree with the proposed ones, which requires re-inputting these variables again in the execution timetable and studying their influence during the project execution and taking the necessary correctional procedures.

b - In large projects, project management department establishes a complete information system to control time, documents, and guarantee the contractor meets the timetable and consequently controlling the project cash flow. This shall be achieved by utilizing the available software packages and computers. The project management department shall also concerned with following the critical tasks and providing early warning about any foreseen obstacles or delays.

9-9-2-4 Testing, preliminary and final delivery services

In these stages, the project management shall be responsible for:

a - Obtaining from the contractor and suppliers all operational and maintenance documents of equipment and systems to be delivered to the owner.

b - Preparing a list of defects and incomplete works (Punch List) and specifying the repair times.

c - Preparing a complete set of as-built drawings approved by the engineer of record.

d - Issuing certificate for preliminary delivery after fulfilling all the previous requirements.

e - Final re-evaluation of final cost of modifications, changes, and bills approved by the owner.

f - Issuing final invoice for contractors and consultants after referring to their contracts.

g - Ensuring the repair of all defects that may appear during the guarantee period before issuing the certificate of final delivery and returning the final letter of guarantee to the contractor.

9 – 10 Security and safety for construction of concrete structures

The project environmental impact evaluation shall be prepared as part of the procedures of obtaining different permits. A record of the project

environmental status shall be kept in accordance with addendum (3) to the Environment law number (4) issued 1994, completing all the information and measurements. The environmental precautions for dealing with materials and basic services such as using electricity, water, equipment and handling solid waste shall be considered. The safety and professional health requirements shall be verified in concrete work execution. These include receiving, preparing and equipping the site; material storage; and the design and construction of scaffolds and forms.

CHAPTER 10
PRESTRESSED CONCRETE

10-1 General

10-1-1 Prestressed concrete elements shall be designed in accordance with the provisions of this chapter.

10-1-2 Many systems are used for concrete prestressing. Concrete may be pre-tensioned or post-tensioned. For post-tensioning, prestressing tendons may be bonded or unbonded. Tendons used for unbonded system may be internal or external. For circular or cylindrical elements, circular prestressing is used. For all cases, prestressing may be either full or partial.

10-1-3 Prestressed members shall be designed to resist applied loads and straining actions in accordance with the requirements of ultimate and serviceability limit states at all stages during the life of the structure from the time prestress is first applied.

10-1-4 Prestressed members shall be designed taking into account effect of adjoining structural members and effect of elastic and plastic deformations, deflections, changes in length or loads due to prestressing. Effects of temperature and shrinkage shall also be included.

10-1-5 Safety against buckling of prestressed member or parts thereof such as thin webs and flanges shall be checked.

10-1-6 In calculating section properties, effect of reduction of area due to open ducts of prestressing cables shall be considered.

10-1-7 Refer to section (1-1 scope of code) for structures for which provisions of this chapter could be applied.

10-2 Prestressed concrete materials

10-2-1 Concrete

10-2-1-1 General

Concrete of prestressed concrete structures is characterized by high compressive strength that makes the concrete section less susceptible to volumetric changes due to shrinkage and creep and hence reduces prestress losses in prestressing steel. Use of high strength concrete allows reduction of member weight which, in most cases, represents high percentage of design load in addition to satisfaction of design limit states.

10-2-1-2 Properties of prestressed concrete constituents

Refer to section (2-2) for the properties of the constituents materials of the prestressed concrete.

10-2-1-3 Characteristic strength

The definition and requirements shall comply with the provisions of section (2-3-2-1). In addition, number of tests less than characteristic strength by a maximum of 4 N/mm² shall not exceed 1%. Table (2-6) gives characteristic strength used for prestressed concrete.

10-2-1-4 Compressive strength of standard concrete cube at prestress transfer

Compressive strength of standard concrete cube at prestress transfer shall not be less than the values given in table (10-10).

10-2-2 Reinforcing steel

Different types of steel are used in prestressed concrete structures. Type used shall satisfy all requirements of relevant standard, and shall be evaluated based on required tests performed by a certified laboratory.

10-2-2-1 Prestressing steel

Steel used (such as high strength steel wires (cold drawn),high strength strands and bars and cables) is produced in different shapes in international market. Cables are formed by grouping wires or strands in one path.

10-2-2-2 Mechanical properties of prestressing steel

Mechanical properties of steel such as tensile strength, proof stress, elongation Percentage, and modulus of elasticity shall be verified in accordance with standard limits. Table (10-1) specifies lower limits of proof stress and Elongation percentage.

Reference values of mechanical properties of prestressing steel in some international standards are given in Appendix (2) in this code.

10-2-3 Cement grout

Cement specifications shall conform to Egyptian standards ‘Normal and early hardened Portland cement E.S 373/1991’.Sand shall conform to Egyptian standard ‘Sand for building mortar E.S 1108’ and shall pass from sieve 1.18mm. Admixtures shall conform to adopted Standard Egyptian and international specifications and may be used if confirmed by tests that it enhances grouting quality provided that it does not contain any chlorides, Nitrates or Sulfates.

The strength of the cubes of cement grout shall not be less than 30N/mm² at 28 days.

The constituents and their percentages are chosen to satisfy specific requirements such as consistency and compressive strength at different ages (or stages).

10-3 Design of prestressed concrete members

10-3-1 Design fundamentals

10-3-1-1 Prestressed members are designed for acting loads according to limit state method based on effects on member. Material strength reduction factors shall be taken as given in chapter three of this code; However γs is replaced by γps for prestressing steel.

10-3-1-2 Main loads and general considerations for design of prestressed concrete members subjected to bending or eccentric forces shall be taken as specified in section (4-2-1) with due consideration of the stress-strain relationship of prestressing steel according to section (10-3-1-3).

10-3-1-3 Stress strain relationship for prestressing steel shall be according to the idealized curve shown in figure (10-1).

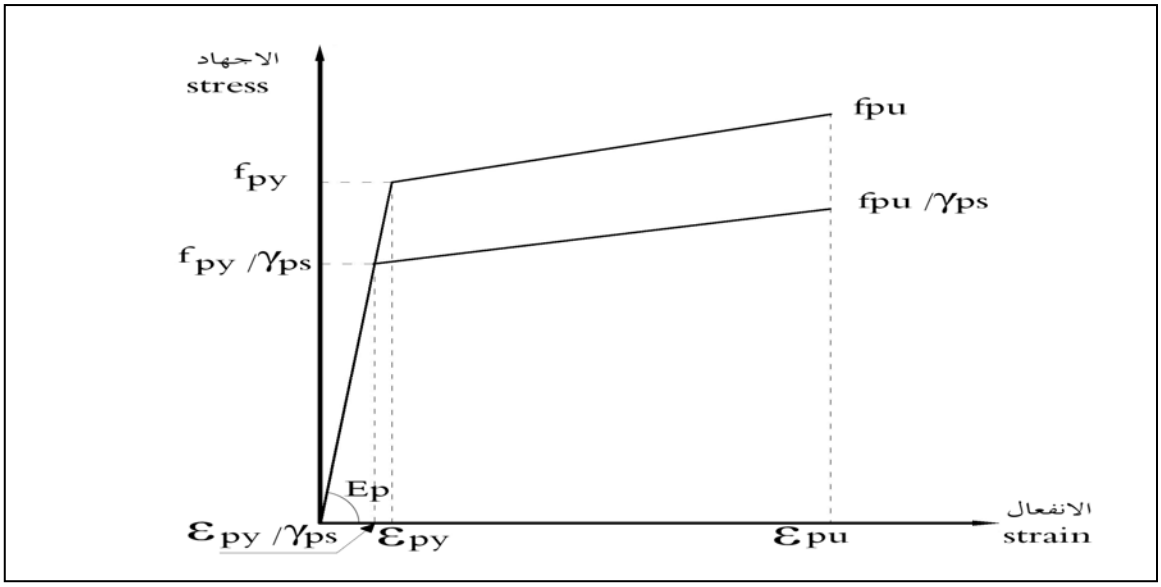


Fig. (10-1) Idealized stress-strain curve for prestressing steel

10-3-1-4 Relationship between ultimate stress (f_{pu}) and yield stress in tension (f_{py}) for prestressing steel shall be taken based on steel type and according to the following relationships:

fpy / fpu = 0.80 for deformed bars (10-1)

fpy / fpu = 0.85 for normal relaxation stress – relieved strands, wires and smooth bars (10-2)

fpy / fpu = 0.90 for low relaxation stress-relieved strands and wires (10-3)

10-3-2 Serviceability limit state requirements

10-3-2-1 Allowable stresses in concrete

10-3-2-1-1 Elastic analysis of section shall be used to check stresses at transfer of prestressing forces to concrete, at service loads, and at cracking load.

10-3-2-1-2 Prestressed concrete members are classified according to the following:

Case (A): "Full prestressing": uncracked members with no tensile stress. The following members shall be designed accordingly;

- Sections subjected to repeated or dynamic loads.
- Sections with tension surface subjected to injurious oxidizing effects causing steel rusting (section four according to table 4-11).

Case (B): Uncracked sections with tensile stresses under the effect of all loads less than the values specified in table (10-2).The following members shall be designed accordingly:

- Solid and flat slabs
- Members with unbonded prestressing steel.
- Members with tension surface subjected to injurious effects(section 3 according to table 4-11)

Case (C): Transient state cracked and uncracked sections in which the tensile stresses shall be less than cracking limit state according to equation (4-61-b) and less than 4N/mm² but shall be higher than maximum permissible tensile stress in case (b).

Case (D): "Partial prestressing" with cracked sections for members with maximum nominal tensile stress under all acting loads higher than cracking limit state but less than $0.85\sqrt{f_{cu}}$ Nominal tensile stresses shall be calculated using full section properties, without considering cracking effects, and neglecting steel reinforcement. Tensile stresses in concrete under permanent loads that include dead loads and sustainable live loads shall be checked to be less than cracking limit state.

For cases (C) and (D), deformed high strength steel reinforcement shall be added to resist tension in concrete section due to service loads.

10-3-2-1-3 Allowable stresses in concrete for sections subjected to flexure and axial compression are as in table (10-2).

Table (10-2) Allowable stresses in concrete (N/mm^2)

Item	Cases	
1	Allowable stresses in concrete due to bending moments immediately after transfer of forces to concrete (and before time-dependant losses) shall not be higher than the following values:-	
	- maximum stress in compression	$0.45f_{cui}$
	- maximum tensile stress except at end of simple beams	$0.22\sqrt{f_{cui}}$
	- maximum tensile stress at end of simple beams	$0.44\sqrt{f_{cui}}$
2	Allowable stress in concrete due to bending moments at service loads (after time-dependant losses) shall not be higher than the following values:-	
	Maximum compression due to prestressing and dead loads	$0.35f_{cu}$
	Maximum compression due to prestressing and all loads	$0.40f_{cu}$
	Maximum tension in pre-compressed tensile zone due to prestressing and all loads	Case (A): zero Case (B): $0.44\sqrt{f_{cu}}$ Case (C): $0.6\sqrt{f_{cu}}$ but not higher than $4N/mm^2$ Case (D): $0.85\sqrt{f_{cu}}$
3	Maximum stresses in concrete due to axial compression	
	Maximum compression	$0.25f_{cu}$

Where;

f_{cui} = Characteristic compressive strength of concrete at transfer of prestressing

f_{cu} = Characteristic compressive strength of concrete at service

10-3-2-2 Allowable stress in prestressing steel

Allowable stresses in prestressing steel shall be taken according to table (10-3)

Table (10-3) Allowable Stresses in prestressing steel

a- Due to jacking force*	$0.9f_{py} \le 0.75f_{pu}$
b- In case of rebars at tensioning	$0.7f_{pu}$
c- Immediately after prestress transfer	$0.8f_{py} \le 0.7f_{pu}$
d- Post-tensioning tendons, at anchorage devices and couplers	$0.8f_{py} \le 0.7f_{pu}$

* shall not exceed recommended design values by manufacturer of cables or anchorages.

10-3-2-3 Limit state of deflection

10-3-2-3-1 In calculating immediate deflection of structural elements:-

1 - Elastic theory with gross moment of inertia Ig shall be used for cases (a), (b), and (c) in section 10-3-2-1

2 - For case (d) in section 10-3-2-1 effective moment of inertia I_e according to equation (4-60) shall be used taking into account effect of prestressing in calculating cracking moment(M_{cr}) according to equation (10-7b).

10-3-2-3-2 Long term deflection shall be calculated taking into account stresses in concrete and prestressing steel (after calculating all losses) under all permanent loads in addition to effects of shrinkage, concrete creep and relaxation of prestressing steel.

10-3-2-3-3 calculated deflections shall not exceed the limits specified in section (4-3-1-2).

10-3-2-3-4 Elastic theory with gross moment of inertia (Ig) shall be used in calculating critical camber. Calculated camber shall not exceed values affecting negatively building usage or its structural or non-structural elements.

10-3-3 Requirements of ultimate limit state

10-3-3-1 Sections subjected to flexure

10-3-3-1-1 Limit state method in section (10-3-1) shall be used to determine ultimate moment limit state for prestressed concrete sections taking into account the distribution of stress on section as shown in figure (10-2).

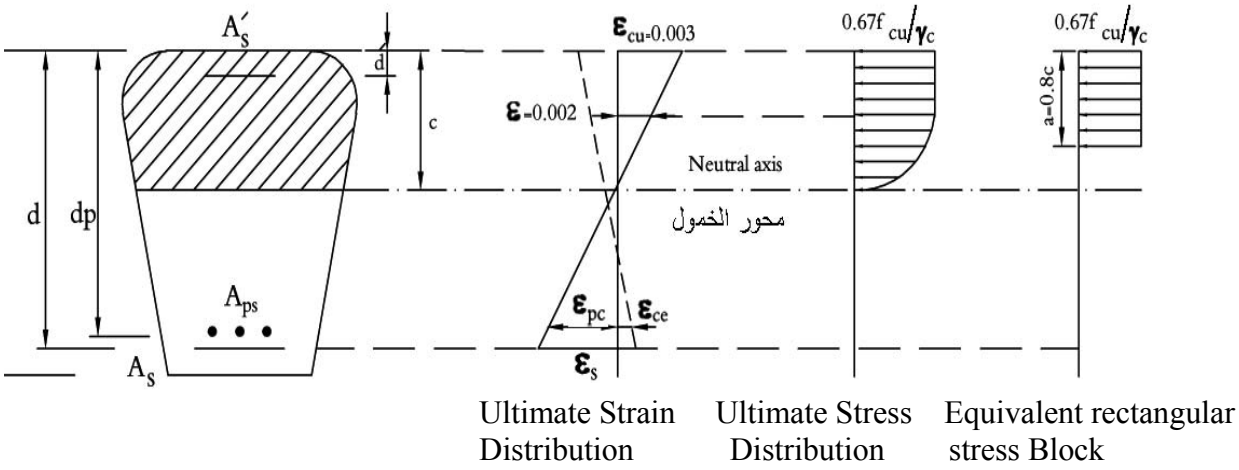


Fig. (10-2) Ultimate Flexural Strength

10-3-3-1-2 Total strain in bonded prestressing steel ε_{ps} shall be calculated from the following equation:

$$\varepsilon_{ps} = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_{pc} \qquad (10\text{-}4)$$

Where:

ε_{ps} = strain in prestressing steel due to prestressing after considering all losses.

ε_{ce} = strain in concrete at level of prestressing after considering all losses.

ε_{pc} = strain in prestressing steel due to strain compatibility at ultimate moment limit state.

10-3-3-1-3 Stress in prestressing steel f_{ps} at ultimate moment limit state of section shall be calculated based on total strain given by equation (10-4) and idealized curve of prestressing steel given in figure (10-1).

10-3-3-1-4 Section with tension prestressing steel reinforcement only:
Ultimate moment limit state for rectangular section with tension prestressing steel reinforcement only shall be calculated using the following equation:-

$$M_u = A_{ps}\left(\frac{f_{ps}}{\gamma_{ps}}\right)\left(d_p - \frac{a}{2}\right) \qquad (10\text{-}5\text{-a})$$

Where d_p is the distance from extreme compression fiber to centroid of prestressing steel.

When using additional normal steel reinforcement in tension side, the ultimate limit moment of section shall be calculated using the following equation:

$$M_u = A_{ps}\left(\frac{f_{ps}}{\gamma_{ps}}\right)\left(d_p - \frac{a}{2}\right) + A_s\left(\frac{f_y}{\gamma_s}\right)\left(d - \frac{a}{2}\right) \qquad (10\text{-}5\text{-b})$$

Effect of steel reinforcement on ultimate moment limit of section shall be considered after determining its stress using equilibrium and strain compatibility.

10-3-3-1-5 Stress in prestressing steel f_{ps} can be calculated using the following approximate equations:

10-3-3-1-5-a As an alternative to the method stated in section (10-3-3-1-2,3), stress in prestressing steel f_{ps} shall be calculated on condition that stress in prestressing steel after considering all losses(f_{pe}) shall not be less than half ultimate stress of prestressing steel ($0.5f_{pu}$).

1 - For sections with bonded prestressing tendons that contains both compression and tension steel reinforcement

$$f_{ps} = f_{pu}\left[1 - \eta_p\left(\mu_p \frac{f_{pu}}{0.8f_{cu}} + \frac{d}{dp}(w - w')\right)\right] \qquad (10\text{-}6)$$

Where:

η_p = coefficient depending on steel type and shall be taken as follows:
0.68 for (f_{py}/f_{pu} not less than 0.8)

0.50 for (f_{py}/f_{pu} not less than 0.85)
0.35 for (f_{py}/f_{pu} not less than 0.90)

μ_p = percentage of prestressing steel in concrete section = $\frac{Aps}{b.dp}$.

ω = tension steel reinforcement ratio in concrete section multiplied by ratio of yield stress of steel reinforcement over characteristic compressive strength of concrete = $\mu \frac{f_y}{0.8f_{cu}}$

ω' = compression steel reinforcement ratio in concrete section multiplied by ratio of yield stress of steel reinforcement over characteristic compressive strength of concrete = $\mu' \frac{As'}{0.8\,f_{cu}}$

Where: $\mu = \frac{As}{b.d}$, $\mu' = \frac{As'}{b.d}$

b = width of section in rectangular sections.

2 - To take into account effect of compression reinforcement in calculating ultimate limit moment M_u, the value calculated from equation (10-7) shall not be less than 0.17 when substituting in equation (10-6). In addition, d' shall not exceed 0.15dp as it shall be assumed that strain in compression steel shall be equal to or exceed yield strain.

$$\left[\left(\frac{\mu_p .f_{pu}}{0.8f_{cu}} + \frac{d}{d_p}(w - w')\right)\right] \geq 0.17 \qquad (10\text{-}7)$$

10-3-3-1-5-b For prestressed members with unbonded prestressing tendons where the span to effective depth ratio shall not exceed 35, stress in pre stressing steel at ultimate limit moment of the section M_u shall be calculated from the following equation:-

$$f_{ps} = f_{pe} + 70 + \left(\frac{0.8f_{cu}}{100\,\mu_p}\right) \qquad N/mm^2 \qquad (10\text{-}8)$$

f_{ps} Shall not exceed f_{py} or (f_{pe}+420) whichever is less; stress is in N/mm^2

10-3-3-1-5-c For prestressed members with unbonded prestressing tendons where span to effective depth ratio is greater than 35, stress in pre stressing steel at ultimate limit moment shall be calculated from the following equation:

$$f_{ps} = f_{pe} + 70 + \left(\frac{0.80 f_{cu}}{300\,\mu_p} \right) \qquad N/mm^2 \qquad (10\text{-}9)$$

f_{ps} Shall not exceed f_{py} or (f_{pe}+200) whichever is less; stress is in N/mm^2

10-3-3-1-6 Upper limit of areas of prestressing steel and normal steel reinforcement.

10-3-3-1-6-a prestressing steel ratio μ_p and normal steel reinforcement ratio μ in the section are taken to satisfy equation (10-10) unless ultimate strength shall be calculated according to section (10-3-3-1-6-b)

$$w_p \le 0.28 \qquad (10\text{-}10a)$$

$$\left[w_p + \left(\frac{d}{d_p} \right)(w - w') \right] \le 0.28 \qquad (10\text{-}10b)$$

$$\left[w_{pw} + \left(\frac{d}{d_p} \right)(w_w - w'_w) \right] \le 0.28 \qquad (10\text{-}10c)$$

Where for rectangular section with width b

$$\omega_p = \mu_p \frac{f_{ps}}{0.8 f_{eu}} = \frac{A_{sp}.f_{ps}}{b.dp.0.8 f_{cu}}$$

w'_w, w_w, w_{wp} are reinforcement coefficients for sections with compression flange similar to w', w, w_p for rectangular sections; with the use of rib width b and reinforcement area sufficient to develop total compression strength of rib.

10-3-3-1-6b when using steel ratios higher than that specified in section (10-3-3-1-6 a), ultimate strength of concrete section shall be calculated using strain compatibility analysis

10-3-3-1-6c Minimum prestressing steel and normal steel reinforcement ratios shall be calculated to insure ultimate strength of section higher than 1.2 the cracking limit calculated using concrete tensile strength f_{ctr} according to section (4-3-1-4). This condition shall not be applyed for the following cases:-

a - Slabs with unbounded post-tensioning tendons

b - Elements subjected to bending moments with shear and moment strengths greater than double the required values.

10-3-3-1-7 Minimum bonded non-prestressed reinforced ratio in members with unbonded prestressing steel.

Bonded non-prestressed steel shall be provided in structural members with unbonded prestressing steel as specified in (10-3-3-1-7 a)and (10-3-3-1-7 b).

10-3-3-1-7a Bonded non-prestressed reinforcement ratio in members with unbonded prestressing steel shall not be less than.

$$A_s = 0.004\ A \qquad (10\text{-}11a)$$

Where;

A = cross sectional area of the part between tension face and center of gravity of gross section. The bonded steel is uniformly distributed as close as possible to the concrete parts subjected to maximum tension due to external loads

10-3-3-1-7-b For two-way slabs and flat slabs with constant depth, the minimum bounded non-prestressed steel is as follows:

1 -In positive moment area, minimum bonded non-prestressed steel in section shall satisfy the following equation

$$A_s = \frac{N_c}{0.5\, f_y} \qquad (10\text{-}11b)$$

Where;

N_c is the tension force resulting from service loads (dead and live). Steel reinforcement shall be uniformly distributed in pre-compressed tensile zone and to be as close as possible to the parts subjected to maximum tensile stresses. f_y shall not exceed 400 N/mm^2.

2 - In negative moment area at columns, bonded steel reinforcement shall be provided not less than:

$$A_S = 0.00075\ t_S\ L \qquad (10\text{-}11c)$$

Where;

t_s = slab thickness

L = span in direction parallel to required steel reinforcement.

These steel reinforcement shall be distributed on slab width equal to $(c + 3t_s)$ where c is the column width. At least 4 bars are provided in each direction. The distance between bars shall not exceed 300 mm.

10-3-3-1-8 In prestressed beams, side bars shall be provided with maximum spacing not exceeding 300 mm.

10-3-3-2 Development length and transfer length for prestressing steel.

The development length (L_d) for (three or seven wire prestressing strands) shall be calculated using the following equation:

$$L_d = L_t + L_a = \left(f_{ps} - \frac{2}{3} f_{pe} \right) \frac{\phi}{7} \quad \text{mm} \qquad (10\text{-}12)$$

Where;

L_t is the transfer length and shall be calculated as follows:

$$L_t = \left(\frac{f_{pe}}{3} \right) \frac{\phi}{7} \quad \text{mm} \qquad (10\text{-}13\text{-}a)$$

Where;

L_a is the length beyond critical section and is given by:

$$L_a = \left(f_{ps} - f_{pe}\right) \frac{\varphi}{7} \quad \text{mm} \quad (10\text{-}13\text{-}b)$$

Where;

ϕ is the tendons diameter in mm, f_{ps} and f_{pe} in N/mm^2

10-3-3-3 Shear

10-3-3-3-1 For prestressed beams in case of direct support under beam where, due to this bearing, compression perpendicular to lower edge of the beam develops, the effective shear stress shall be permitted to be calculated at a distance equal to half depth of the beam (t/2) from internal face of support, or at first change of web width, whichever is more critical.

10-3-3-3-2 Nominal shear strength

a - Ultimate shear stress shall be calculated from the following relationship:

$$q_u = \frac{Q_u}{bd_p} \quad (10\text{-}14)$$

Where;

Q_u is the ultimate shear force due to permanent and live loads. The effective depth d_p is the distance from extreme compression fiber to centroid of prestressing steel or 0.8 t whichever is greater. The effect of openings in the element shall be considered.

b - The nominal shear stress for prestressed concrete members subjected to shear forces with or without torsional moment shall not exceed the following value:

$$q_{u\max}{}^{*} = 0.75\sqrt{\frac{f_{cu}}{\gamma_c}} \quad (10\text{-}15)$$

With upper limit $4.50\ N/mm^2$.

10-3-3-3-3 Nominal shear strength provided by concrete

a - In members with effective prestressing force exceeding 40% of strength of flexural reinforcement, and when no more detailed

calculations according to item (b) are performed, q_{cu} can be approximately calculated as follows:

$$q_{cu} = \left(0.045\sqrt{\frac{f_{cu}}{\gamma_c}} + \frac{3.6 Q_u . d_p}{M_u} \right) \ N/mm^2 \qquad (10\text{-}16)$$

The value of q_{cu} shall not be less than $0.24\sqrt{f_{cu}/\gamma_c}$ and shall not be greater than $0.375\sqrt{f_{cu}/\gamma_c}$; also the value of $Q_u . d_p / M_u$ shall not exceed one where M_u is the ultimate moment at critical section in shear.

b - Nominal shear strength provided by concrete q_{cu} shall be calculated according to items (b-1), (b-2). The value of q_{cu} is the lower of the two values $q_{ci}, q_{c\omega}$.

b-1 Shear strength q_{ci} shall be calculated from the following equation:

$$q_{ci} = 0.045 \sqrt{\frac{f_{cu}}{\gamma_c}} + 0.80\left(q_d + q_i \frac{M_{cr}}{M_{max}} \right) N/mm^2 \qquad (10\text{-}17a)$$

q_{ci} Shall not be less than $0.24\sqrt{f_{cu}/\gamma_c}$ where f_{cu} in N/mm^2.

Where:

M_{max} = ultimate moment at section due to externally applied loads.

q_i = stress due to ultimate shear force at section due to externally applied loads occurring simultaneously with M_{max}.

q_d = shear stress due to service dead loads i.e. without using load factors.

M_{cr} = the bending moment causing first cracking in concrete and shall be determined from following relation:

$$M_{cr} = \left(\frac{I}{y_t}\right) (0.45\sqrt{f_{cu}} + f_{pce} \quad f_{cd}) \qquad (10\text{-}17b)$$

Where:

f_{cd} =stress due to un-factored dead loads at extreme fiber of section where tensile stress is caused by externally applied loads.
f_{pce} =compressive stress in concrete due to effective prestress forces only (after allowance for all prestress losses) at extreme fiber of section where tensile stress is caused by externally applied loads.
y_t =distance from centroidal axis of gross section, neglecting normal steel and prestressing steel reinforcement, to extreme fiber in tension.
I =moment of inertia of full concrete section, neglecting effect of normal steel and prestressing steel reinforcement

b-2 Shear strength q_{cw} shall be calculated from the following equation:

$$q_{cw} = 0.24\left(\sqrt{\frac{f_{cu}}{\gamma_c}} + f_{pcc}\right) + q_{pv} \ \text{N/mm}^2 \qquad (10\text{-}18)$$

Where;

q_{pv} = shear stress due to vertical component of pre-stress force after allowance for all pre-stress losses = $\dfrac{f_{pe} A_p \sin\beta}{\gamma_{ps} A_c}$

f_{pcc} = compressive stress in concrete (after allowance for all prestress losses) at centroid of section or at the junction of web and flange when the centroid lies inside the flange.

β = angle of inclination of tendon along longitudinal axis of the beam.

As an alternative, q_{cw} can be taken equal to the stress due to shearing force due to dead and live loads causing principal tensile stresses equal to $0.25\sqrt{f_{cu}}$ at axis of element or at junction of web and flange when centroid of element lies inside flange.

b-3 For pre-tensioned members, in which, distance t/2 from face of support is less than transfer length calculated from equations (10-13), the corresponding prestressing force when calculating q_{cw} shall be taken on the basis that prestressing force

increases linearly from zero at tendon end to maximum force at distance equal to ℓ_t.

b-4 In pre-tensioned members where bonding of cables does not extend to end of member, reduced value of prestressing force shall be considered when calculating shear stress according to equation (10-16a), (10-18). The prestressing force shall be assumed to vary linearly from zero at cable end to maximum value at distance equal to transfer length ℓ_t.

b-5 For calculating shear strength of webs containing grouted ducts with diameter ϕ exceeding $\frac{b_w}{8}$ where b_w is the web width, the effective web width shall be assumed equal to $(b_w - 0.5\sum\varphi)$ where $\sum\varphi$ is the sum of duct diameters at the level with maximum number of ducts.

10-3-3-3-4 Shear strength provided by shear reinforcement

If calculated ultimate shear stress q_u due to applied shear force on section due to section (10-3-3-3-2) exceeds the nominal concrete strength q_{cu}, then web reinforcement shall be used according to section (4-2-2-1-4). The contribution of web reinforcement shall be calculated according to equation (10-19)

$$q_{su} = q_u - 0.50\, q_{cu} \qquad (10\text{-}19)$$

10-3-3-4 Torsion

10-3-3-4-1 Critical sections for torsion shall be specified according to section (4-2-3-1)

10-3-3-4-2 Ultimate shear stress due to torsional moment q_{tu} shall be specified according to section (4-2-3-2).

10-3-3-4-3 Effect of torsion may be neglected if shear stress due to torsion is less than that calculated from the following equation:

$$q_{tu} = 0.06\sqrt{\frac{f_{cu}}{\gamma_c}}\sqrt{1+\left(\frac{f_{pcc}}{0.25\sqrt{f_{cu}}}\right)} \quad N/mm^2 \qquad (10\text{-}20a)$$

10-3-3-4-4 Concrete dimensions of prestressed concrete sections subjected to shear and torsion and reinforced with longitudinal steel shall satisfy the following relation:

In solid sections $$\sqrt{(q_u)^2 + (q_{tu})^2} \leq q_{u\,max}^{*} \qquad (10\text{-}20b)$$

In box sections $$q_u + q_{tu} \leq q_{u\,max}^{*} \qquad (10\text{-}20c)$$

Where $q_{u\,max}$ is the permissible ultimate shear stress in prestressed concrete sections calculated from equation (10-15)

10-3-3-4-5 Reinforcement required to resist shear stresses due to torsion and shear.

a. If stresses q_{tu} calculated from section(10-3-3-4-2) exceed the value calculated from equation (10-20) section (10-3-3-4-3) and the concrete dimensions of the section satisfy the requirements of section (10-3-3-4-4), then reinforcement to resist torsion consisting of closed stirrups and longitudinal steel shall be used in addition to any reinforcement required for bending moments, axial forces and shear forces according to table (4-5).

b. Lateral reinforcement steel area shall be required to resist torsion in the form of closed stirrups or welded mesh. The area of the stirrup leg in section shall be calculated as follows:

$$A_{str} = \frac{M_{tu} \cdot s}{2\,A_o \left(\frac{f_{yst}}{\gamma_s}\right) \cot\theta} \qquad (10\text{-}21)$$

In rectangular section, equation (10-21) becomes:

$$A_{str} = \frac{M_{tu} \cdot s}{1.7\,(x_1\ y_1) \left(\frac{f_{yst}}{\gamma_s}\right) \cot\theta} \qquad (10\text{-}22)$$

All remarks in section (4-2-3-5) shall be considered.

The angle θ is as follows:

θ = 45 for cases where effective prestressing force is less than 40 % of the tensile strength of flexural reinforcement.

θ = 37.5 for cases where effective prestressing force exceeds 40 % of the tensile strength of flexural reinforcement.

c. Additional longitudinal steel A_{sl}

Area of additional longitudinal steel shall be determined from:

$$A_{sl} = \left(\frac{A_{str} \cdot p_h}{s}\right)\left(\frac{f_{yst}}{f_y}\right)\cot^2\theta \qquad (10\text{-}23)$$

Area of additional longitudinal steel shall not be less than:

$$A_{sl\,min} = \frac{0.4\sqrt{\frac{f_{cu}}{\gamma_c}}\,A_{cp}}{f_y/\gamma_s} - \left(\frac{A_{str}}{s}\right)p_h\left(\frac{f_{yst}}{f_y}\right) \qquad (10\text{-}24)$$

The value of $\frac{A_{str}}{s}$ shall not be less than $\frac{1}{6}\frac{b}{fy_{st}}$

Where f_{cu}, fy, fy_{st} in N/mm^2.

And A_{cp}, p_h as in section (4-2-3-6). In addition, all requirements of section (4-2-3-5) shall be considered.

10-3-3-4-6 In statistically indeterminate structures where torsion is not necessary for equilibrium and is due to compatibility, the ultimate torsional moments in prestressed beams can be reduced to the following value:

$$M_{tu} = 0.316\left(\frac{A^2_{cp}}{p_{cp}}\right)\sqrt{\frac{f_{cu}}{\gamma_c}}\sqrt{1+\left(\frac{f_{pcc}}{0.25\sqrt{f_{cu}}}\right)} \qquad (10\text{-}25)$$

Where P_{cp} as in section (4-2-3-6)

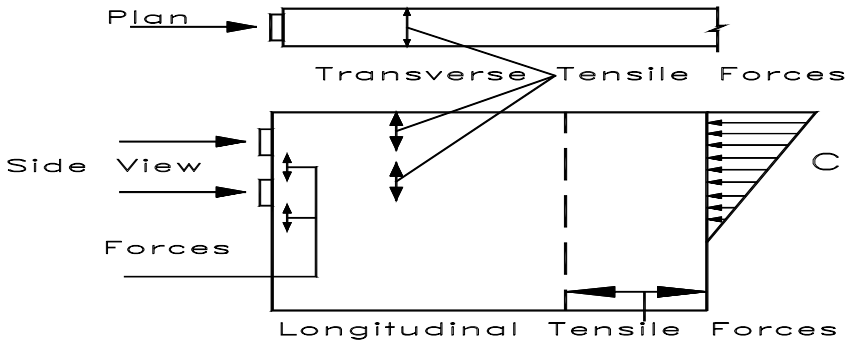
10-3-3-5 Design of anchorage zone

10-3-3-5-1 Anchorage zone

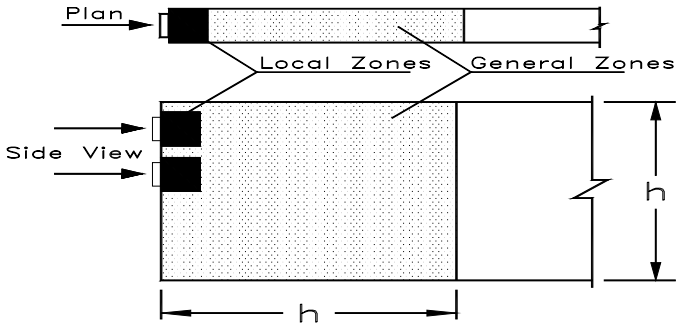
Anchorage zone consists of two zones:

a. Local zone: is the rectangular prism (or equivalent rectangular prism for circular or elliptical anchorage) for concrete surrounding anchorage and any confining anchorage (figure 10-3)

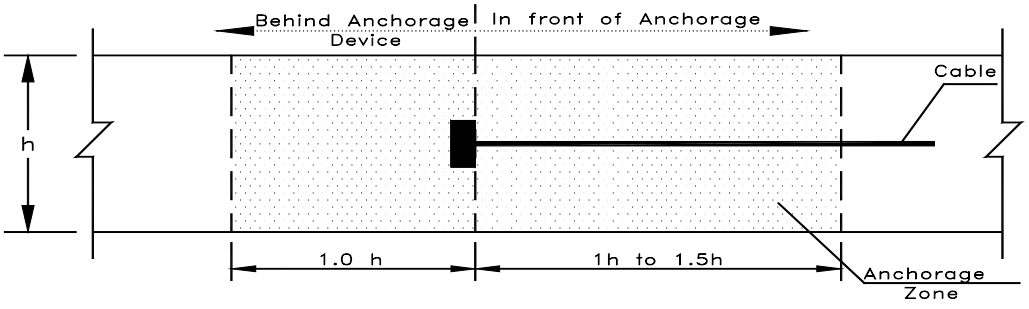
b. General zone: is part of the element through which concentrated prestressing forces are transferred to concrete with more uniform distribution and its length is longer than the greatest cross sectional dimension (figure 10-3).



A- Tension Zones



B- Local and General Zones



C- General Anchorage Zones away from Beam Ends

Fig. (10-3) Anchorage Zones

10-3-3-5-2 Design requirements

a. Design shall be performed using prestressing force equal 1.2 the jacking force and capacity reduction factor according to section (10-3-1-1).

b. Characteristic compressive concrete strength at tensioning (f_{cui}) shall not be less than the values specified in table (10-2)

10-3-3-5-3 Design methods

10-3-3-5-3-1 Local zone

Local zone shall be designed for bearing strength according to section (4-2-4)

10-3-3-5-3-2 General zone

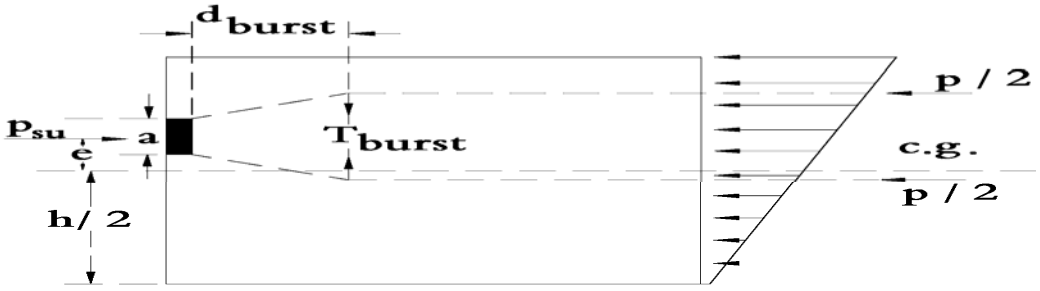
1. Ultimate near uniform compressive stresses at end of general zone at tensioning shall not exceed ($0.56 f_{cui} / \gamma_c$)

2. General zone can be designed by any of the two following methods:

   a. Elastic theory using finite element method or equivalent

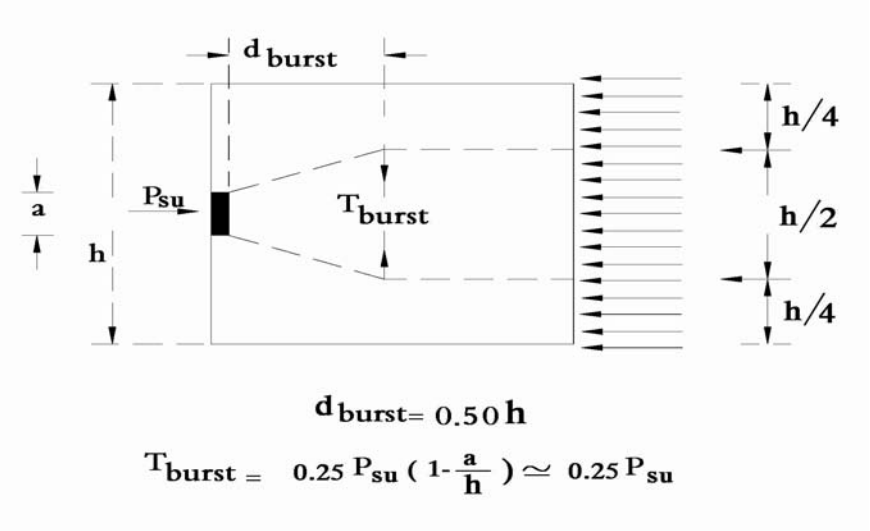
   b. Strut and tie model method according to section (6-11)

3. For developing strut and tie model, all forces acting on general zone shall be considered. In cases where vertical forces can be neglected, strut and tie model can be developed for prestressing forces only, provided that shear requirements of section (10-3-3-3) shall be satisfied. Figure (10-4) shows illustrative strut and tie models for different cases, from which lateral tensile force (T_{burst}) due to prestressing and its location (d_{burst}) can be calculated. Steel reinforcement to resist T_{burst} shall be used. The sufficient anchorage length as well as steel end bends shall be checked for safe transfer of force.



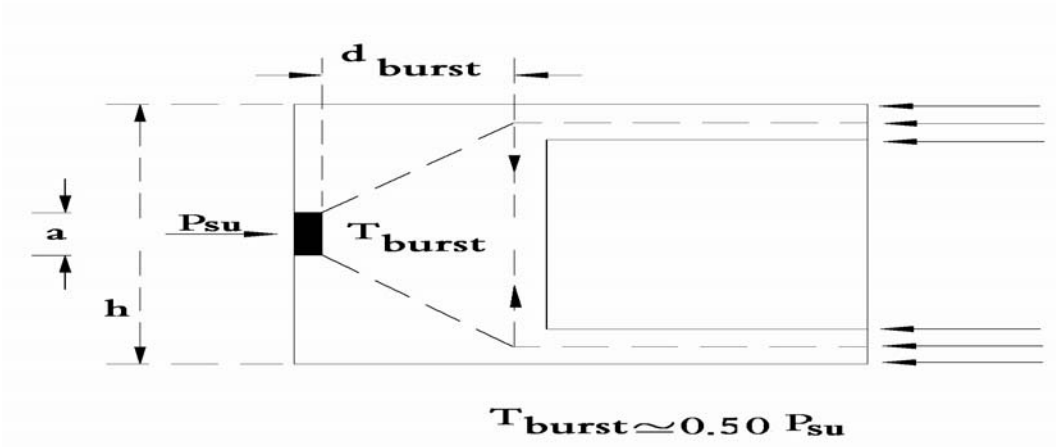
$$T_{burst} = 0.25\, P_{su}\, (1 - \frac{a}{h})$$
$$d_{burst} = 0.50\,(h - 2e)$$

(A) Rectangular Beams with Eccentric Compressive Force



$$d_{burst} = 0.50\,h$$
$$T_{burst} = 0.25\, P_{su}\, (1 - \frac{a}{h}) \simeq 0.25\, P_{su}$$

(B) Rectangular Beams with Eccentric Compressive Force



$$T_{burst} \simeq 0.50\, P_{su}$$

(C) I- Shaped Beam with Rectangular section at anchorage zone

Fig. (10-4) Typical Cases for Strut -Tie Models for Anchorage Zones

Where:

P_{su} = Sum of the prestressing forces
a = Width of the anchorage plate
e = Eccentricity of the prestressing force
h = Thickness of the section

10-3-3-6 Post-tensioned tendon anchorage zone

a. Dimensions of end plates in post-tensioned elements shall be determined to satisfy ultimate limit state of bearing according to section (4-2-4)

b. Steel reinforcement shall be provided in anchorage zones to resist splitting and spalling forces resulting due to tendons anchorage and shall be calculated from elastic theory.

c. For calculating forces and reinforcement in anchorage zone, maximum jacking force shall be considered.

10-3-3-7 Sections subject to concentric forces and bending moments.

Sections subjected to eccentric forces shall be designed using limit state method according to section (4-2-1). Design shall satisfy equilibrium and strain compatibility. Stresses in prestressing steel shall be considered.

10-3-4 Prestress losses

10-3-4-1 General

10-3-4-1-1 Loss of prestress affects the behavior of prestress members under service loads. The loss of prestress is divided into two groups.

a. Immediate loss of prestress due to:
    1. Anchorage slip at transfer
    2. Elastic shortening of concrete
    3. Friction

b. Time dependant losses due to:
    1. Shrinkage of concrete
    2. Creep of concrete

Relaxation of prestressing steel stress.

10-3-4-1-2 When prestressing is applied in stages, losses due to different prestressing stages shall be considered up to final prestressing stage.

10-3-4-2 Immediate loss of prestress

10-3-4-2-1 Anchorage slip losses

Effect of slip of prestressing steel at end anchorages shall be considered in calculating prestress loss Δp. Reference shall be made to documented data of prestressing system supplier when calculating this loss or its extended effect (x_o) along member length (Figure 10-5)

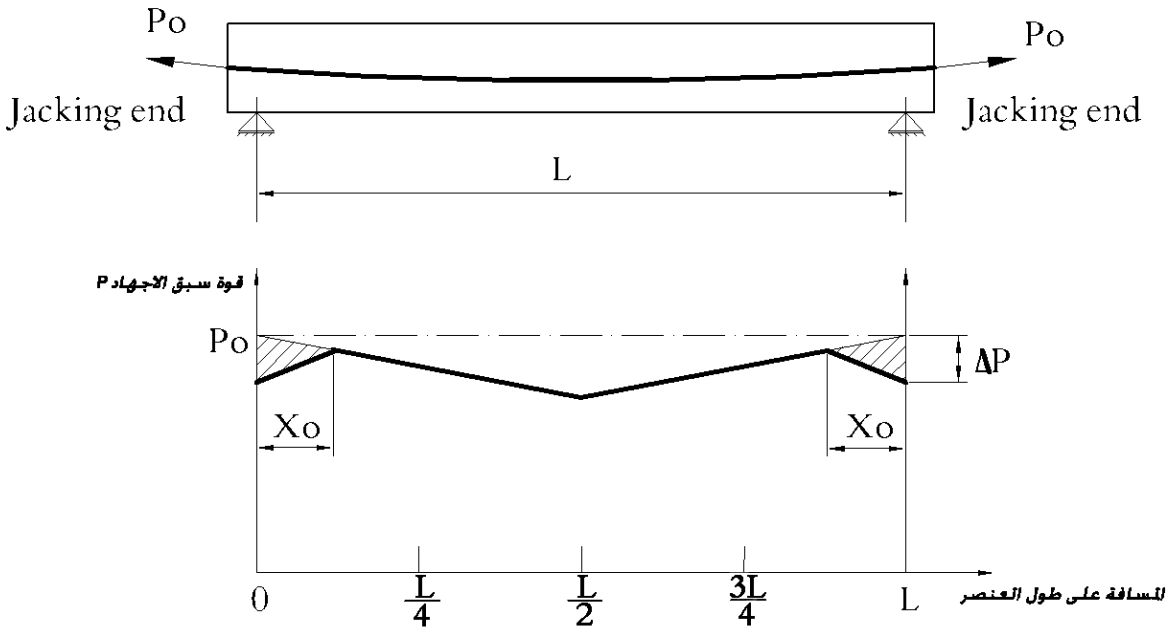


Figure (10-5) Anchorage Slip Losses of Prestressing forces at Anchorage Ends

10-3-4-2-2 Elastic shortening losses

For calculating pre-tensioning and post-tensioning prestress losses, effect of elastic shortening of concrete elements shall be considered as follows:

a. In pre-tensioning, loss in prestress shall be calculated from the following equation:

$$\Delta f_{pe} = \frac{E_p}{E_{ci}} f_{pci} \qquad (10\text{-}26)$$

Where

Δf_{pe} = loss of prestress due to elastic shortening

E_p = modulus of elasticity for prestressing steel.

E_{ci} = modulus of elasticity of concrete at prestressing.

f_{pci} = Initial stresses in concrete at level of prestressing steel before occurrence of time-dependent losses.

b. In post-tensioning when prestressing is performed at once in one stage, the loss of prestress is equal to zero. For sequential prestressing, Effect of sequence of prestressing can be taken into consideration, approximately, by the following equation:-

$$\Delta f_{pe} = \frac{1}{2} \frac{E_p}{E_{ci}} f_{pci} \qquad (10\text{-}27)$$

10-3-4-2-3 Friction losses

10-3-4-2-3-1 Jack internal frictional losses

Effect of internal friction of the jack used in prestressing shall be considered in calculating prestress losses. The value of this loss shall be calculated based on certified data from jack manufacturer.

10-3-4-2-3-2 Wobble friction losses

Prestress loss due to wobble in ducts of prestressing steel shall be calculated from the following equation.

$$P_x = P_o \, . \, e^{-kx} \qquad (10\text{-}28)$$

Where:

P_o = Prestressing force at jacking end.

X = distance from jacking end in meters (figure 10-6)

P_x = Tension force in prestressing steel at distance x from jacking end.

K = Wobble coefficient per meter of prestress; it depends on type, texture of internal surface of ducts, forms erection method, and intensity of vibration during casting.

It can be assumed as follows:-

0.0033 per meter of cable for normal cases

0.0017 per meter of cable for rigid ducts tightly fixed to the forms.

10-3-4-2-3-3 Curvature friction losses

a. Prestress loss due to friction of prestressing steel with enclosing ducts due to curvature of ducts shall be calculated from the following equation:-

$$P_x = P_o \cdot e^{\frac{-\mu \cdot x}{r_{ps}}} \qquad (10\text{-}29)$$

Where:

r_{ps} = radius of curvature of ducts enclosing prestressing steel

μ = coefficient of friction which can be assumed as follows:-

0.55 for friction of steel with hardened concrete.

0.30 for friction of steel with steel.

0.25 for friction of steel with lead.

b. For cases satisfying the following condition

$$\left(\frac{\mu.x}{\gamma_{ps}}\right) \le 0.2$$

Equation (10-29) can be simplified to :

$$P_x = P_o \left(1 - \frac{\mu \cdot x}{r_{ps}}\right) \qquad (10\text{-}30)$$

c. For cases satisfying the following condition:-

$$\left(kx + \frac{\mu.x}{r_{ps}}\right) \le 0.2$$

The loss in prestressing due to wobble and curvature of ducts enclosing prestressing steel shall be calculated from following simplified equation

$$P_x = P_o \left[1 - \left(kx + \frac{\mu . x}{r_{ps}} \right) \right] \qquad (10\text{-}31)$$

Friction coefficients μ,k assumed in design shall be verified during tensioning.

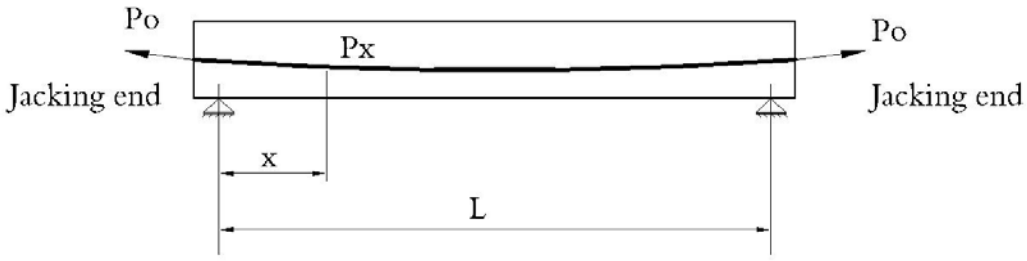


Fig. (10-6) Friction losses of Prestressing force

10-3-4-3 Time-dependent losses

10-3-4-3-1 Residual shrinkage losses

a. Prestress loss for concrete elements shall be calculated based on modulus of elasticity of prestressing steel and strain due to concrete shrinkage ε_{sh}

b. Strain due to shrinkage of concrete ε_{sh} shall be determined from table (2-7). If enough environmental data are not available, strain values may be taken from table (10-4).

c. For stage construction of prestressed element (when accurate data is not available), it may be approximately assumed that half of the strain due to shrinkage occurs in the first month and that three quarters of that strain occurs in the first six months after casting.

Table (10-4) Shrinkage Losses of Concrete , ε_{sh}

Prestressing System	Shrinkage strain , ε_{sh}
Pre-tensioning (after 3-5 days of concrete pours)	300×10^{-6}
Post-tensioning (after 7-14 days of pours))	200×10^{-6}

d. For members with pre-tensioning, loss due to shrinkage shall be calculated from the following relationship:-

$$\Delta f_{psh} = \varepsilon_{sh} \cdot E_p \qquad (10\text{-}32)$$

For members of post-tensioning, the loss shall be calculated from equation (10-32) considering only effective shrinkage that occurs after transfer of prestressing force.

10-3-4-3-2 Creep losses

a. Prestress loss for concrete elements shall be calculated based on modulus of elasticity of prestressing steel and strain due to concrete creep ε_{cr}

b. Values of creep coefficient required to calculate strain due to concrete creep shall be taken from table (2-8-b). If enough environmental data are not available, strain due to concrete creep may be taken from table (10-5)

Table (10-5) Creep Stain of concrete , ε_{cr}

ε_{cr} for each N/mm^2 of working stresses		Prestressing System
Initial strength of concrete at transfer of prestressing force, f_{ci} N/mm^2		
$f_{ci} \le 40$	**$f_{ci} > 40$**	
$10^{-6} \times (40/f_{ci}) \times 48$	**$10^{-6} \times 48$**	(Pre-tensioning) After 3-5 days of Concrete casting
$10^{-6} \times (40/f_{ci}) \times 36$	**$10^{-6} \times 36$**	(Post-tensioning) After 7-14 days of Concrete casting

c. If working stresses at any section in the concrete element exceed 1/3 of characteristic compressive concrete strength (f_{cu}), the strain values given in table (10-5) shall be increased by multiplying them by the factor α shown in the figure (10-7)

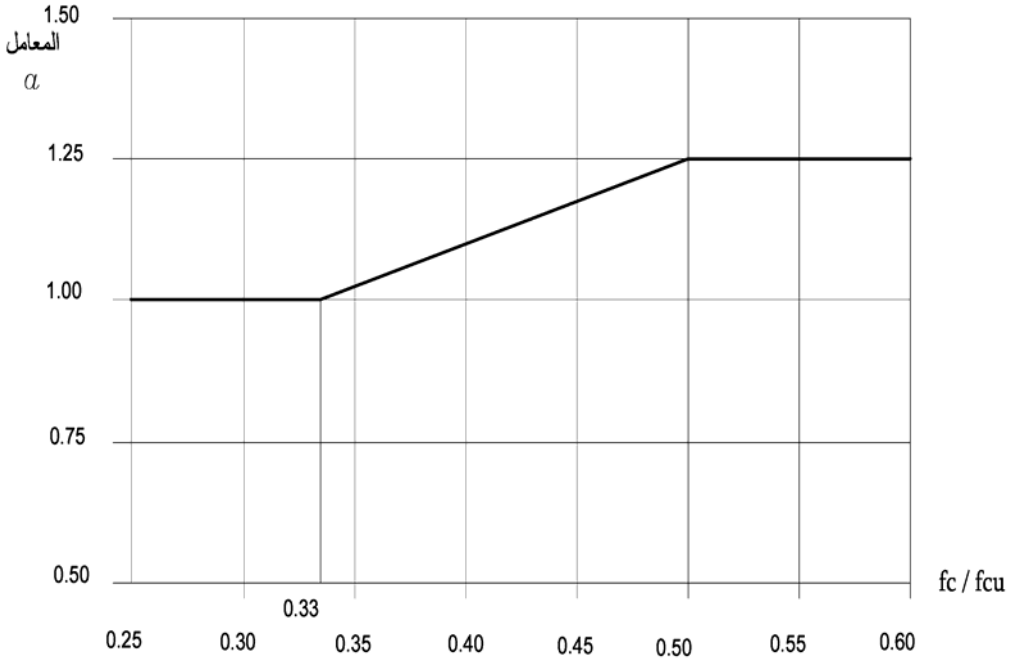


Fig. (10-7) Variations of the Values ,α as a Function of Working Stresses

d. for stage construction of prestressed element (when accurate data is not available), it may be assumed that half of the strain due to creep occurs in the first month and that three quarters of that strain occurs in the first six months after casting.

e. For elements with bonded prestressing steel, the prestress loss due to creep shall be taken as follows:-

$$\Delta f_{pcr} = \frac{\phi . E_p}{E_c} f_{cs} \qquad (10\text{-}33)$$

Where ϕ is the creep coefficient and shall be calculated as follows:-

$$\phi = \frac{\varepsilon_{cr}}{\varepsilon_{el}} \qquad (10\text{-}34)$$

Where ε_{el} is the elastic strain. Value of ε_{cr} shall be taken from table (10-5) or section 10-3-4-3-2-c. The value of creep

coefficient φ for elements with pre-tensioning may be taken equal to 2.0 while for elements with post-tensioning, creep coefficient may be taken equal to 1.6. The value of f_{cs} shall be calculated from the following equation:

$$f_{cs} = f_{cs}^{*} - f_{csd}^{*} \qquad (10\text{-}35)$$

Where f^{*}_{cs} = stress in concrete at level of prestressing steel due to prestressing force at transfer.

f^{*}_{csd} = stress in concrete at level of prestressing steel due to "near" permanent loads at transfer of prestressing force to concrete.

10-3-4-3-3 Steel relaxation losses

a. Effect of steel relaxation shall be considered. when calculating prestress loss,

b. Effect of relaxation of prestressing steel may be neglected if this steel is prestressed to stress exceeding maximum tensile stresses during prestressing for a period to be determined by design engineer.

c. Loss due to prestress relaxation shall be calculated from the following equation:-

$$\Delta f_{PR} = \frac{f_{pi}\ (\log t)}{k_1}\left(\frac{f_{pi}}{f_{py}} - 0.55\right) \qquad (10\text{-}36)$$

Where Δf_{PR} = prestress loss due to prestressing steel relaxation

t = time from tensioning in hours (with maximum value of 1000 hours).

f_{pi} = initial stresses in prestressing steel after immediate loss in prestressing and before time-dependent losses.

f_{py} = tensile yield stress of prestressing steel

k_1 = coefficient depends on type of prestressing steel as follows:

k_1 = 10 for normal relaxation stress-relieved steel

k_1 = 45 for low relaxation stress-relieved steel

10-3-5 External prestressing

10-3-5-1 External post-tensioning of cables shall not be permitted except for necessarily critical cases such as retrofitting, repair or enhancing serviceability. Necessary precautions for protection of external prestressing steel and anchorages from rust shall be implemented.

10-3-5-2 All details for protection shall be shown in working drawings. Proper protection against environmental conditions during member life shall be applied according to section (10-7-6-3).

10-3-5-3 In calculating flexural strength, prestressing steel cables shall be considered unbonded.

10-3-5-4 Necessary precautions to ensure required eccentricity of external prestressing cable to concrete centroid shall be taken for all expected cases along concrete element. External prestressing cables shall be fixed to concrete section in many locations along member between anchorages to balance targeted load and achieve required profile.

10-3-5-5 Accurate structural analysis methods to calculate strength and deformation at anchorages and deflecting locations of external prestressing steel shall be used for different load cases. Critical cases for change of cable eccentricity due to member deformation under load shall be considered.

10-3-5-6 Effect of fatigue on both concrete section and external prestressing cables shall be considered with increasing upper limit and decreasing lower limit of cyclic load by 5 %

10-4 Analysis of Prestress Structures

Structural analysis and design of prestressed concrete elements of statically determinate and indeterminate structures shall be performed to satisfy requirements of ultimate limit state and serviceability limit states.

10-4-1 Statically indeterminate structures

10-4-1-1 Theory of elasticity shall be used to evaluate behavior under service loads considering reactions, bending moments, shearing force, axial forces due to prestressing forces, creep, shrinkage, thermal change,

axial deformation, restraint of attached structural elements, and foundation settlement.

10-4-1-2 Ultimate moments on section shall be calculated as the sum of bending moment due to prestressing force multiplied by load factor of one in addition to bending moments due to other loads multiplied by load factors according to section (3-2-1)

10-4-1-3 Approximate methods shall not be used in calculating internal forces.

10-4-2 Moment redistribution

Moments calculated according to theory of elasticity due to proper arrangement of ultimate loads on spans shall be permitted to be redistributed on condition that:

- For each load case, equilibrium between internal and external forces shall be maintained
- Allowable reduction in bending moments according to theory of elasticity (for all load cases), shall not exceed 10%
- Ductility requirements at sections where moments are redistributed shall be satisfied.

10-4-3 Prestressed slabs

10-4-3-1 Equivalent frame method may be used to determine bending moments and shearing forces according to section (6-2-7-4)

10-4-3-2 More elaborate methods may be used to calculate internal stresses

10-4-3-3 Flexural strength for any section in prestressed slabs shall not be less than strength required according to section (10-3-3).

10-4-3-4 Punching shear strength in prestressed slabs

10-4-3-4-1 Critical section of punching shear stresses in prestressed slabs is located at d/2 from perimeter of concentrated load or reaction

10-4-3-4-2 Nominal Strength of Punching Shear in Slabs

10-4-3-4-2-a Ultimate punching shear stress shall be calculated according to section (4-2-2-3) taking into account effect of bending according to section (6-2-7-7).

10-4-3-4-2-b Nominal punching shear strength in slabs for any section shall not be less than that specified in section (4-2-2-3)

10-4-3-4-2-c In slabs satisfying requirements a, b, and c in this section and is prestressed in both directions, nominal punching shear strength shall be calculated from following equation:-

$$q_{cup} = \beta_p \sqrt{\frac{f_{cu}}{\gamma_c}} + 0.1 f_{pcc} + q_{pv} \qquad (10.37)$$

Where β_p is the lower of 0.275 or $0.8\left(\frac{\alpha d}{b_o} + 0.15\right)$

f_{ppc} = average compressive stress in concrete at perimeter of critical section (after all prestress losses) at slab section centerline.

q_{pv} = shear stress due to vertical components of prestressing forces (after all losses) of all cables intersecting critical section perimeter.

$$q_{pv} = \frac{f_{pe} \sum (A_p \, SinBi)}{\gamma_{ps} \, b_o d} \qquad (10\text{-}38)$$

The following conditions shall be satisfied:-

a. Punching shear strength shall be calculated from equation (10-37) only for internal columns or for cases where critical section perimeter is closed.

b. f_{cu} used in this section shall not exceed 40 N/mm^2

c. compressive stress at slab section centerline f_{ppc} shall not be less than 0.9 N/mm^2 in both directions and not to exceed 3.5 N/mm^2

10-4-3-5 Serviceability limit states in slabs shall be satisfied

10-4-3-6 Slab reinforcement details

- For distributed and normal live loads, the distance between tendons or tendon groups in one direction shall not exceed six times slab thickness or 1.5 m or the larger dimension of duct (for non circular ducts)
- Tendons are located such that minimum average stress of prestressing forces in one tendon after all prestress loss shall be higher than 0.9 N/mm^2 for slab section attributed to one tendon or group of tendons.
- Number of tendons in shear section over column shall not be less than two in each direction.

10-5 Detailing of prestressing systems

10-5-1 General

in addition to the following requirements, reference shall be made to structural details of reinforced concrete in chapter seven.

10-5-2 Ultimate limit of cable area in concrete section

Refer to section (10-3-3-1-6).

10-5-3 Concrete tendon cover

Concrete tendon cover is generally determined according to requirements of durability, fire resistance, and design according to chapters two and four and Egyptian code for fire Protection .

10-5-3-1 Bonded tendons

10-5-3-1-1 General

concrete cover for bonded tendons shall satisfy requirements of sections (4-3-2-3) and (9-7) in addition to requirements of section (10-5-3-1-2) for protection of rebars from rusting and sections (10-5-3-1-3) for fire protection and requirements shown in figure (10-8).

In pretensioning systems, tendon ends do not need concrete cover. They are, preferably, cut at concrete member end and insulted with anti rust paint.

10-5-3-1-2 Concrete cover for rust protection

The concrete cover for rust protection shall be determined based on: the environmental conditions specified in table (4-11), and mix design and constituents as shown in table (10-6). Recommendations concerning

concrete materials and mixtures detailed in chapter two of this code shall be applied on data of table (10-6) and requirements of prestressed concrete in section (10-2). The cement content in the mix shall not be less than 350 kg per cubic meter of concrete in addition to satisfying requirements of table (2-13).

Table (10-6) Minimum Thickness of Concrete Cover

		Thickness of Concrete Cover (mm)			
Characteristic Strength of Concrete, f_{cu} (N/ mm2)		Less than 35	40	45	More than 50
Exposure Class	First	25	25	25	25
	Second	-	40	30	30
	Third	-	50	40	40
	Fourth	-	-	60	50
Free water/ cement Ratio		0.5	0.45	0.4	0.35
Minimum cement Contents (Kg/ m^3)		350	400	425	450

10-5-3-1-3 Concrete cover for fire protection

Recommendations of Tables (2-14-a), and (2-14-b) of chapter two for the protection of structures against fire and those of the Egyptian Code for Fire Protection of Structures shall be implemented. Values given in Table (10-7) shall be considered as minimum.

10-5-3-2 Concrete cover of straight ducts (non curved)

Concrete cover measured from external of ducts shall not be less than 50 mm or concrete cover specified in section (10-5-3-1) and tables (10-6) and (2- 14-B) plus diameter of stirrup, or those shown in figures (10-9) and (10-10), whichever is greater and with due consideration to use concrete cover made of dense concrete. As for curved cables the requirements of section (10-5-5) shall be observed.

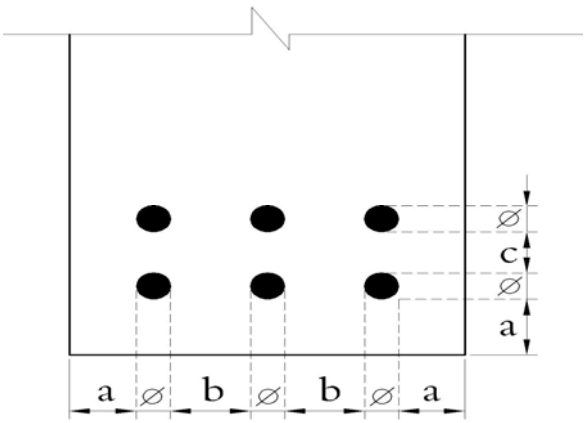


Fig. 10-8 Minimum thickness of concrete cover and the distances between wires and strands for prestressing systems

Where,

a≥ 2.5 ϕ b ≥ nominal diameter of aggregate + 5 mm

≥ concrete cover + diameter of stirrups ≥2 ϕ

≥ 20 mm

C > nominal diameter of aggregate

ϕ ≥2

≥ 10 mm

Distances a, b and c shall not be less than those specified by the cable suppliers.

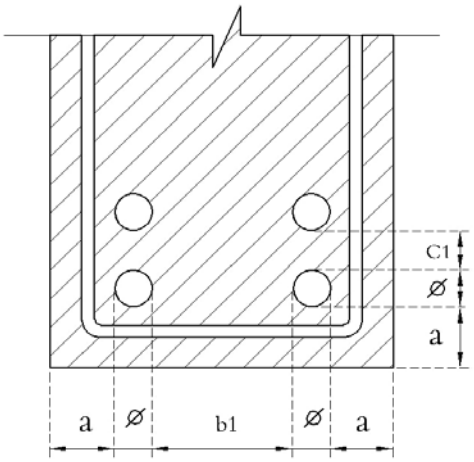
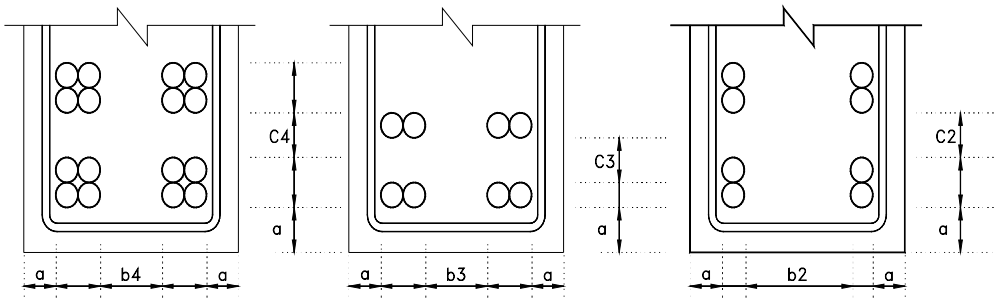


Fig. 10-9 **Minimum thickness of concrete cover and the distances between cable ducts for post-tensioning systems (Separate Cables)**

Where,

a≥ minimum concrete cover **in accordance with section (10-5-3-1)+ diameter of stirrups**
≥ **Diameter of duct,** ϕ for, ϕ≤ 80 mm
≥ **0.75 Diameter of duct**, ϕ for, ϕ> 120 mm
≥ 50 mm

b≥ Diameter of Duct or 40 mm
c≥ Diameter of Duct or 50 mm



a ≥ 1.5 Duct diameter	a ≥ 1.5 Duct diameter	a ≥ shown in Fig(10-8.)
b4 ≥ 1.5 Duct diameter	b3 ≥ 1.5 Duct diameter	b2 ≥ 1.5 Duct diameter
c4 ≥ 1.2 Duct diameter	c3 ≥ 1.2 Duct diameter	c2 ≥ 1.2 Duct diameter
ϕ ≥ 100 mm	ϕ ≥ 50 mm	ϕ ≥ 50 mm

With due consideration of the requirements of Tables (10-9) & (10-10)

Fig. (10-10) Minimum concrete cover and the distances between cable ducts for post-tensioning systems (Bundled Cables)

10-5-3-3 External tendons

In case of protecting external tendons with a concrete cover, dense concrete with a minimum stress of 40 N/mm2 shall be used. The thickness of the concrete cover shall not be less than the cover needed if the cables are inside the structural concrete section under similar circumstances. The concrete cover shall be tied using steel reinforcement in the pre-stressed element verifying that cracks are controlled according to the requirements mentioned in Chapter 4.

10-5-4 Spacing between pre-stressed cables

10-5-4-1 General

The spacing between the cables or group of cables shall satisfy the following requirements and by any means not less than the specifications of the producing companies.

10-5-4-2 Cable spacing in pre-tensioning systems

The cable spacing shall be determined according to Figure (10-8). For pre-tensioned elements where the steel adheres to the concrete by bonded tendons, the spacing between wires or strands ends shall satisfy sections (10-3-3-2) and (10-3-3-5). If these cables are placed apart in 2 or more sets, the occurrence of longitudinal splitting in the structural element shall be considered. Reinforcement and stirrups should be added to prevent this splitting.

10-5-4-3 Cable spacing between in post-tensioning system

The minimum net spacing between the ducts or between the ducts and the other cables, according to Figures (10-9) and (10-10) or not less than the following values, whichever is greater:

- The maximum aggregate nominal size plus 5 mm.
- In the vertical direction: the inside vertical dimension of the duct.
- In the horizontal direction: the inside horizontal dimension of the duct.

Precautions shall be taken to allow enough spacing between the ducts to allow the movement of the internal vibrators if used. If two or more rows of the ducts are needed, the spacing between the ducts shall be vertically continuous, as much as possible, to facilitate the construction works. The additional requirements stated in section (10-5-5) regarding curved cables

shall be taken into consideration. For slabs, the requirements of section (10-4-3-6) shall be considered.

10-5-5 Curved cables

10-5-5-1 General

When using curved cables in the construction of post-tensioning systems, the three-dimensional coordinates of the ducts of the cables shall be defined. The sequence of tensioning the cables shall be defined to avoid the following:

- Breakage of the side concrete cover perpendicular to the plane of the curvature of the ducts.
- Breakage of the cover in the plane of the curvature of the ducts.
- Break in the concrete separating the ducts in the plane of curvature or perpendicular to it.

In addition, the specifications mentioned in sections (10-5-5-2) and (10-5-5-3) shall be considered and the concrete cover and the cable spacing are not less than those shown in sections (10-5-3) and (10-5-4).

10-5-5-2 Concrete cover

To avoid the breakage of the concrete cover perpendicular to the plane of curvature of the cables and in their plane, the cover thickness shall be chosen according to Table (10-7). In this case, the movement of the ducts that may result in radial forces perpendicular to the visible concrete surface shall be prevented using stirrups fixed inside the structural element.

10-5-5-3 Spacing between ducts

a. The spacing between the ducts in the cable curvature plane shall not be less than the spacing shown in Table (10-8) or the spacing specified according to section (10-5-4-3), whichever is greater.

b. The spacing between the ducts perpendicular to the cable curvature plane shall not be less than the spacing specified according to section (10-5-4-3).

10-5-5-4 Decreasing the spacing between ducts

It is possible to decrease the spacing between the ducts than what is stated in section (10-5-5-3) in some special cases, according to the approval of the design engineer, if the smaller diameter cable is tensioned and injected

followed by the tensioning and injection of the next diameter cable after 48 hours.

10-5-6 Tendon anchorage zone

Figure (10-11) shows the requirements for the spacing between tendon anchorages.

10-5-7 Ducts and couplers sizes

10-5-7-1 Duct Sizes

The inside duct diameter shall exceed that of the cable diameter by at least 6-mm when using a single cable inside the duct. The area of the duct void shall not be less than twice the cross-sectional area of a group of cables inside the duct (preferably 2.5 times). Table (10-9) shows the minimum allowable inner dimensions and thickness for the ducts. For the ducts used in post-tensioning cables, a minimum straight distance of 50 cm shall exist before the curvature starts inside the duct.

Table (10-7) Minimum concrete cover for cables with curved ducts measured from the side of curvature center

القطر الداخلى للجراب (مم)															نصف قطر انحناء الجراب
170	160	150	140	130	120	110	100	90	80	70	60	50	40	30	19	
القوة الموجودة بالكابل (كيلو نيوتن)																
13200	11248	10338	9424	8640	7200	6019	5183	4320	3360	2640	1920	1337	960	387	296	
		مم أنصاف أقطار غير شائعة الاستخدام	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	متر
										445	320	220	155	55	50	2
	مم						420	350	265	205	145	100	70	50		4
مم				460	375	310	265	220	165	125	90	65	50			6
		395	360	330	270	220	185	150	115	95	75	55				8
	330	300	275	250	205	165	140	120	100	85	65	50				10
315	260	240	215	200	165	145	125	110	90	75	60					12
260	215	200	185	170	150	130	115	100	85	70	55					14
225	205	190	175	160	140	125	110	95	80	65	55					16
215	190	180	165	150	135	115	105	90	75	65	50					18
205	180	170	155	145	125	110	100	85	70	60						20
195	175	160	150	140	120	105	95	80	70	55						22
185	165	155	145	130	115	100	90	80	65	55						24
180	160	150	135	125	110	100	85	75	65	50						26
170	155	145	130	120	105	95	85	75	60							28
165	150	140	130	120	105	90	80	70	60							30
160	145	135	125	115	100	90	80	70	55							32
155	140	130	120	110	100	85	75	65	55							34
150	140	125	115	105	95	85	75	65	55							36
150	135	125	115	105	90	80	70	60	50							38
145	130	120	110	100	90	80	70	60	50	50	50	50	50	50	50	40

Notes

(1) The cable force mentioned in table is the maximum force existing in cables placed in ducts with the sizes shown in table (taken as 75% of the cable characteristic strength).

(2) If the duct contains profilers or spacers between the cables and the use of these profilers or spacers will cause the concentration of the radial forces, the values shown in table shall be increased.

(3) It is possible to decrease the given cover versus the inside diameter of the duct and the radius of curvature shown in Table by the ratio of the square root of the cable force if it is less than the value given in Table on condition of satisfying clauses (10-5-3-1-2) and (10-5-3-1-3).

Table (10-8) Minimum Spacing Between The Ducts Axes In The Plane Of The Curved Ducts

الفطر الداخلى للجراب (مم)																نصف قطر انحناء الجراب
170	160	150	140	130	120	110	100	90	80	70	60	50	40	30	19	
القوة الموجودة بالكابل (كيلو نيوتن)																
13200	11248	10336	9424	8640	7200	6019	5183	4320	3360	2640	1920	1337	960	387	296	
مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	مم	متر
الاستخدام	غير شائعة	أنصاف أقطار								960	700	485	350	140	110	2
							940	785	610	480	350	245	175	70	55	4
				1045	870	730	630	525	410	320	235	165	120	60	38	6
		940	855	785	655	545	470	395	305	240	175	125	90			8
	815	750	685	630	525	440	375	315	245	195	140	100	80			10
800	680	625	570	525	435	365	315	265	205	160						12
785	585	535	490	450	375	315	270	225	175	140						14
600	510	470	430	395	330	275	235	195	160							16
535	455	420	380	350	290	245	210	180								18
480	410	375	345	315	265	220	200									20
435	370	340	310	285	240											22
400	340	315	285	265												24
370	320	300	280	260												26
345																28
340																30
																32
																34
																36
																38
340	320	300	280	260	240	220	200	180	160	140	140	100	80	60	38	40

Notes:

(1) The cable force mentioned in table is the maximum force existing in cables placed in ducts with the sizes shown in table (taken as 75% of the cable characteristic cable strength).

(2) Spacing between ducts must not be less than double the inside diameter of the duct.

(3) If the duct contains profilers or spacers between the cables and the use of these profilers or spacers will cause the concentration of the radial forces, the values shown in table should be increased and if necessary use steel reinforcement between the ducts. Steel reinforcement may be used between ducts, if necessary.

(4) It is possible to decrease the shown spacing versus the inside diameter of the duct and the radius shown in Table by the ratio of the forces in the cable if it is less than the values shown in Table on condition of satisfying clause (10-5-4-3).

10-5-7-2 Couplers

Couplers shall only be used in the locations shown in the drawings or as approved by the design engineer of record. It shall not be permitted to have couplers in more than 50% of the cables at the same section. In addition, no other couplers shall be allowed (for uncoupled cables) except for distance greater than 1.5 meters measured in the longitudinal direction of the cables relative to beams less than 2-meters height or 3.0 meters for beams more than 2-meters height. The couplers shall be chosen to satisfy the ultimate resistance stated for pre-tensioning steel without exceeding the expected deformation of the coupler or for pre-tensioning steel. The couplers shall not reduce the extensibility of the cables and shall be placed in ducts that allow movement during tensioning and be provided with the means that allow complete injection for all the coupler components.

10-5-8 Construction documents

10-5-8-1 Presentation of the construction documents

The contractor shall present the construction documents to be used during the work to the design engineer before the work begins by sufficient time for revision and approval. It should be noted that the approval of the design or checking engineer on these drawings does not alleviate the contractor from the responsibility of preparing them.

10-5-8-2 Documents including the construction documents

The execution documents referred to in the previous item include the following:

a. Complete details of the system used including the specifications of the used cables, anchors, ducts, used equipment, cable tensioning method, working stresses, anchoring stresses and cable elongation under loads due to tension.

Symbols:

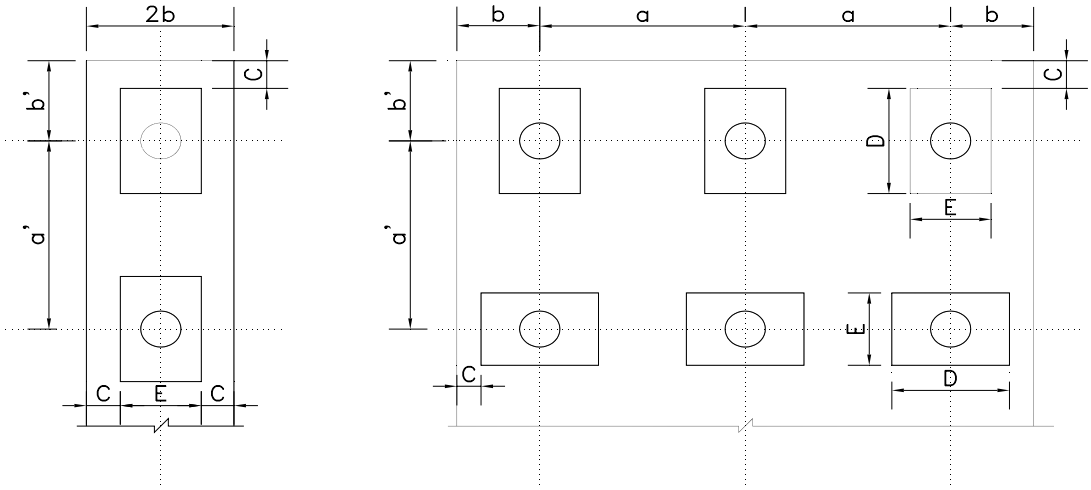
E = smaller dimension of the tendon anchorage from the manufacturer's catalogue.

D = larger dimension of the tendon anchorage from the manufacturer's catalogue.

a_o = minimum allowable spacing between tendon anchorage axes (taken from the manufacturer's catalogue).

a_o > (D or E) + 30 millimeter

b_o = minimum allowable spacing between the axis of the tendon anchorage and the edge of the concrete (taken from the manufacturer's catalogue).
a = actual horizontal spacing between the tendon anchorage axes.
b = actual horizontal spacing between the tendon anchorage axis and the edge of the concrete.
a' = actual vertical spacing between the tendon anchorage axes.
b' = actual vertical spacing between the tendon anchorage axis and the edge of the concrete.
c = spacing between the tendon anchorage and the edge of the concrete (according to the following table):



Anchorage tendons distributed over several horizontal and vertical lines Consider the following: $a > a_o$, $a' > a_o$ $b > b_o$, $b' > b_o$ $ba' > 1.6\ b_o^2$ $b'a > 1.6\ b_o^2$ $aa' > 1.5$ original tensile force/fcu	Anchorage tendons distributed on a single vertical line Consider the following $b' > 1.5\ b_o$ $2ba' >$ original tensile force/fcu $ba' > 1.6\ b_o^2$

Original tensile force (kN)	500	500-1000	1500-3000	3000-4000	> 4000
Spacing c (mm)	30	50	70	80	100

Figure (10-11) Spacing between anchorage tendons

Table (10-9) Minimum inside dimensions and minimum allowable thickness for ducts*

Type of pre-stressed steel	Number of wires or braids forming the cable	Corrugated steel thin ducts**	Corrugated steel thin ducts**	Rigid Steel ducts***,****	Rigid Steel ducts***,****
		Internal Diameter (mm)	Thickness (mm)	Internal Diameter (mm)	Thickness (mm)
7 mm diameter Wires	9 14 18 22 30 54 84	40 46 50 60 65 90 110	0.4 0.4 0.4 0.4 0.4 0.6 0.6	--- --- --- --- 76 89 108	--- --- --- --- 2 2 2
Braids: nominal diameter 12.5 mm or 12.9 mm	7 12 18 31 55	50 65 80 105 140	0.4 0.4 0.6 0.6 0.6	55 76 84 108 139	2 2 2 2 2
Braids: nominal diameter 15.2 mm or 15.7 mm	5 8 12 19 37	50 65 80 95 130	0.4 0.4 0.6 0.6 0.6	55 76 84 101 139	2 2 2 2 2

* In cases not mentioned in the table, use the nearest equivalent value

** The duct curvature diameter is not less than 100 times the inner diameter or the value specified by the manufacturer, whichever is bigger

*** The minimum curvature diameter for the duct is not less than 3 meters - used under special conditions for cables with small radii or for external tendon ducts

**** In case of using plastic ducts, the duct internal diameter should be according to the table and the minimum duct thickness is 3 mm.

b. The structural calculations prepared by the contractor based on the system he shall follow, illustrating the differences between the preliminary design done by the design engineer and the contractor design regarding the dimensions of the concrete sections, number and location of cables as well as the steel reinforcement. The calculations shall be clearly written showing the items of the code of practice on which the design was based.

c. The executive drawings shall be drawn to a suitable scale, sufficient to show all the details needed for construction, clearly showing all cables, their types, locations, and three-dimensional coordinates (relative to the center of the cables section). Additionally, the locations and specifications of the anchorage and tying tendons. Comprehensive details about the steel reinforcement and the concrete section shall be shown illustrating the locations of any other parts that may be in the concrete section along the complete length of the element and at the fixation areas such as supports or anchorages such that these drawings guarantee there is no conflict between the pathways and locations of these parts. The values of the design friction coefficients u, k shall be shown on the drawings.

10-6 Inspection and quality control

The provisions of Chapter 8 shall be applied to pre-stressed concrete works. Attention shall be given to the quality of the concrete including its strength when transferring the pre-stressed force and the quality of the steel reinforcement. Assurance the pre-stressing force, injection quality, vapor treatment quality (if found) and safety during the construction such as the process of cable tensioning. The following additional items shall be considered.

10-6-1 Concrete quality

a. A sufficient number of concrete cubes shall be poured for compressive strength testing during the transfer of the pre-stressing force. This is to ensure the characteristic strength at the times required by the consulting engineer. Samples shall be taken on each pouring day or when the element differs such that it shall not exceed 100 m^3 of concrete in continuous working durations.

b. The concrete compressive strength shall be tested before applying tension to the cables. The compressive strength test results shall satisfy the required compressive strength during the transfer of the pre-stressing force. Any test result shall not be less than 85% of the

required compressive strength. If the required strength shall not be satisfied, the compressive strength shall be tested at a later time.

c. If the requirements in items (8-9-3) and (10-2-1-3) are satisfied, the concrete shall be considered to meet the characteristic strength grade required while loading during construction.

10-6-2 Supervision and quality control of the injection mortar

a. The quality control steps and requirements shall be applied on the materials of the injection mortar including the cement, filler materials, admixtures, and water. The consistency slump test shall be performed on fresh mortar over suitable durations, not less than 3 times each day. Continuous visual inspection shall be maintained during the day.

b. The compressive strength of mortar shall be carried out according to the standard specifications. Samples shall be taken at suitable intervals during the day and for different injected elements.
c. The mortar compressive strength shall satisfy the required strength such that the result of any test is not less than 85% of the required strength.

10-6-3 Inspection and quality control of pre-stressed steel

In addition to the testing and inspection certificates accompanying the pre-stressed steel, quality control tests shall be performed. The pre-stressed steel shall meet the requirements of the international standard specifications according to which it was manufactured. The wires and braids shall be inspected after unfolding from the rolls it was brought on to ensure that they are straight and free from deformation and bending. All steel must be free from pits, adhering materials such as dust or oils. If the pre-stressed steel is left in ducts without stressing for more than 5 weeks, the steel shall be inspected again for rust.

10-6-4 Inspection of ducts and cables

a. The ducts shall be inspected upon receipt. Any ducts with constrictions or holes shall be excluded. Ducts shall be inspected after installation according to drawings and the strength and rigidity of the duct supports shall be checked. Inspection of the quality of the insulation of the ducts at the edges and couplers so mortar would not get inside and affect tensioning the cables shall be made. Compressed air - with maximum pressure of 2 N/mm2 and 1 N/mm2 for horizontal and vertical ducts, respectively, while monitoring the air pressure - shall be used to ensure the ducts are not clogged.

b. Tension shall be gradually applied to each cable using the required design force. This shall be achieved by defining the actual elongation of cable in site and comparing it with the computed elongation. Any displacement in the cable end connectors shall be considered. The minimum accuracy for elongation readings shall be 2 mm. The tensile force shall be measured from one end using a calibrated device (reading accuracy not more than 1.5%).

c. For short elements or when using braids with 19 or more wires, it is preferable to check the tensile force in the cables using a force meter.

10-6-5 Calibration of equipment for tensioning cables

Elongation measurement and cable tensile force measuring devices are calibrated before use and every 6 months thereafter under normal conditions or more frequent according to the consulting Engineer. The error in the accuracy of these devices shall not exceed the allowable limits stated in the Egyptian Standards for equipment calibration.

10-6-6 Inspection of concrete elements after load and element transfer

a. Ensure that the concrete element shall be free from deformations or cracks after the load transfer. The maximum element camber shall be measured and compared with the allowable limits.

b. For pre-cast units, ensure the element shall be free from deformations or cracks after transfer to its location in the structure. Inspections of the connectors between the pre-stressed elements to attached and/or carrying elements shall be made.

10-6-7 Concrete tests

Refer to sections (8-9-4) and (8-9-5).

10-6-8 Durability tests for elements and concrete structures

Section (8-9-6) shall apply to pre-stressed structures and concrete elements.

10-7 Construction

10-7-1 General

10-7-1-1 Reference shall be made to Chapter Nine in addition to the requirements mentioned in this chapter.

10-7-1-2 Construction shall be carried by contractor well experienced with the prestressing system used, and his experience shall be approved by the consultant and the designer before commissioning. The contractor shall

insure that all personnel dealing with prestressing, grouting and anchoring the tendons are highly skilled professionals in their areas and upon to the approval of the consultant based on their training and experience certificates.

10-7-1-3 Contractor shall submit all documents mentioned in section (10-5-8) and the quality assurance plan according to section (10-7-9). Construction shall follow these documentations after receiving the approval from the consultant and the designer of record.

10-7-1-4 Materials used shall comply with the specifications and requirements given in Chapter Two, and shall be tested before construction and periodically according to requirements stated in Chapter Eight.

10-7-1-5 The following documents shall be submitted before construction for the approval of the consultant and the designer:

a -Prestressing system assurance certificate approved from authorized parties (original certificate).

b -Experience certificate for working personnel.

10-7-2 Prestressing program

10-7-2-1 Prestressing shall not be applied until the concrete achieves suitable compressive strength capable of safely withstanding the acting forces, taking into consideration locations of force application. Table (10-10) gives the allowable minimum compressive strength for concrete at the time of prestressing application. Compressive and tensile stresses in concrete shall be calculated and shall not be more than the limit values given in section (10-3).

Table (10-10) Minimum allowable compressive strength for prestressing application

Concrete Grade	Minimum Compressive Strength (N/mm2)
30	26
35	30
40	32
45	36
50	40
55	44
60	48

10-7-2-2 Prestressing shall be conducted according to a schedule which shows the sequence of prestressing the tendons in addition to the value of the prestressing force, direction and location. The values of friction factors, slipping values, the time for form removal as well as stresses induced due to prestressing shall be verified in the schedule. Contractor shall submit this schedule prior to the commencement of prestressing.

10-7-2-3 Prestressing might be applied earlier in stages in special cases with the condition that the compressive strength of the concrete is not less than 75% of the values given in section (10-7-2-1). The compressive strength of the concrete shall be verified through standard compressive strength testing. The prestressing force acting on the tendons for each stage shall not exceed 35% from the allowable stresses. Also, the stresses on concrete shall not exceed the values stated in table (10-2), where, fcui is the characteristic strength of concrete at the stage of transferring the prestressing force to the concrete measured through standard testing of specimens at the time of prestressing application. If the compressive strength is higher than the expected values, the prestressing force might be proportional by increased.

10-7-2-4 If prestressing shall be applied in stages, losses that occur in each stage shall be accounted for until the final prestressing.

10-7-3 Tendons

10-7-3-1 All precautions shall be exercised during storage and handling of tendons to prevent any damage. As a minimum requirement tendons shall be stored above the ground with protection against humidity, weather, and any materials that might react with the tendons and from welding sparks. The materials used in covering and protecting the tendons shall be chemically stable with the tendons, and provide full protection.

10-7-3-2 Welding and heat treatment such as galvanization shall not be performed on tendons and without violating section (10-7-3-5) with respect to cutting.

10-7-3-3 The outer surface of the tendons and the interior surface of the ducts shall be clean from rust, dust, oil, grease, paintings and or any harmful materials.

10-7-3-4 Wires and strands shall be delivered in such a way to ensure its straightness during unrolling. Minor straightening might be used on site if required and under full supervision. Also, bars shall be straight and in case of minor twisting they might be manually straightened on site under full supervision. Bars with twisted screwed ends shall not be used and shall be cold straightened.

10-7-3-5 Cutting tendons to required length and finishing their ends shall be conducted using high speed rotating discs or friction saw or any other mechanical method that shall not negatively affect tendon properties.

10-7-4 Fixing tendons and ducts

10-7-4-1 Prestressing tendons and ducts are to be accurately fixed in locations specified in drawings. The tolerance in position for tendons, ducts and duct former shall not exceed ± 5mm. In case of slabs and sections with thickness less than or equal to 300mm the tolerance shall not exceed +2 mm.

10-7-4-2 Tendons and ducts and their components shall be fixed in a way to prevent their movement due to prolonged or over vibration or concrete weight during casting or worker and construction movements. Fixation shall be in such a way as not to increase friction between tendons during tensioning process.

10-7-4-3 Ducts splices (or couplers) shall be securely locked to prevent mortar or concrete leakage inside. Also, the ducts ends shall be closed and protected after prestressing and injection. Splices (or couplers) in adjacent ducts shall be apart by more than 300mm.

10-7-4-4 Ducts shall be equipped with ventilation openings at all high points, and grouting openings at all lower positions, unless the duct curvature is small or the duct is horizontal. Table (10-11) gives the minimum inner diameter for grouting and ventilation pipes.

Table (10-11) Minimum inner diameter for grouting and ventilation* pipes

Prestressing Steel Type	Number of Wires or Strands in Tendon	Minimum Diameter for Grouting Pipe (mm)	
		Grouting Pipes	Ventilation Pipes
7mm wires	9-30 54	20 26	20 20

	84	33	26
Strands with nominal diameter 12.5mm or 12.9mm	7 12 18 31 55	20 20 26 33 40	20 20 20 26 33
Strands with nominal diameter 15.2mm or 15.7mm	5 8 12 19 37	20 20 26 33 40	20 20 20 26 33

* Values in the table shall be used if the duct length is equal to or less than 1200 times its inner diameter. In case the duct is longer, the inner diameter for the grouting and ventilation pipes shall be taken the next following value (corresponding to the bigger number).

10-7-5 Tensioning Process

10-7-5-1 General

10-7-5-1-1 Wires and strands which shall be used in a single process and shall be taken from the same consignment. The tendon shall be labeled at its end with the consignment number type and number of wires in the tendon. Twisted tendons or loose strands shall not be permitted.

10-7-5-1-2 All necessary precautions shall be taken to protect personnel, property, and equipment from sudden energy release from the prestressed tendons in case of any damage.

10-7-5-1-3 The following requirements shall be applied at the time of prestressing the tendons:

1 -Tendons shall be securely fixed in the tensioning jack.

2 -In case of prestressing more than one tendon at the same time they shall be equal in length measured from the fixation point to the elongation gauge.

10-7-5-1-4 Personnel in charge of tensioning shall confirm that tensioning process is designed and executed in such a way it is securely fixed and tension force is applied gradually without any secondary stresses in tendons, anchorage or concrete.

10-7-5-1-5 Force in tendons shall be measured during tensioning wither directly by load measuring devices, or indirectly by measuring the compression in jacks Elongation measuring devices shall be available to measure elongation in tendons or any movement of tendons in the gripping

devices. Calibration of load and elongation measuring devices shall be carried out according to section (10-6-5).

10-7-5-2 Pre-tensioning

10-7-5-2-1 In pre-tensioning process, necessary methods shall be used to completely hold the tension force in the tendons during the period of tensioning and transferring the force. Transfer of stress shall be gradually done.

10-7-5-2-2 In case of straight tendons and using longitudinal lines to apply pre-tensioning force, special pieces shall be distributed through the tendon path to prevent tendon movement during casting. These pieces shall permit longitudinal movement of tendons to allow force transfer to concrete through the whole length. In case of using single mould, the mould shall be rigid enough to transfer pre-tensioning force without any torsion to the mould.

10-7-5-2-3 In case of using deflected tendons and single tendons, the fixing pieces shall have a diameter more than five times the wire tendon diameter, and more than ten times the strand tendon diameter, and the angle of curvature shall not exceed 15 degrees.

10-7-5-3 Post-tensioning

10-7-5-3-1 Tendons arrangement

1 - Tendons shall be arranged in such a way not be sharply bent or curved which will result in breaking the tendons, or the concrete or the tendon ducts.

2 - In case it shall not be possible to apply post-tensioning to wires and strands at the same time, consideration shall be taken that spacing elements are stiff enough not to be moved during simultaneous post-tensioning.

10-7-5-3-2 Anchorages

1 - Anchorages shall comply with international standards for prestressing system. Their design and fixation shall allow uniform stress distribution over the concrete at the end of the concrete element and to preserve the prestressing force acting under permanent and changing loadings and impacts.

2 - Anchorages shall be of the split wedge and barrel type. They shall be manufactured from materials which shall not allow the strain in the barrel to move the wedge until these wedges yield sufficient lateral force to strongly hold the tendons, or the wedges will cause extra force on the tendons at or before it reaches its maximum path.

3 - Anchorages suitable to the prestressing system shall be used taking into consideration full compliance with regulations and recommendations of the manufacturer with respect to installation in concrete elements and the necessity to maintain the cleanliness of the anchorage holding surfaces before applying tension force. Tension force shall be applied evenly and gradually to avoid sudden stress to tendons or anchorage.

4 - Tendons slippage during gripping shall be in accordance to the instructions of the supervising party. The actual slip for each tendon shall be recorded, and after gripping the tendon the tension force acting by the loading device shall be reduced gradually.

5 - Anchorages shall be protected from rust by all necessary means.

10-7-5-3-3 Deflected tendons for external prestressing

The radius of the deflector connected to tendons shall not be less than 50 times the tendon diameter. The angle of shaping the tendon shall not exceed 15 degrees. If the radius of the deflector is less than 50 times the tendon diameter or the shaping angle is more than 15 degree, a test shall be conducted to calculate the loss in the force and the necessary correction shall be applied accordingly.

10-7-5-3-4 Tendons tensioning

1 - Load shall be applied to tendons until the required elongation and/or load is reached. Slipping of tendons at the non-jacking end shall be taken into consideration. Measurement shall not start until no sagging in the tendons is confirmed. Force in tendons calculated from the measured elongation and that measured from the loading jack gauge shall not differ by more than 6% of the lower value. If the difference is more than 6% all precautions shall be considered to reduce the difference.

2 - In pre-tensioning process all measurements shall be recorded in a log book which will at least include value of pre-tensioning force, its direction and location. Irregular measurements shall also be recorded

and brought to the attention of the designer engineer and the supervising team to make necessary corrections. If deviation is more than the required theoretical stress by 5%, all necessary precautions shall be considered to keep the deviation from increasing.

3 - In case of applying tension or construction at several stages, the designer engineer shall determine the tension stages and the value of load for each stage.

10-7-6 Protection and bonding of tendons using injection

10-7-6-1 General

Tendons shall be protected from damage, corrosion and fire and in addition they shall be tied to the structure using injection.

10-7-6-2 Protection of inner tendons

Inner tendons shall be tied to the concrete element by injection of cement grout or cement mortar (cement and sand) according to injection precautions stated in section (10-7-8).

10-7-6-3 Protection of external tendons

External tendons shall be protected from mechanical damage and corrosion by encasing the tendons with dense concrete or dense mortar with sufficient cover. Other corrosion resistant materials and with sufficient hardness might be used for protecting the tendons. Relative movement between the concrete element and the protective casing shall be taken into consideration in choosing the protection system. This movement shall be due to change in forces, stresses, creep, relaxation, drying shrinkage and thermal shrinkage.

10-7-7 Protection of anchorage

All necessary precautions for protecting anchorage shall be applied. High strength mortar shall be cast between the anchorage and the concrete element.

10-7-8 Grouting

10-7-8-1 General

In post-tension prestressing systems, injection is used to protect tendons from corrosion and ensure the transfer of stresses between tendons and concrete element.

10-7-8-2 Inspection of ducts

Ducts shall be manufactured from corrosion resistant materials. Ducts shall be strong enough to resist stresses and concrete weight. Sudden change in diameter and path of ducts shall not be allowed. Ducts shall be equipped with openings for ventilation and grouting with the minimum dimensions stated in table (10-11), and spacing between openings shall not exceed 15m. Before casting, ducts shall be inspected to check the couplers integrity especially those at the anchorages. Any inspection method might be used after the approval of the supervising engineer.

10-7-8-3 Injection process

Ducts shall be injected as soon as the tensioning process is complete to prevent the corrosion of the tendons. Injecting mortar shall be used within 30 minutes after its mixing, except in case of using retarding admixtures. Injection shall be performed in such a way to ensure complete filling of the ducts. Injecting pumps with adequate power shall be used with an injection rate of 6 cubic meters to 12 cubic meters per minute for horizontal ducts so that the injection shall be steady, slow and continuous to avoid segregation especially in congested cross sections. Ventilation openings shall be closed simultaneously during the filling of the duct while maintaining a pressure of 0.5 N/mm2 for five minutes after the closure of the last ventilation opening. For vertical ducts an injection pump with an injection rate of 2 meters to 3 meters per minute with a pressure not more than 2N/mm2 shall be used.

10-7-9 Quality assurance for prestressing works

The contractor shall submit in writing a detailed quality assurance plan including all the previously stated sections in addition to the following:

1 -Execution steps to all works included in the scope of the contractor job in the project in the area of prestressing or supplementary works.

2 - In case of executing the structure or tendons tensioning in stages, the contractor shall submit plan and detailed drawings to show the structure segment to be cast, the tendons to be tensioned, the tensioning force and the elongation for each stage, as well as the calculations to ensure that the structure did not de-bond from the forms during the tensioning process.

3 - A list of equipment including its accuracy and the system and calibration certificate for the equipments used in calibration.

4 - Test results on materials used from certified laboratories approved by supervising party.

5 -Site safety regulations especially during tendons tensioning process.

6 - Storage of materials on site to protect them from moisture or damage or corrosion.

7 - Detailed plan for testing concrete and mortar during casting and regulations to be followed in case results did not comply with requirements.

8 - Detailed plan of regulations to be followed to ensure the concrete cover of the ducts, and the ducts position before and during casting. Also, checking the fixation of ducts and tendons in position during installation of reinforcing steel and casting of concrete or form erecting shall be included in the detailed plan.

9 - Non destructive test on duct path shall be conducted by the contractor to confirm the full injection of the duct with grout.

10 - Detailed plan on casting procedure, timing, equipment, precautions to protect plastic concrete from cracking, and precautions during casting highly congested reinforcement sections, as well as alternate power sources to use on site.

11 - Detailed plan on concrete curing including procedures that will be used to protect anchorage from curing water.

APPENDIX (I)

(SI) SYSTEM – METRIC SYSTEM (KG.CM) CONVERSIONS

A) UNITS OF (SI) SYSTEM.

B) CONVERSION FROM METRIC TO (SI) SYSTEM.

C) CODE EQUATIONS IN METRIC SYSTEM (KG.CM).

(A)

UNITS OF (SI) SYSTEM

Quantity	Unit	International System
Length	Meter	m
	Centimeter	cm
	Millimeter	mm
	kilometer	km
Mass	Gram	g
	Kilogram	kg
	Ton	t
	Milligram	mg
Time	Second	s
	Minute	min
	Hour	h
	day	d
Planar Angle	Degree	o
	Minute	/
	Second	//
Volume	Liter	L
	Milliliter	mL
	Meter cube	m^3
Area	Meter square	m^2
	Millimeter square	mm^2
Force	Newton	N
	Kilo Newton	KN
Stress	Newton / Millimeter square (Mega Pascal Mpa)	N/mm^2
	Kilo Newton / meter square	KN/m^2
Temperature	Degree	c^o

B) CONVERSION COEFFICIENTS FROM METRIC SYSTEM TO (SI) SYSTEM:

metric System		SI system
kilogram force	=	9.81 Newton
kilogram force . meter	=	9.81 Newton / meter
kilogram force / meter	=	9.81 Newton / meter
kilogram force / centimeter square	=	0.098 Newton / millimeter square
kilogram force / meter square	=	9.81 Newton / meter square
kilogram force / meter cube	=	9.81 Newton / meter cube
0.102 kilogram force	=	1.00 Newton
0.102 kilogram force . meter	=	1.00 Newton . meter
0.102 kilogram force / meter	=	1.00 Newton / meter
10.2 kilogram force / centimeter Square	=	1.00 Newton / millimeter square
0.102 kilogram force / meter Square	=	1.0 Newton / meter square
0.102 kilogram force / meter cube	=	1.0 Newton / meter cube

Note:

For simplicity:

(1.0 kilogram force = 10 Newton) is considered when converting equations in the Code.

C) CODE EQUATIONS BY METRIC SYSTEM (KG.CM):

Table (4-5)

Lateral reinforcement for tension and shear resistance

	$q_{tu} \le 0.19\sqrt{\frac{f_{cu}}{\gamma_c}}\left(kg/cm^2\right)$	$q_{tu} > 0.19\sqrt{\frac{f_{cu}}{\gamma_c}}\left(kg/cm^2\right)$
$q_u<q_{cu}$	Minimum shear reinforcement according to item (4-2-2-1-6)	Reinforcement to resist q_{tu}.
$q_u>q_{cu}$	reinforcement to resist $(q_u - q_{cu}/2)$	reinforcement to resist both q_{tu}, $(q_u - q_{cu}/2)$

Table (5-1)

Allowable (working) stresses in concrete and steel

Stress Type			Symbol	Working stress based on characteristic strength of concrete (kg/cm^2)		
Characteristic strength of concrete			f_{cu}	200	250	300
Axial compression (e = e_{min})			f_{co}^{*}	50	60	70
Flexure or compression with large eccentricity.			f_{c}^{**}	80	95	105
Shear ***						
Concrete shear resistance						
without reinforcement in slabs and footings			q_c	8	9	9
without reinforcement in other members			q_c	6	7	7
With web reinforcement in all members						
(shear and torsion).			q_2	17	19	21
Punching shear			q_{c_p}	8	9	10
Steel ****						
1. mild steel	2400/3500			1400	1400	1400
2. steel	2800/4500			1600	1600	1600
3. steel	3600/5200		f_s	2000	2000	2000
4. steel	4000/6000			2200	2200	2200
5. welded	4500/5200	plain		1600	1600	1600
		or deformed		2200	2200	2200

* Represents the maximum axial compressive stress on section under working loads.

** The stresses are for beams and slabs deeper than 20 cm. Allowab stresses shall be reduced than the given values by 15, 20, 25 and 30 kg/cm^2 for slabs with thickness 20, 12, 10 and 8 cm respectively.

*** Items (4-5) and (5-5) shall be taken into account.

**** Steel stresses shall be reduce to fulfill cracking limit state (item 4-3-2) if applicable.

Table (5-2)

Lateral reinforcement for torsion and shear resistance

	$q_t \le 0.13\sqrt{\frac{f_{cu}}{\gamma_c}}\left(kg/cm^2\right)$	$q_t > 0.13\sqrt{\frac{f_{cu}}{\gamma_c}}\left(kg/cm^2\right)$
$q \le q_c$	min. shear reinforcement ratio (item 4-2-2-1-6)	Reinforcement to resist q_t
$q > q_c$	Reinforcement to resist $(q - q_{c/2})$	Reinforcement to resist both q_t, $(q - q_{c/2})$

Table (10-2)

Allowable chesses in concrete (kg/cm^2)

Item	Case		
1	Allowable stresses in concrete due to bending moments immediately after transfer of prestressing to concrete (before occurrence of time dependent losses).		
	Maximum compressive stress		0.45 f_{cui}
	Maximum tensile stresses except at ends of simply supported beams.		0.7 $\sqrt{f_{cui}}$
	Maximum tensile stresses at end of simply supported beams.		1.4 $\sqrt{f_{cui}}$
2	Allowable stresses in concrete due to bending moments under working loads (after all time dependent losses).		
	Maximum compressive stresses due to prestressing plus permanent loads.		0.35 f_{cu}
	Maximum compressive stresses due to prestressing plus all loads.		0.40 f_{cu}
	Maximum tensile stresses in precompassed tensile zone due to prestressing plus all loads.	Case a Case b Case c Case d	Zero 1.4 $\sqrt{f_{cu}}$ 1.9 $\sqrt{f_{cu}} \ngeq 40$ kg/cm^2 2.7 $\sqrt{f_{cu}}$
3	Allowable stresses in concrete due to axial compressive stress		
	Maximum compressive stress		0.25 fcu

APPENDIX II

VALUES OF MECHANICAL PROPERTIES OF PRESTRESSING STEEL IN ACCORDANCE WITH INTERNATIONAL CODES

Table (1)

Mechanical properties of stress – relived Wires in American standard ASTM A 421

Nominal diameter (mm)	Tensile Strength N/mm^2		Yield strength (N/mm^2) **	
	WA type	BA type	WA type	BA type
4.88	1725	-	1465	-
4.98	1725	1655	1465	1407
6.35	1655	1655	1407	1407
7.01	1620	1620	1377	1377

Table (2)

Mechanical properties of stress – relived 7-Wires standards *

in American Standard ASTM A 416

Nominal diameter (mm	Nominal area (mm^2	Tensile Strength (KN)		Yield strength (KN)	
		Grade 250	Grade 270	Grade 250	Grade 270
6.35	23.22	40.0	-	34.0	-
7.94	37.42	64.5	-	54.7	-
9.53	51.61	89.0	102.3	75.6	87.6
11.11	69.68	120.1	137.9	102.3	117.2
12.70	92.90	160.1	183.7	136.2	156.1
15.24	139.35	240.2	260.7	204.2	221.5

* Relaxation percentage is determined after 1000 hour in wires and low relaxation strands.

** Yield strength shall not be less than 90 % of tensile strength in case of wires and low relaxation strands.

Table (3)
Mechanical Properties Of Prestressing Cold drawn Wires in British Standards* Bs 5896

nominal diameter (mm	Tensile strength (N/mm^2	Yield strength (N/mm^2) (proof stress 0.10%)
4.0	1670	1386
4.0	1770	1469
4.5	1620	1345
5.0	1670	1386
5.0	1770	1469
6.0	1670	1386
6.0	1770	1469
7.0	1570	1303
7.0	1670	1386

- Relaxation percentage in determined after 1000 hour for both normal and low relaxation wires .

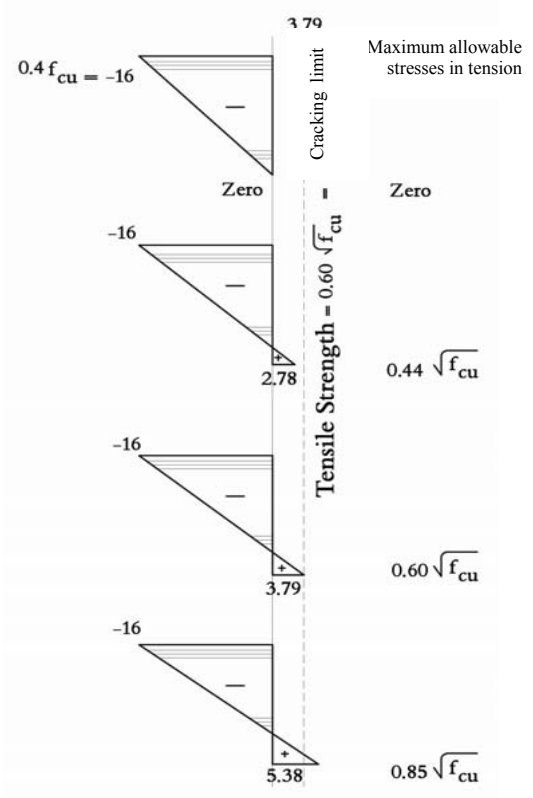
Table (4)
Mechanical Properties of Prestressing strands in British Standards*

Strand type	nominal diameter (mm)	nominal Area (mm^2)	tensile strength (KN)	proof strength 0.10% (KN)	Standard
Standard 7-wire strands	9.3	52.3	92.0	78.0	Bs 5896
	11.0	71.0	125.0	106.0	
	12.5	94.2	164.0	139.0	
	15.2	138.2	232.0	197.0	
Super 7-wire strands	8.0	38.0	70.0	59.0	Bs 5896
	9.6	55.0	102.0	89.0	
	11.3	75.0	139.0	118.0	
	12.9	100.0	186.0	158.0	
	15.7	150.0	265.0	225.0	
Drawn 7-wire strands	12.7	112.0	209.0	178.0	Bs 5896
	15.2	165.0	300.0	255.0	
	18.0	223.0	380.0	323.0	
19 wires strand	18.0	210.0	370.0	319.5	Bs 4757
	25.4	423.0	659.0	560.15	
	28.6	535.0	823.0	699.55	
	31.8	660.0	979.0	832.15	

- Relaxation percentage in determined after 1000 hour for both normal and low relaxation strands .

Table (5)

Allowable stresses for different cases of prestressed concrete

Case	Maximum allowable stresses on section after all based due to prestressing plus all loads	Classification	Comments
a		Full prestressing	$f_{c_t} = 0$ all elements subjected to repeated or dynamic loads and elements of section four according to table (4-11)
b		Uncracked section	$f_{ct} \leq 0.44 \sqrt{f_{cu}} \leq 4.0$ Mpa solid and flat slabs. Elements of section three table (4-11) Prestressed elements with unbounded reinforcement
c		Transition	$0.44 \sqrt{f_{cu}} < f_{ct} \leq 0.60\sqrt{f_{cu}} < 4.0$ MPa section (4-3-1-3), Equation 4-64 $\rightarrow f_{ct}$ = $0.6\sqrt{f_{cu}}$ for $f_{cu} = 40$ MPA $\rightarrow f_{ct}$ = 3.79 Mpa
d		Partial Prestressing	$0.6 \sqrt{f_{ct}} < f_{ct} \leq 0.85\sqrt{f_{cu}}$

APPENDIX III

NOTATION

NOTATION

a

a	rectangular stress Depth of the equivalent block
a	Short side of rectangular bearing surface
a	concentrated load Distance between the and face of the support
a	Shorter effective span of slab
a	Nominal max. Diameter of bars ϕ_{max} or one and .half max. nominal size of aggregates or (max size of aggregates + 15 mm) Which is bigger
a	height of Fixing plate in the considered direction
a	Actual horizontal distance between axes of Fixing plats
A	Long length of footing
A	Area of that part of cross section between flexural tension face and center of gravity of gross section
a′	Effective depth of cross-section corresponding to bending moment M_x
A'_s	Area of Secondary reinforcement
a,b,c	Distance as shown in fig.(10-8)
a′	Actual vertical distance between axes of Fixing plates.
a_1	-Suspended short span of slab
A_1	-Loaded bearing area
A_2	the Maximum area of the portion of supporting surface that is geometrically similar to and concentric with the loaded area
A_c	concrete -Cross- sectional area of
Ac	-connection surface area
A_{cef}	Effective concrete area in tension
A_{cp}	Area enclosed by outside perimeter of concrete cross section including area of openings
A_{creq}	Area of concrete section required by calculation
A_f	reinforcement Area of main flexural steel in corbels
A_g	Gross area of concrete section
A_h	Horizontal reinforcement in corbels and deep beams

A_j	- Area of effective section in splice sector in a plane parallel to the Steel plane perpendicular to Shear plane
A_k	Area of concrete core enclosed by the spiral stirrups
a_{max}	Maximum depth of the equivalent rectangular stress block
A_n	Area of tensile force reinforcement required
A_o	area enclosed by shear flow path
a_o	Min. allowable distance between Fixing plates axes.
A_{oh}	Area enclosed by centerline of the outermost closed transverse torsion reinforcement
A_{ps}	Area of prestressing reinforcement in tension zone
A_s	Area of non-prestressed tension reinforcement
A_s	Area of non – prestressed bonded Steel in members where unbonded prestressed Steel is used.
A_{s1}	Required area of longitudinal reinforcement for torsion resistance
A_{sb}	Area of bent bars
A_{sb}	Cross section area of Stirrup or bent bars
A_{sc}	-Area of longitudinal steel bars in section subject to compressive forces
A_{sf}	- Area of reinforcing Steel perpendicular to Shear plane.
A_{sf}	-Area of shear- friction reinforcement
A_{sl}	-Additional, longitudinal reinforcement area
A_{slmin}	-Minimum amount of additional longitudinal reinforcement area
A_{sm}	- Area of reinforcement in the zone with concentrated reinforcement of footing
A_{smax}	- Max. allowable reinforcing area in reinforced Sections in tension Side only.
A_{smin}	- Man. allowable reinforcement ratio in sections Subjected to bending moments.
A_{sp}	- Cross section area of Spiral reinforcing

	Stirrup.
$A_{sprovied}$	- Area of actual existing reinforcing Steel.
$A_{srequired}$	- Calculated required area of reinforcing Steel.
A_{ss}	- Area of Steel profile cross Section.
A_{st}	-Area of stirrups resisting shearing forces
A_{st}	- Cross section area of Stirrups branches
A_{st}	- Total cross section area of Stirrups including perpendicular branches Within the distance S and perpendicular to dimension y_1.
A_{stmin}	- Min. area of web reinforcement in beams.
A_{str}	-Area of one leg of stirrups resisting torsion moments
A_t	- Area of Steel section.
A_v	-Vertical web reinforcement in deep beams
B	Nominal section dimension
b	-Width of a rectangular section , web, ribs, or box section or width of web section in form of T or L.
B	-Width of compression flange of T-section
b	-long side of rectangular bearing surface
b	- Sum of breadths of webs in box Section.
B	- Breadth of flange
B	- Breadth of web
b	- Max. Slab dimension.
b	- Effective long span
b	- Breadth of webs.
b	- Min dimension of torsion element.
b	- Dimension of rectangle column.
B	- Breadth of horizontal wall support
B	Short dimension of footing or length of column Section
b	- Width of strip

b	- Max. Diameter of bars ϕ max or one and half max. Nominal Size of gravel, which is bigger
B	-Bottom reinforcement
b	- Breadth of Section in case of rectangular Section.
b	- Actual Horizontal distance between axis of Fixing plate and concrete edge.
b′	- Actual vertical distance between axis of Fixing plate and concrete edge.
b_1	-Suspended long span
b_1	-Length of punching shear critical section measured in the loaded span direction
b_1	- One of the dimensions of the rectangle Steel column.
b_2	-Length of punching shear critical section measured perpendicular to the loaded span direction
b_2	- One of the dimensions of the rectangle Steel column.
b_c	-Width of compression face of beam
b_c	- Dimension of the column measured perpendicular to the beam.
b_e	- Effective width of flat slab transferring negative bending moments
b_e	- Breadth of Strip transferring bending moment.
b_o	- Perimeter of critical Section.
b_o	-Perimeter of critical section for punching shear

b_o	- Min. allowable distance between axis of Fixing plate and concrete edge.
f_{bu}	- limit bond Stress between concrete and reinforcing Steel.
b_v	- Breadth of connection between precast part and the part casted in Site.
b_w	- Breadth of effective Section of connection
b_w	- Breadth of web.
c	-Distance from extreme compression fiber to neutral axis
C	- torsion constant
c	-Thickness of solid floor cover
c	- Concrete cover for bars.
c	- Distance between fixing plate edge and concrete edge
c_1	-Dimension of rectangular, or equivalent rectangular column measured in the direction of the span of flat slabs for which moments are being determined
c_1	-Dimension of column Section in analysis direction.
c_2	-Dimension of column Section in direction perpendicular to analysis direction.
c_2	- Connection depth
C_{AB}	- Dimension defined in Figure (6-15)
$c_{balanced}$	- Depth of the part applied to compressive stresses equivalent to balanced reinforcement of the Section.
C_{CB}	- Dimension defined in Figure (6-15)
c_{max}	-Maximum allowable distance from extreme compression surface to neutral axis in singly reinforced sections subject to flexure
D	-Dead loads
d	-Effective depth of cross-section
D	-Diameter of the largest circle that can be drawn inside column cross section
d	- Effective depth of slab
D	-Diameter of circular column
D	-Diameter of Steel circular column
d	- Effective depth of wall cross- section

d	- Thickness of footing
d	- Depth of beam
d	- Total depth of composite element
d	-Distance from extreme compression fiber to centroid of compression reinforcement
D	- Max dimension of fixing plate
D_k	-Diameter of the concrete core enclosed by the centerline of spiral stirrups
d_p	-Distance from extreme compression fiber to centroid of prestressed reinforcement
d_p	- Effective depth of prestressing Steel
E	- Nominal value of loads due to lateral pressures or internal Forces due to them.
e	-Eccentricity of compression force
e	-Eccentricity of axial load
e	-Clear distance between webs
E	-Modulus of elasticity
e	-Eccentricity of prestressed force
E	- Min. dimension of Fixing plate
E_c	-Modulus of elasticity of concrete
$E_c.I$	-Flexural stiffness
E_{ci}	-Modulus of elasticity of concrete at time of initial prestress
E_{ct}	- Modulus of elasticity of concrete at beginning of loading
e_{min}	- Min limit for the value of eccentricity of axial load
e_{min}	-Minimum eccentricity
E_p	-Modulus of elasticity of prestressed reinforcement
E_s	-Modulus of elasticity of steel reinforcement
E_{soil}	-Modulus of elasticity of soil
f_c	- Compression stress
f_c	-Allowable working stress in compression of concrete sections subject to bending moments, or eccentric compressive forces with large eccentricity
f_{cd}	-Stresses due to permanent loads without using load factors at the section edge where tensile stresses exist under the action of external loads

f_{ci}	-Strength of concrete at time of initial prestress
f_{co}	-Allowable axial compressive working stresses for e < 0.05t.
f_{co}	-Allowable working stresses in axial compression of little eccentricity
f_{cs*}	-Stress in concrete at the level of prestressing steel at time of transfer
f_{csd*}	-Stress in concrete at level of prestressing steel due to permanent loads at time of transfer
$f_{ct(M)}$	-Tensile stresses due to bending moment
$f_{ct(N)}$	-Tensile stresses in concrete due to axial ----- tensile forces
f_{ctr}	-Cracking-limit tensile - stresses of concrete
f_{ctr}	-Cracking-limit tensile - stresses of concrete
f_{cu}	-Characteristic strength of concrete
f_{cui}	- Characteristic strength of concrete at time of initial prestress
f_{m}	-Target mean strength
f_{o}	Axial stress
f_{pcc}	-Compressive stress in concrete at section centroid or at the flange bottom face when the section centroid lies inside the flange
f_{pce}	-Compressive stresses in concrete due to effective prestressing force only at the section face where tensile stresses exist under the action of external loads
f_{pci}	- Initial Stresses in concrete which is contact to prestrssing Steel before occurs of losses depending on time
f_{pe}	-Effective stress in prestressed reinforcement (after allowance for all prestress losses)
f_{pi}	- Initial Stresses in prestrssing Steel after occurs of immediate losses in prestressing immediately and before occurs of losses depending on time.
f_{ppc}	- Average compressive strength in concrete on circumference of critical Section (after occurs of all prestressing losses) at middle of Slab Section.
f_{ps}	-Stress in prestressing reinforcement

f_{pu}	-Specified tensile strength of prestressing reinforcement
f_{py}	-Specified yield strength of prestressing reinforcement
f_s	-Stress in prestressing in the Tension Side of the Section after cracking and which is calculated according to a cracked Section under effect of working loads.
f_s	- Allowable working Stresses in Steel.
f_s	- Allowable working Stresses in Steel used in Stirrups
f_s	- Allowable working stress of steel reinforcement
f_{sr}	-The stress in the tension reinforcement calculated on the basis of a cracked section under the loading conditions causing first cracking
f_t	- Allowable direct Tension Stress in concrete
f_y	- yield strength or proof strength of reinforcement.
f_y	- Yield Strength of reinforcing Steel
f_{yp}	-Yield strength of spiral stirrups.
f_{ysc}	- Yield Strength of reinforcing Steel
f_{yss}	- Yield Strength of Steel Section
f_{yst}	-Yield strength of stirrups
f_{yst}	- Yield Strength of Steel Stirrups carrying torsion moment not exceeding 400 N/mm^2
g	-Effective depth or span of a deep beam, whichever is smaller
G	- Shear modules of rigidity
g	-Uniformly distributed working dead loads
g	-Uniformly distributed working dead load in unit area
g	- Uniformly distributed dead load in unit length
G C	- Torsion rigidity of rectangular Section
g_u	-Ultimate uniformly distributed dead loads
h	-Height of column
H	-Clear height of wall
H	- Clear wall height between Supports
h	- Total thickness of section in the considered

	direction
H_b	- Total height of the building over Foundation Surface
He	-Buckling length or effective height of column in the considered direction
He	effective height of wall
h_L	-Height of lower column
H_o	-Clear height of column
h_u	-Height of upper column
h_w	height of wall
I	Moment of inertia (Rigidity)
i	-Radius of gyration of column cross section
I	-Moment of inertia of gross concrete section about centroidal axis, neglecting reinforcement
I_b	-Moment of inertia of beam cross- section
I_B	-Moment of inertia of foundation or foundation frames and shear walls per unit strip width
I_{cr}	-Moment of inertia of cracked concrete section
I_e	-Effective moment of inertia
I_{ec}	-Equivalent moment of inertia of column
I_g	-Moment of inertia of gross concrete section about centroidal axis, neglecting reinforcement
Ig	-Moment of inertia of gross concrete section about centroidal axis, neglecting reinforcement
I_L	Moment of inertia of lower column cross section
I_{sc}	-Moment of inertia for longitudinal reinforcement
I_{ss}	-Moment of inertia of Steel Section
I_t	-Moment of inertia of Steel Section
Iu	-Moment of inertia of upper column cross section
J_{cy}وJ_{cx}	-Property of the assumed punching shear critical section analogous to polar moment of inertia
K	-Coefficient for bending moment calculation
K	-Dynamic loads
k	- Value to be From Fig. (6-2)
K	-Effective length factor for wall
K	- Modulus of sub grade reaction (Winkler coefficient)
K	-Wobble friction coefficient of prestressing

	tendons per meter length
K_1	-A coefficient which takes into account bond properties of the reinforcing bars
k_1	-Coefficient depending on kind of prestressing Steel
K_2	-A coefficient which takes into account the strain distribution
K_b	-Stiffness of beam
K_{ec}	- Bending Stiffness of equivalent column
K_j	-Coefficient of grade of confinement depending on Surrounding beams
K_L	-Stiffness of lower column
K_m	-Bending Coefficients in continuous beams
K_q	-Coefficient for shear force for beams
K_r	-Relative rigidity coefficient
K_u	-Stiffness of upper column
L	-Live loads
L	-Distance between point of Inflection for beams and slabs or Length of cantilever
L	-effective Span for one way slabs
L	-Longest Span dimension of slab
L	-effective length in simple Span slabs or distance between point of Inflection in continuous slabs
L	-Length of beam Span between axes of supports
L	-Distance between joints
L	- Longitudinal anchorage
L	-Length of mechanical joint or Length of welded joint
L	-Distance between axes of supports
L	- span Length in the direction parallel to the required calculated Steel reinforcement
x_1	-Breadth of rectangle reinforcing Stirrup measured between the two axes of the Stirrup
L_1	- span used for calculating moment M_1
L_1	-Length of span in the direction where moments are being determined, measured center to center of supports
L_{1a}	-Average length of the two spans adjacent to the column in the analysis plane

L_2	- span used for calculating moment M_2
L_2	-Breadth of the span in the direction perpendicular to the considered Span direction measured between axes of columns
L_2	-length of Span in direction perpendicular to direction of analysis
L_2	-Distance between two points of Inflection
L_{2a}	-Average length of the two spans adjacent to the column perpendicular to the analysis plane
L_a	-Additional length of reinforcing bars at supports or at points of inflection
L_a	-Length after critical Section
L_b	-Span of beam
L_c	-As defined in Fig. (6-41)
L_d	-Development length
L_{f1}	-Horizontal distance between lateral support and free end
L_{f2}	-Average horizontal distance between lateral supports
L_n	-Clear span of beam
L_n	-Clear span
L_n	-Clear span between faces of supports
L_o	-Distance with heavy stirrups in columns of seismic- resistant frames
L_o	-Distance with heavy stirrups in columns of seismic- resistant frames
L_o	- Distance having more Stirrups
L_t	-Transfer length in prestressed concrete
L_w	-Wall Length
L_x	-Shorter Length of Span measured From Columns axes
L_y	-Longer Length of Span measured From Columns axes
ΣM_c	-Sum of ultimate bending moments of columns in plane of analysis in the area of beam – column connection
ΣM_g	-Sum of ultimate bending moments of beams in the plane of analysis in the area of beam– column connection
M	-Safety margin of concrete mix design
M_{cr}	-Minimum cracking moment of concrete

M'_u	- negative bending moment resistance for section
M'_x	-Effective uniaxial design moment about the x-x axis for the case of biaxial bending
M'_y	-Effective uniaxial design moment about the y-y axis for the case of biaxial bending
M_1	-Negative moment calculated for one of the two Slabs
M_1	-Min. edge bending moment in the column.
M_1	-Difference between bending moments at axis of Support and at Face of Support for Slabs Supported on Walls or beams poured
M_2	-Negative moment of adjacent Slab
M_2	-Max. edge bending moment in the column
M_1 , M_2	-Bending moments at column ends resulting from structural analysis
M_a	-Maximum value of bending moment in member at the stage of computing deflection
m_a	-Ratio of length between points of inflection in a strip of the Slab Loaded in span direction a to total span length a
M_a	-Bending moments in Slabs in both directions
M_{add}	-Additional bending moment induced by buckling of column which accounts for the slenderness of column
m_b	-Ratio of effective length between inflection points of loaded span to total span length in direction b
M_b	-Bending moments in Slabs in both directions
M_c	-Bending moments between two adjacent Slabs in case of different values of negative bending moments on two Sides of contact line
M_{cr}	- Min. bending moment causing cracks in concrete
M_f	-Total moments transferred to column
M_f	-Edge bending moment in beam at its framed connection with exterior column assuming the beam to be totally fixed at both ends
M_i	-Primary moment
M_{max}	-Max bending moment in Section due to external Loads

M_{min}	-Min negative bending moments at mid Span of continuous beams loaded with heavy live loads
M_{min}	-Negative middle moments in internal Spans
M_o	-Maximum bending moment in simply supported beam
M_{pr}	-Probable moment when plastic hinge is Formed
M_t	-Torsion moments
M_{tu}	-Ultimate torsion moment
M_{tu}	-Torsion moment on edge beam
M_u	-Max. limit bending moment for Section Strength
M_u	-Resisting strength of Section for positive bending moments
M_u	-Ultimate bending moment
M_u	-value of max moment at the critical Section in shear
M_{umax}	-Maximum admissible value of ultimate bending moments in singly reinforced sections
M_{-ve}	-Negative bending moments
M_x	-Bending moment about the x-x axis
M_y	-Bending moment about the y-y axis
n	- Modular ratio
N	-Summation of vertical loads
n	-Number of stories
n	-Number of column per floor
N_c	-Value of tension Forces resulting From Working loads (Dead and live)
N_u	-Ultimate tensile force
p	-Pitch of spiral stirrups
P	-Centric working load
P	-Distributed live load
P	Concentrated load
P	-Uniformly distributed live load per unit area
p	- Uniformly distributed live load per unit length
P	-anchorage on perimeter
P_{a1}, P_{b1}	-Loads in directions a and b, respectively
p_b	-Load of balancing compression of Section
P_c	-Perimeter of concrete section exposed to drying
p_{cp}	-Outside perimeter of concrete cross section
PH	

P_h	-Perimeter of the center line of outermost closed transverse torsion reinforcement
P_o	-Prestressing tendon force at jacking end
p_{su}	
γ_{ps}	-Coefficient of reducing max Strength of prestressing Steel
P_u	-Max Strength of column Section in compression
P_u	-Max centric force applied on the Section
P_u	-Uniformly distributed live load
P_x	-Prestressing tendon force at a distance x from jacking end.
q	- Nominal shear stress in beams
Q	-Shear force
Q	-Design Shear force transfer to column when the adjacent spans are loaded with the total design Load
$q^*_{u\ max}$	-Max allowable shear stresses in pestressed concrete sections
q_c	- Allowable working concrete shear strength
q_{cp}	-Allowable Working Stresses in concrete for punching shear
q_{cu}	-Concrete ultimate shear strength
q_{cu}	-Max concrete strength in shear
q_{cup}	-Concrete ultimate punching shear strength
q_d	-Shear Strength resulting from working permanent loads without using coefficient of increasing Loads
Q_e	
q_i	Stresses resulting from max shearing forces due to external loads accompanying max bending moment - M_{max}
q_p	-Punching Shear Stress
q_{pv}	-Shear stress due to vertical components of prestressing forces after all losses of prestressing
q_{st}	-Nominal shear stress provided by stirrups
q_{su}	-Nominal shear strength provided by shear reinforcement
q_{sub}	-Nominal shear strength provided by bent bars
q_{suh}	-Sharing of vertical web reinforcement in max.

	stresses of shear strength in deep beams
q_{sus}	-Nominal shear strength provided by stirrups
q_{suv}	Sharing of horizontal web Steel rein for cement in max stresses of shear strength in deep beams
q_{tu}	-value of Shearing Stresses resulting from torsion moment on which the effect of max torsion moment which causes leases stresses may be neglected
q_{tu}	-Nominal shear stress due to torsion
q_u	-Nominal ultimate shear stress
Q_u	-Max shear stresses resulting from dead and live loads
q_{uc}	-Shear Strength of concrete section
Q_{uhr}	-Max horizontal design shear strength
q_{umax}	-Max allowable shear stress in reinforced concrete sections
q_{umax} *	-Nominal shear stresses for prestressed concrete elements subjected to shear forces
q_{up}	
Q_{up}	-maximum Punching shear force
Q_{ur}	-Maximum shear force for beams with variable depth
q_x	-Punching shear strength resulting from moment M_x and considering γ_{qx} coefficient of moments transferred by torsion
q_y	-Punching shear strength resulting from moment My and considering γ_{qy} coefficient of moments transferred by torsion
r	-Aspect ratio for rectangular (rectangulartity coefficient)
R_b	-Non- dimensional value used to calculate β as in figure (6-26-b)
R_{max}	-Ultimate flexural strength coefficient for singly reinforced sections in tension
r_{ps}	-Radius of curvature of ducts containing prestressing reinforcement
S	- Standard deviation
S	Max value for force resulting from earthquakes or internal forcing resulting from them
s	-Spacing between stirrups in axis direction
S	-Spacing between stirrups

S_1	-Initial loaded width for the uniform load equivalent to a concentrated load in the direction perpendicular to the main reinforcement
S_2	-Initial width uniformly loaded for an equivalent concentrated load in the direction parallel to the main reinforcement
S_h	-Spacing between horizontal web reinforcement in deep beams
s_o	-Spacing between stirrups in distance Lo
s_o	-Spacing between stirrups
s_o	-Maximum stirrups spacing in seismic resisting columns
s_{rm}	-Coefficient used in calculating w_k depending on strain in steel reinforcement and other factors
s_v	-Spacing between web vertical reinforcement in deep beams
T	- Nominal value of loading due to effect of temperature, creep, shrinkage differential settlement and internal forces produced from them
t	-Total depth section in eccentricity direction
t	-Overall thickness of concrete member
t	thickness of slab
t	Longer dimension of torsion element
t	-Longer dimension of rectangular cross-section
t	-Thickness of steel section covering the concrete section
t	thickness of wall
T	-Transverse anchorage
T	Upper reinforcement
t	-Time in hours starting at tensioning of prestressing reinforcement
t_v	-Virtual thickness of cross- section
t'	-Side length in buckling plane
T.D.S	-Total dissolved salts
t_1	-Loaded width in direction perpendicular to main reinforcement
t_2	-Loaded width in direction parallel to main reinforcement
t_e	-Thickness of the wall of the box section

	equivalent to rectangular section
t_f	-Thickness of the flange for T and L Sections
t_{min}	-Minimum thickness of slab
t_s	Minimum thickness of compression slab
t_s	thickness of slab
U	-Ultimate load in case of limit state or internal forces produced from them
u_v	-Percentage of steel for vertical anchorages
V	-Coefficient of variation
V	-vertical anchorage
V_{sp}	-volume of spiral steel reinforcement for unit length of column
V_{sp}	-Ratio of volume of spiral stirrups to stirrup pitch
W	-Nominal value of loads due to wind pressure or internal forces produced from it
w	-Uniformly distributed load acting on slabs
w	-Total load for unit area of span
w	-Uniformly distributed slab load on unit area
w	-Mechanical percentage of tension steel reinforcement in concrete section
w`	- Mechanical percentage of compressive steel reinforcement in concrete section
w_k	- Coefficient of assurance of achieving cracking limit
w_{kmax}	-Max allowable value for coefficient W_k
w_p	-Mechanical percentage of prestressed steel
w_u	-Ultimate uniformly distributed load acting on slabs
w_u	-Ultimate loads
x	-Distance from jacking end along prestressing tendons
x_o	-Distance of extension of losses effect on prestressing
y_1	-Distance of column core measured from Stirups axes
y_1	-Length of rectangle reinforcing stirrup measured between the axes of the stirrup
y_{ct}	- Lever arm
f_{yst}	-Yield strength for steel of stirrups resisting torsion moment with maximum limit of 400

	N/mm^2
y_t	- Distance between extreme fiber in tension to neutral axis of gross section ignoring the presence of normal and prestressing reinforcement

APPENDIX IV

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